PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

(p) Do a visual examination of the mating face of the Boss for Engine Anti-Ice (EAI), Fuel Tube, Firewall and Kit Engine Supports (Figure 608 (Sheet 1)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting

a) Permitted up to 0.0007 in. (0.02 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

3) Tears/cracks.

a) Not permitted.

(q) Do a visual examination of the mating face of the Surface Air Cooled Oil Cooler (SACOC) Bosses (Figure 608 (Sheet 2) and Figure 608 (Sheet 3)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting

a) Permitted up to 0.0007 in. (0.02 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

3) Tears/cracks

a) Not permitted.

(r) Do a visual examination of the SACOC sealing surfaces (Figure 608 (Sheet 2) and Figure 608 (Sheet 3)).

1) Fretting

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

2) Dents, nicks, scratches

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

3) Tears/cracks.

a) Not permitted.

(s) Do a visual examination of Dia CV (8 locations) (Figure 608 (Sheet 3)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting

a) Permitted if Dia CV is between 0.74725 in. (18.980 mm) and 0.74881 in. (19.020 mm).

3) Tears/cracks

a) Not permitted.

(t) Do a visual examination of the mating faces of the Assembly Bosses (Figure 608 (Sheet 4)).

1) Dents, nicks, scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth.

2) Fretting/wear on mating surfaces

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

a) Permitted up to 0.003 in. (0.1 mm) in depth.

3) Tears/cracks

a) Not permitted.

(u) Do a visual examination of all other surfaces.

1) Dents, nicks, scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth.

2) Fretting

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(v) Do a visual examination of the rivets, floating nuts, pins and bushings of the Shroud Fan Frame (Figure 608 (Sheet 5)).

1) Bent, torn, worn, rivets, floating nuts, pins and bushings

a) Not permitted.

2) Loose or missing rivets, floating nuts, pins and bushings

a) Not permitted.

(w) Do a visual examination of the inserts of the Shroud Fan Frame (Figure 608 (Sheet 5)).

1) Bent, torn, worn, cut inserts

a) Not permitted.

b) Maximum repairable limit: any amount (TASK 72-23-04-300-801-G00).

2) Loose or missing inserts

a) Not permitted.

b) Maximum repairable limit: any amount (TASK 72-23-04-300-801-G00).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-23-00-211-038-G00

(7) Examine the fan frame shroud [6] (Figure 607).

(a) Do a visual examination of the V-groove (Figure 607 (Sheet 5)).

1) Fretting on the sealing surface

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

2) Dents, nicks, scratches on the sealing surface

a) Not permitted.

3) Dents, nicks, scratches on the other surfaces

a) Not permitted.

4) Fretting on base metal of the varnish surface (Figure 641).

NOTE: Fretting area starting at 6 and 12 o'clock, 0.3937 in. (10 mm) from the edge of the flange.

a) V-Groove bottom (area B) (Figure 607 (Sheet 6)).

<1> Permitted up to 0.0098 in. (0.25 mm) in depth.

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<2> Permitted if dim. X1 is greater than 0.1405 in. (3.57 mm).

b) Forward flange (area A).

<1> Permitted up to 0.0039 in. (0.10 mm) in depth.

c) Afterward flange (area C).

<1> Permitted up to 0.0118 in. (0.3 mm) in depth.

<2> Permitted if dim. X2 is between 0.1882 in. (4.78 mm) and 0.2275 in. (5.7785 mm).

5) Missing varnish PTFE on varnish surface

a) Permitted.

b) Maximum repairable limit: any amount (TASK 72-23-04-300-802-G00).

6) Distortion of the forward flange

a) Not permitted.

7) Tears and cracks

a) Not permitted.

(b) Do a visual examination of all other surfaces.

1) Dents, nicks, scratches

a) Not permitted.

2) Fretting

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(c) Do a visual examination of the nuts, floating nuts, rivets, screws, handling pins, handling pin bushings, strut 5 spacers, platform spacers and identification bracket (Figure 608 (Sheet 6)).

1) Bent, torn, worn, cut part

a) Not permitted.

2) Loose or missing parts

a) Not permitted.

(d) Do a visual examination of the inserts of the Shroud Fan Frame (Figure 608 (Sheet 5)).

1) Bent, torn, worn, cut inserts

a) Not permitted.

b) Maximum repairable limit: any amount (TASK 72-23-04-300-801-G00).

2) Loose or missing inserts

a) Not permitted.

b) Maximum repairable limit: any amount (TASK 72-23-04-300-801-G00).

(e) Examine the Engine Transfer Gearbox (ETGB) mounts (Figure 608 (Sheet 7)).

1) Do a visual examination of all areas.

a) Tears/cracks

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

<1> Not permitted.

b) Nicks, dents, scratches

<1> Not permitted.

c) Fretting on mating faces

<1> Not permitted.

2) Do a visual examination of inserts, nuts and screws of ETGB mounts.

a) Bent, torn, worn, cut parts\

<1> Not permitted.

b) Loose or missing parts

<1> Not permitted.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-23-00-211-011-G00

(8) Examine the spigot [7] (Figure 609).

(a) Do a visual examination of Dia CB1 and Dia CB2.

1) Dents, nicks, scratches

a) Not permitted.

2) Fretting

a) Permitted if Dia CB1 and Dia CB2 are between 2.04606 in. (51.970 mm) and 2.04684 in. (51.990 mm).

3) Tears/cracks

a) Not permitted.

(b) Do a visual examination of all other surfaces of the spigot [7].

1) Tears/cracks

a) Not permitted.

2) Dents, nicks, scratches

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

3) Fretting

a) Permitted up to 0.0007 in. (0.02 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-23-00-211-039-G00

(9) Examine the spigot [7] (Figure 609 (Sheet 2)).

(a) Do a visual examination of Dia E.

1) Tears/cracks

a) Not permitted.

2) Nicks, dents, scratches

a) Not permitted.

3) Fretting on diameter E

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

a) Permitted if Dia E is between 2.04606 in. (51.970 mm) and 2.04684 in. (51.990 mm).

(b) Do a visual examination of all other surfaces.

1) Tears/cracks

a) Not permitted.

2) Dents, nicks, scratches

a) Not permitted.

3) Fretting

a) Not permitted.

(c) Do a visual examination of inserts, nuts and rivets.

1) Bent, torn, worn, cut parts

a) Not permitted.

2) Loose or missing parts

a) Not permitted.

SUBTASK 72-23-00-211-040-G00

(10) Examine the Forward Engine Mount (Figure 633).

(a) Do a visual examination of Dia A and Dia B.

1) Tears/cracks

a) Not permitted.

2) Nicks, dents, scratches

a) Not permitted.

3) Fretting on diameter A

a) Permitted if Dia A is between 0.87607 in. (22.252 mm) and 0.87684 in. (22.272 mm).

4) Fretting on diameter B

a) Permitted if Dia B is between 1.18736 in. (30.159 mm) and 1.18814 in. (30.179 mm).

(b) Do a visual examination of surface C and surface D.

1) Tears/cracks

a) Not permitted.

2) Nicks, dents, scratches

a) Not permitted.

3) Fretting on surface C

a) Permitted if Dim C is between 0.4686 in. (11.90 mm) and 0.4763 in. (12.10 mm).

4) Fretting on surface D

a) Permitted if Dim D is between 0.08808 in. (2.237 mm) and 0.09303 in. (2.363 mm).

(c) Do a visual examination of all other surfaces.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

1) Tears/cracks
   a) Not permitted.
2) Dents, nicks, scratches
   a) Not permitted.
3) Fretting
   a) Not permitted.

(d) Do visual examination of bushings, nuts and rivets.
1) Bent, torn, worn, cut parts
   a) Not permitted.
2) Loose or missing parts
   a) Not permitted.

SUBTASK 72-23-00-211-043-G00

(11) Examine the Hold Open Rod Right Bracket and the Hold Open Rod Left Bracket (Figure 634).
(a) Do a visual examination of all surfaces.
1) Nicks and scratches
   a) Not permitted on area A.
   b) Permitted up to 0.003 in. (0.07 mm) in depth on all other faces.
2) Dents and pits
   a) Not permitted on area A.
   b) Permitted up to 0.024 in. (0.6 mm) in depth on all other faces.
3) Fretting
   a) Not permitted.
4) Tears/cracks
   a) Not permitted.

(b) Do a visual examination of inserts, rivets and floating nuts.
1) Bent, torn, worn cut inserts, rivets, floating nuts
   a) Not permitted.
2) Loose or missing inserts, rivets, floating nuts
   a) Not permitted.

SUBTASK 72-23-00-211-044-G00

(12) Examine the Upper Oil Tank Bracket and the Rear Oil Tank Bracket (Figure 635).
(a) Do a visual examination of all surfaces.
1) Nicks and scratches
   a) Not permitted on area A.
   b) Permitted up to 0.002 in. (0.05 mm) in depth on all other faces.
2) Dents and pits
   a) Not permitted on area A.
   b) Permitted up to 0.020 in. (0.5 mm) in depth on all other faces.

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3) Fretting
   a) Not permitted.
4) Tears/cracks
   a) Not permitted.

(b) Do a visual examination of the inserts.
1) Bent, torn, worn cut inserts
   a) Not permitted.
2) Loose or missing inserts
   a) Not permitted.

SUBTASK 72-23-00-211-045-G00

(13) Examine the Fuel Supply Bracket (Figure 636).
(a) Do a visual examination of all surfaces.
1) Nicks and scratches
   a) Permitted up to 0.003 in. (0.08 mm) in depth.
2) Dents and pits
   a) Permitted up to 0.024 in. (0.6 mm) in depth.
3) Fretting
   a) Not permitted.
4) Tears/cracks
   a) Not permitted.

SUBTASK 72-23-00-211-046-G00

(14) Examine the Fire Sealing Ring Spring (Figure 637).
(a) Do a visual examination of all surfaces.
1) Nicks and scratches
   a) Not permitted.
2) Dents and pits
   a) Not permitted.
3) Fretting
   a) Not permitted.
4) Tears/cracks
   a) Not permitted.

SUBTASK 72-23-00-211-047-G00

(15) Examine the Front Lug Hold and the Lateral Lug Hold (Figure 638).
(a) Do a visual examination of all surfaces.
1) Nicks and scratches
   a) Not permitted.
2) Dents and pits
   a) Not permitted.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

3) Fretting
   a) Not permitted.
4) Tears/cracks
   a) Not permitted.

(b) Do a visual examination of damper.
1) Bent, torn, worn damper
   a) Not permitted.
2) Loose or missing damper
   a) Not permitted.

SUBTASK 72-23-00-211-048-G00

(16) Examine the OTBD MDOS Bracket and the INBD MDOS Bracket (Figure 639).
(a) Do a visual examination of all surfaces.
1) Nicks and scratches
   a) Not permitted.
2) Dents and pits
   a) Not permitted.
3) Fretting
   a) Not permitted.
4) Tears/cracks
   a) Not permitted.

SUBTASK 72-23-00-211-049-G00

(17) Examine the Strut 1 Bracket (Figure 640).
(a) Do a visual examination of all surfaces.
1) Nicks and scratches
   a) Not permitted.
2) Dents and pits
   a) Not permitted.
3) Fretting
   a) Not permitted.
4) Tears/cracks
   a) Not permitted.

(b) Do a visual examination of the inserts, rivets, floating nuts, pins and seals.
1) Bent, torn, worn cut inserts, rivets, floating nuts, pins and seals
   a) Not permitted.
2) Loose or missing inserts, rivets, floating nuts, pins and seals
   a) Not permitted.

PTF ALL

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SUBTASK 72-23-00-211-012-G00

(18) Examine the fan frame hub [8] (Figure 610).

(a) Do a visual examination of the sealing surfaces (Figure 610 (Sheet 2))

1) Fretting

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

2) Dents, nicks, scratches

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

3) Tears/cracks

a) Not permitted.

(b) Do a visual examination of the sealing surfaces (Figure 610 (Sheet 3)).

1) Dents, nicks, scratches

a) Not permitted.

2) Fretting

a) Not permitted.

3) Tears/cracks.

a) Not permitted.

(c) Do a visual examination of the conical flange (Figure 610 (Sheet 1)).

1) Dents, nicks, scratches

a) Not permitted.

2) Tears/cracks

a) Not permitted.

3) Fretting on the mating surface A10

a) Permitted up to 0.0007 in. (0.02 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

4) Fretting on the mating surface A11

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

5) Fretting on diameter Y

a) Permitted if Dia Y is between 40.1535 in. (1019.90 mm) and 40.1613 in. (1020.10 mm).

(d) Do a visual examination of the thrust link clevis (Figure 610 (Sheet 4), Figure 610 (Sheet 9) and Figure 610 (Sheet 10)).

1) Dents, nicks, scratches

a) 0.011 in. (0.3 mm) in depth.

2) Tears/cracks

a) Not permitted.

3) Fretting/Wear

a) Permitted on external ball's surface of the spherical bearings up to 0.0007 in. (0.02 mm) in depth.

(e) Do a visual examination of the ribs thrust link clevis (Figure 610 (Sheet 4)).

1) Dents, nicks, scratches

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a) Not permitted.

2) Tears/cracks

a) Not permitted.

(f) Do a visual examination of the diameters U and V (Figure 610 (Sheet 5)).

1) Fretting

a) Permitted if dim U is between 0.31304 in. (7.951 mm) and 0.31389 in. (7.973 mm) and dim V is between 0.3308 in. (8.40 mm) and 0.3393 in. (8.62 mm).

2) Dents, nicks, scratches

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(g) Do a visual examination of the diameters AN, AO1 and AO2 (Figure 610 (Sheet 6) and Figure 610 (Sheet 7)).

1) Dents, nicks, scratches

a) Not permitted.

2) Fretting

a) Permitted as follows:

<1> Dia AN is between 0.55118 in. (14.000 mm) and 0.55306 in. (14.048 mm)

<2> Dia AO1 is between 2.32283 in. (59.000 mm) and 2.32464 in. (59.046 mm)

<3> Dia AO2 is between 2.32283 in. (59.000 mm) and 2.32464 in. (59.046 mm)

3) Tears/cracks

a) Not permitted.

(h) Do a visual examination of all other surfaces of the hub.

1) Tears/cracks

a) Not permitted.

2) Dents, nicks, scratches

a) Not permitted.

(i) Do a visual examination of the spherical bearings (Figure 610 (Sheet 8)).

1) Missing

a) Not permitted.

2) Worn

a) Not permitted.

(j) Do a visual examination of the inserts (Figure 610 (Sheet 8)).

1) Missing

a) Not permitted.

b) Maximum Repairable limit: Any number.

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c) Repair Method: Refer to Fan Frame Hub Inserts Replacement, TASK 72-23-00-900-801-G00.

2) Looseness

a) Not permitted.

b) Maximum Repairable limit: Any amount.

c) Repair Method: Refer to Fan Frame Hub Inserts Replacement, TASK 72-23-00-900-801-G00.

3) Bent, worn, torn threads fully or not

a) Not permitted.

b) Maximum Repairable limit: Any amount.

c) Repair Method: Refer to Fan Frame Hub Inserts Replacement, TASK 72-23-00-900-801-G00.

4) Recessed or protruding

a) Not permitted.

b) Maximum Repairable limit: Any amount.

c) Repair Method: Refer to Fan Frame Hub Inserts Replacement, TASK 72-23-00-900-801-G00.

SUBTASK 72-23-00-211-013-G00

CAUTION

AFTER YOU EXAMINE THE OUTLET GUIDE VANE MADE OF ALUMINUM, REPAIR THE OXIDE LAYER. IF YOU DO NOT OBEY, CORROSION AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

(19) Examine the outlet guide vanes OGV [9] (Figure 611).

(a) Visually examine the inner platform fillet (but not Leading Edge (LE) and Trailing Edge (TE)) and outer mounting flange fillets (Figure 611 (Sheet 1)).

1) Dents, nicks, and scratches

a) Not permitted.

2) Corrosion

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(b) Visually examine the suction side flowpath surface (but not the LE, TE, and fillets) (Figure 611 (Sheet 2)).

1) Dents, nicks, and scratches

a) Not permitted.

2) Corrosion

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(c) Visually examine the pressure side flowpath surface (but not the LE, TE, and fillets) (Figure 611 (Sheet 4) and Figure 611 (Sheet 5)).

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1) Dents, nicks, and scratches
   a) Not permitted.
2) Corrosion
   a) Not permitted.
3) Tears/cracks
   a) Not permitted.
4) Porosities in the glue around the bonded cover (bonded OGV only).
   a) Permitted.

(d) Do a visual examination of OGV aft conical flange (Figure 611 (Sheet 2)).
1) Tears/cracks
   a) Not permitted.
2) Corrosion
   a) Not permitted.
3) Missing surface protection on mating faces (but not the non-coated Titanium OGV)
   a) Any amount is permitted if there is no corrosion and if base metal is not affected.
   b) Maximum repairable limit: In the limits of the applicable repair.
      <1> If it is necessary, do this task: Outlet Guide Vanes Paint Primer Local Repair, TASK 72-21-03-370-802-G00.
4) Fretting/wear on base metal of the mating faces
   a) Permitted up to 0.0019 in. (0.05 mm) in depth.
5) Thread marks in holes
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.
6) Dents, nicks, and scratches
   a) Not permitted.

(e) Visually examine the inner platform flowpath surface (fillets not included) (Figure 611 (Sheet 1)).
1) Dents, nicks, and scratches
   a) Not permitted.
2) Corrosion
   a) Not permitted.
3) Tears/cracks
   a) Not permitted.

(f) Visually examine the inner mounting flange (fillets not included) (Figure 611 (Sheet 1) and Figure 611 (Sheet 6)).
1) Dents, nicks, and scratches
   a) Not permitted.
2) Missing surface protection in and around holes (but not the non-coated Titanium OGV)
   a) Any amount is permitted if there is no corrosion and if base metal is not affected.

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b) Maximum repairable limit: In the limits of the applicable repair.

<1> If it is necessary, do this task: Outlet Guide Vanes Paint Primer Local Repair, TASK 72-21-03-370-802-G00.

3) Corrosion

a) Not permitted.

4) Tears/cracks

a) Not permitted.

5) Fretting/wear on the mating surfaces

a) Permitted up to 0.0011 in. (0.03 mm) in depth.

6) Thread marks inside two forward mounting holes with base metal visible (but not the Titanium OGV)

a) Permitted up to 0.0059 in. (0.15 mm) in depth.

(g) Visually examine the outer mounting flange (fillets not included) (Figure 611 (Sheet 1), Figure 611 (Sheet 2), and Figure 611 (Sheet 6)).

1) Dents, nicks, and scratches

a) Not permitted.

2) Missing surface protection in and around holes (but not the non-coated Titanium OGV)

a) Any amount is permitted if there is no corrosion and if base metal is not affected.

b) Maximum repairable limit: In the limits of the applicable repair.

<1> If it is necessary, do this task: Outlet Guide Vanes Paint Primer Local Repair, TASK 72-21-03-370-802-G00.

3) Corrosion

a) Not permitted.

4) Tears/cracks

a) Not permitted.

5) Fretting/wear on the mating surfaces

a) Permitted up to 0.0011 in. (0.03 mm) in depth.

6) Thread marks on the inner four mounting holes that show base metal (but not the Titanium OGV)

a) Permitted up to 0.0059 in. (0.15 mm) in depth.

(h) Visually examine the LE, the LE fillet, and the TE (but not the bottom of TE and the outer mounting flange fillets) (Figure 611 (Sheet 1), Figure 611 (Sheet 2), and Figure 611 (Sheet 4)).

1) Dents, nicks, and scratches

a) Not permitted.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do these tasks: Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair, TASK 72-21-03-320-801-G00 and Titanium Outlet Guide Vane Leading Edge Blending Repair, TASK 72-21-03-320-803-G00.

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2) Local distortion

a) Permitted up to 0.0196 in. (0.50 mm) deformation and 0.196 in. (5.0 mm) length.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do these tasks: Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair, TASK 72-21-03-320-801-G00 and Titanium Outlet Guide Vane Leading Edge Blending Repair, TASK 72-21-03-320-803-G00.

3) Tears/cracks

a) Not permitted.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do these tasks: Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair, TASK 72-21-03-320-801-G00 and Titanium Outlet Guide Vane Leading Edge Blending Repair, TASK 72-21-03-320-803-G00.

4) Erosion and/or missing surface protection

a) Any amount is permitted if there is no corrosion and if base metal is not affected.

b) Maximum repairable limit: In the limits of the applicable repair.

<1> If it is necessary, do these tasks:

- Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair, TASK 72-21-03-320-801-G00
- Titanium Outlet Guide Vane Leading Edge Blending Repair, TASK 72-21-03-320-803-G00
- Outlet Guide Vanes Coating Local Repair, TASK 72-21-03-370-801-G00
- Outlet Guide Vanes Paint Primer Local Repair, TASK 72-21-03-370-802-G00.

5) Corrosion pitting

a) Not permitted.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do these tasks: Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair, TASK 72-21-03-320-801-G00 and Titanium Outlet Guide Vane Leading Edge Blending Repair, TASK 72-21-03-320-803-G00.

(i) Do a visual examination of the OGV bottom trailing edge (Figure 611 (Sheet 4)).

1) Tears/cracks

a) Not permitted.

2) Corrosion

a) Not permitted.

3) Erosion and/or missing surface protection (but not the non-coated Titanium OGV)

a) Any amount is permitted if there is no corrosion and if base metal is not affected.

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b) Maximum repairable limit: In the limits of the applicable repair.

<1> If it is necessary, do these tasks: Outlet Guide Vanes Coating Local Repair, TASK 72-21-03-370-801-G00 and Outlet Guide Vanes Paint Primer Local Repair, TASK 72-21-03-370-802-G00.

4) Dents, nicks, and scratches

a) Not permitted.

(j) Do a visual examination of the sealing surfaces (Figure 611 (Sheet 4)).

1) Fretting

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

2) Dents, nicks, and scratches

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

3) Tears/cracks

a) Not permitted.

(k) Visually examine the pressure side flowpath, suction side flowpath, inner platform flowpath, and inner platform fillet surfaces and outer mounting flange fillets (but not the LE, LE fillets, the TE, and TE fillets) (Figure 611 (Sheet 1), Figure 611 (Sheet 2) and Figure 611 (Sheet 4)).

1) Erosion and/or missing surface protection

a) Missing paint and anodized coating are permitted up to 53.47 in^2 (345 cm^2) of the surfaces if there is no corrosion and if base metal is not affected.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do these tasks: Outlet Guide Vanes Coating Local Repair, TASK 72-21-03-370-801-G00 and Outlet Guide Vanes Paint Primer Local Repair, TASK 72-21-03-370-802-G00.

2) Corrosion

a) Not permitted.

(l) Visually examine the inner platform forward face of aluminum OGVs (Figure 611 (Sheet 3)).

1) Nicks, dents and scratches

a) Permitted up to 0.0078 in. (0.20 mm) in depth.

b) Number of indications: two.

2) Tears and cracks

a) Not serviceable.

3) Corrosion

a) Not serviceable.

(m) Visually examine the area A on the inner platform aft face of aluminum OGVs, (boundaries not included) (Figure 611 (Sheet 3)).

1) Nicks and dents

a) Permitted up to 0.0078 in. (0.20 mm) in depth.

b) Minimum distance between two indications: 0.12 in. (3 mm).

c) Number of indications: two.

2) Scratches

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a) Not serviceable.

3) Tears and cracks

a) Not serviceable.

4) Corrosion

a) Not serviceable.

(n) Do a visual examination of the sealing cap (Figure 611 (Sheet 1)).

1) Bent, torn, worn, cut sealing cap

a) Not permitted.

2) Loose or missing sealing cap

a) Not permitted.

(o) Do a visual examination of the rivets and floating nuts (Figure 611 (Sheet 4)).

1) Bent, torn, worn, rivets and floating nuts

a) Maximum Serviceable Limit: Not Permitted.

b) Maximum Repairable Limit: Any Amount.

c) Repair method: Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement, TASK 72-21-03-960-801-G00.

2) Loose or missing rivets and floating nuts

a) Maximum Serviceable Limit: Not Permitted.

b) Maximum Repairable Limit: Any Amount.

c) Repair method: Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement, TASK 72-21-03-960-801-G00.

(p) Visually examine all other surfaces.

1) Tears/cracks

a) Not permitted.

2) Dents, nicks, and scratches

a) Not permitted.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0171

SUBTASK 72-23-00-210-002-G00

(20) Examine the outlet guide vanes OGV [9] (Figure 611 (Sheet 1)).

(a) Do a visual examination of the OGV plugs.

1) Missing plugs

a) Permitted up to 14.

PTF ALL

SUBTASK 72-23-00-211-014-G00

(21) Examine the OGV strut 1 [10] (Figure 612).

(a) Visually examine the inner platform fillet, (but not the LE, and outer mounting flange fillet) (Figure 612 (Sheet 1) and Figure 612 (Sheet 2)).

1) Dents, nicks, and scratches

a) Not permitted.

2) Erosion

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a) Not permitted.

3) Tears/cracks

a) Not permitted.

(b) Visually examine the suction side flowpath surface (but not the LE and LE fillets) (Figure 612 (Sheet 2)).

1) Dents, nicks, and scratches

a) Permitted up to 0.0098 in. (0.25 mm) in depth.

2) Erosion

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(c) Visually examine the pressure side flowpath surface (but not the LE and LE fillets) (Figure 612 (Sheet 1), and Figure 612 (Sheet 3)).

1) Dents, nicks in area A

a) Permitted as follows:

<1> Not serviceable if the pressure side flowpath surface was repaired.

<2> Any amount up to 0.0039 in. (0.10 mm) in depth.

<3> Minimum distance between two indications must be two and half times the length of the longest indication.

2) Scratches in area A

a) Not permitted.

3) Dents, nicks, and scratches in other areas

a) Not permitted.

4) Erosion

a) Not permitted.

5) Tears/cracks

a) Not permitted.

(d) Do a visual examination of the LE and LE fillets (Figure 612 (Sheet 1)).

1) Dents, nicks, and scratches

a) Permitted up to 0.0098 in. (0.25 mm) in depth.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

2) Local distortion

a) Permitted up to 0.0196 in. (0.50 mm) deformation and 0.1968 in. (5.00 mm) in length.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

3) Tears/cracks

a) Not permitted.

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b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

4) Erosion

a) Not permitted.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

(e) Visually examine the inner platform flowpath surface (fillets not included) (Figure 612 (Sheet 1) and Figure 612 (Sheet 2)).

1) Dents, nicks, and scratches

a) Permitted up to 0.0098 in. (0.25 mm) in depth.

2) Erosion

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(f) Do a visual examination of the sealing surfaces (Figure 612 (Sheet 2)).

1) Fretting

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

2) Dents, nicks, and scratches

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

3) Tears/cracks

a) Not permitted.

(g) Do a visual examination of all other surfaces.

1) Tears/cracks

a) Not permitted.

2) Dents, nicks, and scratches

a) Not permitted.

(h) Do a visual examination of the sealing cap, inserts, rivets, floating nuts and pins. (Figure 612 (Sheet 1) and Figure 612 (Sheet 2)).

1) Bent, torn, worn, cut sealing cap, inserts, rivets, floating nuts and pins

a) Not permitted.

2) Loose or missing sealing cap, inserts, rivets, floating nuts and pins

a) Not permitted.

SUBTASK 72-23-00-211-015-G00

(22) Examine the OGV strut 5 [11] (Figure 613).

(a) Visually examine the outer platform fillet and outer mounting flanges fillets and inner mounting flanges fillets (LE not included) (Figure 613 (Sheet 2) and Figure 613 (Sheet 3)).

1) Dents, nicks, and scratches on outer platform fillet

a) Not permitted.

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2) Erosion on outer platform fillet

a) Not permitted.

3) Tears/cracks on outer platform fillet

a) Not permitted.

(b) Visually examine the suction side flowpath surface (but not the LE and fillets) (Figure 613 (Sheet 1) and Figure 613 (Sheet 3)).

1) Dents, nicks, and scratches

a) Permitted up to 0.0098 in. (0.25 mm) in depth.

2) Erosion

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(c) Visually examine the pressure side flowpath surface (but not the LE and LE fillets) (Figure 613 (Sheet 3), and Figure 613 (Sheet 4)).

1) Dents, nicks in area B

a) Permitted as follows:

<1> Not serviceable if the pressure side flowpath surface was repaired.

<2> Any amount up to 0.0039 in. (0.10 mm) in depth.

<3> Minimum distance between two indications must be two and half times the length of the longest indication.

2) Scratches in area B

a) Not permitted.

3) Dents, nicks, and scratches in other areas

a) Not permitted.

4) Erosion

a) Not permitted.

5) Tears/cracks

a) Not permitted.

(d) Do a visual examination of the LE and LE fillets (Figure 613 (Sheet 2) and Figure 613 (Sheet 3)).

1) Dents, nicks, and scratches

a) Permitted up to 0.0098 in. (0.25 mm) in depth.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

2) Local distortion

a) Permitted up to 0.0196 in. (0.50 mm) deformation and 0.1968 in. (5.00 mm) in length.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Tears/cracks

a) Not permitted.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

4) Erosion

a) Not permitted.

b) Maximum repairable limit: In the limits of the applicable repair.

If it is necessary, do this task: Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair, TASK 72-21-03-320-802-G00.

(e) Visually examine the outer platform flowpath surface (fillets not included) (Figure 613 (Sheet 2)).

1) Dents, nicks, and scratches

a) Permitted up to 0.0098 in. (0.25 mm) in depth.

2) Erosion

a) Not permitted.

3) Tears/cracks

a) Not permitted.

(f) Do a visual examination of all other surfaces.

1) Tears/cracks

a) Not permitted.

2) Dents, nicks, and scratches

a) Not permitted.

(g) Do a visual examination of the sealing cap, inserts, rivets, floating nuts and pins. (Figure 613 (Sheet 2) and Figure 613 (Sheet 3)).

1) Bent, torn, worn, cut sealing cap, inserts, rivets, floating nuts and pins

a) Not permitted.

2) Loose or missing sealing cap, inserts, rivets, floating nuts and pins

a) Not permitted.

SUBTASK 72-23-00-211-016-G00

(23) Examine the outer OGV platforms [12] (Figure 614).

(a) Do a visual examination of the platforms.

1) Cracks

a) Permitted.

2) Broken or missing outer OGV platforms [12]

a) Permitted for all areas and up to 3 platforms affected and up to 2 consecutive platforms.

3) Erosion

a) Not permitted.

(b) Do a visual examination of the seals.

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1) Non through cracks, crazing on seal surface
   a) Permitted up to 9.44 in. (240 mm) length for all platform seals.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION USE THE ADAPTED TOOL TO CUT THE SEAL. BE CAREFUL NOT TO CAUSE DAMAGES TO THE ADJACENT PARTS. IF YOU DO NOT OBEY, IT CAN CAUSE DAMAGE TO THE PARTS IN WORK AND ADJACENT PARTS.

2) Through cracks, tears, disbond, bend, missing material.
   a) In case of disbonded seal, cut the floating part and do as follows:
      <1> Serviceable if a maximum of three platforms or less are impacted and if the cumulative length is less than 15.74 in. (400 mm) per platform.
      <2> Serviceable if more than three platforms are impacted and if the cumulative length is less than 3.14 in. (80 mm) per platform.

(c) Do a visual examination of the dampers.
   1) Bent, worn, torn, cut dampers
      a) Not permitted.
      b) Maximum Repairable Limit: Any amount.
      c) Repair Method: Outer Outlet Guide Vane Platform - Damper Replacement, TASK 72-21-09-960-802-G00.
   2) Loose or missing dampers
      a) Not permitted.
      b) Maximum Repairable Limit: Any amount.
      c) Repair Method: Outer Outlet Guide Vane Platform - Damper Replacement, TASK 72-21-09-960-802-G00.

(d) Do a visual examination of the screws.
   1) Surface oxidation on screw head
      a) Permitted.

SUBTASK 72-23-00-211-017-G00

(24) Examine the OGV shims [13] and OGV shims [14] (Figure 615).

(a) Do a visual examination of the shims.
   1) Dents, nicks, and scratches
      a) Permitted up to 0.0118 in. (0.30 mm) in depth.
   2) Fretting on mating surfaces
      a) Not permitted.
   3) Corrosion
      a) Not permitted.
   4) Tears/cracks
      a) Not permitted.

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SUBTASK 72-23-00-211-018-G00

(25) Examine the cups [15] (Figure 616).

(a) Do a visual examination of the cups.

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

3) Tears/cracks

a) Not permitted.

4) Fretting/wear on mating surfaces

a) Permitted up to 0.0019 in. (0.05 mm) in depth in area around the hole.

SUBTASK 72-23-00-211-019-G00

(26) Examine the gaskets [16] (Figure 617).

(a) Do a visual examination of the gaskets.

1) Cracks, crazing on gasket

a) Not permitted.

2) Missing parts

a) Permitted up to 0.39 in. (10 mm) in length at the forward end of the gasket and up to 0.59 in. (15 mm) in length at the aft end of the gasket or up to 0.59 in. (15 mm) in cumulative missing length.

SUBTASK 72-23-00-211-020-G00

(27) Examine the VBV supports [17] (Figure 618).

(a) Do a visual examination of the top of the upper lug area.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(b) Do a visual examination of the top of the lower lug area.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(c) Do a visual examination of the rub pad area.

1) Wear on the rub surface

a) Permitted.

2) Nicks and scratches

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a) Permitted.

3) Dents

a) Permitted.

4) Cracks

a) Permitted.

5) Loss of the rub pad.

a) Permitted.

(d) Do a visual examination of area A.

1) Cracks.

a) Not permitted.

2) Dents

a) Not permitted.

3) Nicks and scratches

a) Permitted with no depth.

SUBTASK 72-23-00-211-021-G00

(28) Examine the VBV actuating ring [18] (Figure 619).

(a) Do a visual examination of the body area.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(b) Do a visual examination of the area A.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Not permitted.

3) Dents

a) Not permitted.

4) Contact marks

a) Not permitted.

(c) Do a dimension check of the actuator ring bushings.

1) Actuator ring bushing diameter limits:

a) Minimum serviceable - 0.3119 in. (7.92 mm) thick.

b) Maximum serviceable - 0.3125 in. (7.94 mm) thick.

SUBTASK 72-23-00-211-022-G00

(29) Examine the Variable Bleed Valve (VBV) large door [19] (Figure 620).

(a) Do a visual examination of all areas

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1) Loss of anodized surface.
   a) Permitted if there is no corrosion.

(b) Do a visual examination of the leading edge area.
1) Cracks
   a) Not permitted.
2) Nicks and scratches
   a) Permitted up to 0.00499 in. (0.127 mm) in depth.
3) Dents
   a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(c) Do a visual examination of the rear of the door.
1) Cracks
   a) Not permitted.
2) Nicks and scratches
   a) Permitted up to 0.00499 in. (0.127 mm) in depth.
3) Dents
   a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(d) Do a visual examination of the area A.
1) Cracks
   a) Not permitted.
2) Nicks and scratches
   a) Not permitted.
3) Dents
   a) Not permitted.
4) Contact marks
   a) Not permitted.

(e) Do a visual examination of the contact surfaces.
1) Contact marks
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(f) Do a dimension check of the VBV door bushings.
1) VBV door upper bushing diameter limits:
   a) Minimum serviceable - 0.3130 in. (7.95 mm).
   b) Maximum serviceable - 0.3141 in. (7.98 mm).
2) VBV door lower bushing diameter limits:
   a) Minimum serviceable - 0.2543 in. (6.46 mm).
   b) Maximum serviceable - 0.2555 in. (6.49 mm).

SUBTASK 72-23-00-211-023-G00

(30) Examine the VBV short door [20] (Figure 621).

(a) Do a visual examination of all areas.
1) Loss of anodized surface

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a) Permitted if there is no corrosion.

(b) Do a visual examination of the leading edge area.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(c) Do a visual examination of the corner.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(d) Do a visual examination of the area A.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Not permitted.

3) Dents

a) Not permitted.

4) Contact marks

a) Not permitted.

(e) Do a visual examination of the contact surfaces.

1) Contact marks

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(f) Do a dimension check of the VBV door bushings.

1) VBV door upper bushing diameter limits:

a) Minimum serviceable - 0.3130 in. (7.95 mm).

b) Maximum serviceable - 0.3141 in. (7.98 mm).

2) VBV door lower bushing diameter limits:

a) Minimum serviceable - 0.2543 in. (6.46 mm).

b) Maximum serviceable - 0.2555 in. (6.49 mm).

SUBTASK 72-23-00-211-024-G00

(31) Examine the VBV rear seal holders [21] (Figure 622).

(a) Do a visual examination of the lug area.

1) Nicks, scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

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2) Dents
   a) Permitted up to 0.01999 in. (0.508 mm) in depth.

(b) Do a visual examination of the leading edge area.
1) Nicks, scratches
   a) Permitted up to 0.00499 in. (0.127 mm) in depth.
2) Dents
   a) Permitted up to 0.01999 in. (0.508 mm) in depth.

(c) Do a visual examination of the seal area.
1) Wear and/or frayed material of the seal
   a) Permitted if raw rubber is not visible.
2) Scratch of the fabric
   a) Permitted up to 0.999 in. (25.4 mm) in length if raw rubber is not visible.
3) Missing rubber seal material
   a) Not permitted.
4) Tear on the rubber seal
   a) Not permitted.

(d) Do a visual examination of area A.
1) Cracks
   a) Not permitted.
2) Nicks and scratches
   a) Not permitted.
3) Dents
   a) Not permitted.
4) Contact marks
   a) Permitted with no depth.

(e) Do a visual examination of the bushings.
1) Contact marks
   a) Permitted if less than 0.0007 in. (0.02 mm) in depth.

(f) Do a dimension check of the bushings.
1) Bushings inner diameter limits:
   a) Minimum serviceable - 0.37500 in. (9.525 mm).
   b) Maximum serviceable - 0.3795 in. (9.64 mm).

SUBTASK 72-23-00-211-025-G00

(32) Examine the VBV lower front seal holders [22] and VBV upper front seal holders [23] (Figure 623).

(a) Do a visual examination of the lug area of the VBV lower front seal holders [22].
1) Cracks
   a) Not permitted.
2) Nicks, scratches

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a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

4) Contact marks

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(b) Do a visual examination of the lug area of the VBV lower front seal holders [22].

1) Cracks

a) Not permitted.

2) Nicks, scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(c) Do a visual examination of the body area of the VBV upper front seal holders [23].

1) Cracks

a) Not permitted.

2) Nicks, scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(d) Do a visual examination of the area A of the VBV lower front seal holders [22] and VBV upper front seal holders [23].

1) Cracks

a) Not permitted.

2) Nicks, scratches

a) Not permitted.

3) Dents

a) Not permitted.

4) Contact marks

a) Not permitted.

SUBTASK 72-23-00-211-026-G00

(33) Examine the VBV bellcranks [24] (Figure 624).

(a) Do a visual examination of all areas.

1) Loss of anodized surface

a) Permitted if no corrosion.

(b) Do a visual examination of the body area.

1) Cracks

a) Not permitted.

2) Nicks, scratches

a) Not permitted.

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3) Dents
   a) Not permitted.
4) Contact marks
   a) Not permitted.
5) Loose bearing
   a) Permitted up to 0.003 in. (0.1 mm) of radial movement.

(c) Do a visual examination of the extremities area.
1) Cracks
   a) Not permitted.
2) Nicks, scratches
   a) Permitted up to 0.00499 in. (0.127 mm) in depth.
3) Dents
   a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(d) Do a visual examination of the top of the bushings.
1) Contact marks
   a) Permitted up to 0.007 in. (0.2 mm) in depth.
2) Wear
   a) Permitted up to 0.007 in. (0.2 mm) in depth.

(e) Do a visual examination of Area 5.
1) Nicks and scratches
   a) Permitted up to 0.007 in. (0.2 mm) in depth.
2) Dents
   a) Permitted up to 0.007 in. (0.2 mm) in depth.
3) Cracks
   a) Not permitted.

(f) Do a visual examination of the top of the safety wire hole.
1) Nicks and scratches
   a) Permitted up to 0.007 in. (0.2 mm) in depth.
2) Dents
   a) Permitted up to 0.007 in. (0.2 mm) in depth.
3) Cracks
   a) Not permitted.

(g) Do a dimension check of the bellcrank bushings.
1) Bellcrank bushings diameter limits:
   a) Minimum serviceable - 0.3154 in. (8.01 mm).
   b) Maximum serviceable - 0.3161 in. (8.03 mm).
2) Bellcrank auto lub bushings diameter limits:
   a) Minimum serviceable - 0.5512 in. (14.00 mm).
   b) Maximum serviceable - 0.5559 in. (14.12 mm).

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(h) Do a visual examination of Area 6.
1) Cracks
a) Not permitted.
2) Nicks and scratches
a) Not permitted.
3) Contact marks
a) Permitted up to 0.00499 in. (0.127 mm) in depth.
4) Dents.
a) Permitted up to 0.00499 in. (0.127 mm) in depth.

SUBTASK 72-23-00-211-027-G00

(34) Examine the VBV actuating bellcranks [25] (Figure 625).
(a) Do a visual examination of all areas.
1) Contact marks
a) Permitted if less than 0.0007 in. (0.02 mm) in depth.
(b) Do a visual examination of the body.
1) Cracks
a) Not permitted.
2) Nicks and scratches
a) Not permitted.
3) Dents
a) Not permitted.
4) Contact marks
a) Not permitted.
5) Loose bearing
a) Permitted up to 0.003 in. (0.1 mm) of radial movement.
(c) Do a visual examination of the extremities area.
1) Cracks
a) Not permitted.
2) Nicks, scratches
a) Permitted up to 0.00499 in. (0.127 mm) in depth.
3) Dents
a) Permitted up to 0.00499 in. (0.127 mm) in depth.
(d) Do a visual examination of the top of the bushings.
1) Contact marks
a) Permitted up to 0.007 in. (0.2 mm) in depth.
2) Wear
a) Permitted up to 0.007 in. (0.2 mm) in depth.
(e) Do a dimension check of the bellcrank bushings.
1) Bellcrank bushings diameter limits:

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a) Minimum serviceable - 0.3154 in. (8.01 mm).

b) Maximum serviceable - 0.3161 in. (8.03 mm).

2) Bellcrank auto lub bushings diameter limits:

a) Minimum serviceable limit - 0.5512 in. (14.00 mm).

b) Maximum serviceable limit - 0.5559 in. (14.12 mm).

SUBTASK 72-23-00-211-028-G00

(35) Examine the VBV front gaskets [26] (Figure 626).

(a) Visually examine all areas (area 1 not included).

1) Wear and/or frayed material

a) Permitted if raw rubber is not visible.

2) Scratch of the fabric

a) Permitted up to 0.999 in. (25.4 mm) in length, if raw rubber is not visible.

3) Missing rubber seal material

a) Not permitted.

4) Tear on the rubber seal

a) Not permitted.

(b) Do a visual examination of area 1: open door seal sides.

1) Wear and/or frayed material

a) Not permitted.

2) Scratch of the fabric

a) Not permitted.

3) Missing rubber seal material

a) Not permitted.

4) Tear on the rubber seal

a) Not permitted.

SUBTASK 72-23-00-211-029-G00

(36) Examine the VBV bellcrank pins [27] (Figure 627).

(a) Do a visual examination of the lug area.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

4) Contact marks

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(b) Do a visual examination of the body axis area.

1) Cracks

a) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Nicks and scratches
   a) Not permitted.
3) Dents
   a) Not permitted.
4) Contact marks
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.
5) Teflon deposit
   a) Permitted.

(c) Do a visual examination of area A.
1) Cracks
   a) Not permitted.
2) Nicks and scratches
   a) Not permitted.
3) Dents
   a) Not permitted.
4) Contact marks
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.
5) Teflon deposit
   a) Permitted.

(d) Do a dimension check of the pin bellcrank diameter.
1) Pin bellcrank diameter limits:
   a) Minimum serviceable - 0.55052 in. (13.983 mm).
   b) Maximum serviceable - 0.55094 in. (13.994 mm).

SUBTASK 72-23-00-211-030-G00

(37) Examine the VBV door turnbuckles [28] and VBV actuator turnbuckles [29] (Figure 628).
(a) Do a visual examination of the area A of the VBV door turnbuckles [28].
1) Cracks
   a) Not permitted.
2) Nicks and scratches
   a) Not permitted.
3) Dents
   a) Not permitted.
4) Contact marks
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.
5) Bearing radial movement
   a) Permitted up to 0.003 in. (0.1 mm) in maximum clearance between the ball and race of the bearing.

(b) Do a visual examination of the body area of the VBV actuator turnbuckles [29].
1) Cracks

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a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

3) Dents

a) Permitted up to 0.00499 in. (0.127 mm) in depth.

(c) Do a visual examination of the lug area of the VBV actuator turnbuckles [29].

1) Cracks and dents

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.00011 in. (0.003 mm) in depth.

(d) Do a visual examination of the area A of the VBV actuator turnbuckles [29].

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Not permitted.

3) Dents

a) Not permitted.

4) Contact marks

a) Not permitted.

5) Bearing radial movement

a) Permitted up to 0.003 in. (0.1 mm) in maximum clearance between the ball and race of the bearing.

SUBTASK 72-23-00-211-031-G00

(38) Examine the pins [30] (Figure 629).

(a) Do a visual examination of all areas.

1) Scratches

a) Not permitted.

2) Cracks, dents, and nicks

a) Not permitted.

3) Contact marks

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(b) Do a dimension check of the ring diameter of the pin.

1) Pin ring diameter limits:

a) Minimum serviceable - 0.31126 in. (7.906 mm).

b) Maximum serviceable - 0.31161 in. (7.915 mm).

SUBTASK 72-23-00-211-032-G00

(39) Examine the spacers [31] (Figure 630).

(a) Do a visual examination of all areas.

1) Nicks and scratches

a) Not permitted.

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2) Dents
   a) Not permitted.
3) Cracks
   a) Not permitted.
4) Contact marks
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(b) Do a dimension check of the spacer diameter.
1) Spacer inner diameter limits:
   a) Minimum serviceable - 0.3130 in. (7.95 mm).
   b) Maximum serviceable - 0.3141 in. (7.98 mm).

SUBTASK 72-23-00-211-033-G00

(40) Examine the bolts [32], bolts [33], bolts [34], bolts [35], axes [36], nuts [37], nuts [38], and washers [39] (Figure 631).

(a) Visually examine all areas (but not the tooling contact surfaces).
1) Scratches
   a) Not permitted.
2) Nicks and dents
   a) Not permitted
3) Cracks or breaks
   a) Not permitted
4) Oxidation, pitting, or corrosion marks
   a) Not permitted.

(b) Do a visual examination of the tooling contact surfaces.
1) Contact marks
   a) Permitted up to 0.0011 in. (0.03 mm) in depth.
2) Nicks, dents, and scratches
   a) Permitted up to 0.0011 in. (0.03 mm) in depth.
3) Cracks or breaks
   a) Not permitted.
4) Oxidation, pitting, or corrosion marks
   a) Not permitted.

SUBTASK 72-23-00-211-034-G00

(41) Examine the anti-icing tube [40] (Figure 632).

(a) Do a visual examination of the anti-icing tube [40].
1) Splits, cracks, or kinks
   a) Not permitted.
2) Wrinkles in bends
   a) Not permitted.
3) Dents
   a) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Flat areas

a) Not permitted.

5) Nicks, scratches, chafing, or scores in straight sections or bends

a) Not permitted.

SUBTASK 72-23-00-370-001-G00

CAUTION AFTER YOU EXAMINE THE OUTLET GUIDE VANE MADE OF ALUMINUM, REPAIR THE OXIDE LAYER. IF YOU DO NOT OBEY, CORROSION AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

(42) After you examine the aluminum outlet guide vanes OGV [9], repair the oxide film coating:

(a) Do the touch-up of the oxide film coating on all permitted indications (TASK 70-43-07-300-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-00-860-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-00-860-006-G00

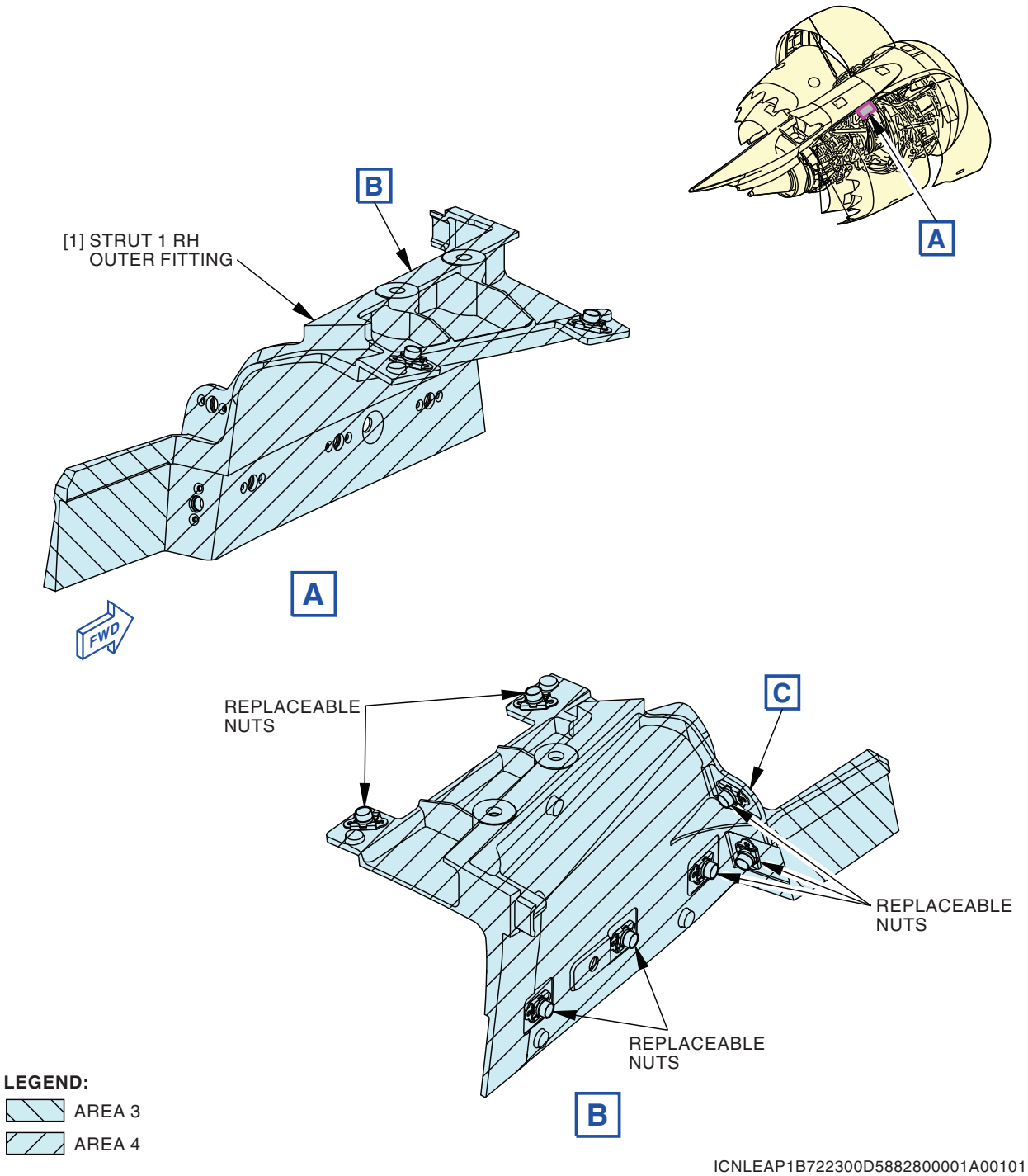
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

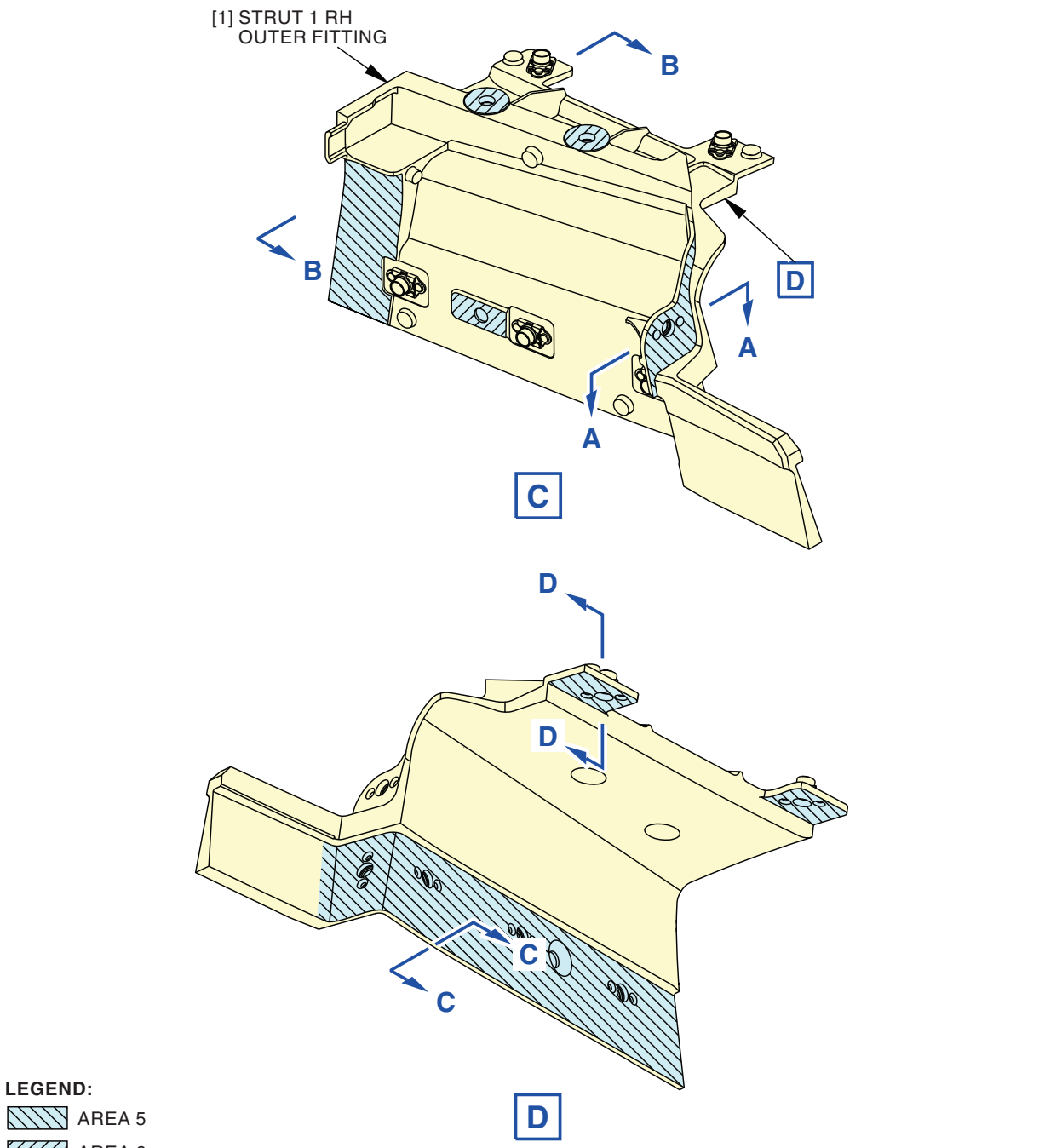
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



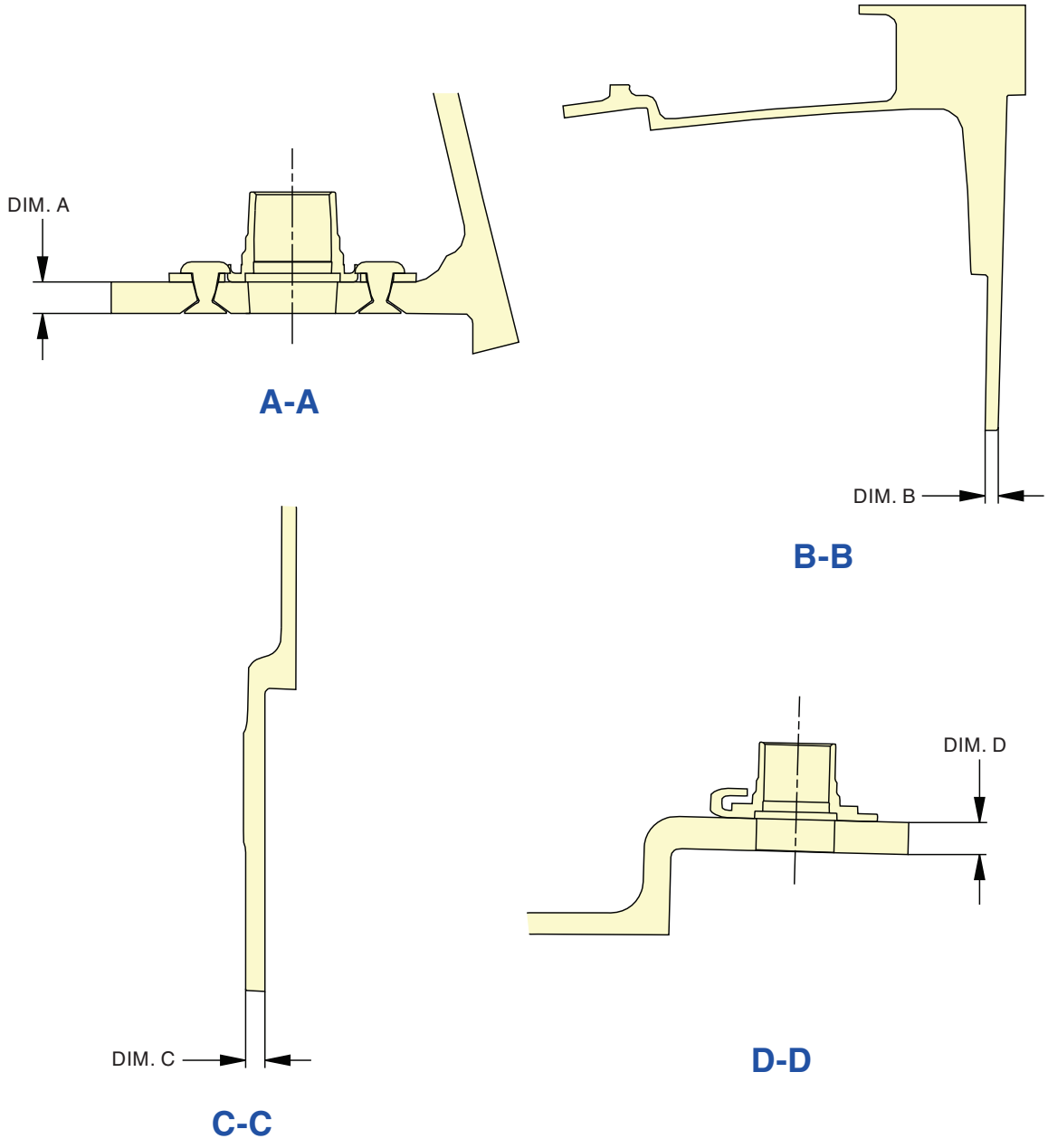
Strut 1 RH Outer Fitting Inspection
Figure 601/72-23-00-990-802-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut 1 RH Outer Fitting Inspection
Figure 601/72-23-00-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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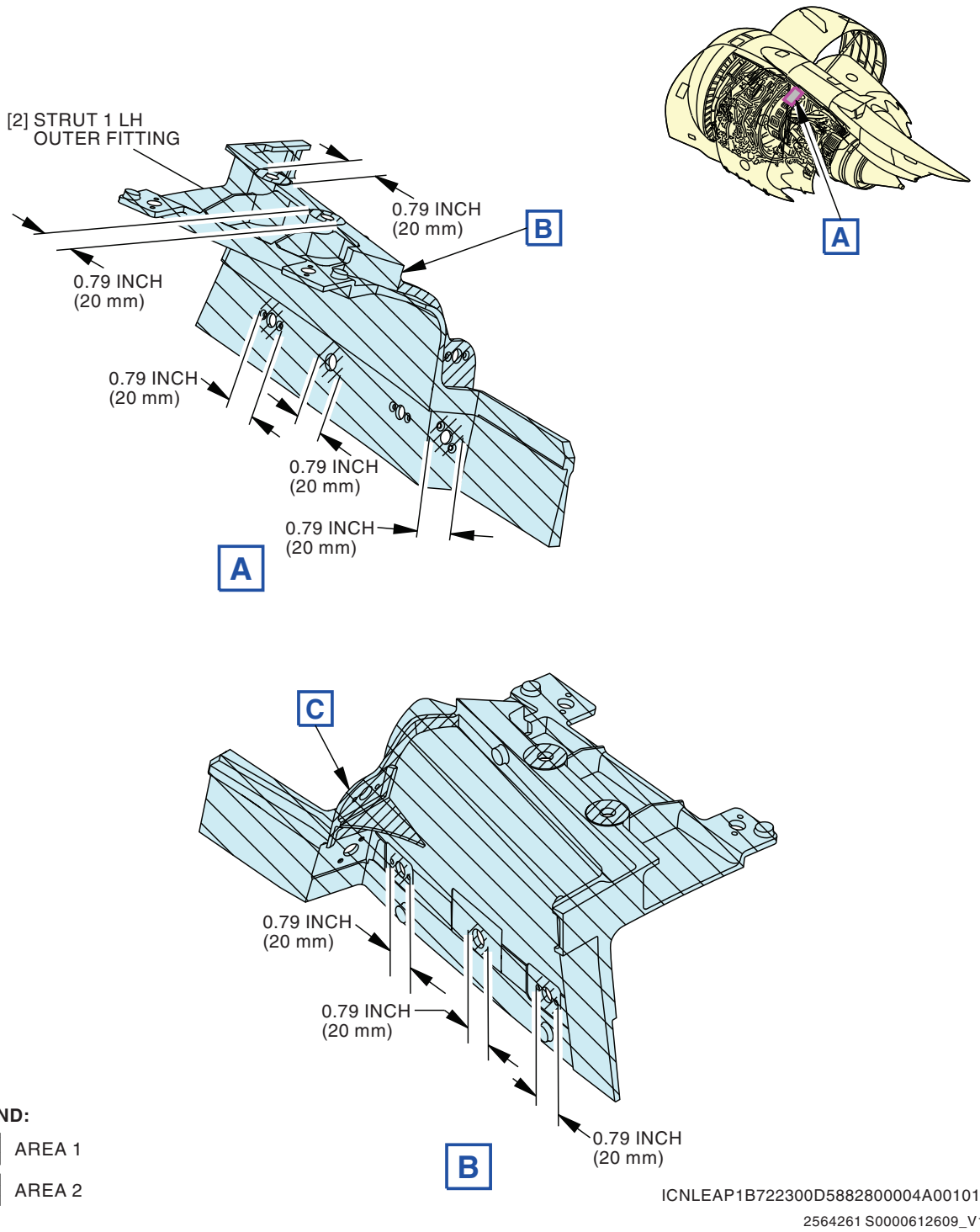
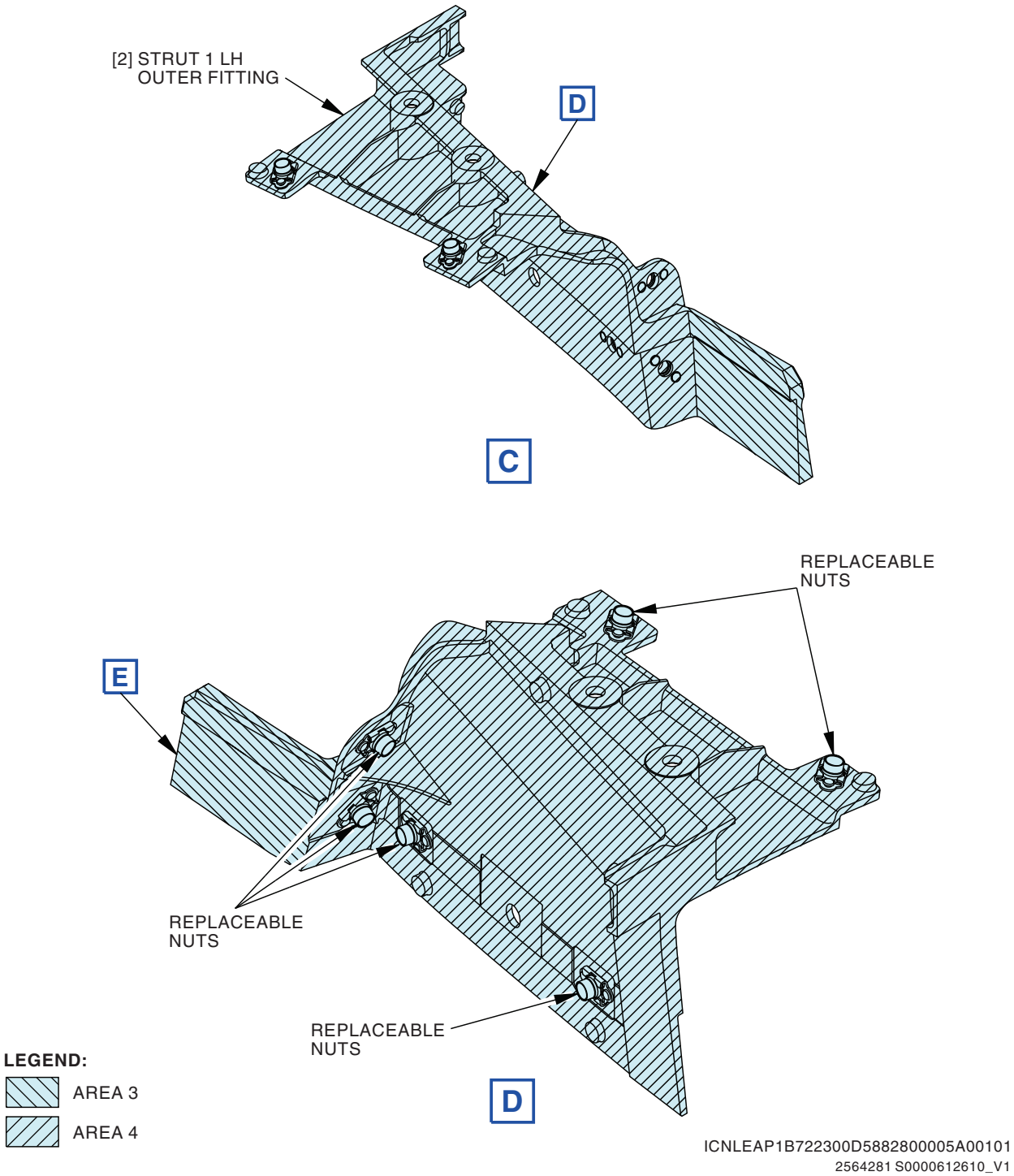
Strut 1 RH Outer Fitting Inspection
Figure 601/72-23-00-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



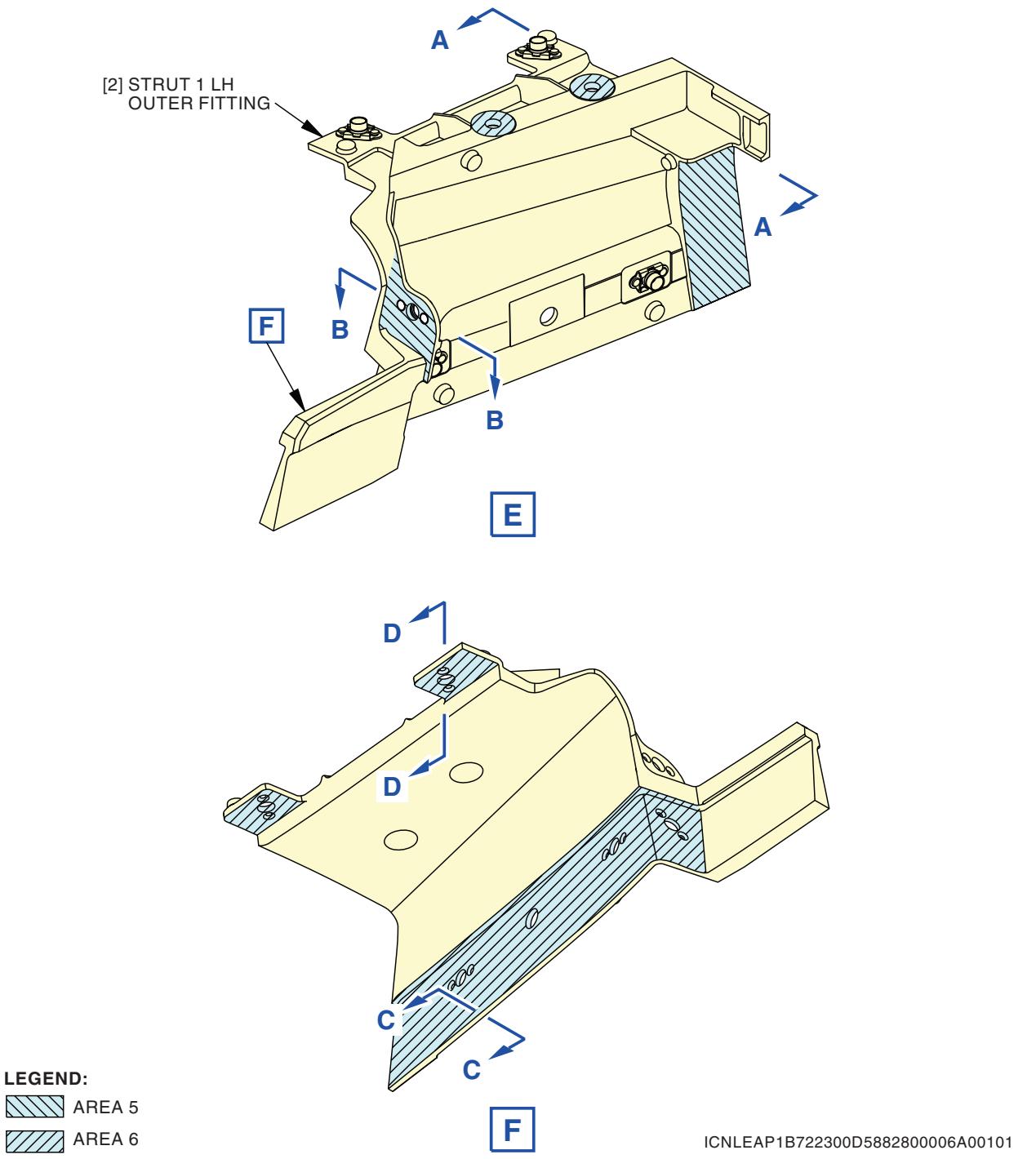
Strut 1 LH Outer Fitting Inspection
Figure 602/72-23-00-990-803-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



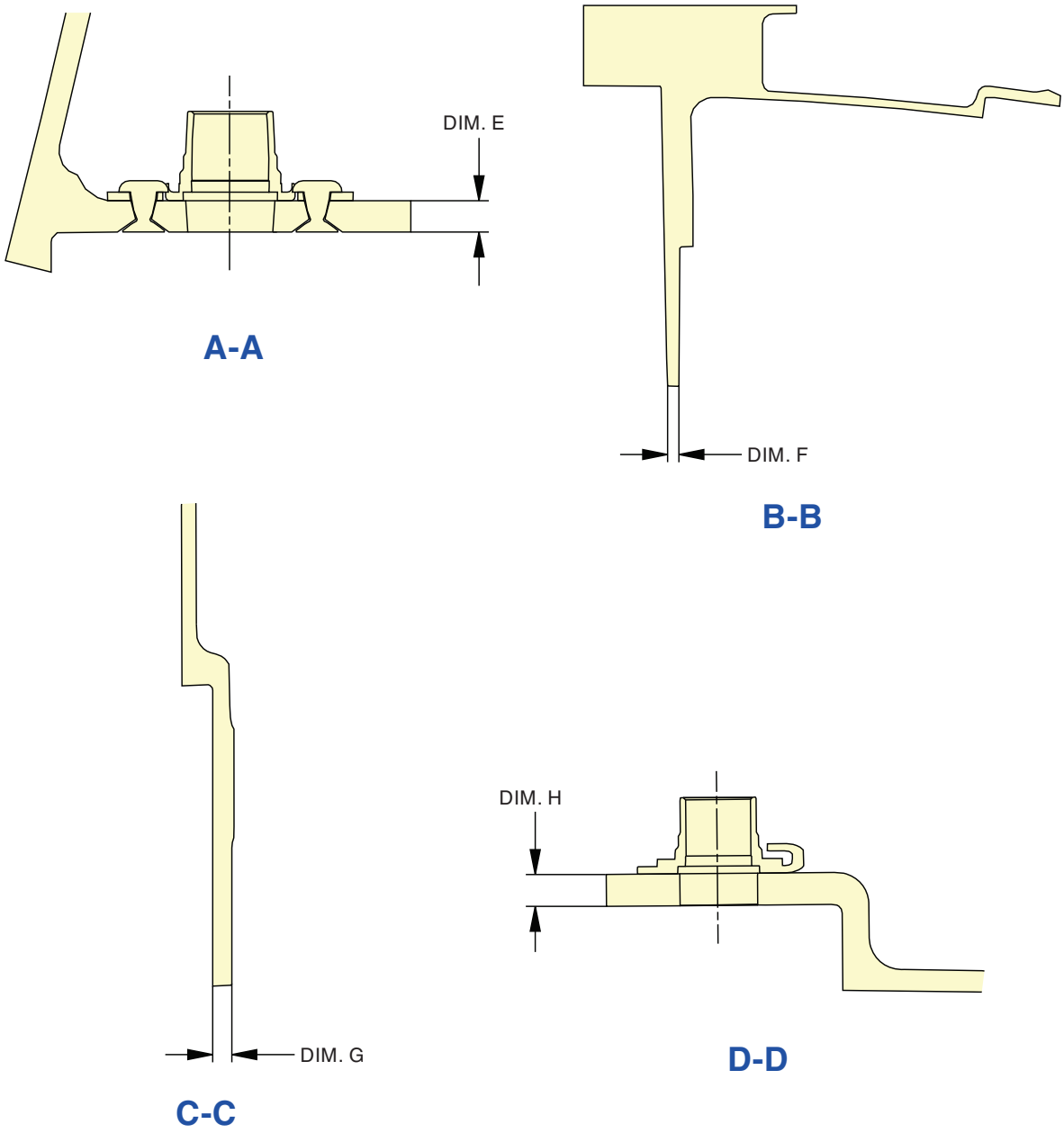
Strut 1 LH Outer Fitting Inspection
Figure 602/72-23-00-990-803-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut 1 LH Outer Fitting Inspection
Figure 602/72-23-00-990-803-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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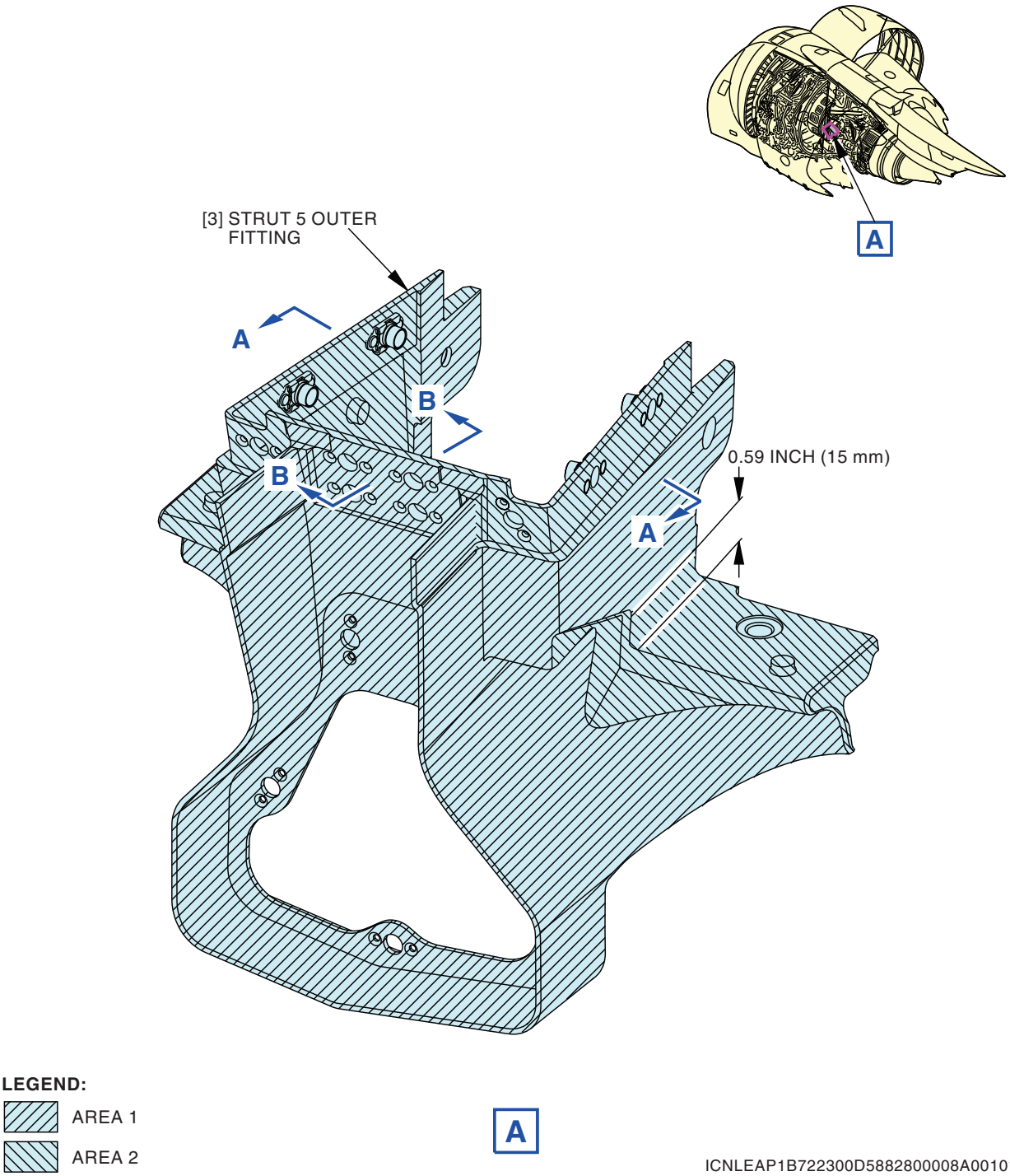
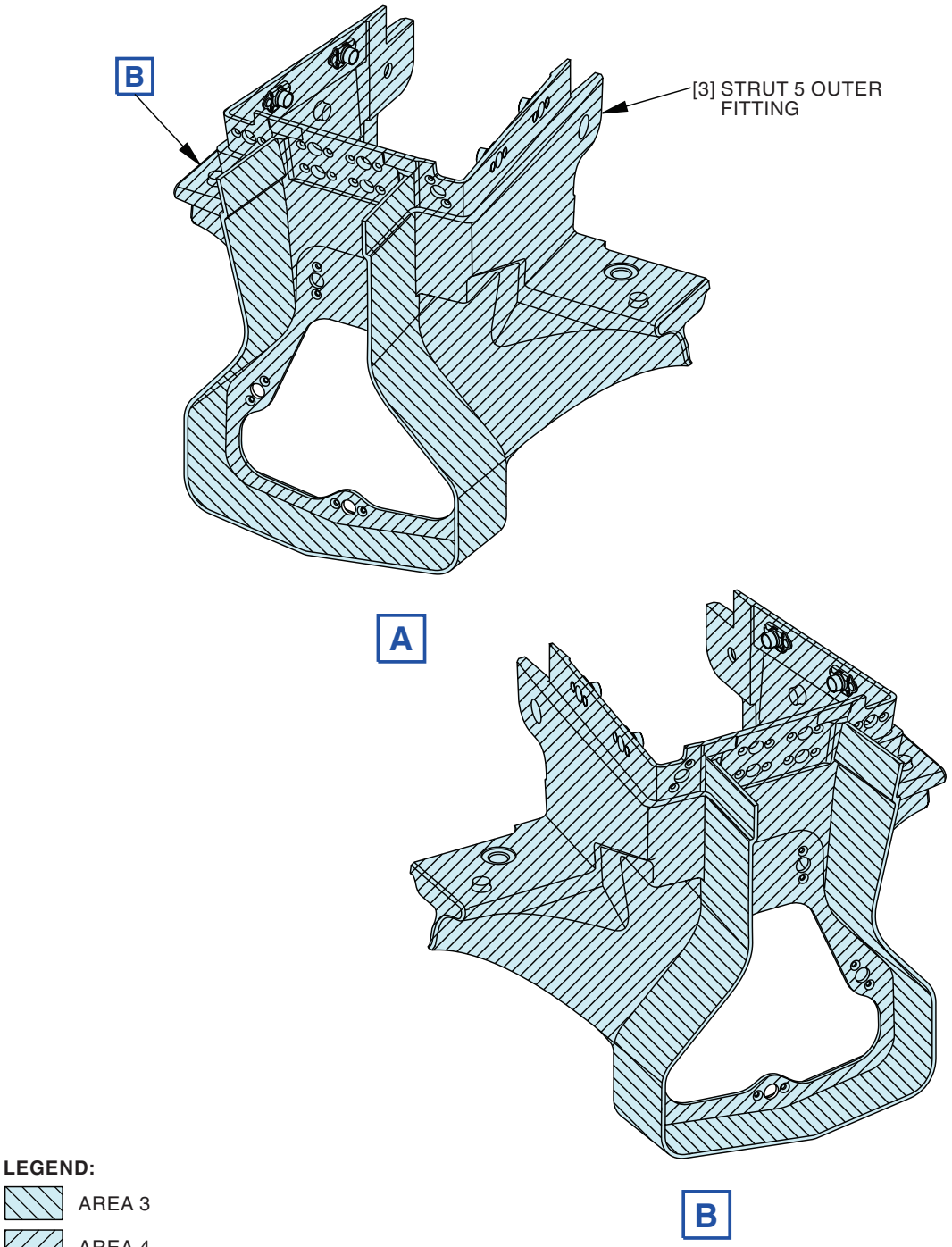
Strut 1 LH Outer Fitting Inspection
Figure 602/72-23-00-990-803-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



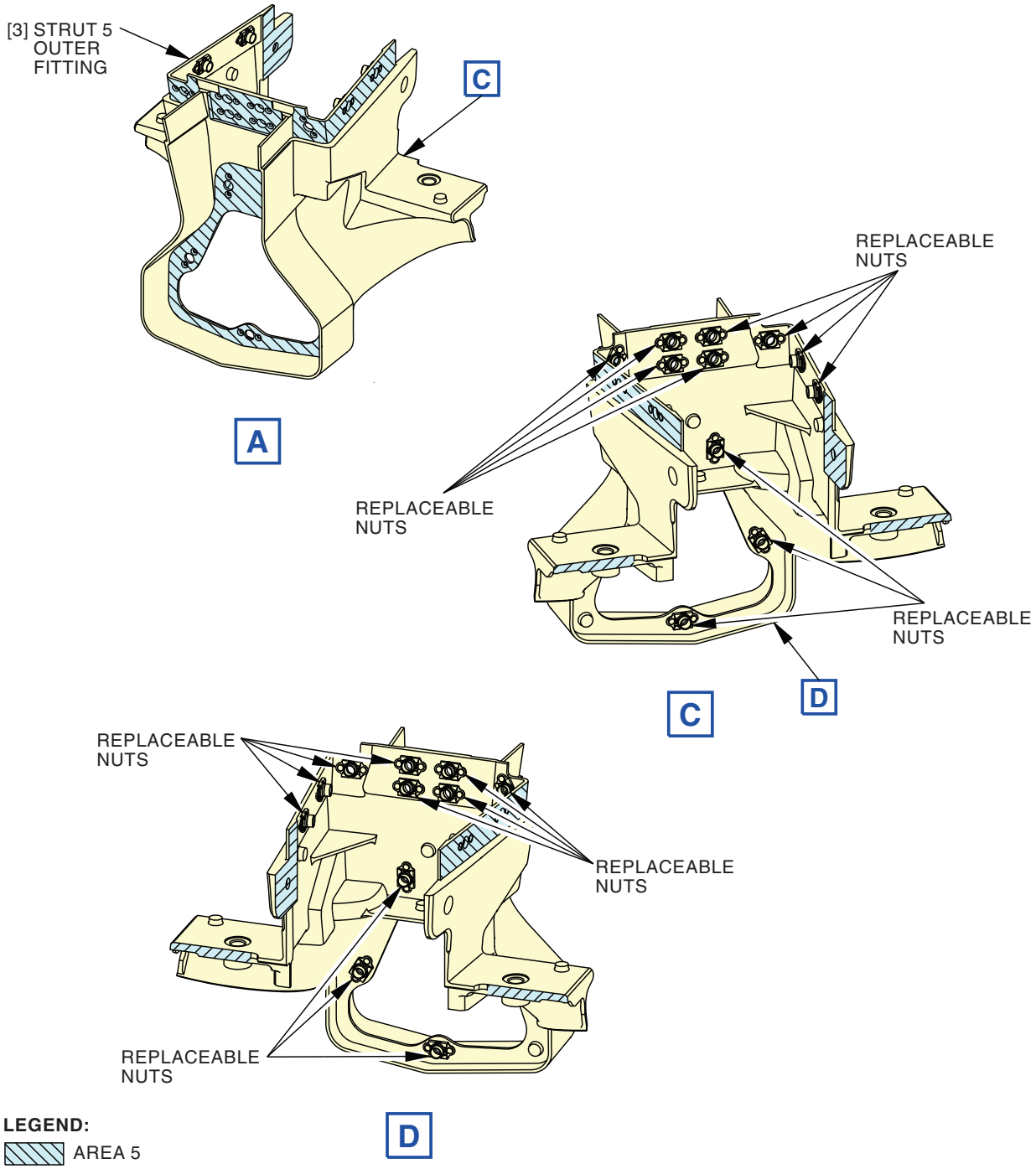
Strut 5 Outer Fitting Inspection
Figure 603/72-23-00-990-804-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut 5 Outer Fitting Inspection
Figure 603/72-23-00-990-804-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut 5 Outer Fitting Inspection
Figure 603/72-23-00-990-804-G00 (Sheet 3 of 4)

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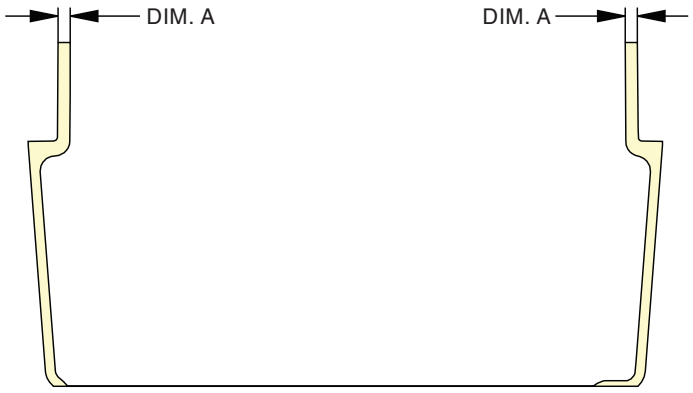

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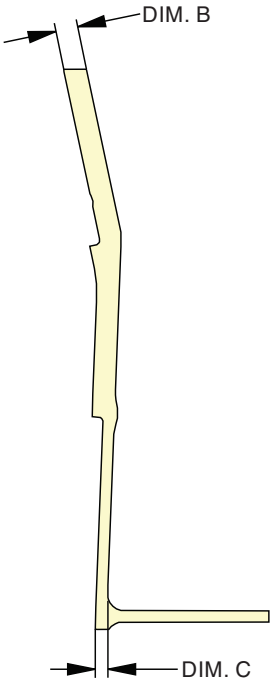

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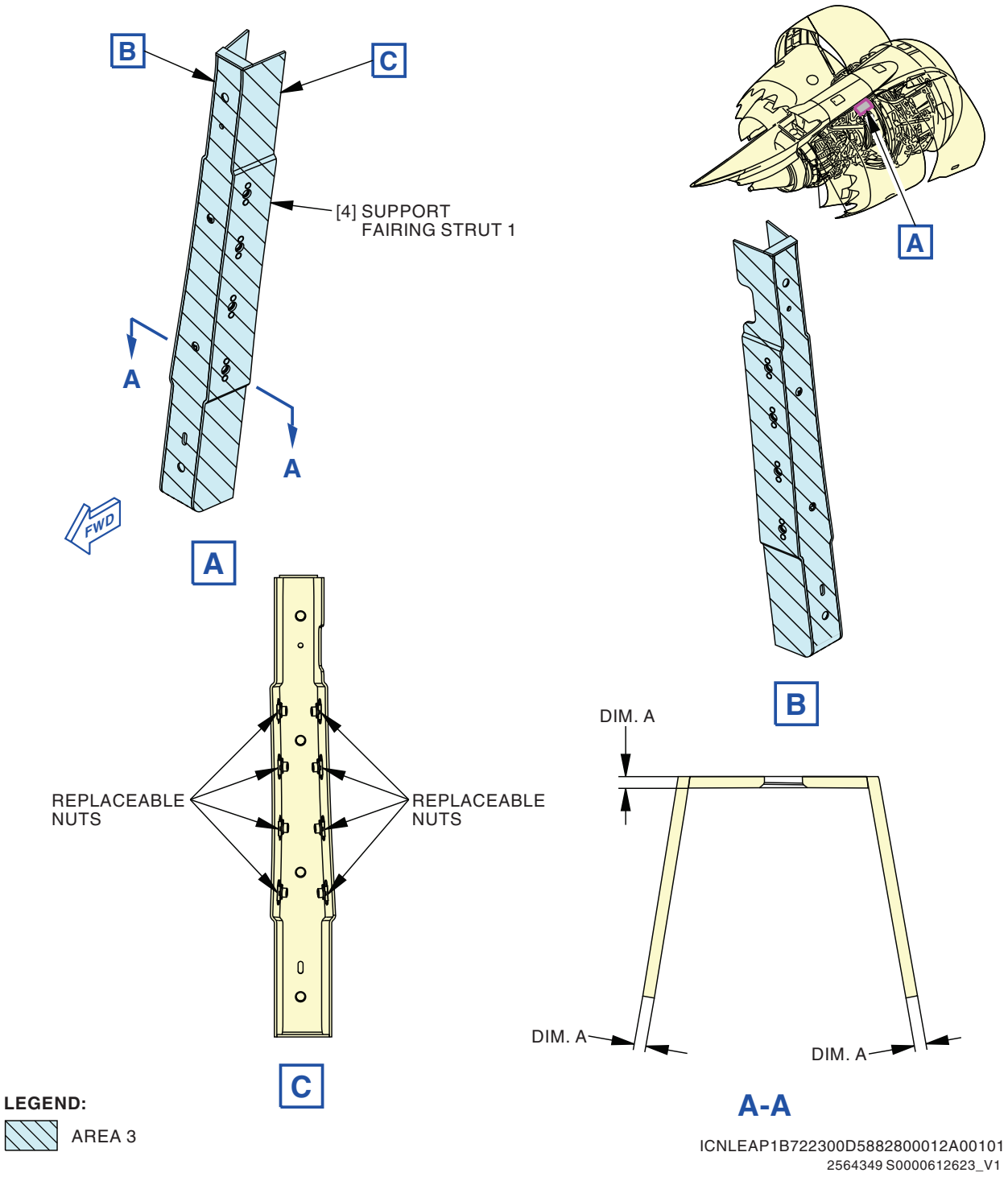
Strut 5 Outer Fitting Inspection
Figure 603/72-23-00-990-804-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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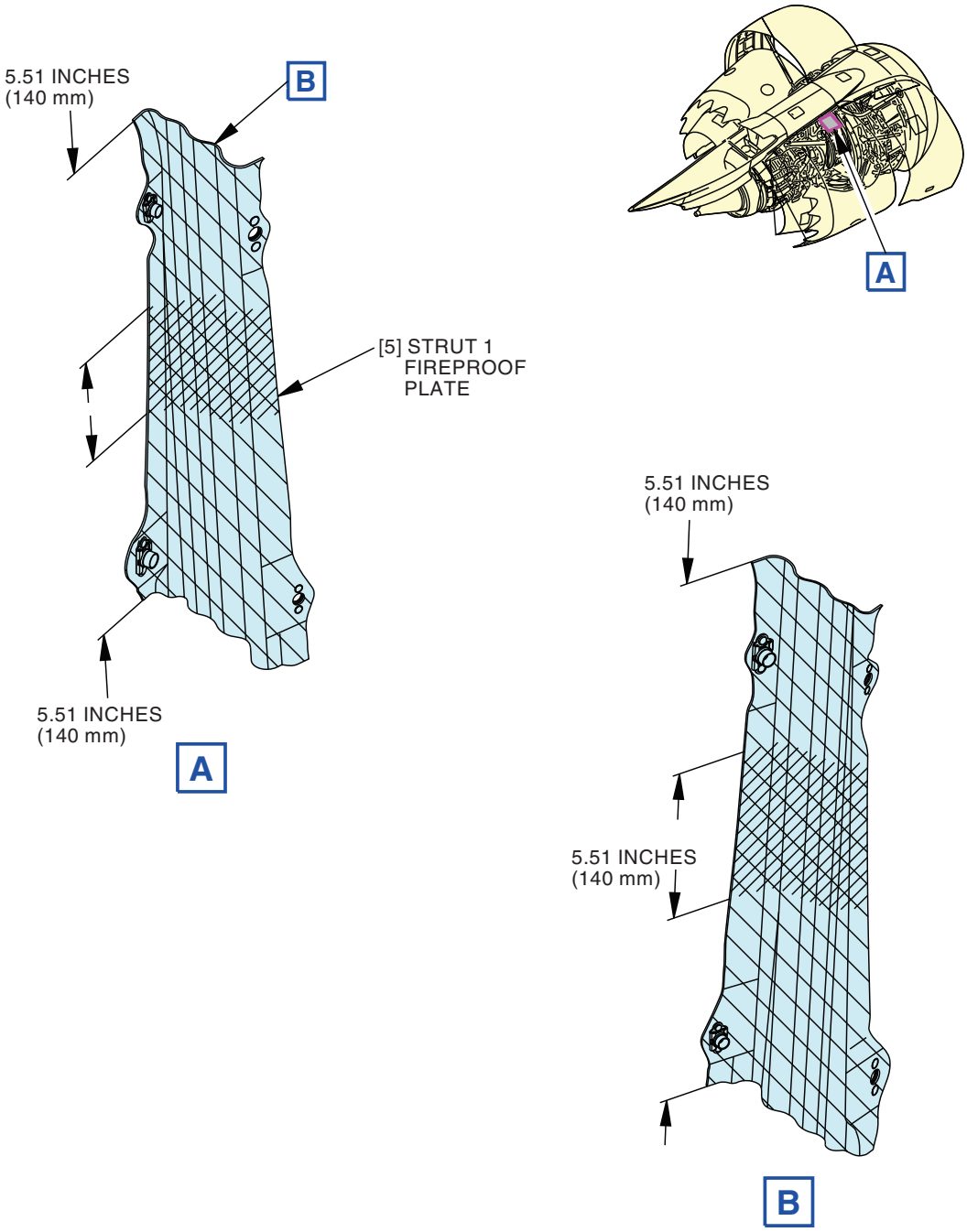
Strut 1 Fairing Support Inspection
Figure 604/72-23-00-990-805-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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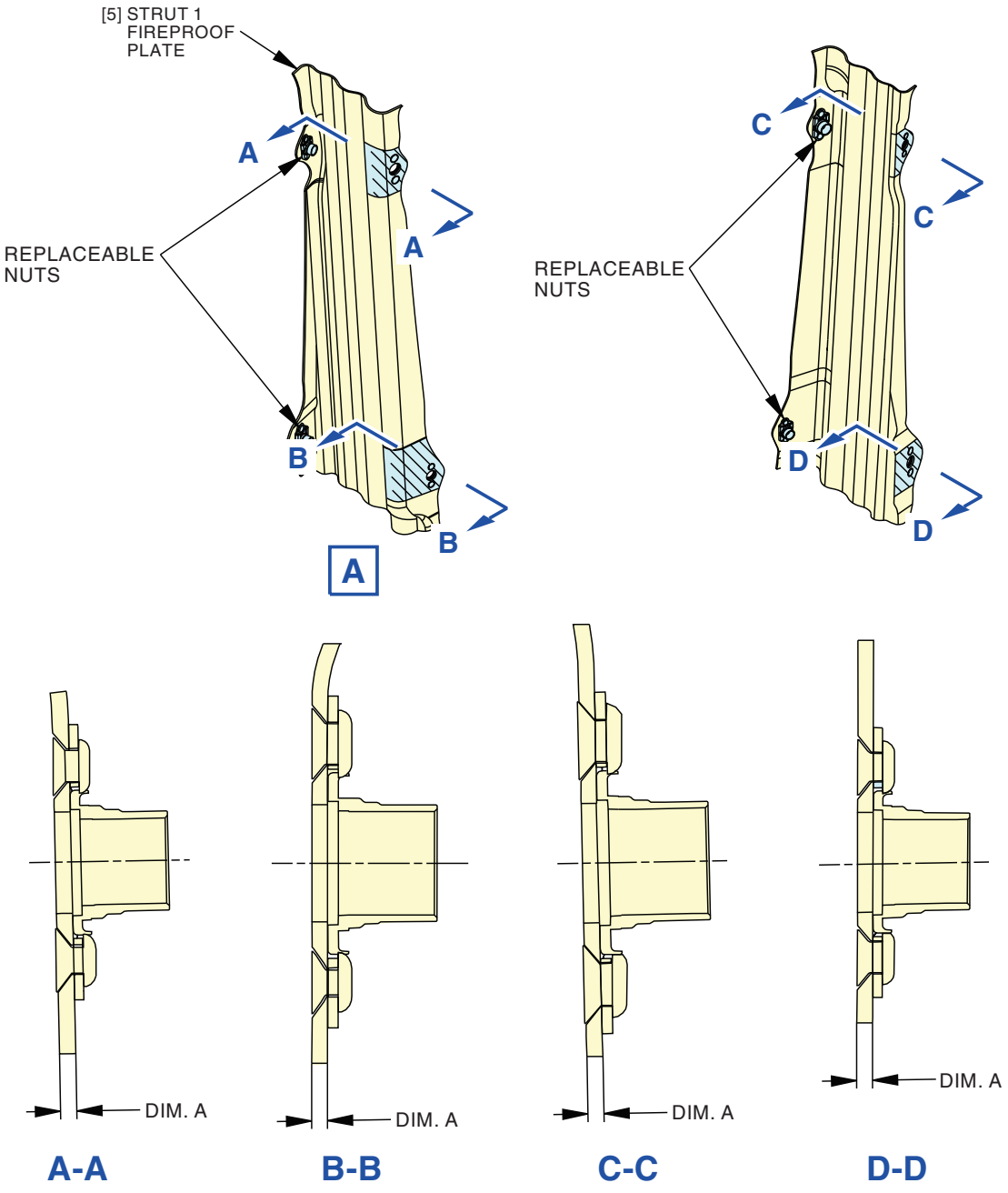
Strut 1 Fireproof Plate Inspection
Figure 605/72-23-00-990-806-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

AREA 3

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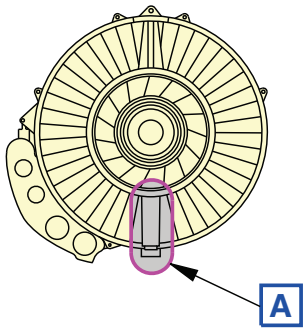
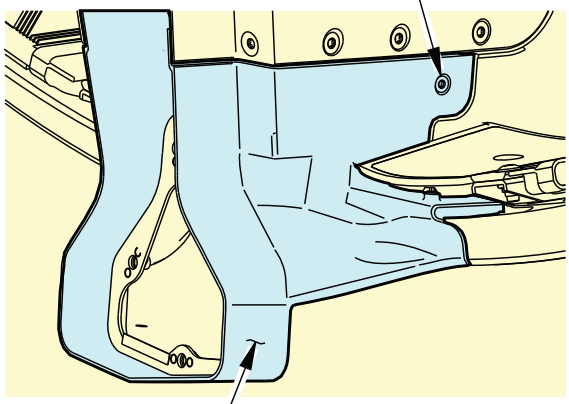
Strut 1 Fireproof Plate Inspection
Figure 605/72-23-00-990-806-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

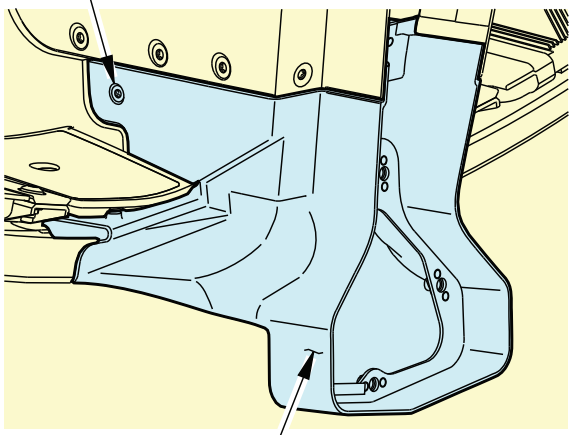


(VIEW IN THE
FORWARD DIRECTION)



(RIGHT SIDE VIEW)

A



(LEFT SIDE VIEW)

A

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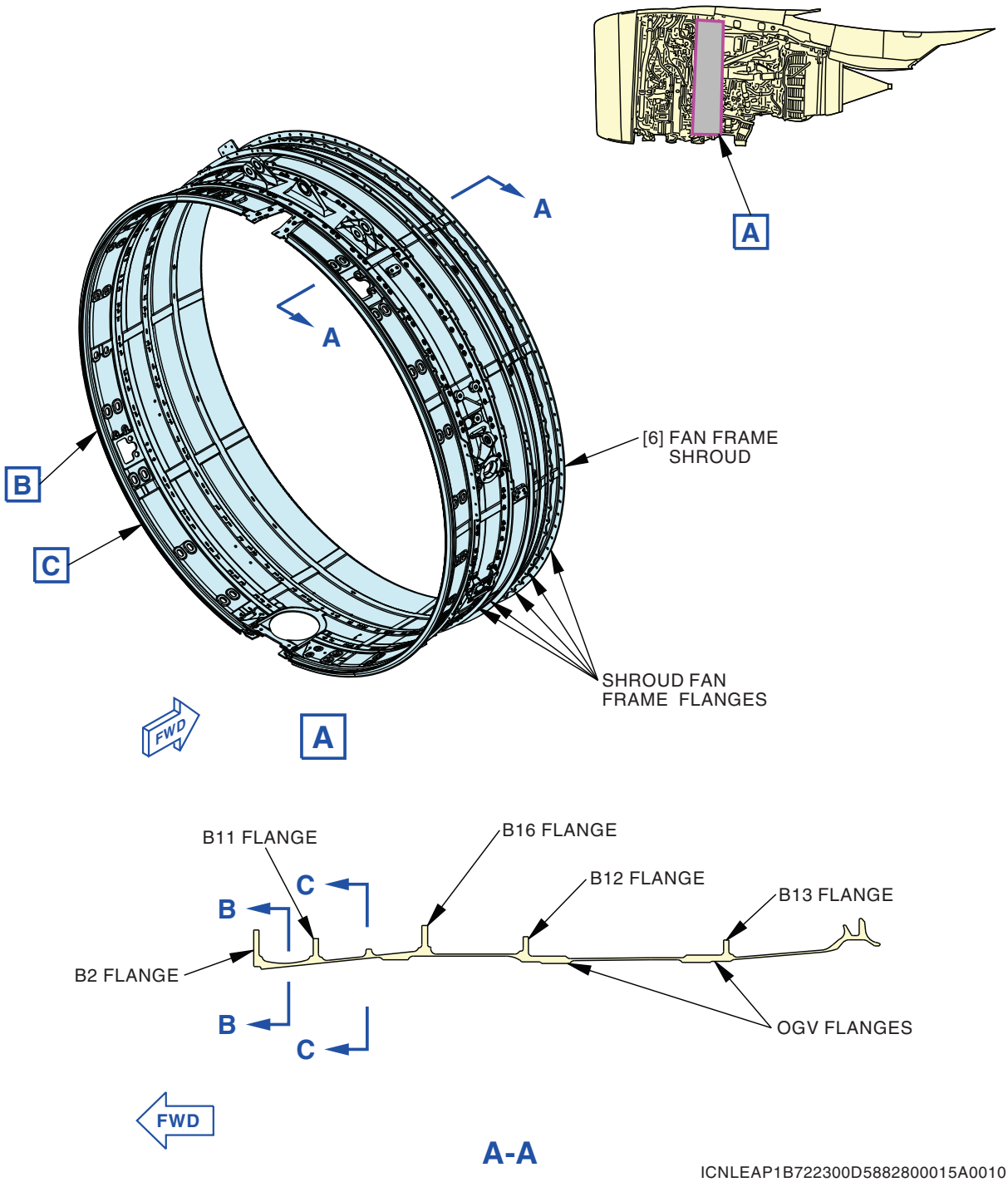
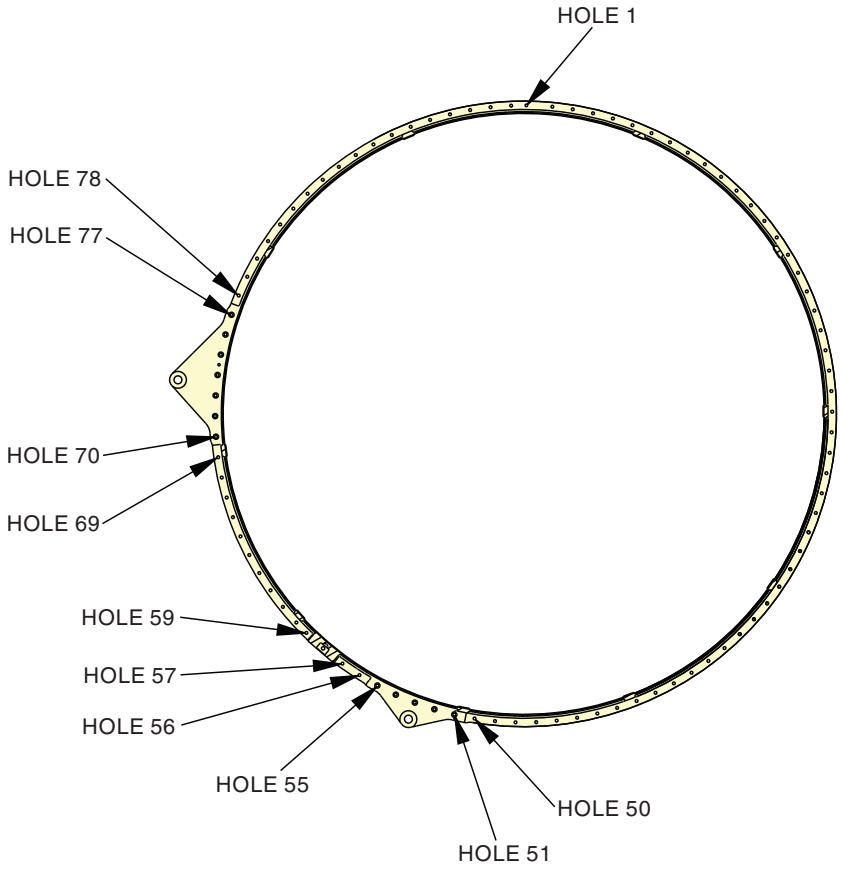
Strut 5 Outer Fitting
Figure 606/72-23-00-990-847-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 1 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B2 FLANGE
AFT LOOKING FORWARD

B-B

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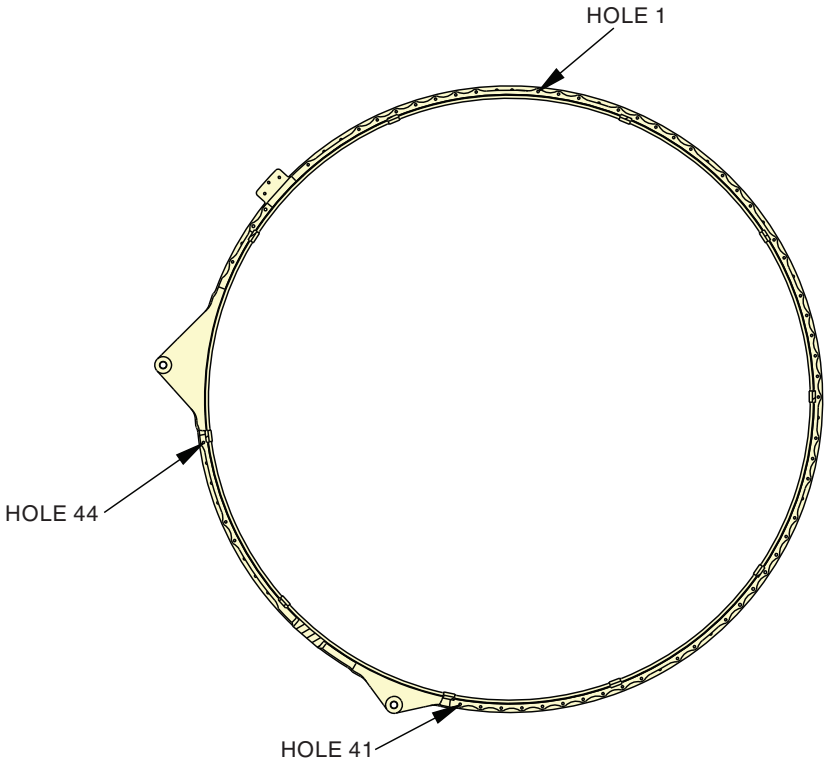
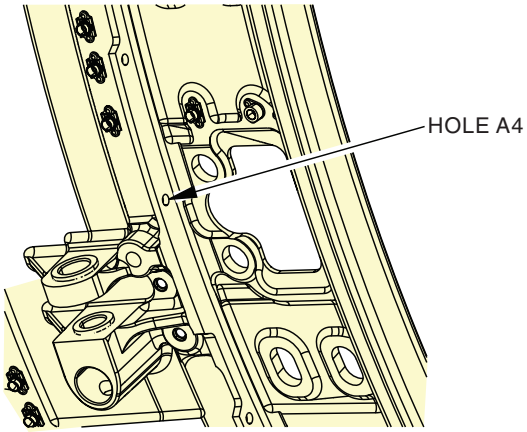
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 2 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B11 FLANGE
AFT LOOKING FORWARD

C-C



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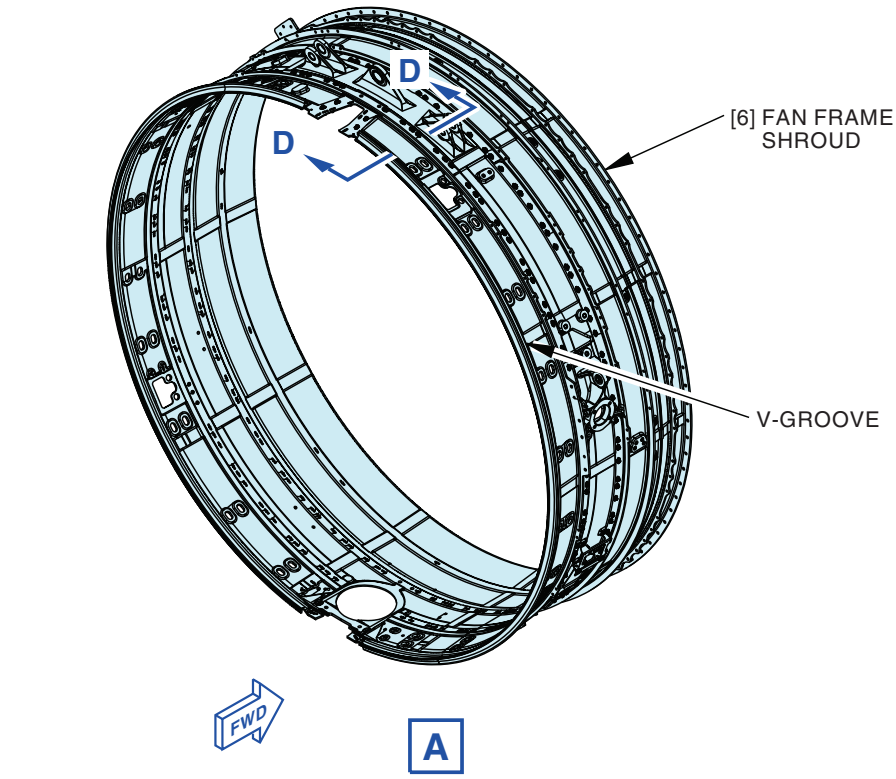
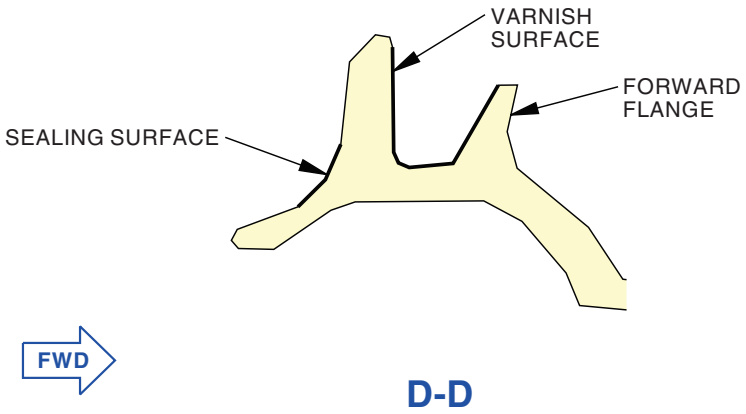
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 3 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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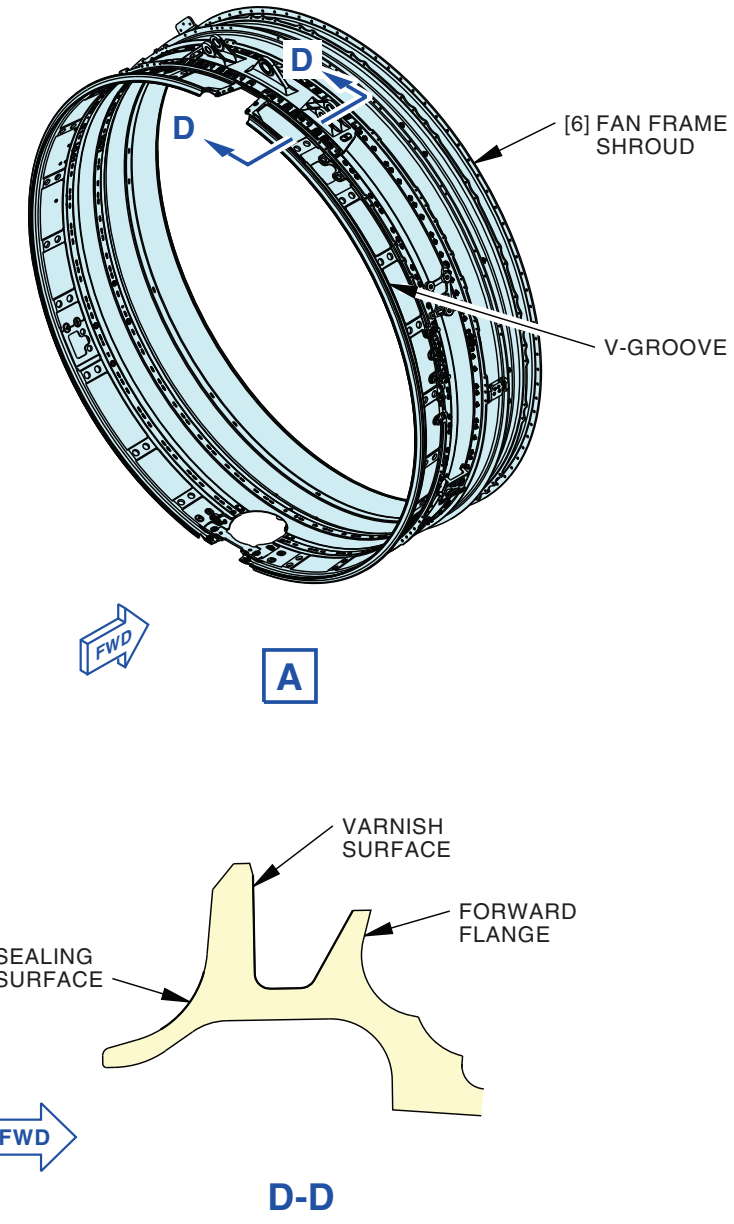
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 4 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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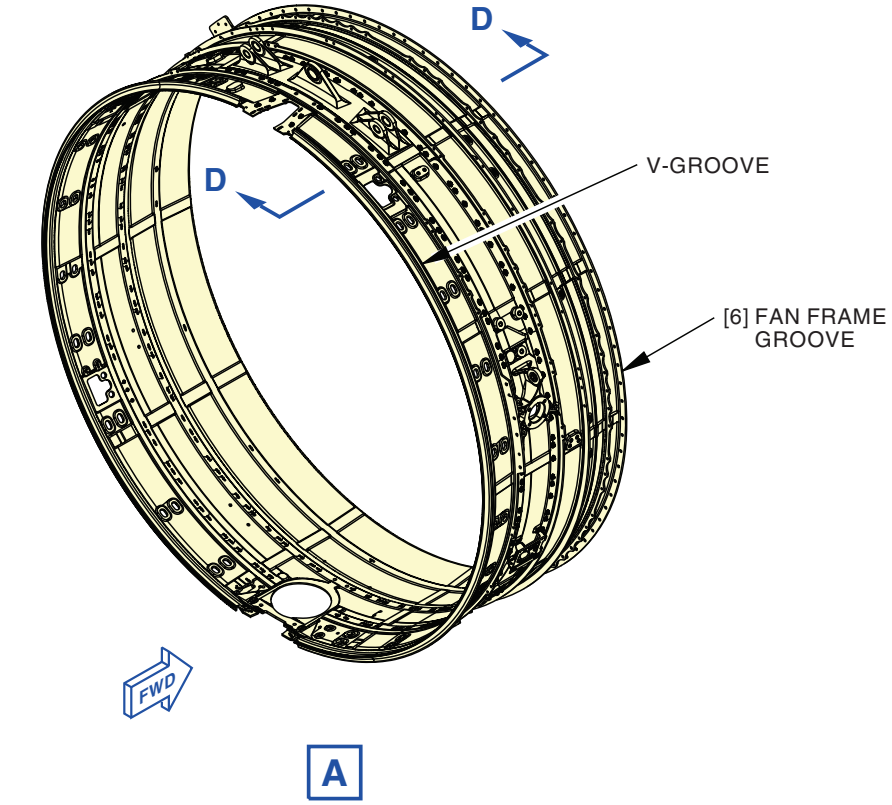
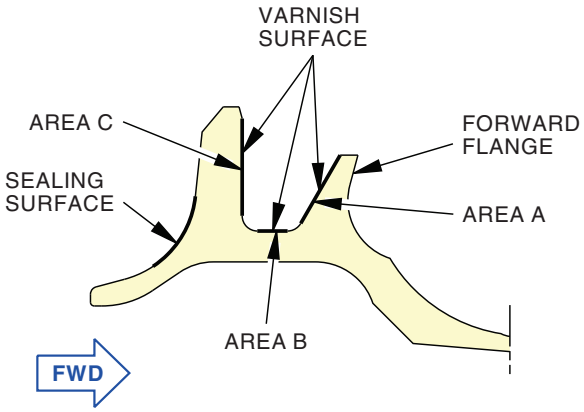
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 5 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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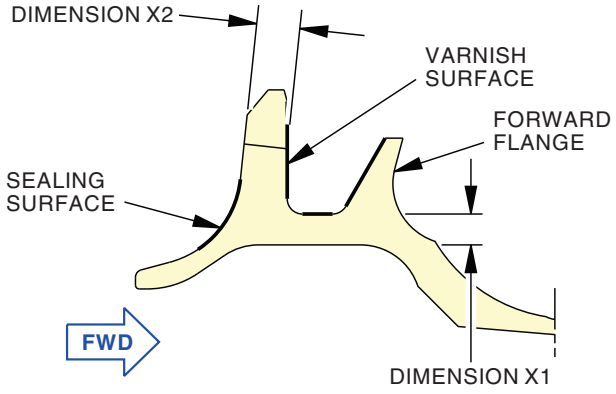

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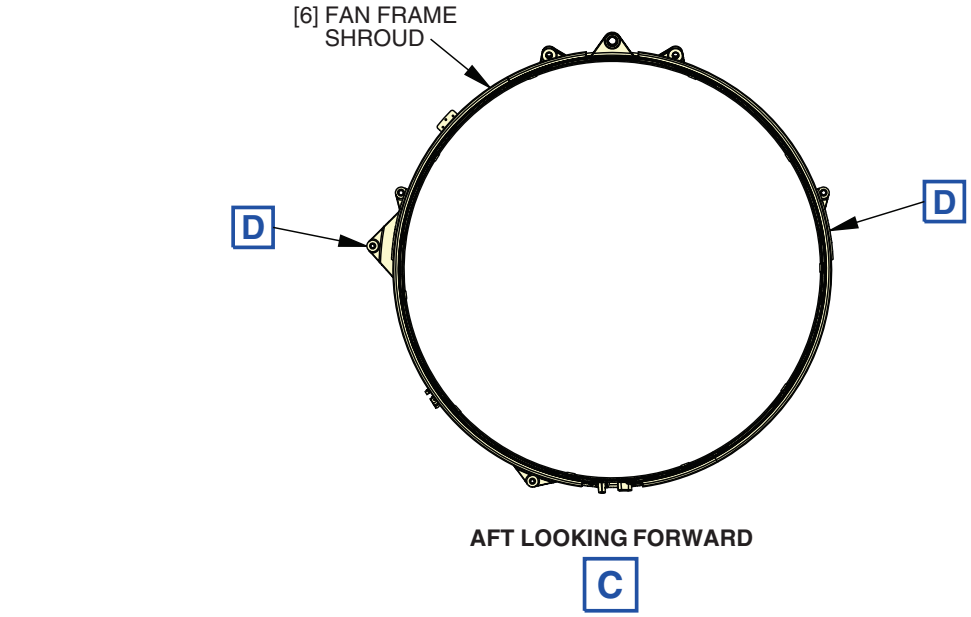
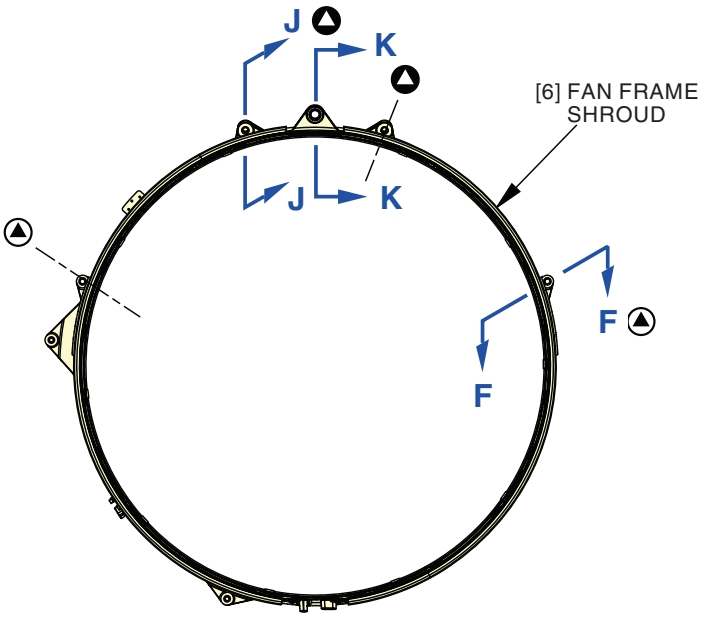
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 6 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



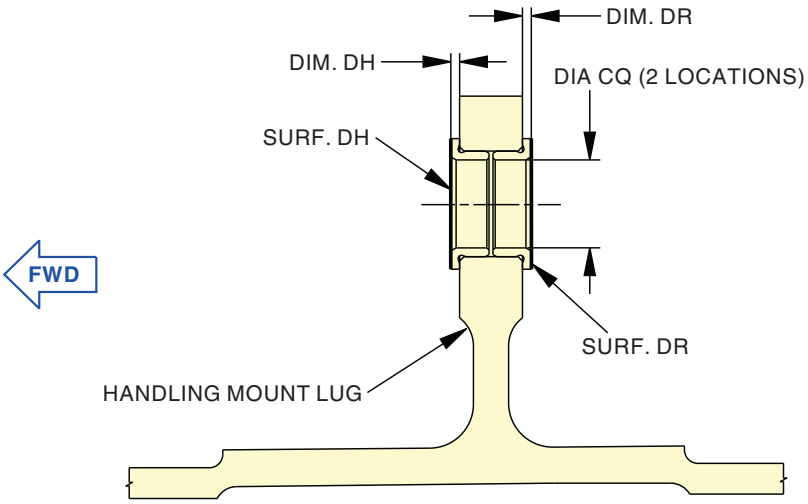
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 7 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AFT LOOKING FORWARD

C

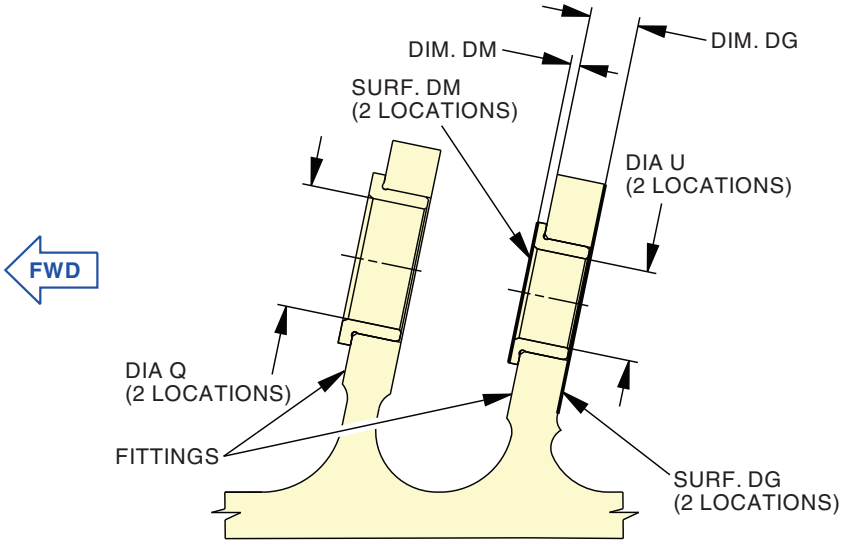


F-F

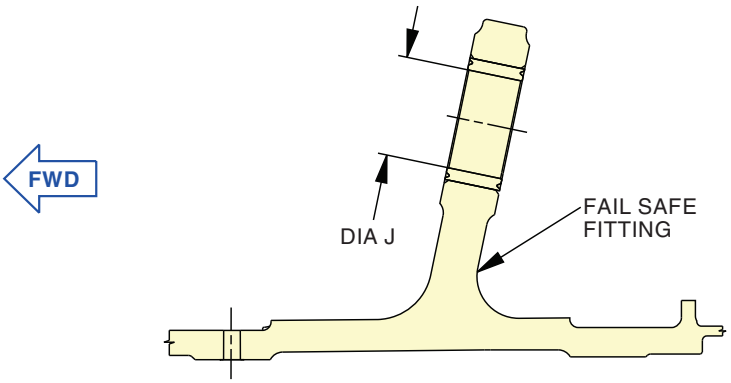
ICNLEAP1B722300D5882800020A00101
2564868 S0000612631_V2

Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 8 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



J-J

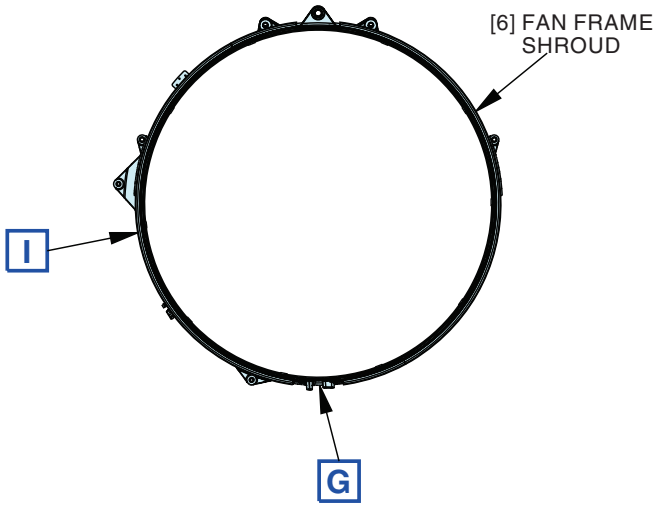


K-K

ICNLEAP1B722300D5882800023A00101
2564909 S0000612634_V1

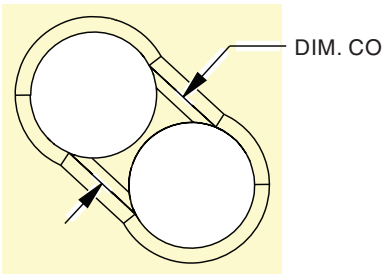
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 9 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

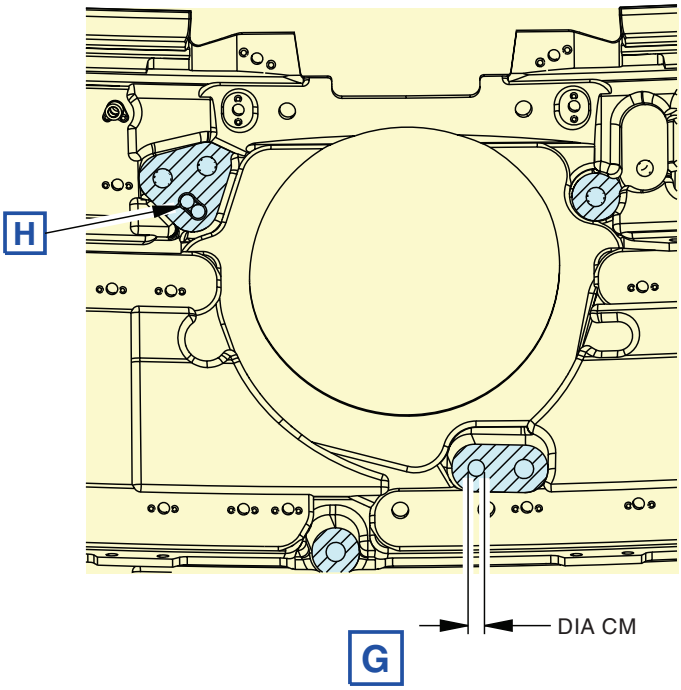


AFT LOOKING FORWARD

C



H



LEGEND:

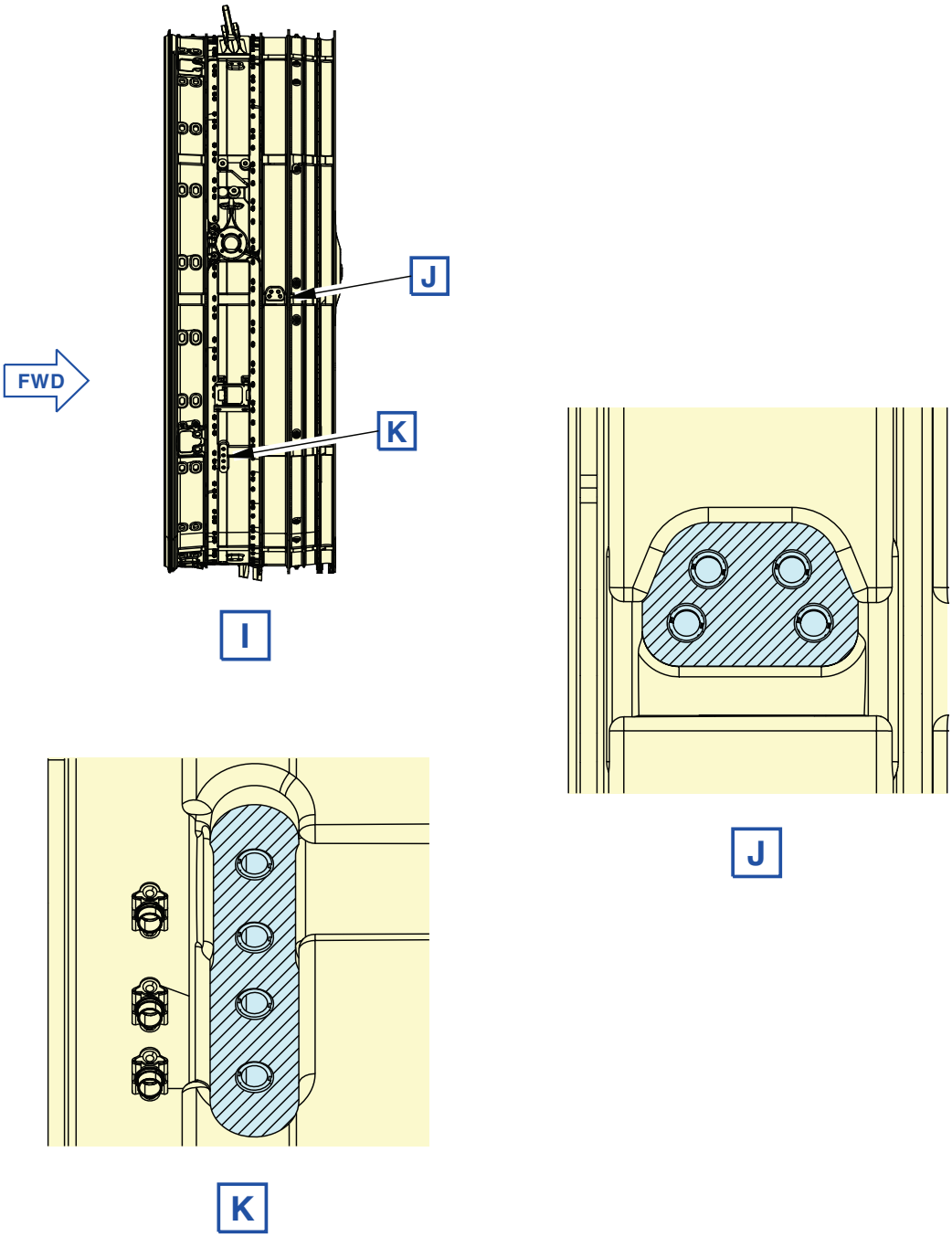
MATING FACES OF TGB MOUNTS

ICNLEAP1B722300D5882800024A00101

2564918 S0000612635_V2

Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 10 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

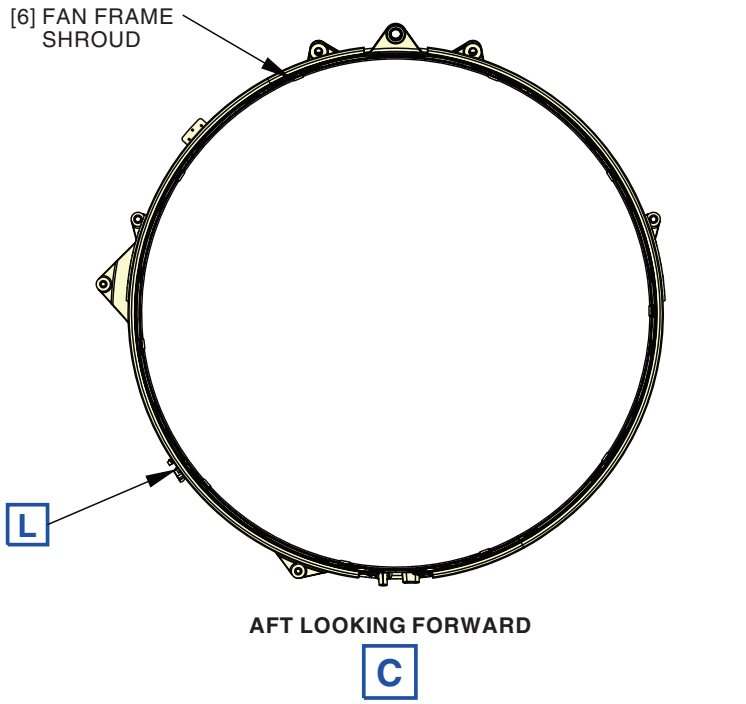
MATING FACES OF OIL TANK MOUNTS

ICNLEAP1B722300D5882800025A00101

2564944 S0000612636_V2

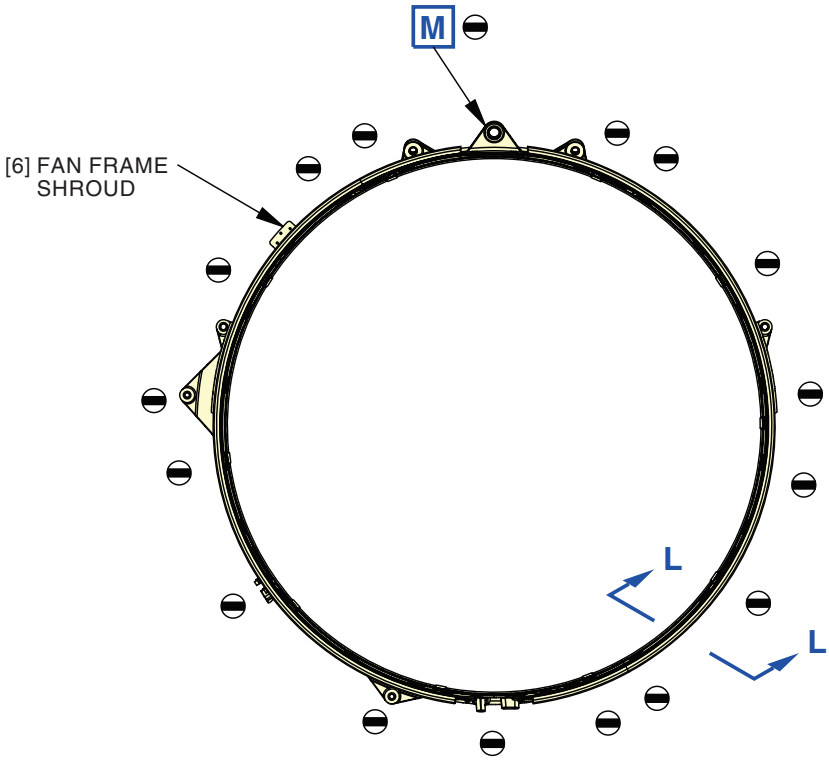
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 11 of 13)

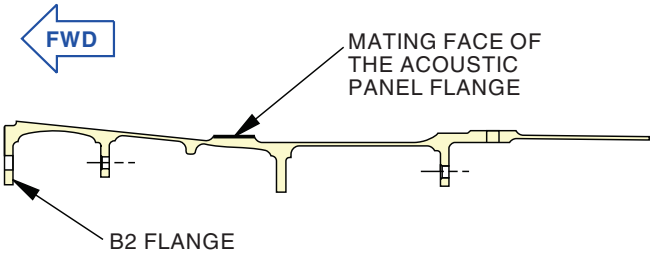
ECCN 9E991 BOEING PROPRIETARY - See title page for details



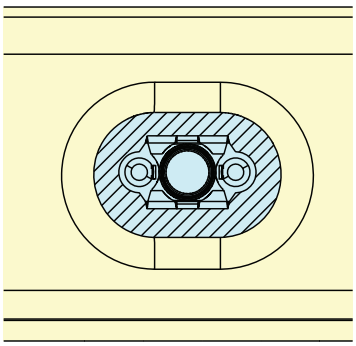
Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 12 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





L-L



M

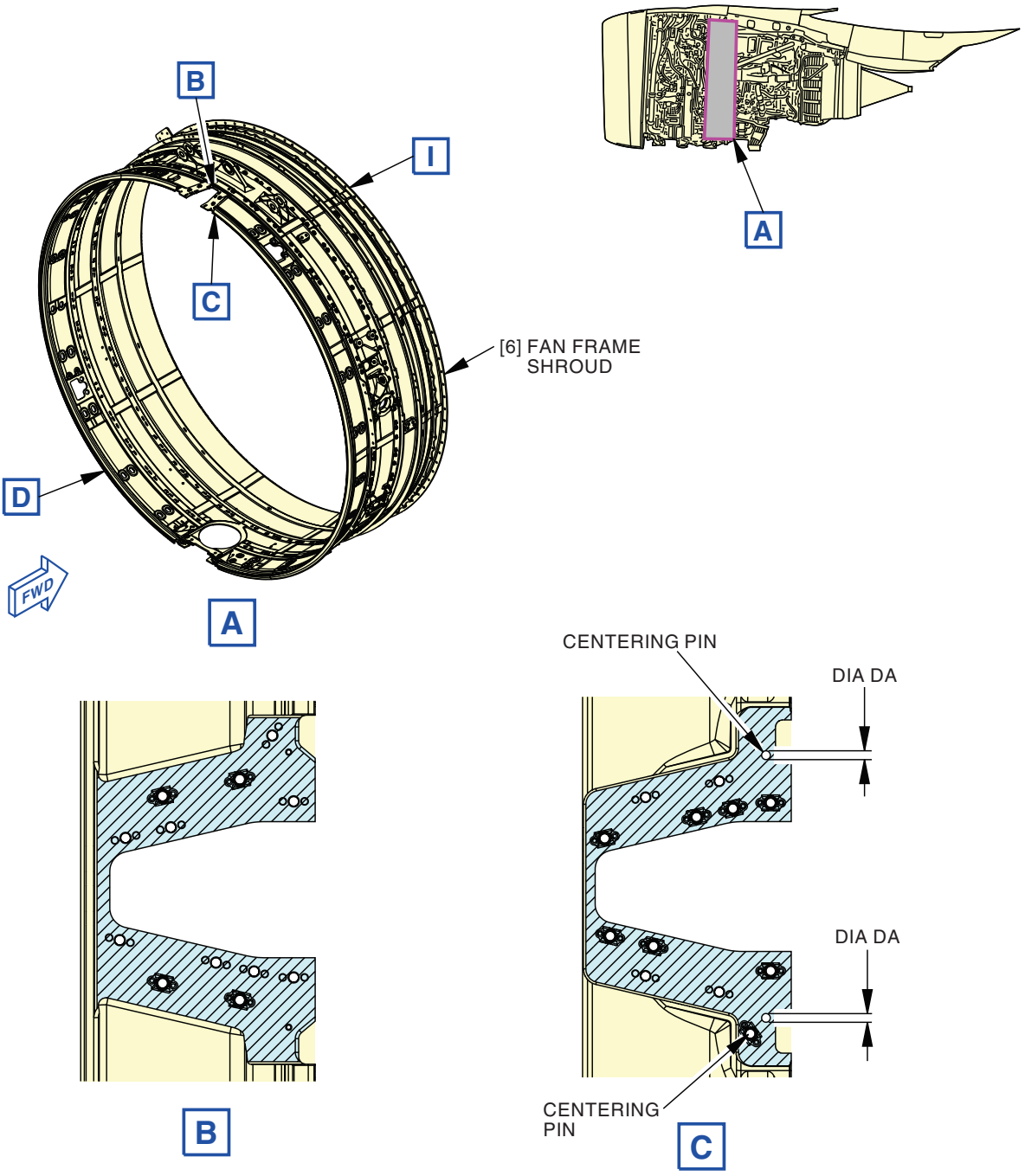
LEGEND:

MATING FACES OF ACOUSTIC PANEL BOSSES

ICNLEAP1B722300D5882800027A00101
2565195 S0000612638_V1

Fan Frame Shroud Inspection
Figure 607/72-23-00-990-807-G00 (Sheet 13 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



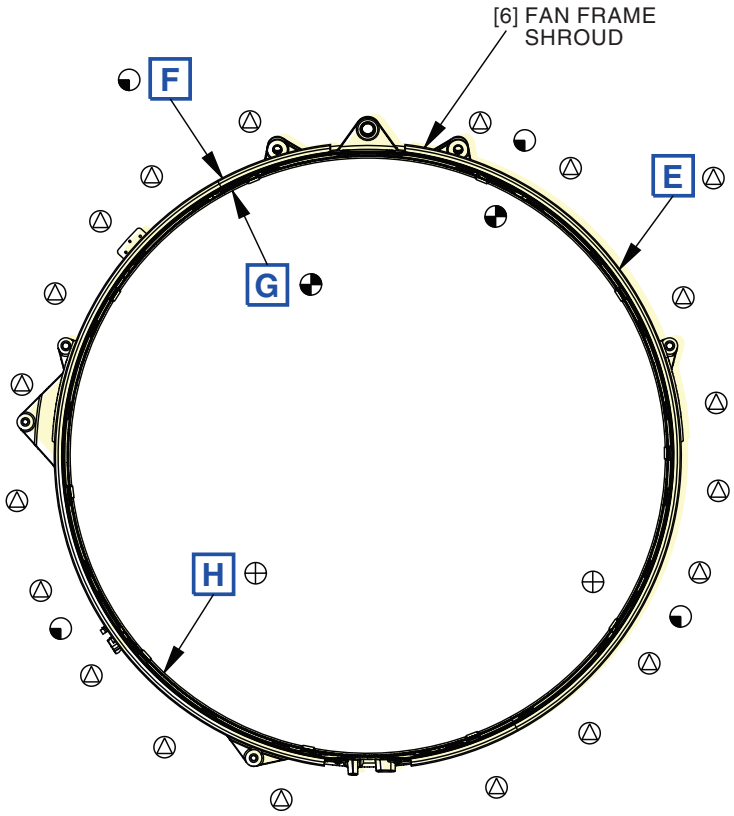
LEGEND:

MATING FACES OF THE BOSS FOR EAI, FUEL TUBE, FIREWALL AND KIT ENGINE SUPPORTS

ICNLEAP1B722300D5882800028A00101
2565200 S0000612639_V1

Fan Frame Shroud Inspection
Figure 608/72-23-00-990-808-G00 (Sheet 1 of 7)

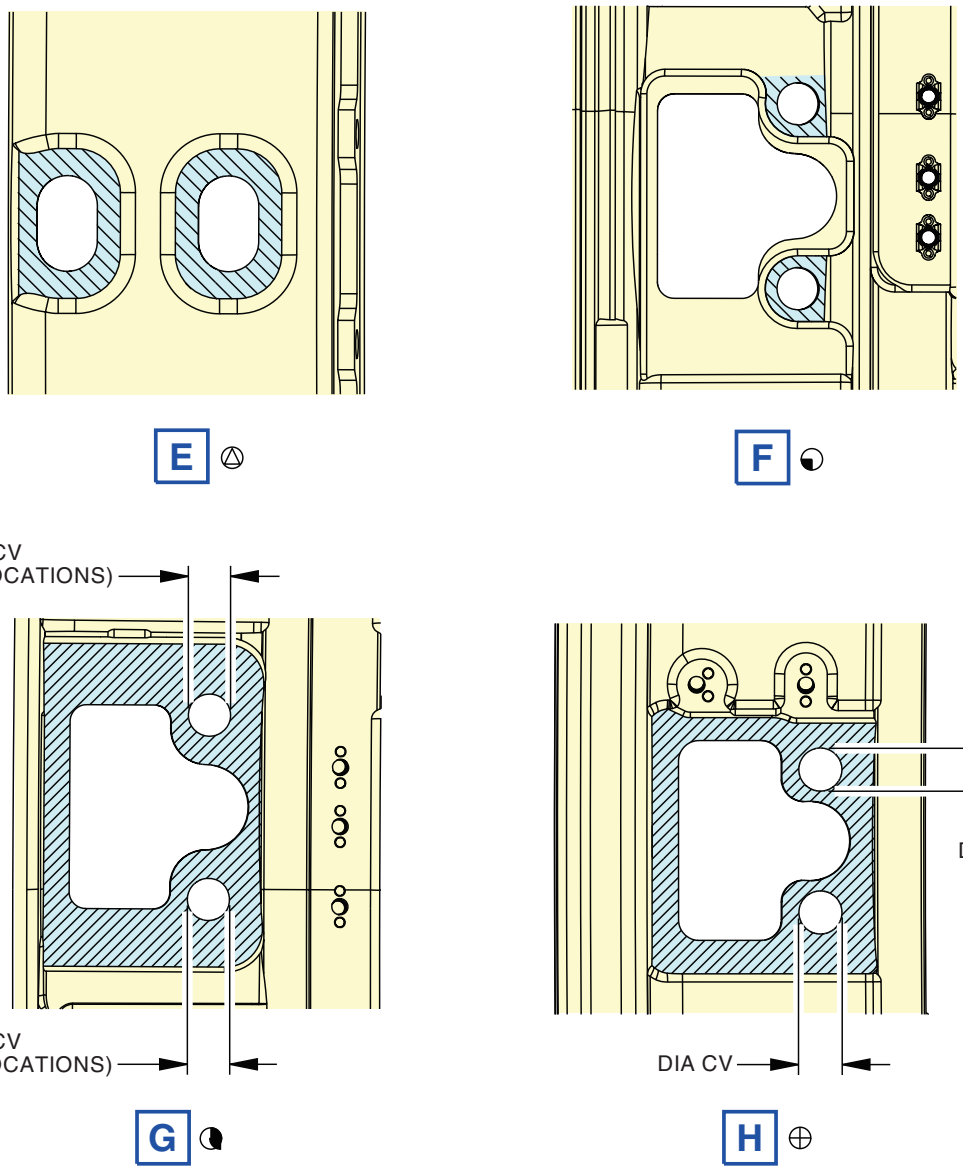
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800029A00101
2565204 S0000612640_V1

Fan Frame Shroud Inspection
Figure 608/72-23-00-990-808-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

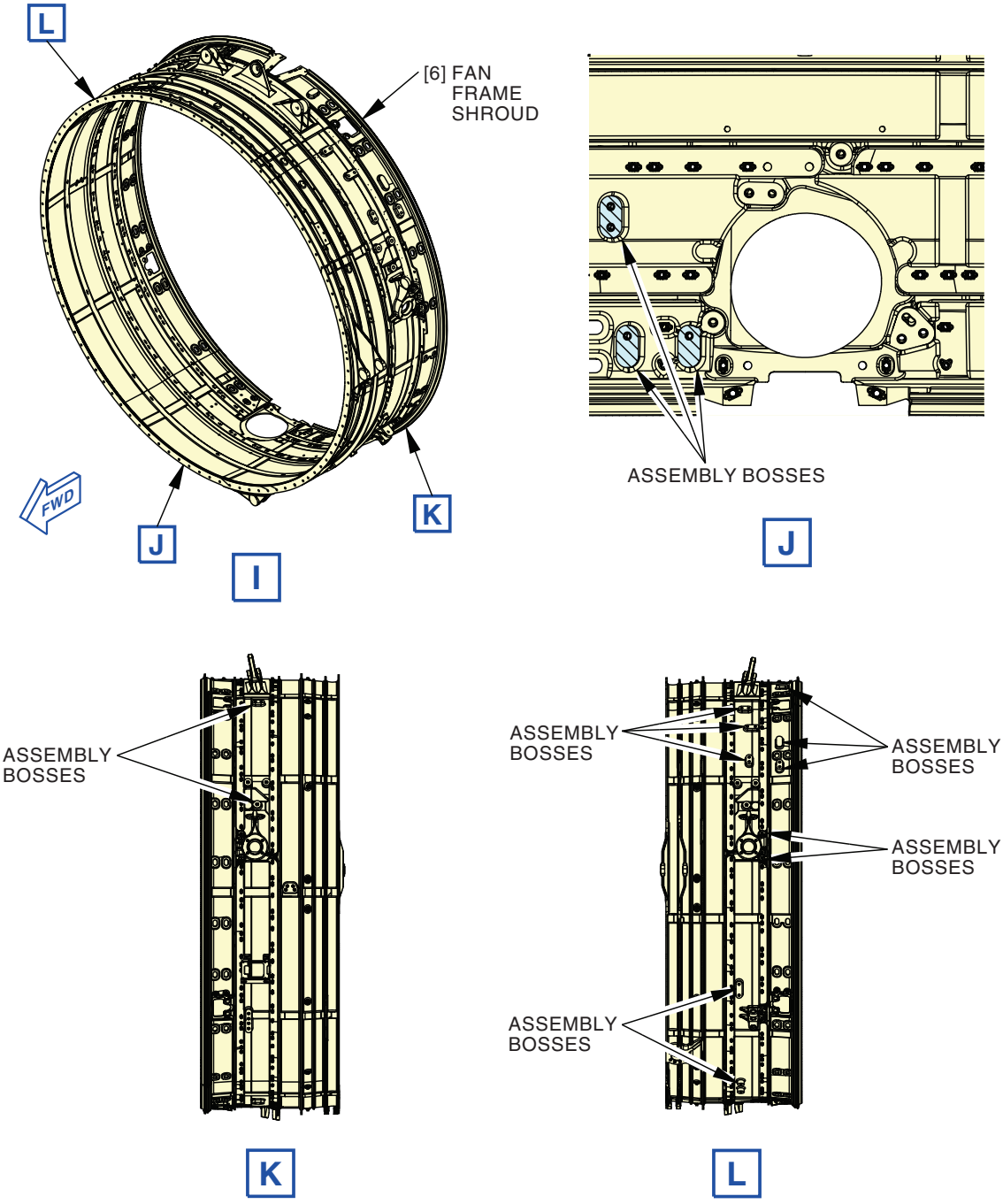
MATING FACES OF SACOC BOSSES

SACOC SEALING SURFACES

ICNLEAP1B722300D5882800030A00101
2565206 S0000612641_V1

Fan Frame Shroud Inspection
Figure 608/72-23-00-990-808-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

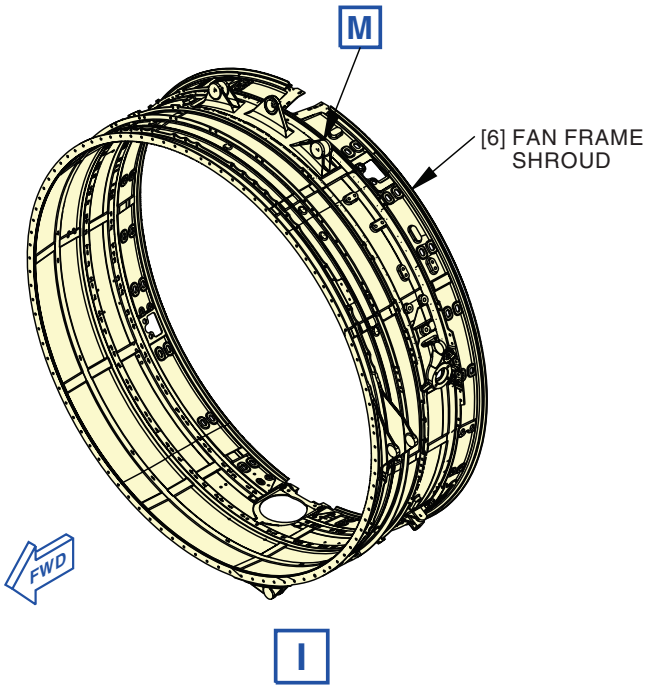
MATING FACES OF ASSEMBLY BOSSES

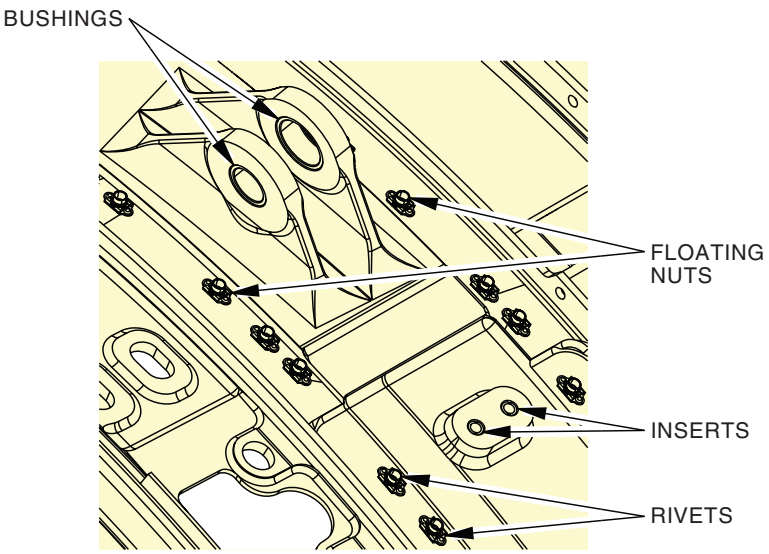
ICNLEAP1B722300D5882800031A00101

2565214 S0000612642_V1

Fan Frame Shroud Inspection
Figure 608/72-23-00-990-808-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





TYPICAL VIEW

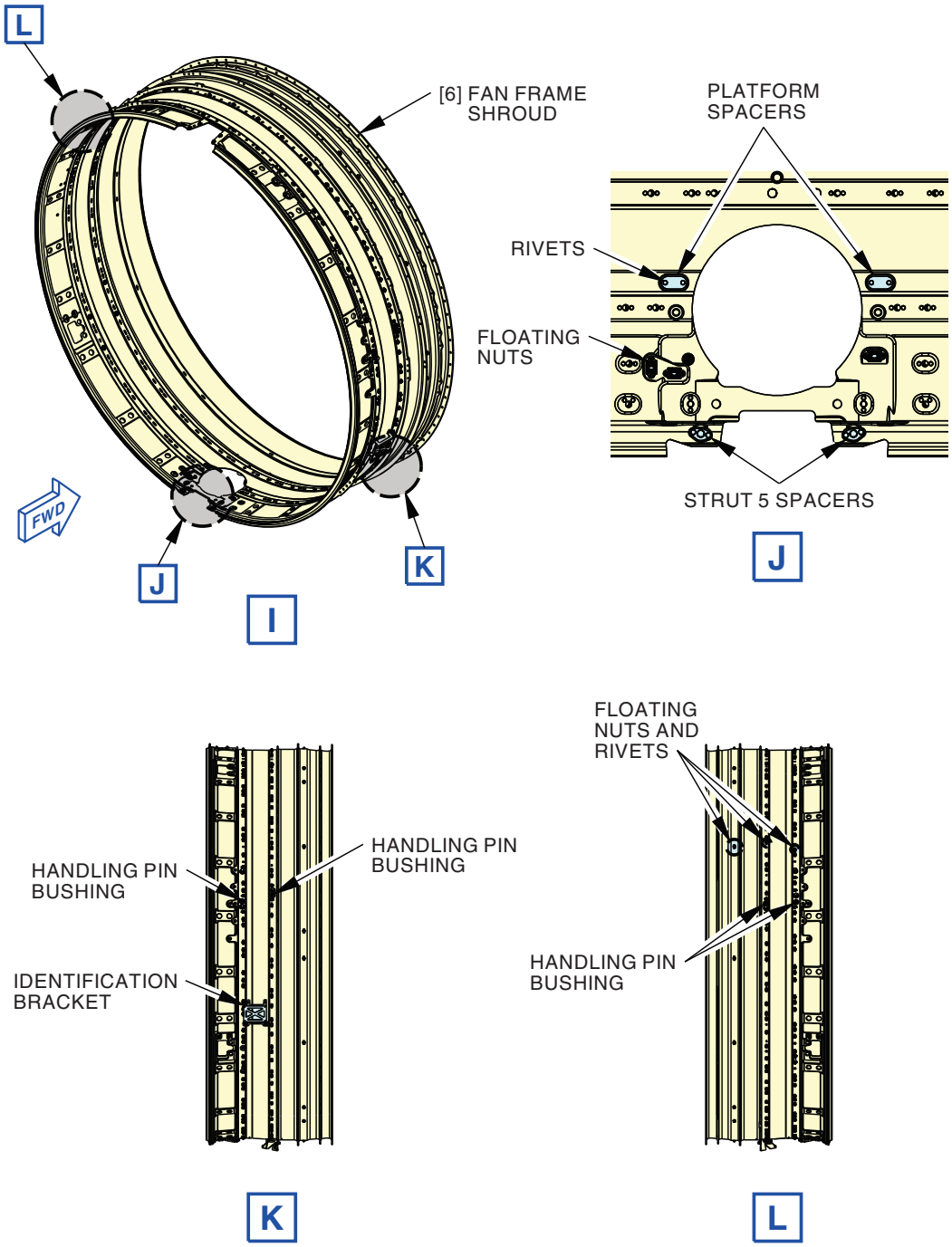
M

ICNLEAP1B722300D5882800032A00101

2565235 S0000612643_V1

Fan Frame Shroud Inspection
Figure 608/72-23-00-990-808-G00 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

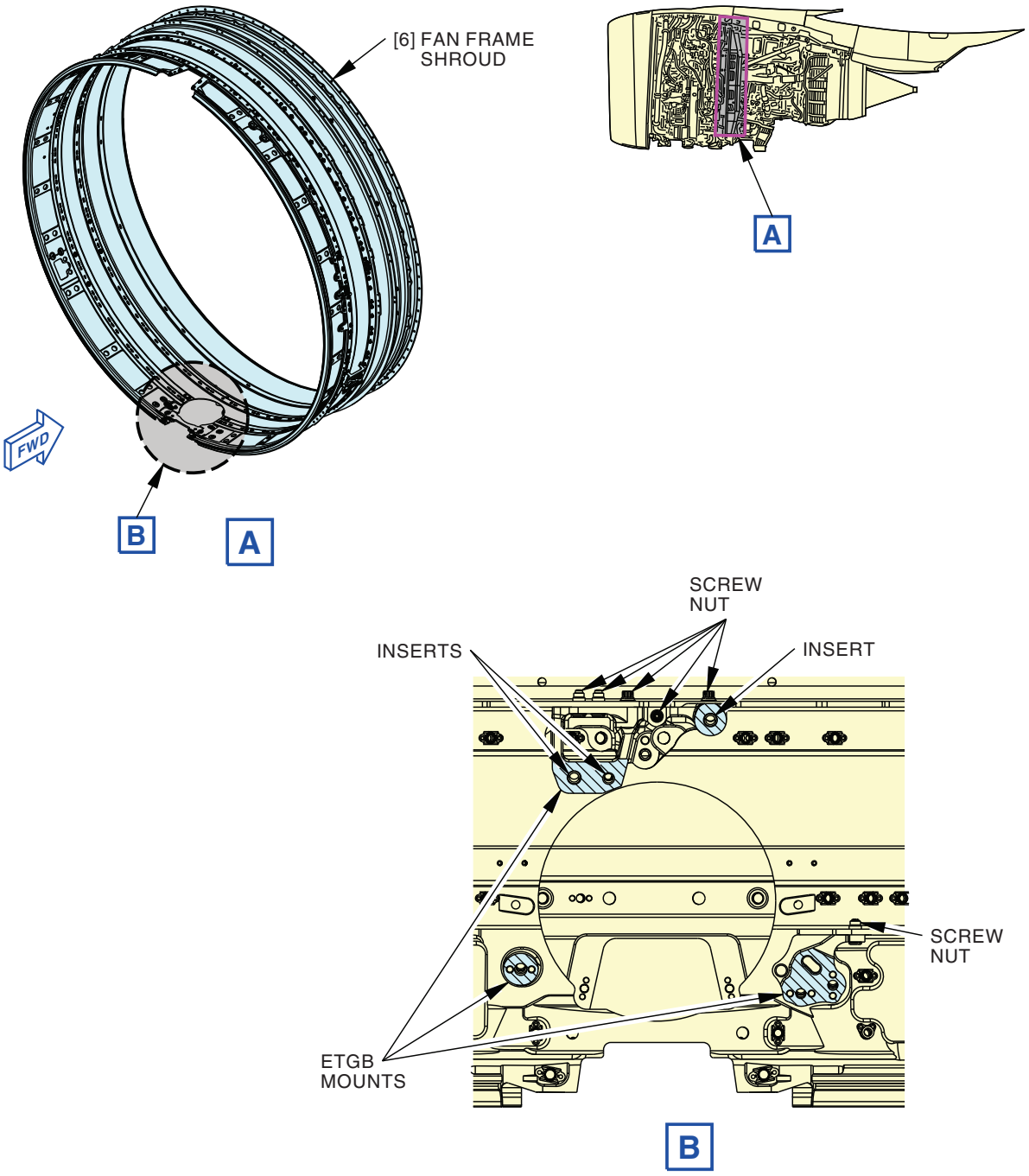


ICN LEAP 1B 722300 D 58828 00089 A001 01

2841218 S0000663019_V1

Fan Frame Shroud Inspection
Figure 608/72-23-00-990-808-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

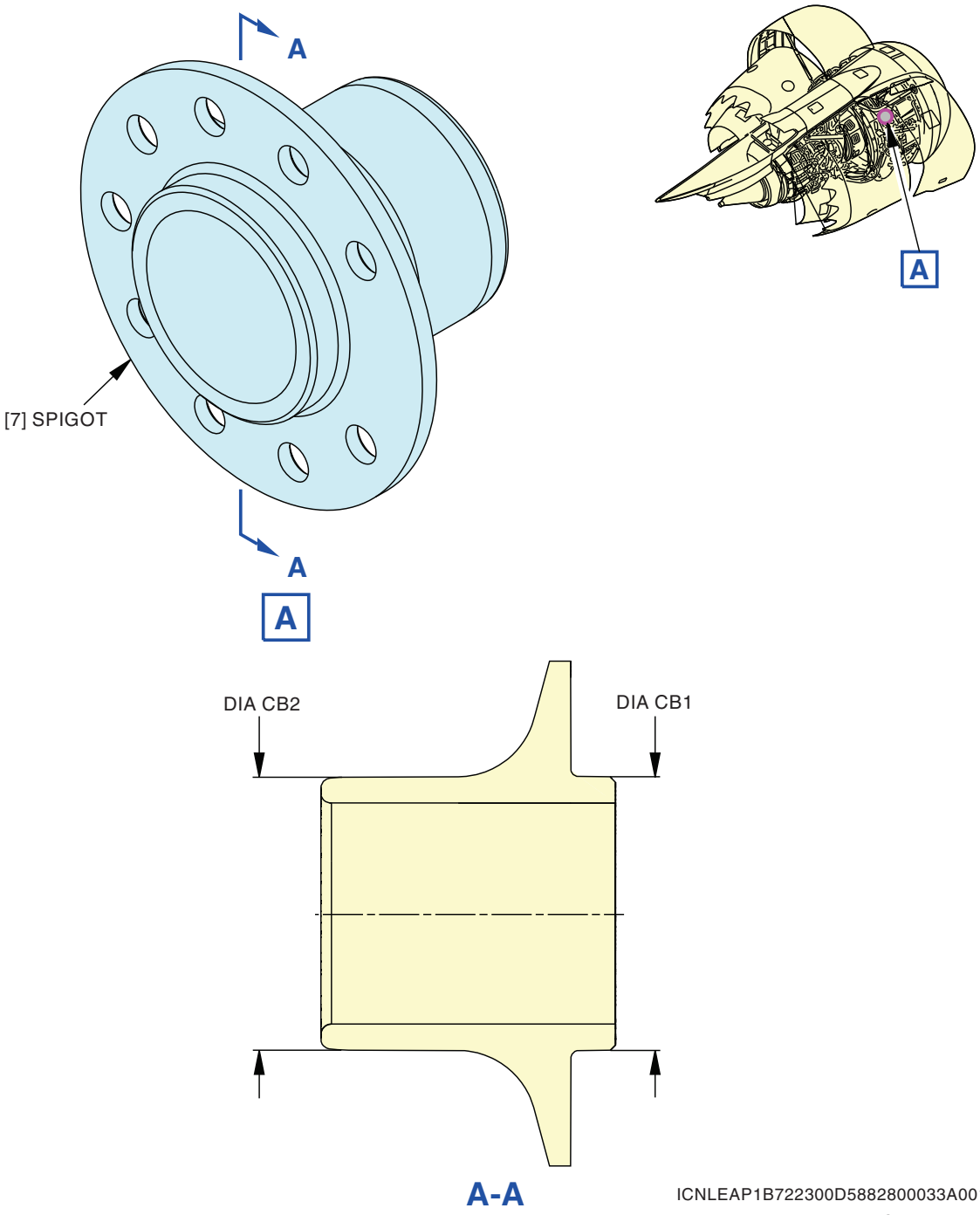
ETGB MOUNTS MATING FACES

ICN LEAP 1B 722300 D 58828 00089 A001 01

2841234 S0000663020_V1

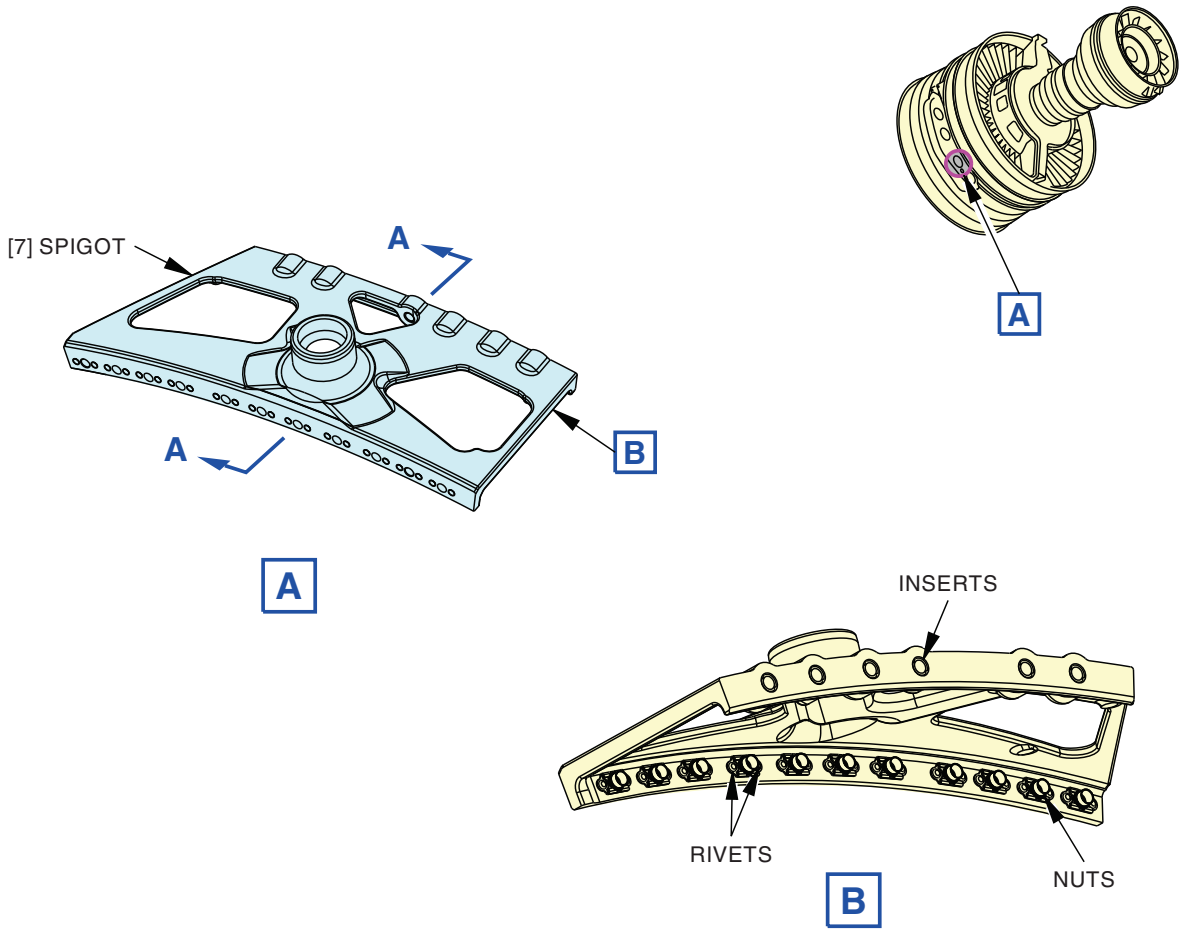
Fan Frame Shroud Inspection
Figure 608/72-23-00-990-808-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Spigot Inspection
Figure 609/72-23-00-990-809-G00 (Sheet 1 of 2)

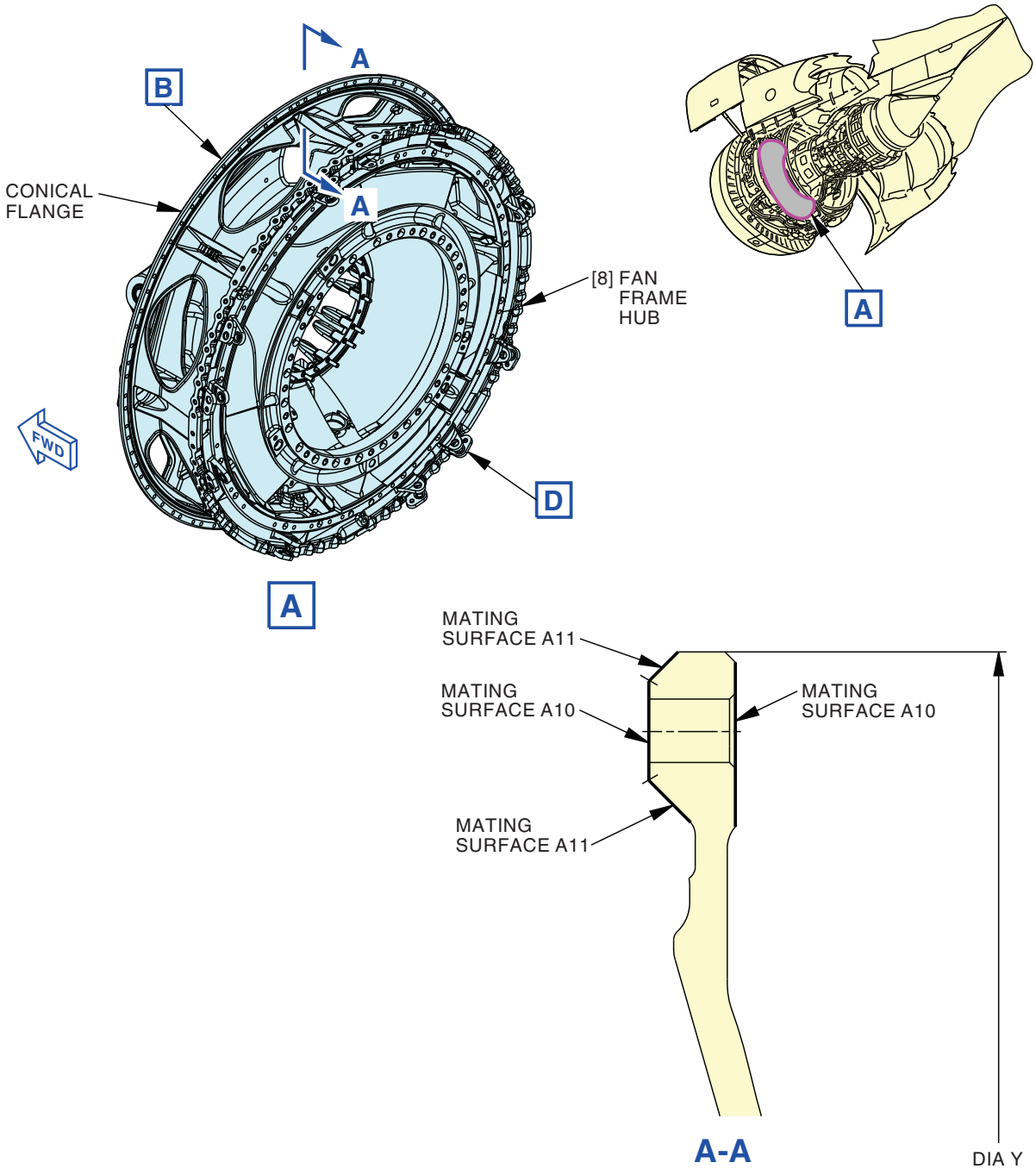
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN LEAP 1B 722300 D 58828 00090 A001 01
2841404 S0000663021_V1

Spigot Inspection
Figure 609/72-23-00-990-809-G00 (Sheet 2 of 2)

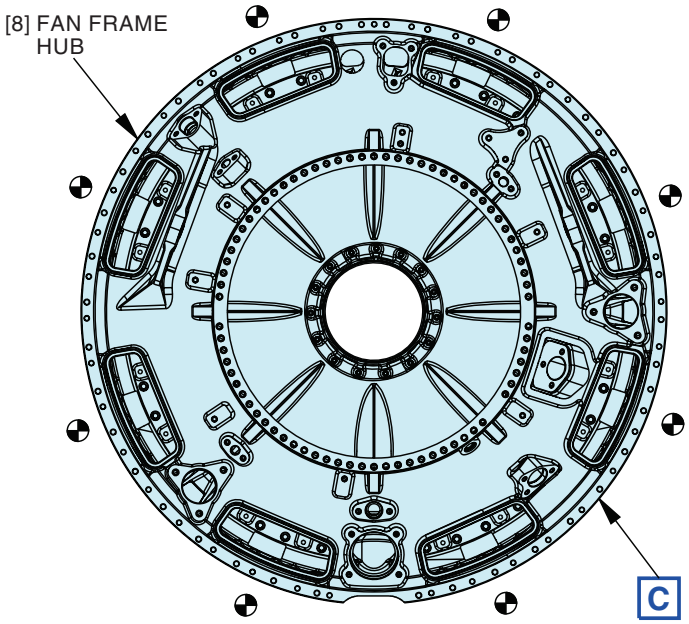
ECCN 9E991 BOEING PROPRIETARY - See title page for details

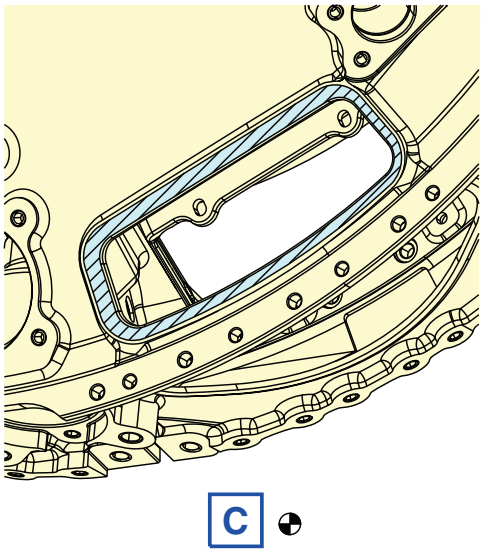


ICNLEAP1B722300D5882800034A00101
2565249 S0000612645_V1

Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 1 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





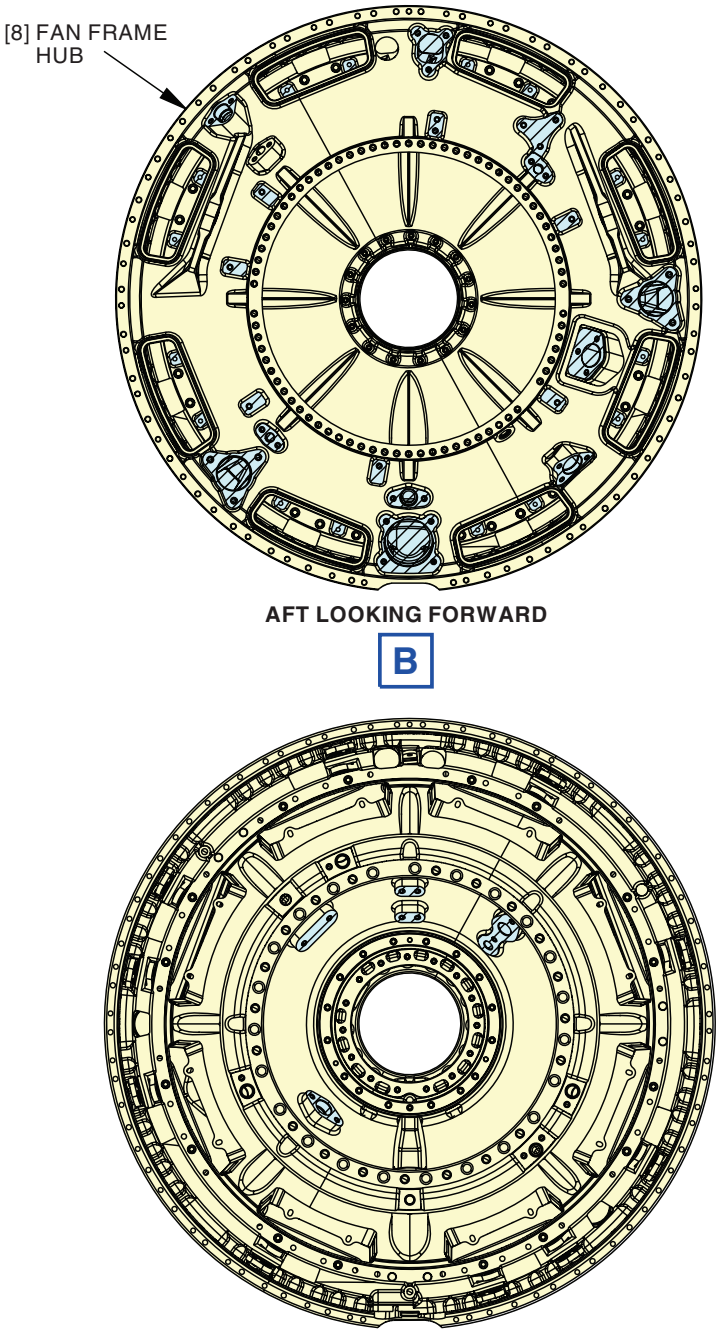
LEGEND:

SEALING SURFACE

ICNLEAP1B722300D5882800035A00101
2565259 S0000612646_V1

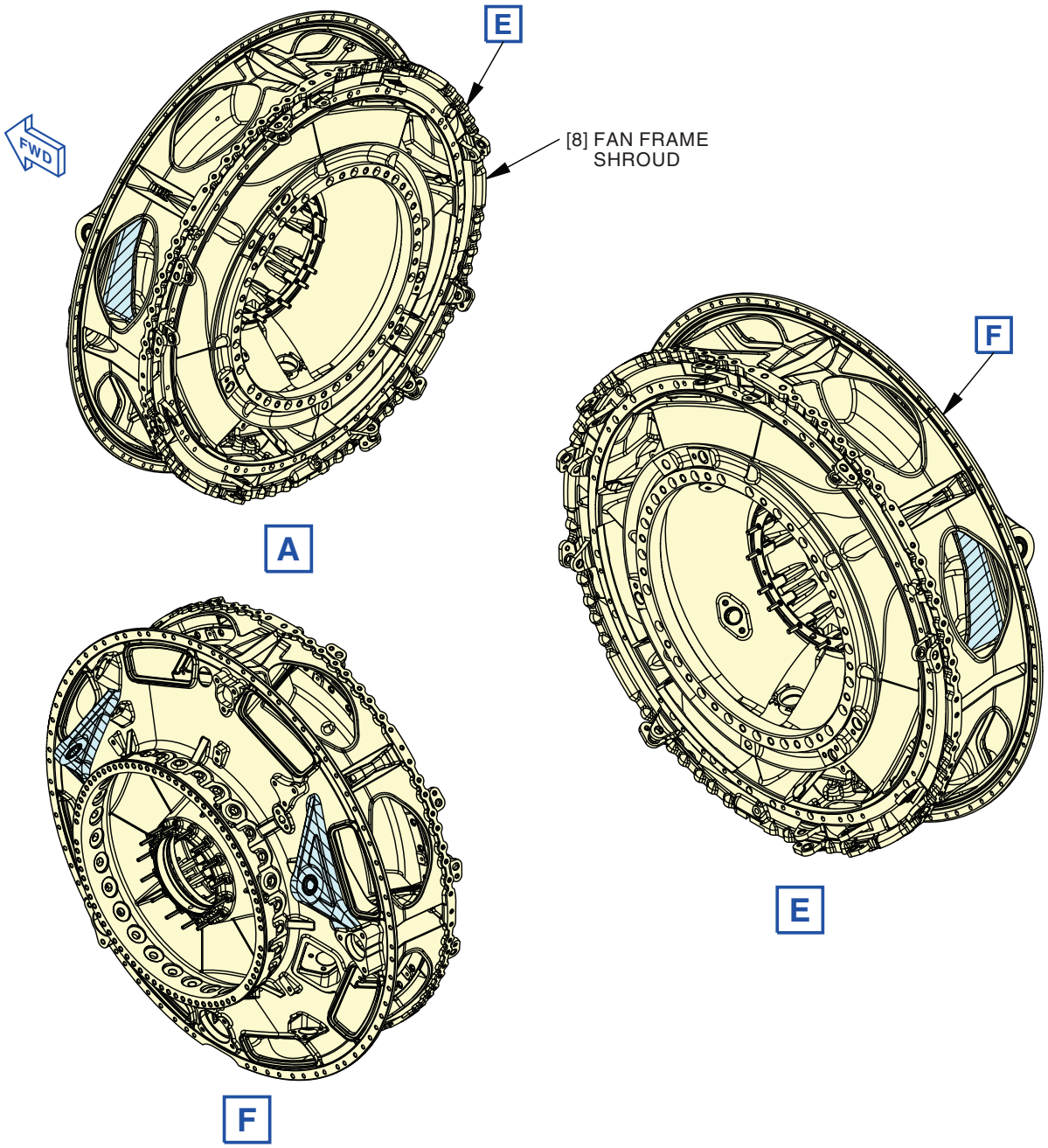
Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 2 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 3 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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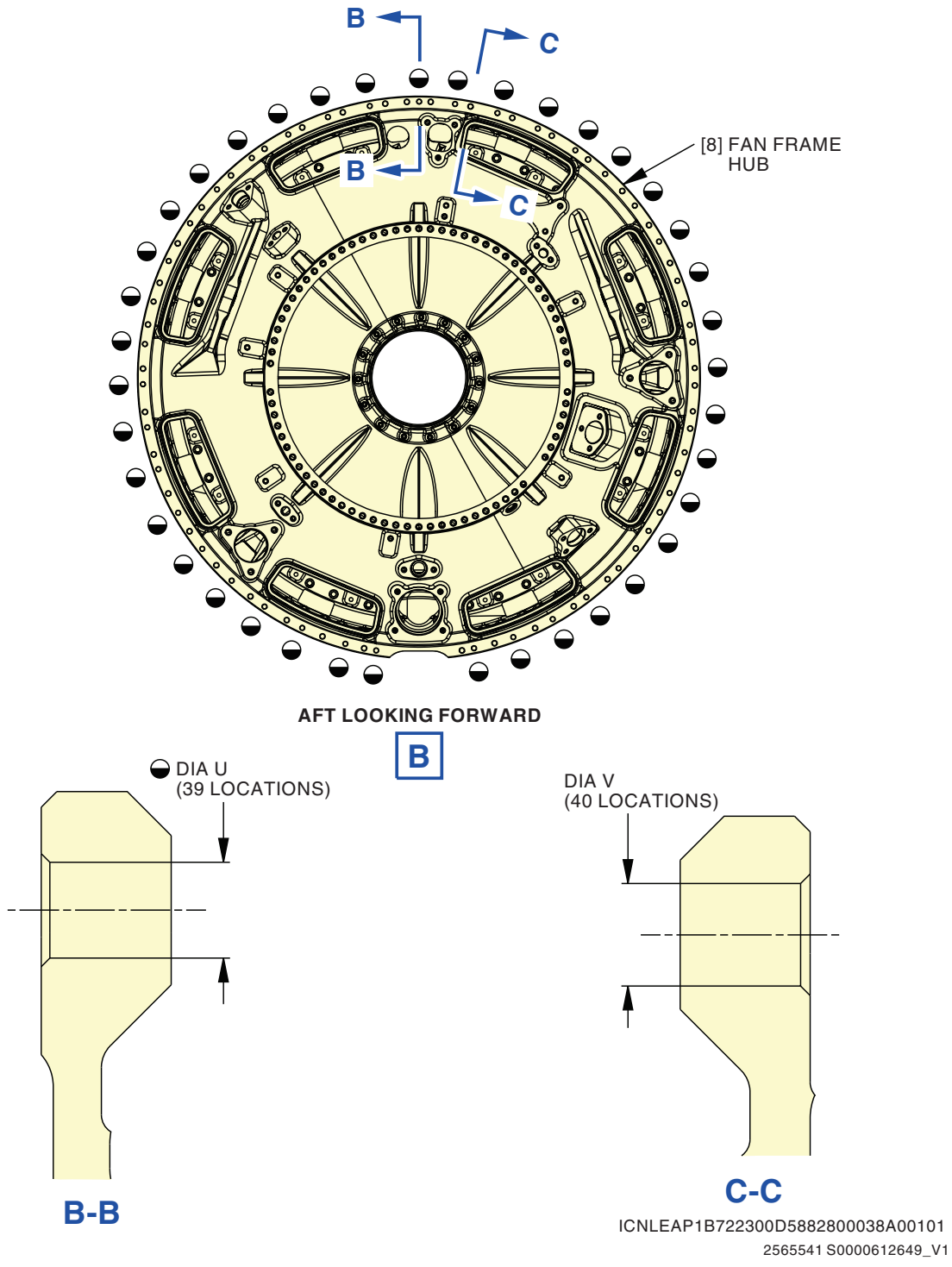
2565320 S0000612648_V1

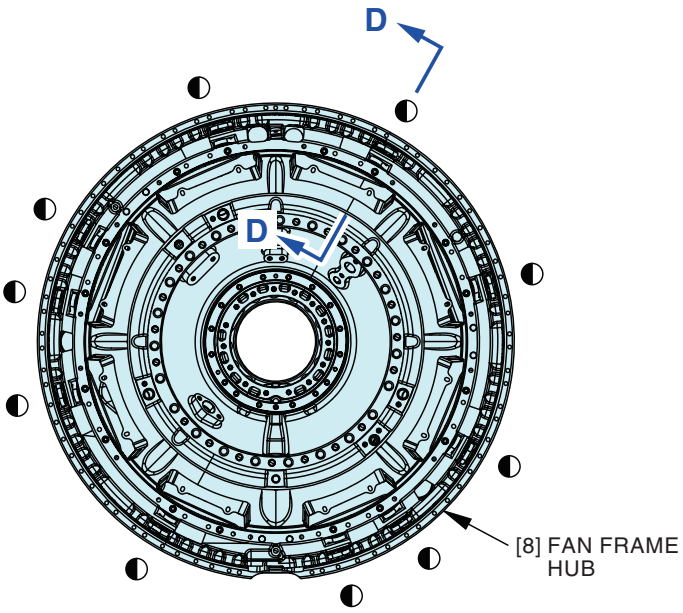
Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 4 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



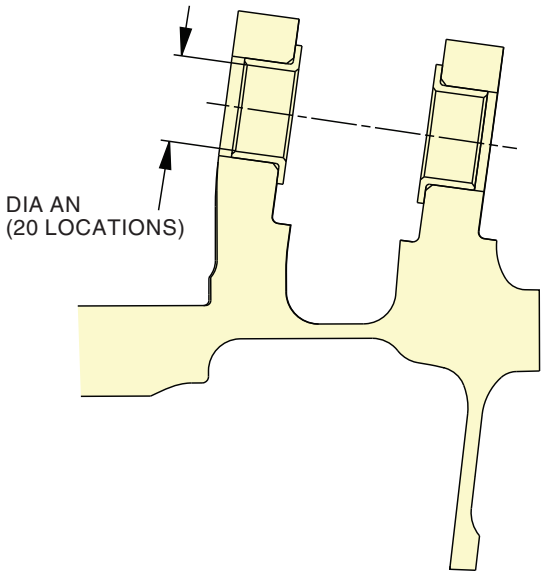
Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 5 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD LOOKING AFT

D



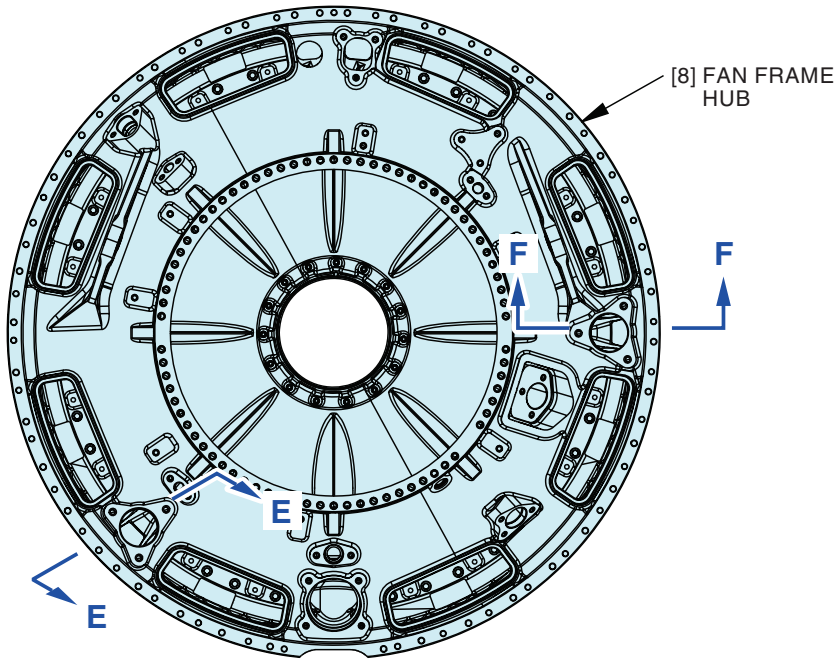
D-D

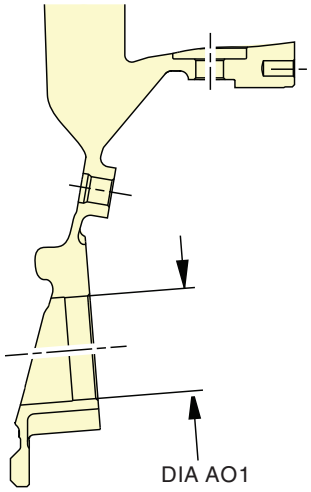
ICNLEAP1B722300D5882800039A00101

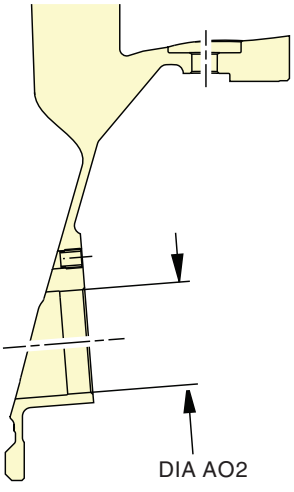
2565543 S0000612650_V1

Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 6 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details







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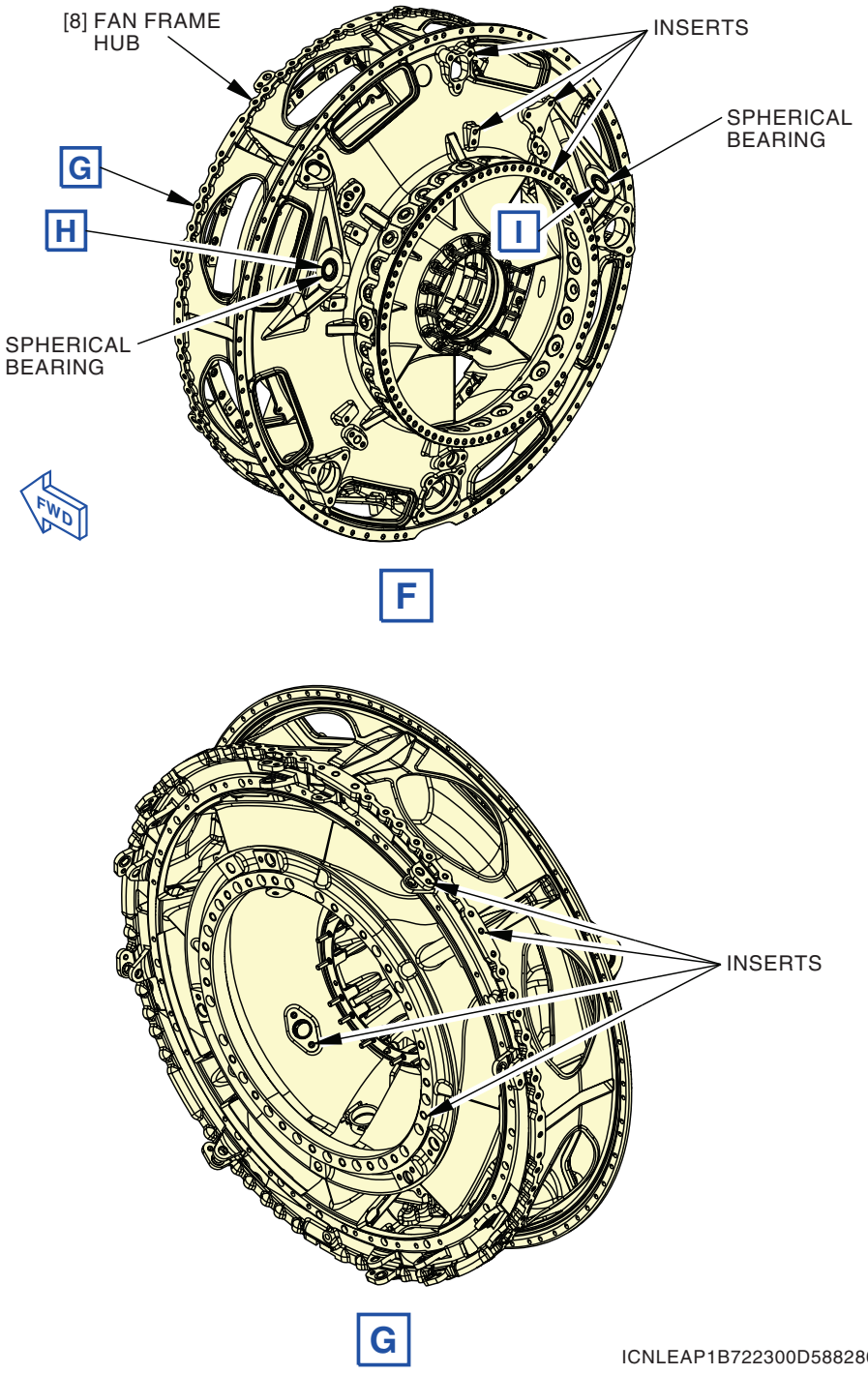
2565549 S0000612651_V1

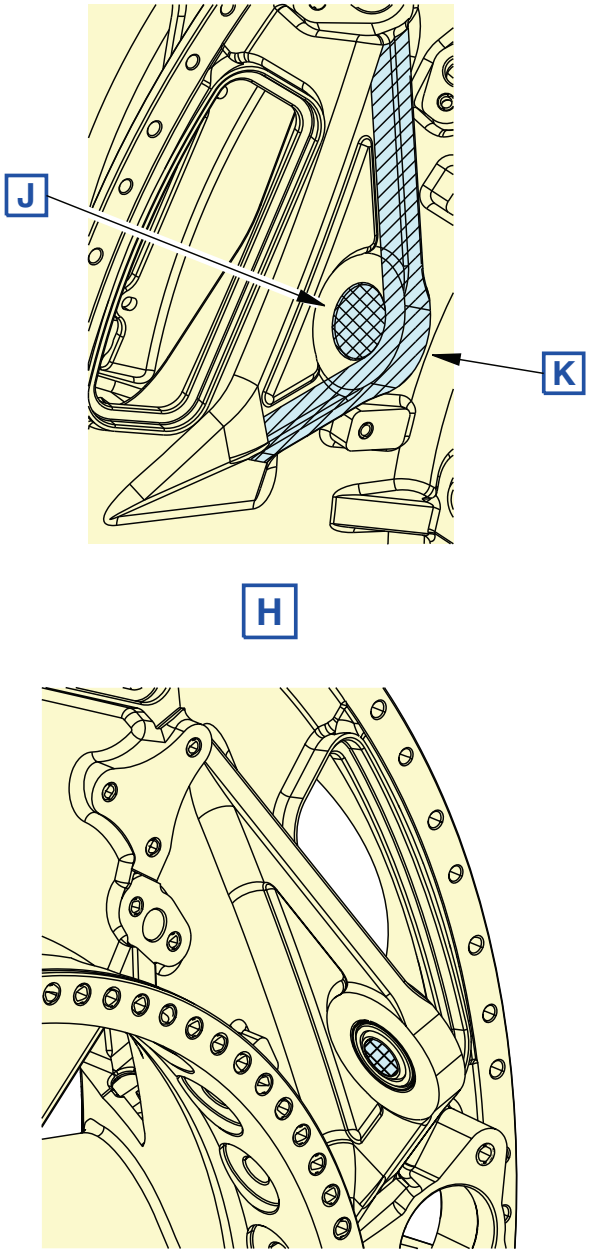
Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 7 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 8 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

EXTERNAL SURFACES OF THE SPHERICAL BEARING BALLS

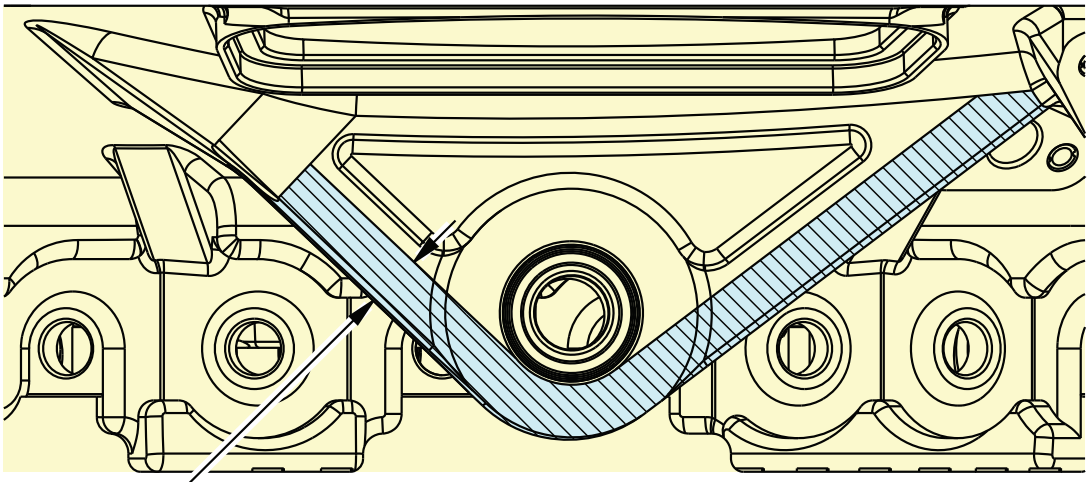
ZONE OF THE CRITERIA ON THRUST LINK CLEVIS

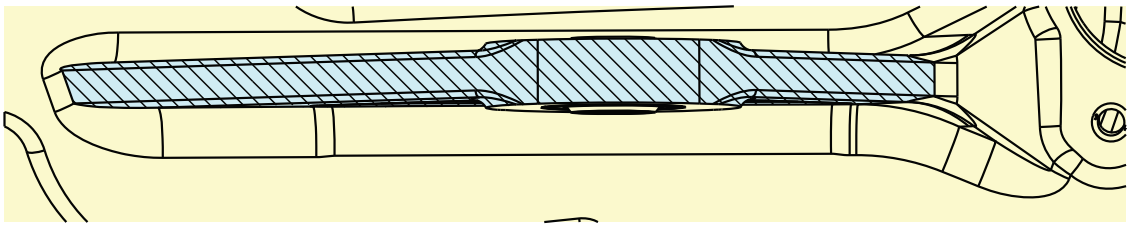
ICNLEAP1B722300D5882800081A00201

2565555 S0000612653_V2

Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 9 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

ZONE OF THE CRITERIA ON THRUST LINK CLEVIS

NOTE:

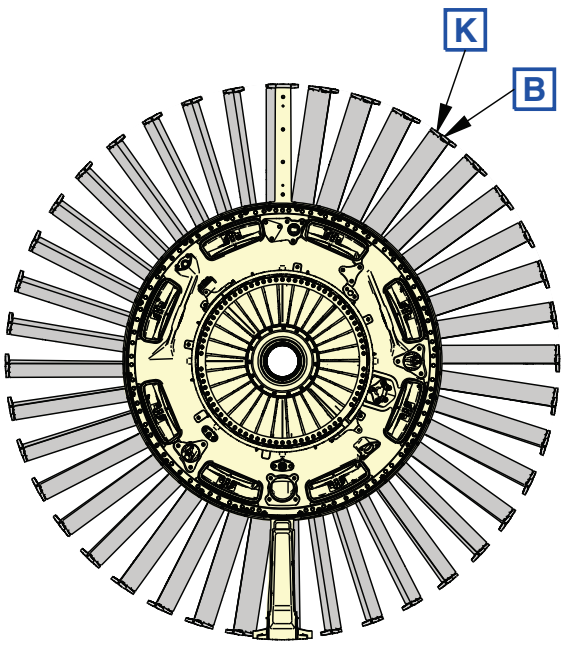
RIGHT HAND IS SHOWN, LEFT HAND IS SYMMETRICAL

ICNLEAP1B722300D5882800126A00101

3083296 S0000838297_V1

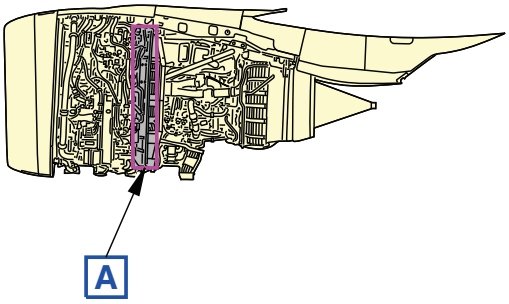
Fan Frame Hub Inspection
Figure 610/72-23-00-990-810-G00 (Sheet 10 of 10)

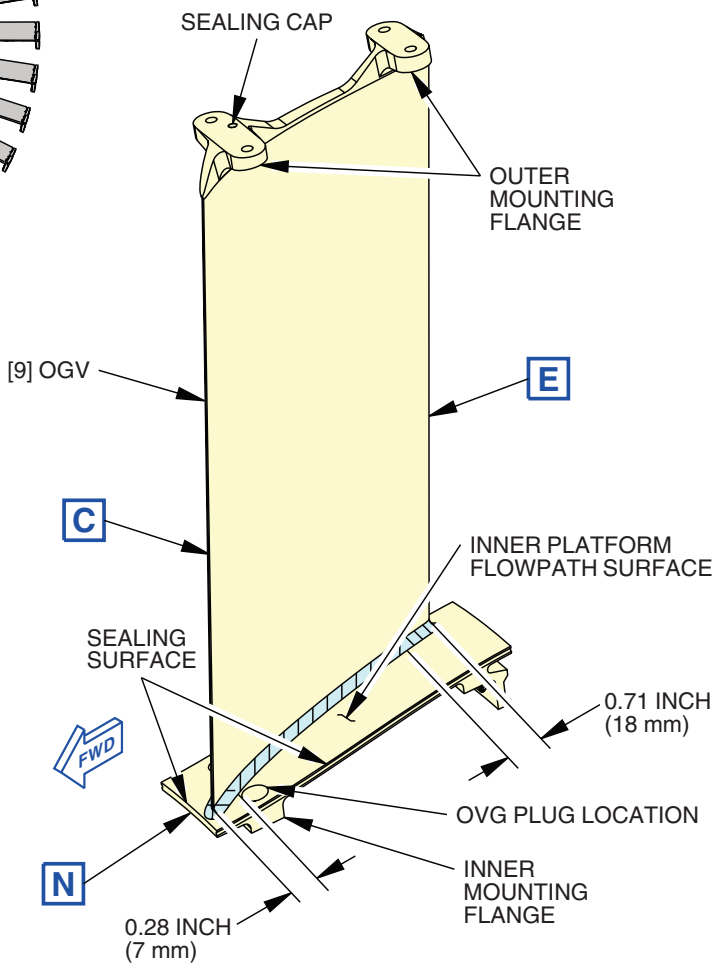
ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE FORWARD DIRECTION)

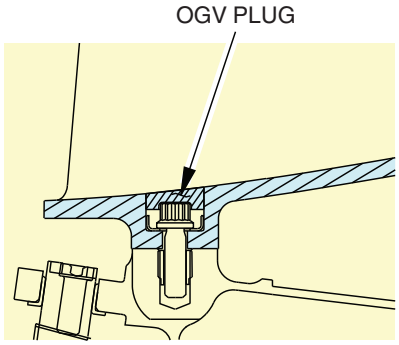
A





TYPICAL VIEW OF AN OGV

B



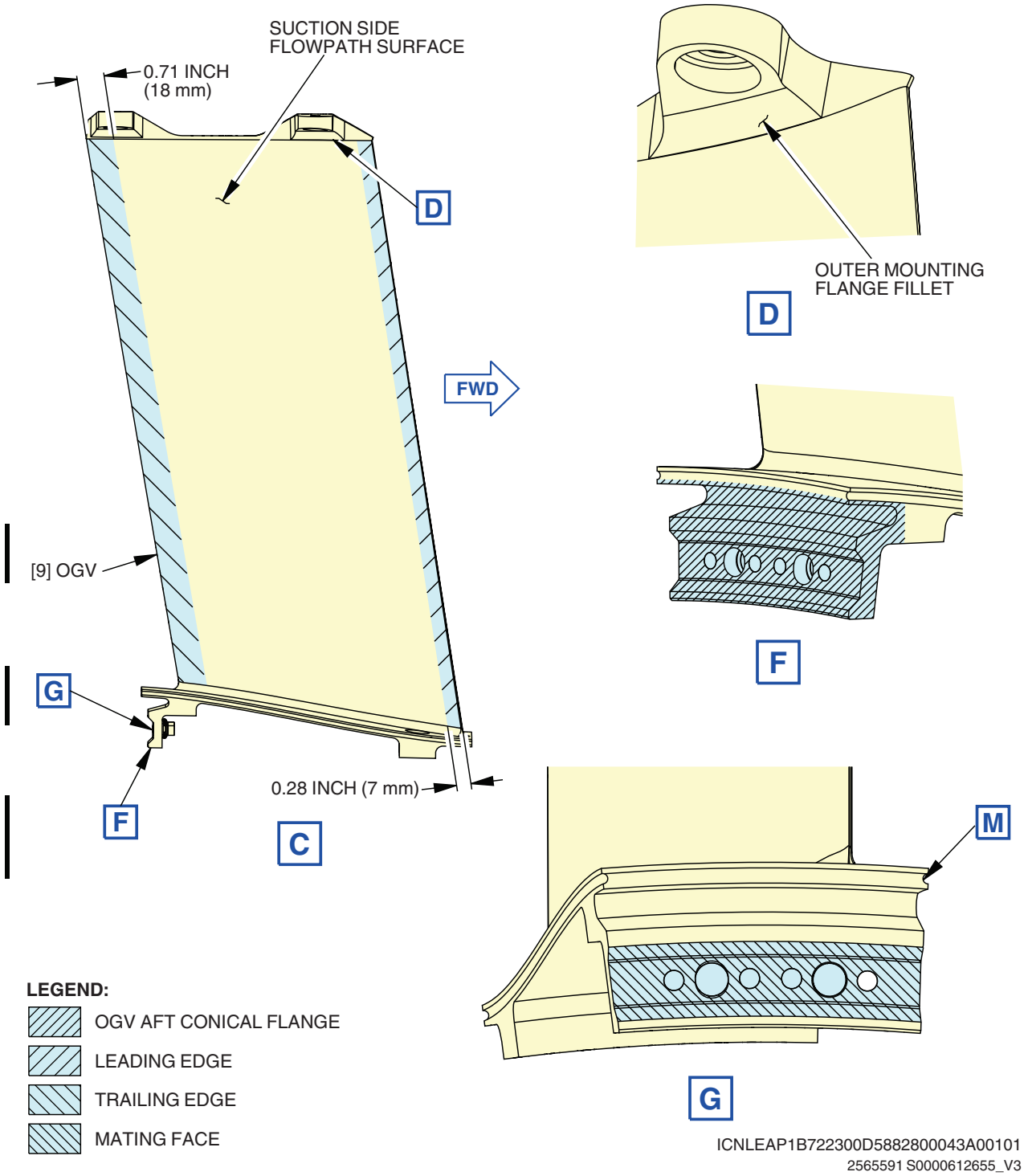
LEGEND:

- INNER PLATFORM FILLET
- LEADING EDGE FILLET
- TRAILING EDGE FILLET

ICNLEAP1B722300D5882800042A00101
2946493 S0000723200_V3

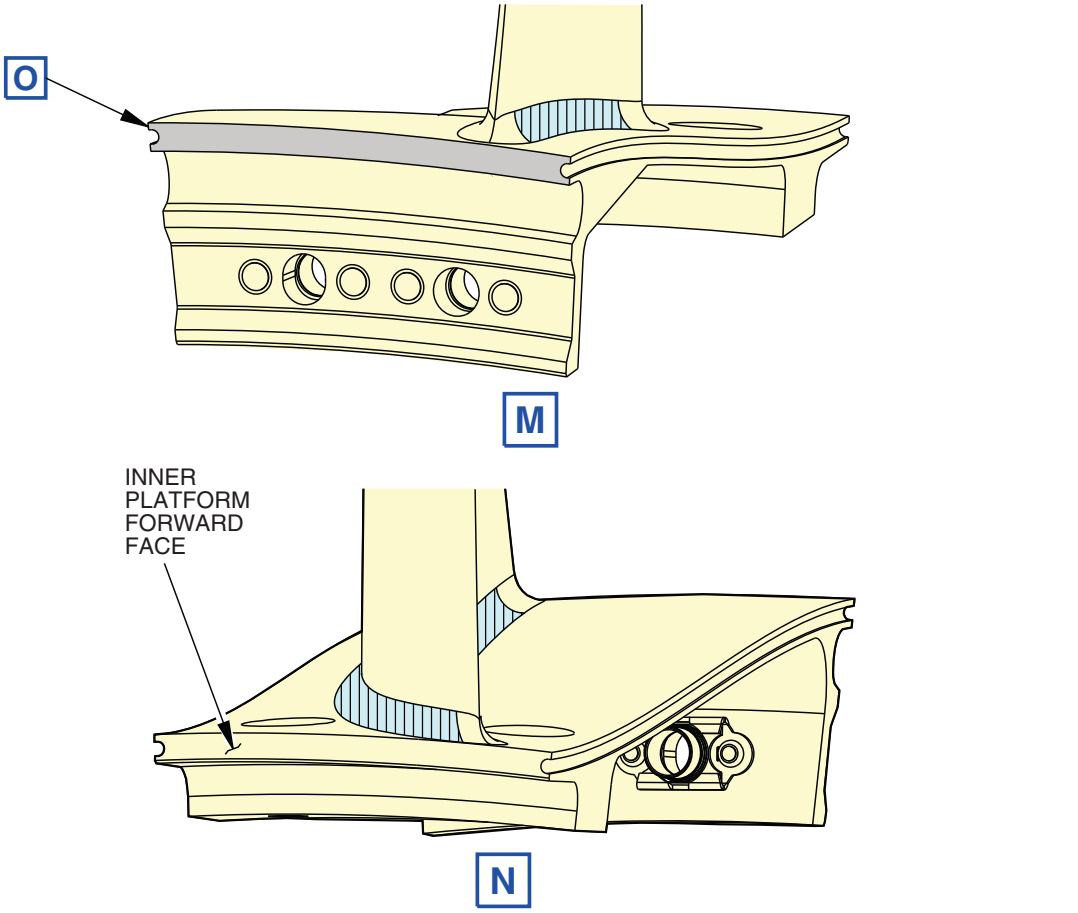
Outlet Guide Vane Inspection
Figure 611/72-23-00-990-811-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outlet Guide Vane Inspection
Figure 611/72-23-00-990-811-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW OF INNER PLATFORM AFT FACE

O

LEGEND:

AREA A

INDICATIONS IN THESE AREAS ARE NOT PERMITTED

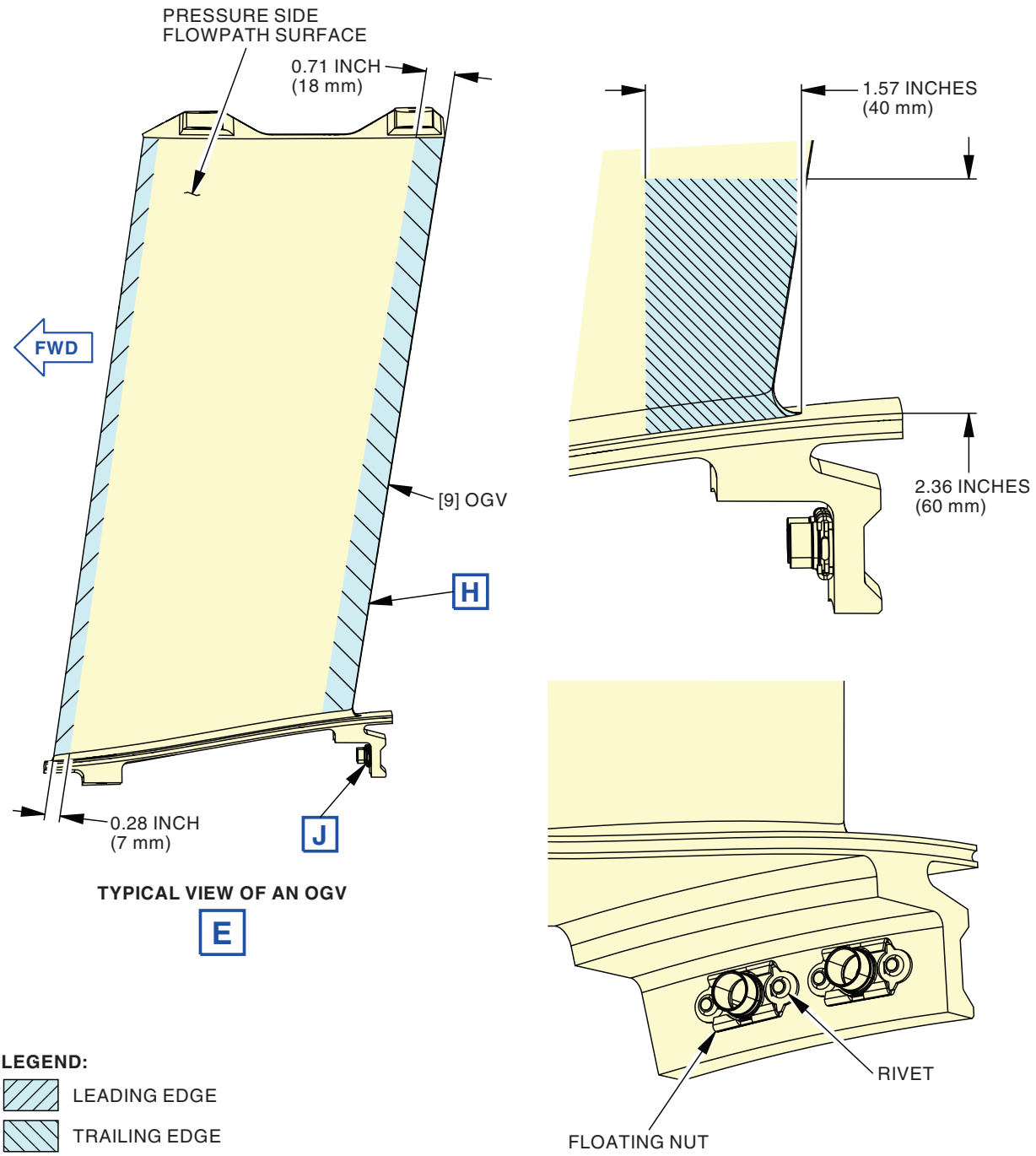
INNER PLATFORM FILLET

ICNLEAP1B722300D5882800138A00101

3123978 S0000857765_V1

Outlet Guide Vane Inspection
Figure 611/72-23-00-990-811-G00 (Sheet 3 of 7)

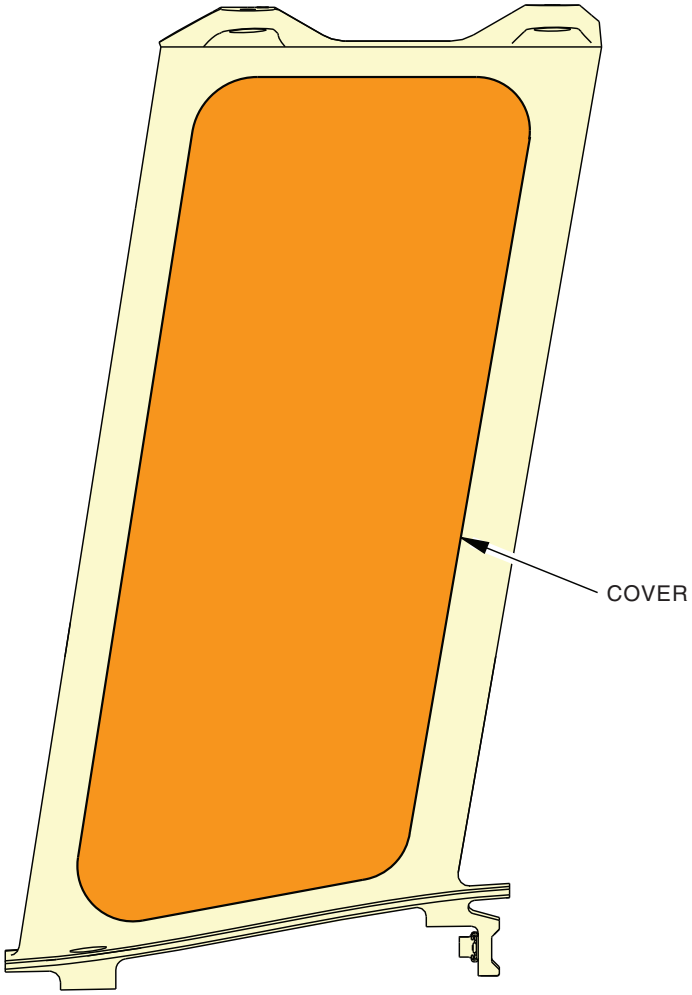
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800044A00101
2565679 S0000612656_V1

Outlet Guide Vane Inspection
Figure 611/72-23-00-990-811-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

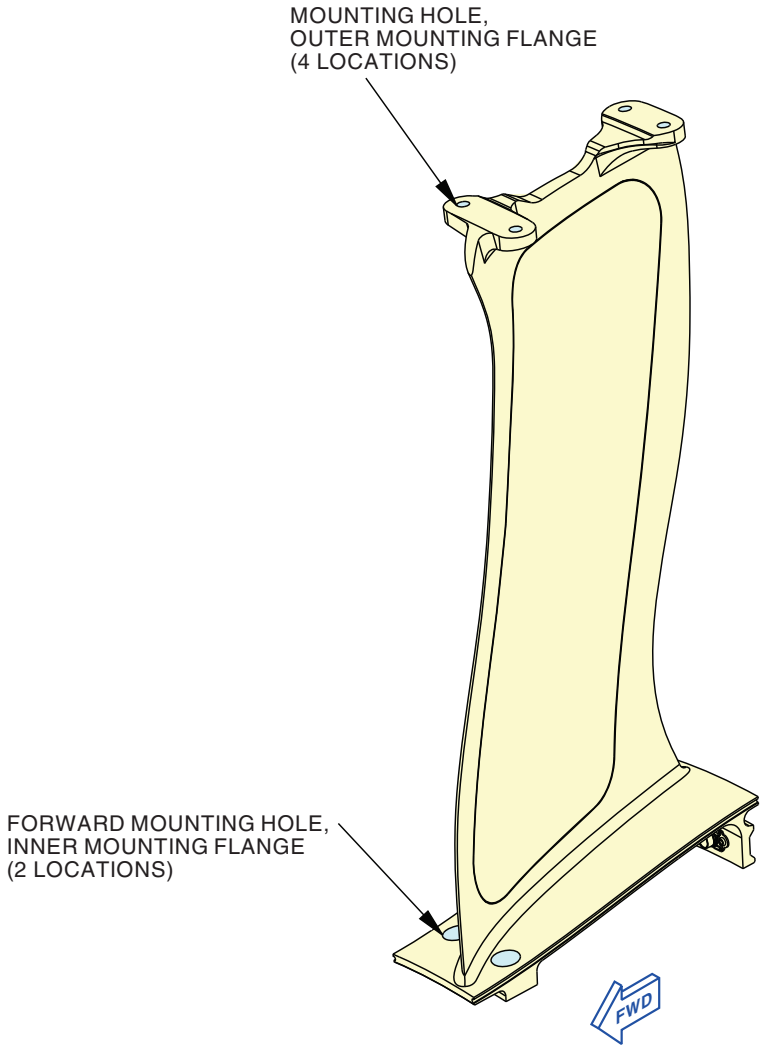


ICNLEAP1B722300D5882800110A00101

2880382 S0000682850_V1

Outlet Guide Vane Inspection
Figure 611/72-23-00-990-811-G00 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

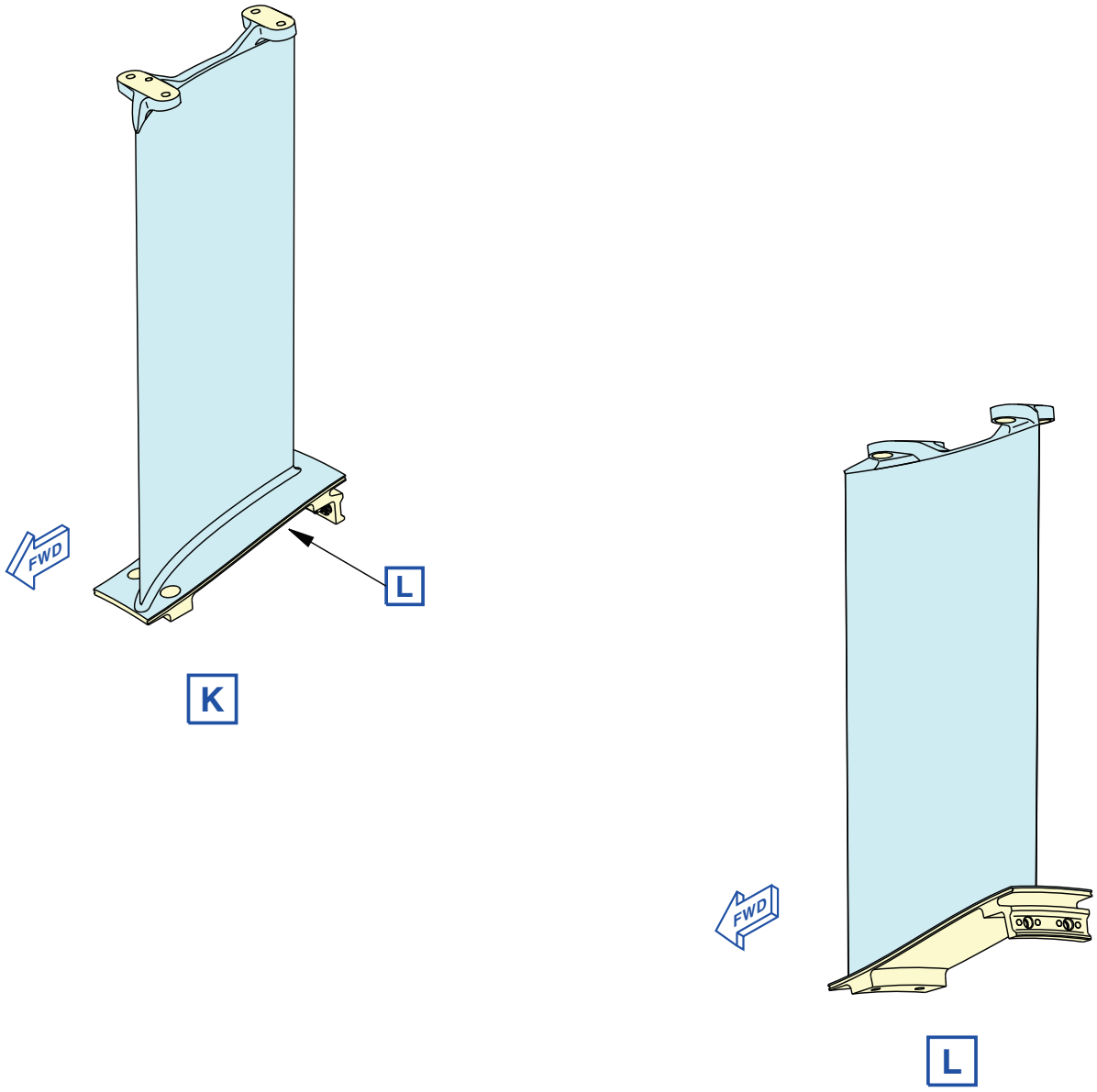


TYPICAL VIEW OF AN OGV

ICNLEAP1B722300D5882800122A00101
3043435 S0000810563_V1

Outlet Guide Vane Inspection
Figure 611/72-23-00-990-811-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

ANTI-CORROSION PAINT (PRIMER)

ANTI-CORROSION PAINT (PRIMER)
+ANTI-EROSION PAINT (MAT BLACK PAINT)

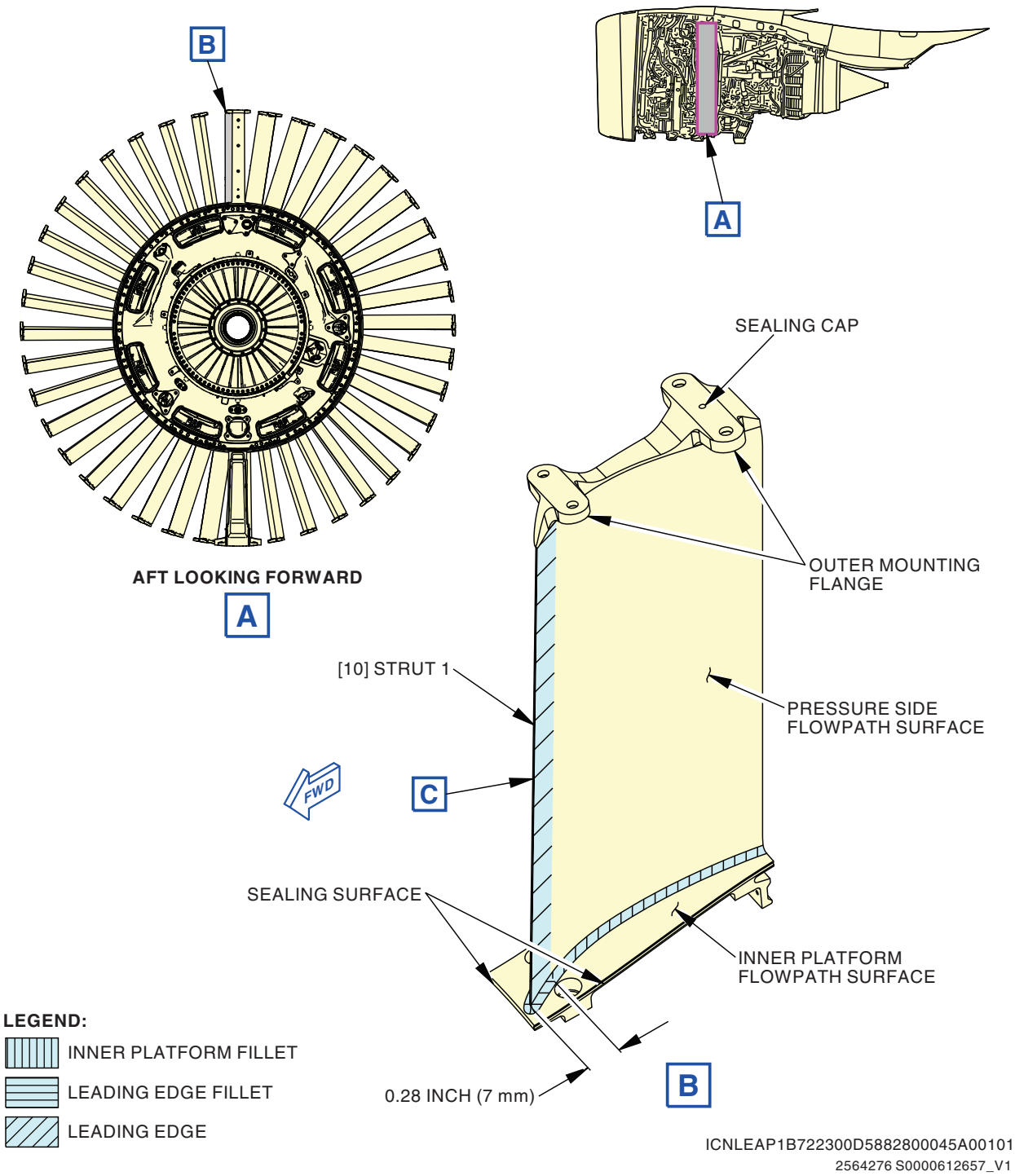
NOTE:

SURFACE PROTECTION AT PRODUCTION (FROM BASE METAL UP TO VISIBLE OUTER SURFACE) FOR ALUMINIUM BLACK-PAINTED OGV

ICNLEAP1B722300D5882800121A00101
3044269 S0000811382_V1

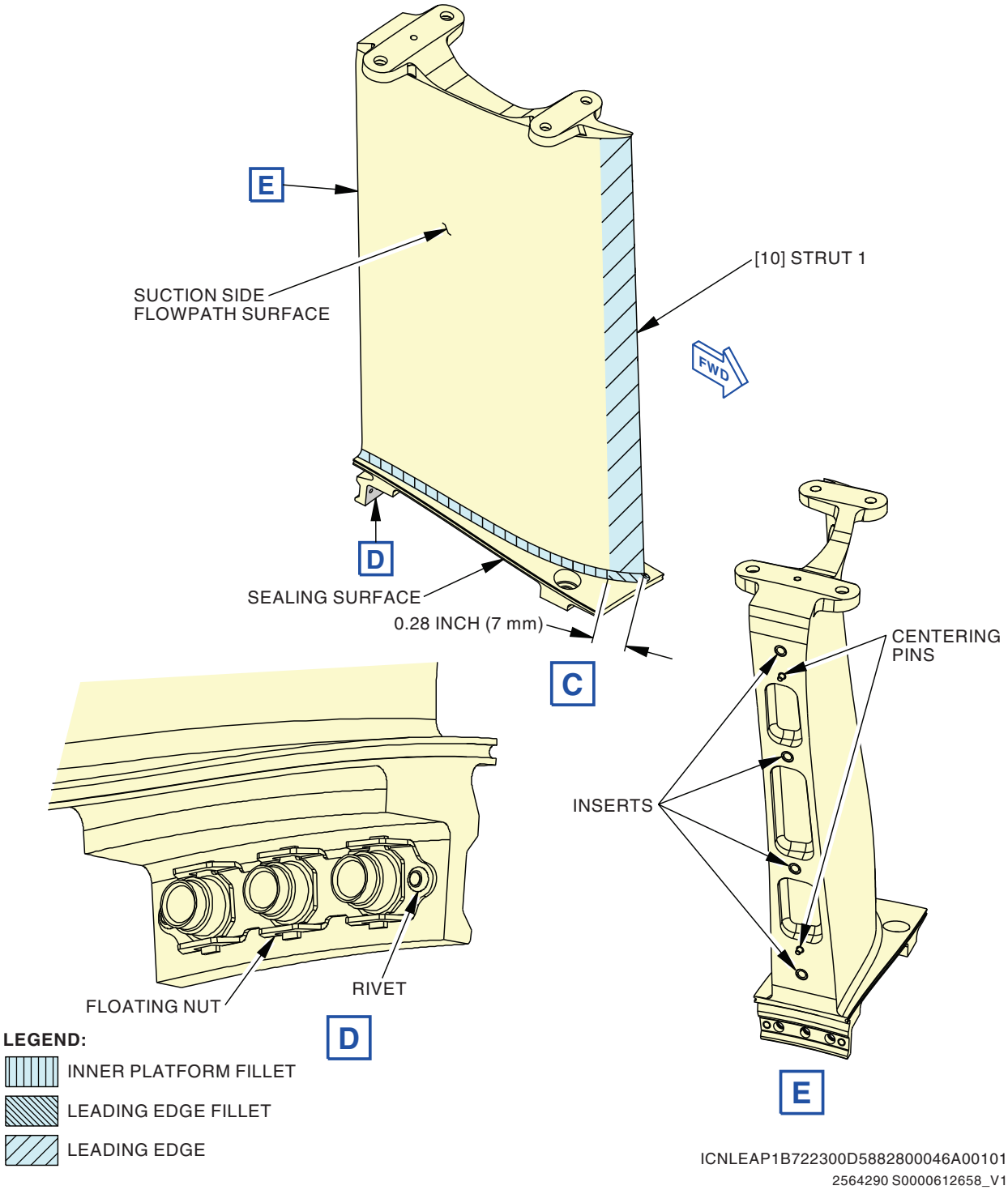
Outlet Guide Vane Inspection
Figure 611/72-23-00-990-811-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



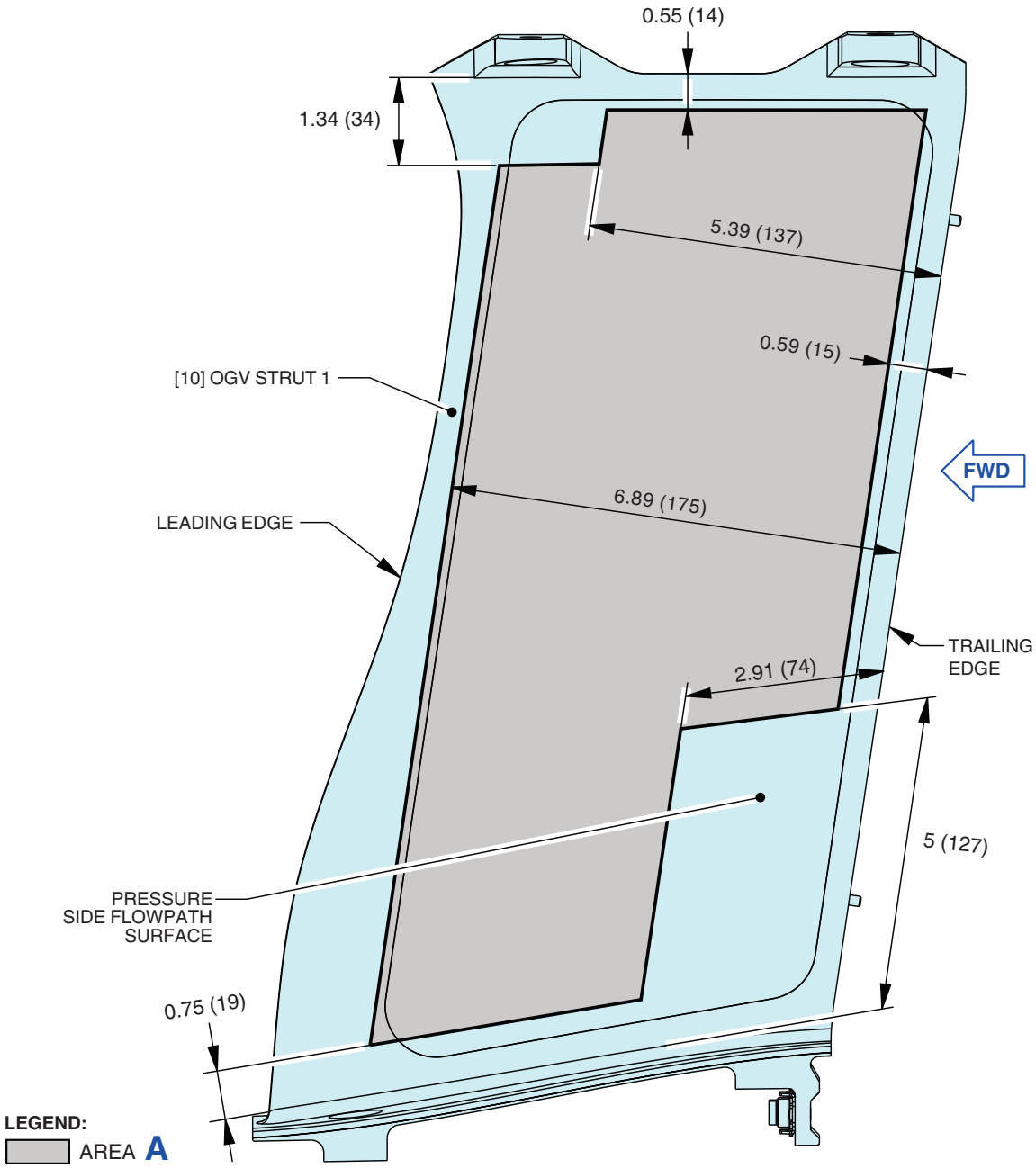
OGV Strut 1 Inspection
Figure 612/72-23-00-990-812-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OGV Strut 1 Inspection
Figure 612/72-23-00-990-812-G00 (Sheet 2 of 3)

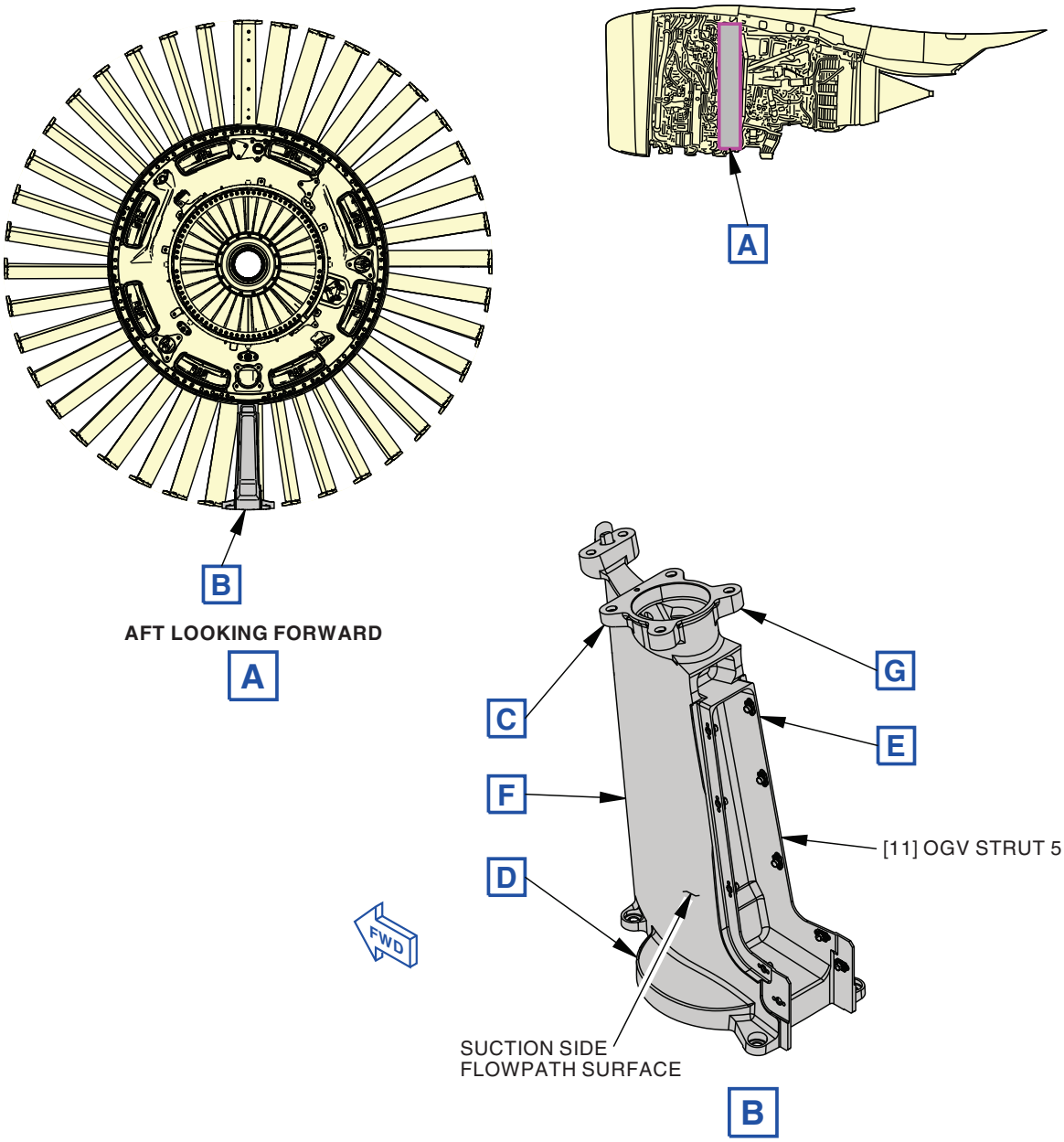
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-722300-D-58828-00132-A-001-01
3113243 S0000851541_V1

OGV Strut 1 Inspection
Figure 612/72-23-00-990-812-G00 (Sheet 3 of 3)

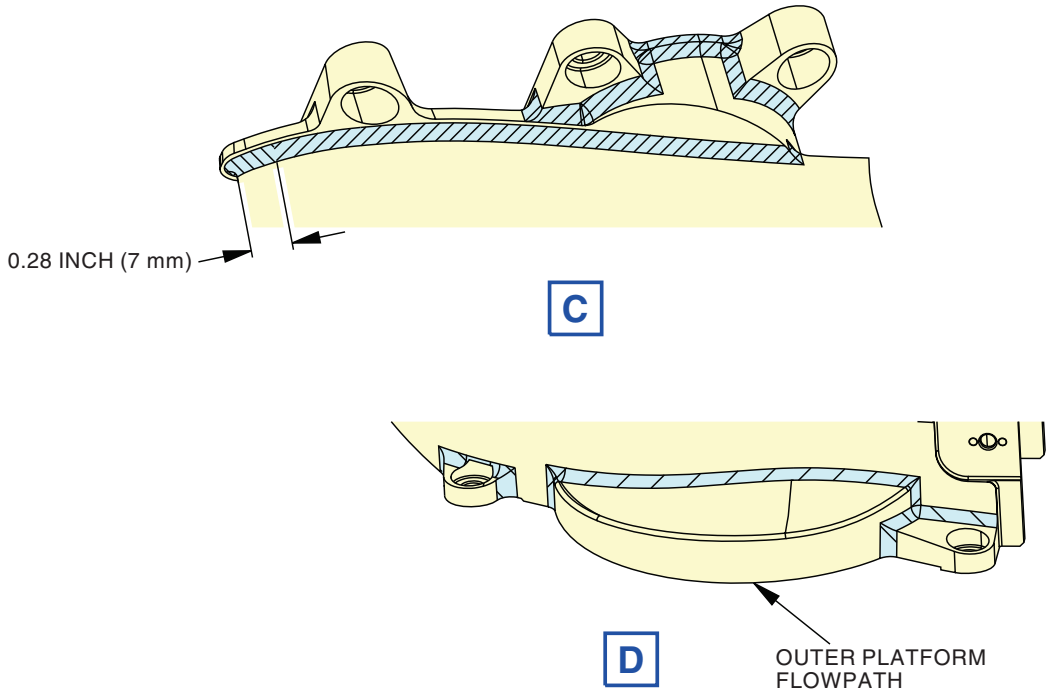
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800047A00101
2564292 S0000612659_V1

OGV Strut 5 Inspection
Figure 613/72-23-00-990-813-G00 (Sheet 1 of 4)

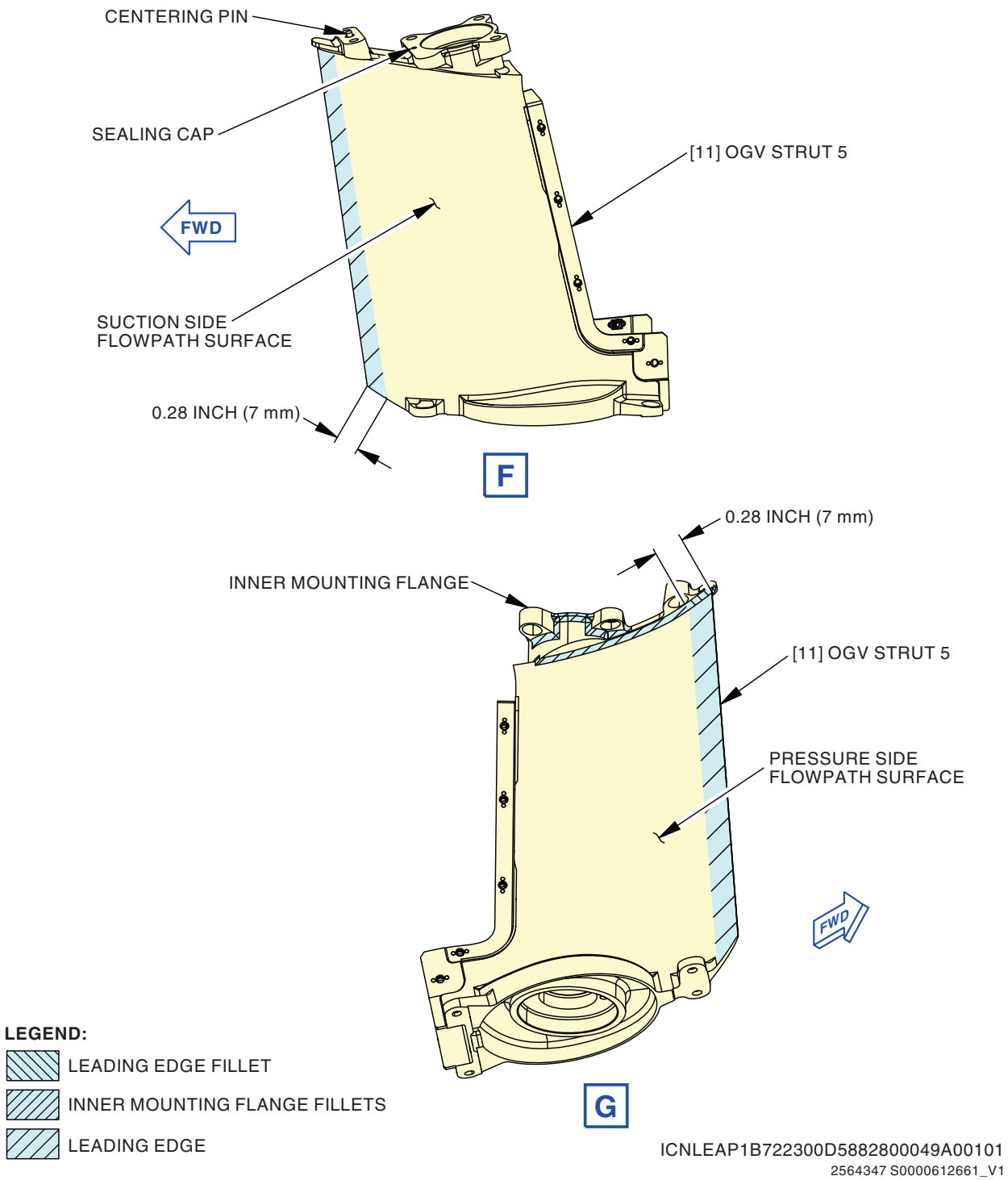
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800048A00101
2564312 S0000612660_V1

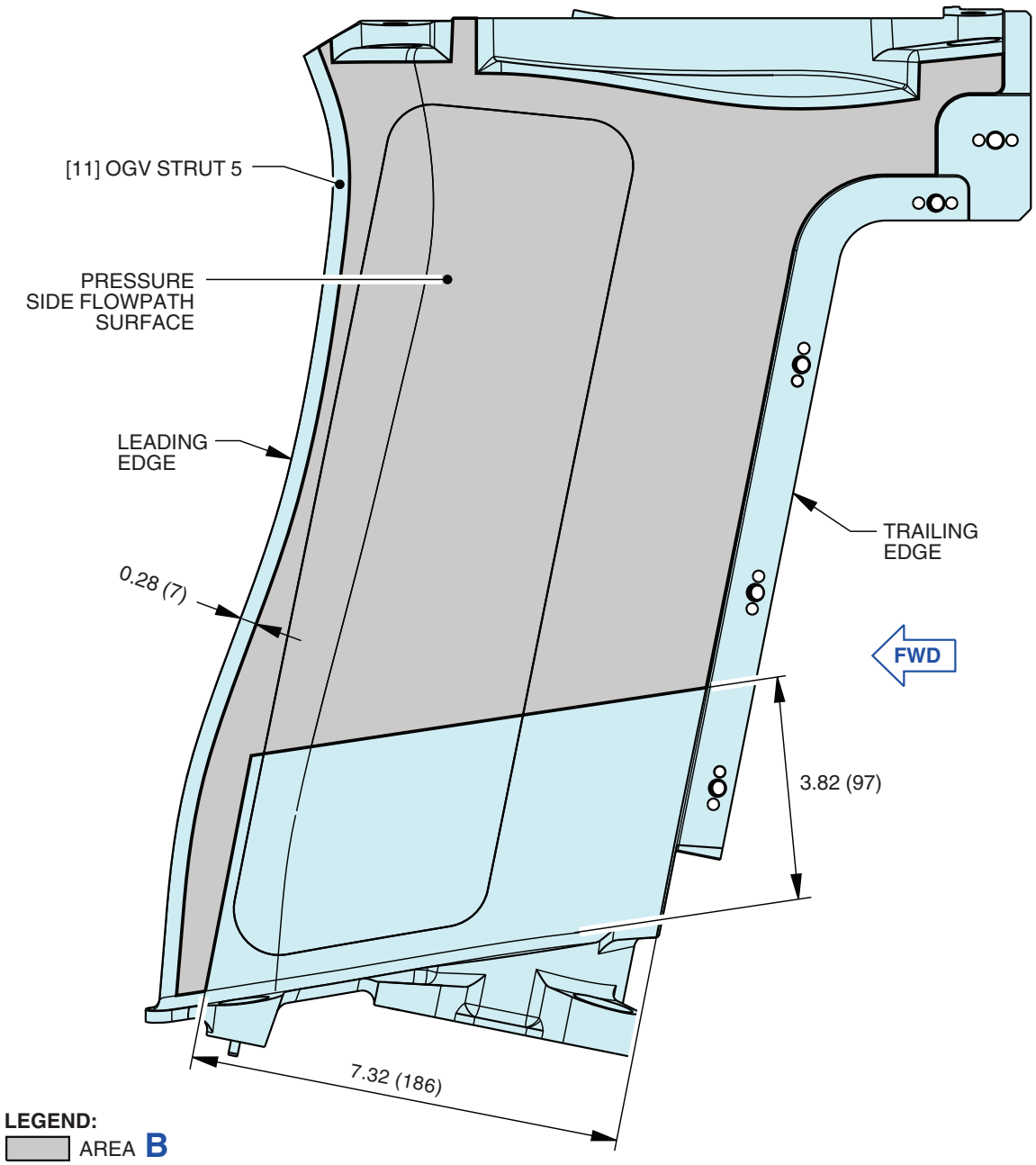
OGV Strut 5 Inspection
Figure 613/72-23-00-990-813-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OGV Strut 5 Inspection
Figure 613/72-23-00-990-813-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



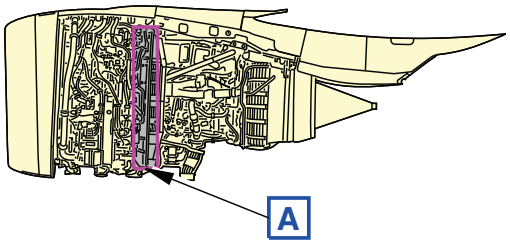
LEGEND:
AREA **B**

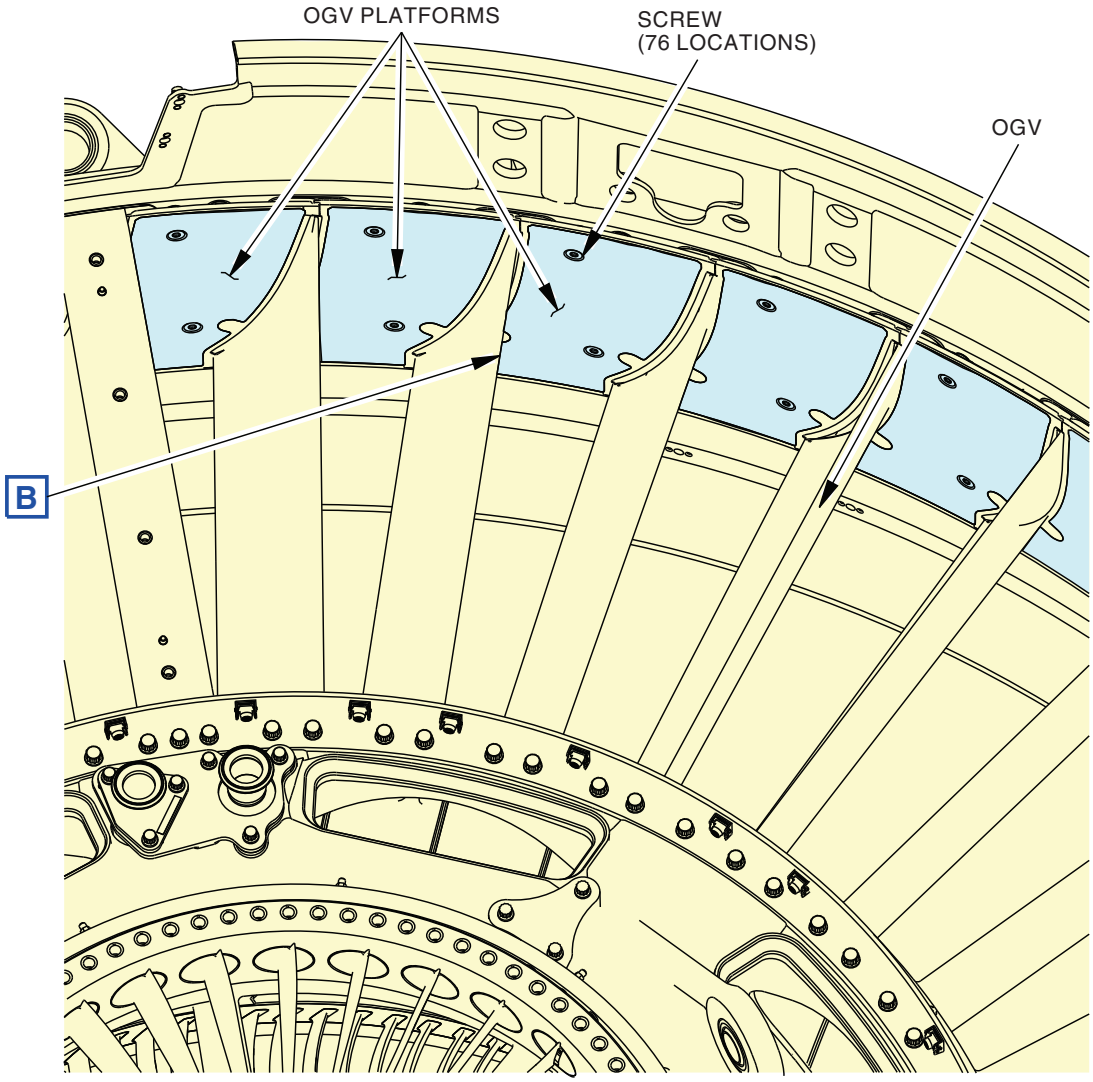
NOTE:
DIMENSIONS ARE IN INCHES WITH MILLIMETERS IN PARENTHESES

ICN-LEAP-1B-722300-D-58828-00133-A-001-01
3113252 S0000851542_V1

OGV Strut 5 Inspection
Figure 613/72-23-00-990-813-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





(VIEW IN THE FORWARD DIRECTION)

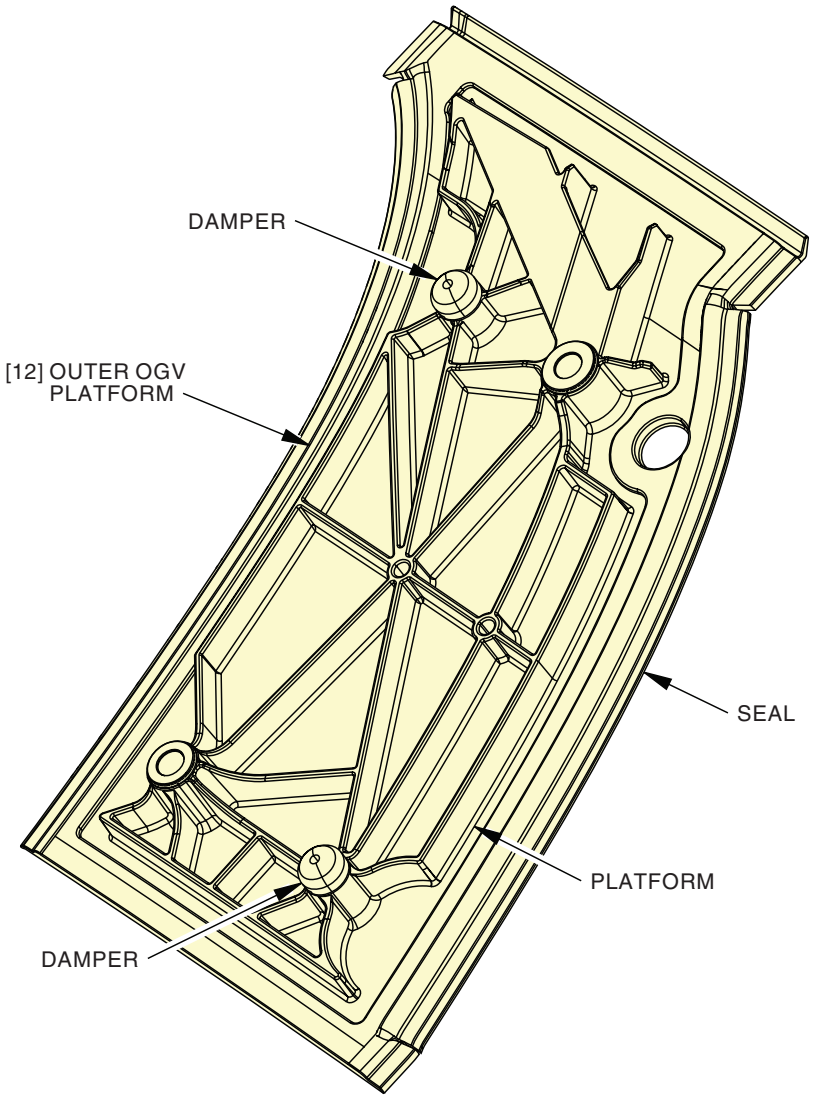
A

ICNLEAP1B722300D5882800111A00101

2946494 S0000723201_V1

OGV Platform Inspection
Figure 614/72-23-00-990-814-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW OF OUTER OGV PLATFORMS

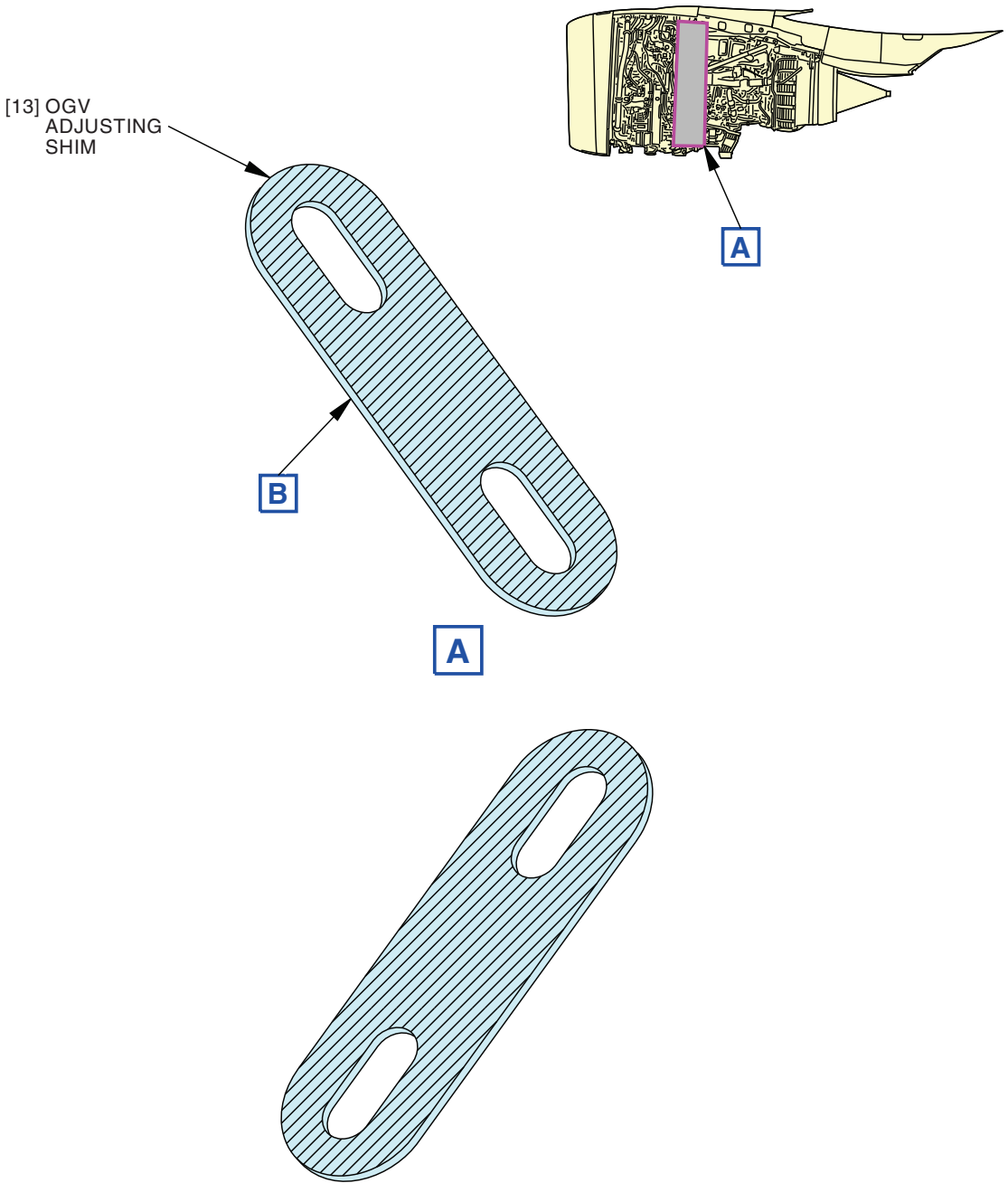
B

ICNLEAP1B722300D5882800050A00101

2565710 S0000612662_V3

OGV Platform Inspection
Figure 614/72-23-00-990-814-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW OF OGV ADJUSTING SHIMS

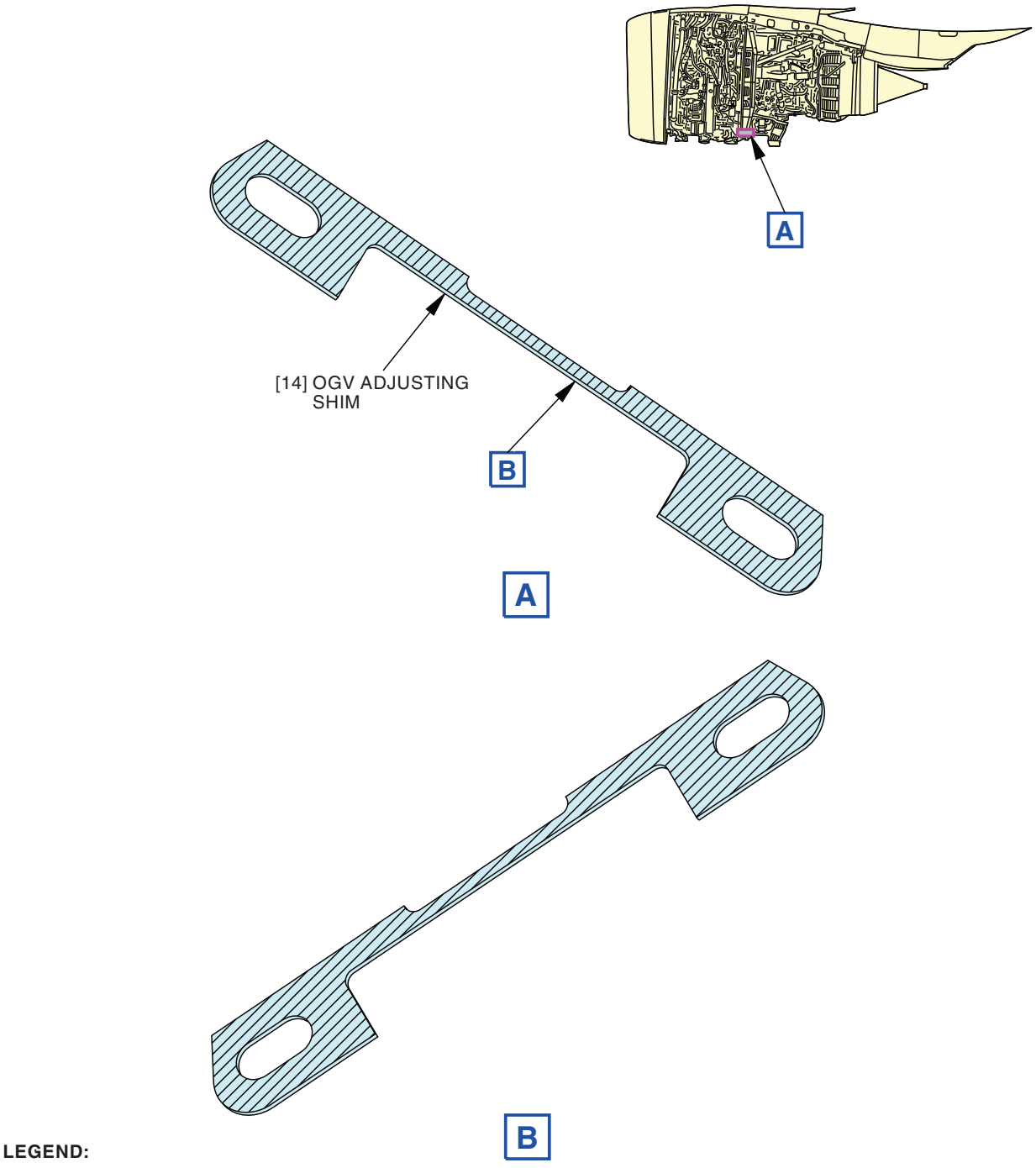
LEGEND:

MATING SURFACE

ICNLEAP1B722300D5882800051A00101
2565713 S0000612698_V1

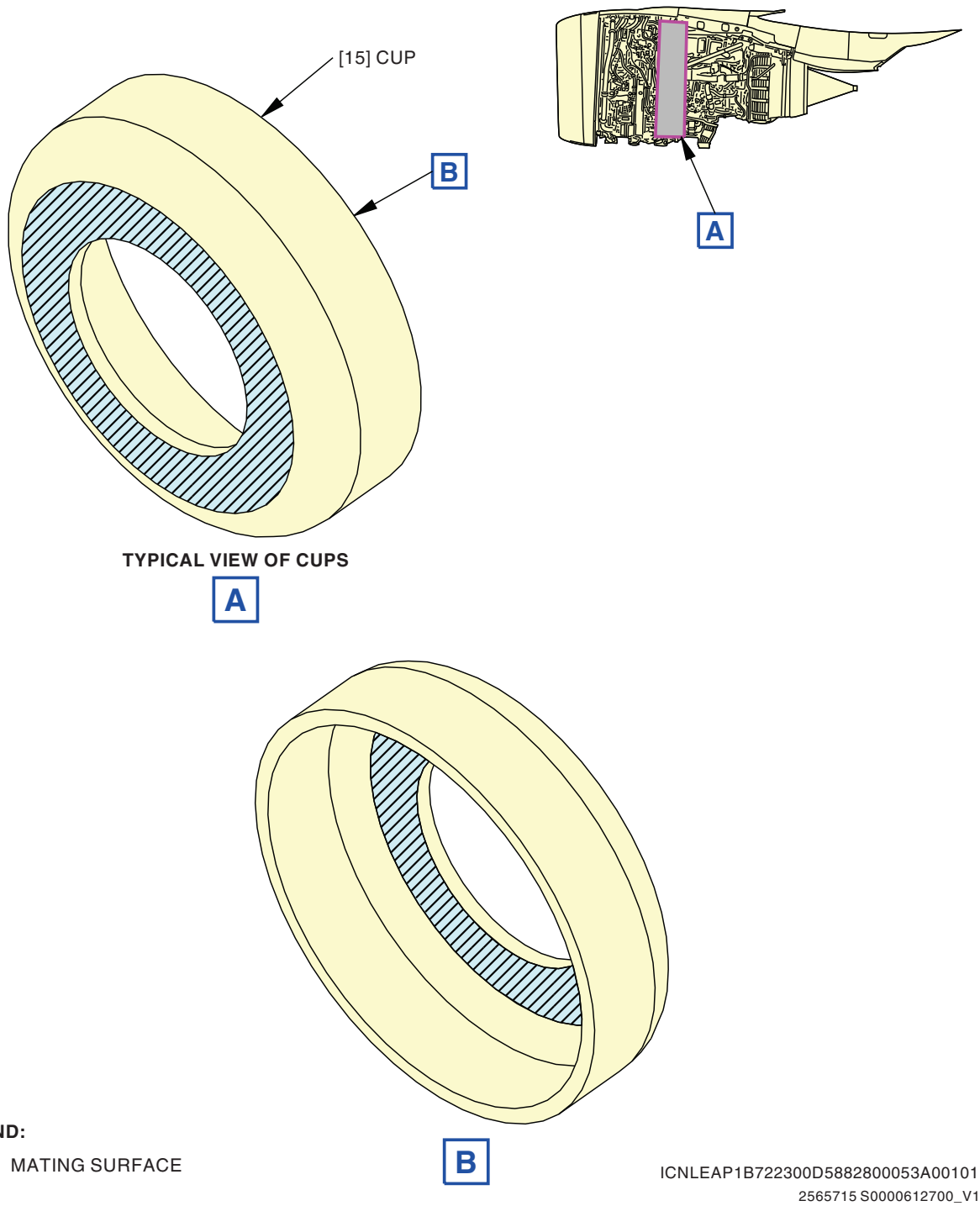
OGV Adjusting Shims Inspection
Figure 615/72-23-00-990-815-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



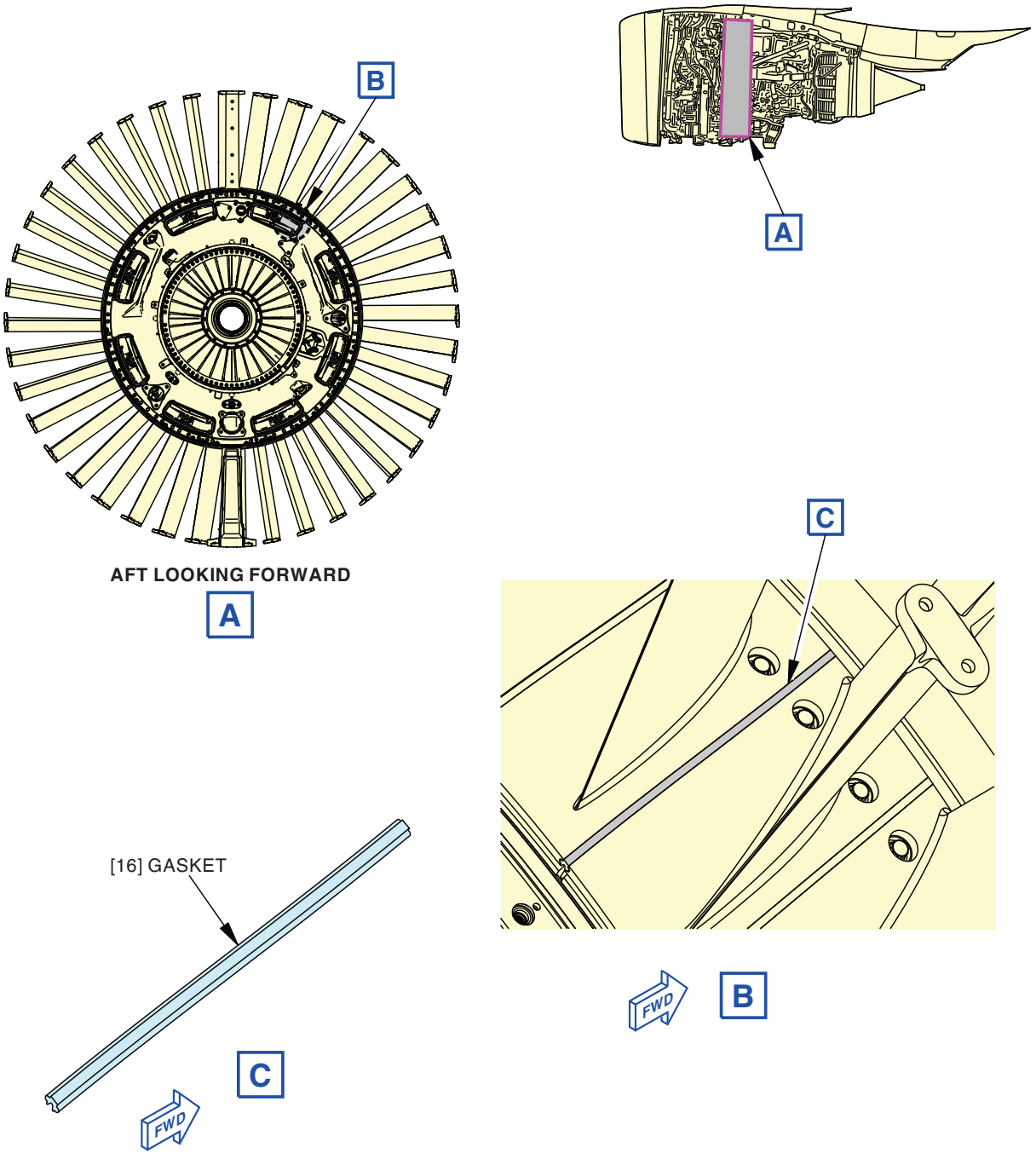
OGV Adjusting Shims Inspection
Figure 615/72-23-00-990-815-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cups Inspection
Figure 616/72-23-00-990-816-G00

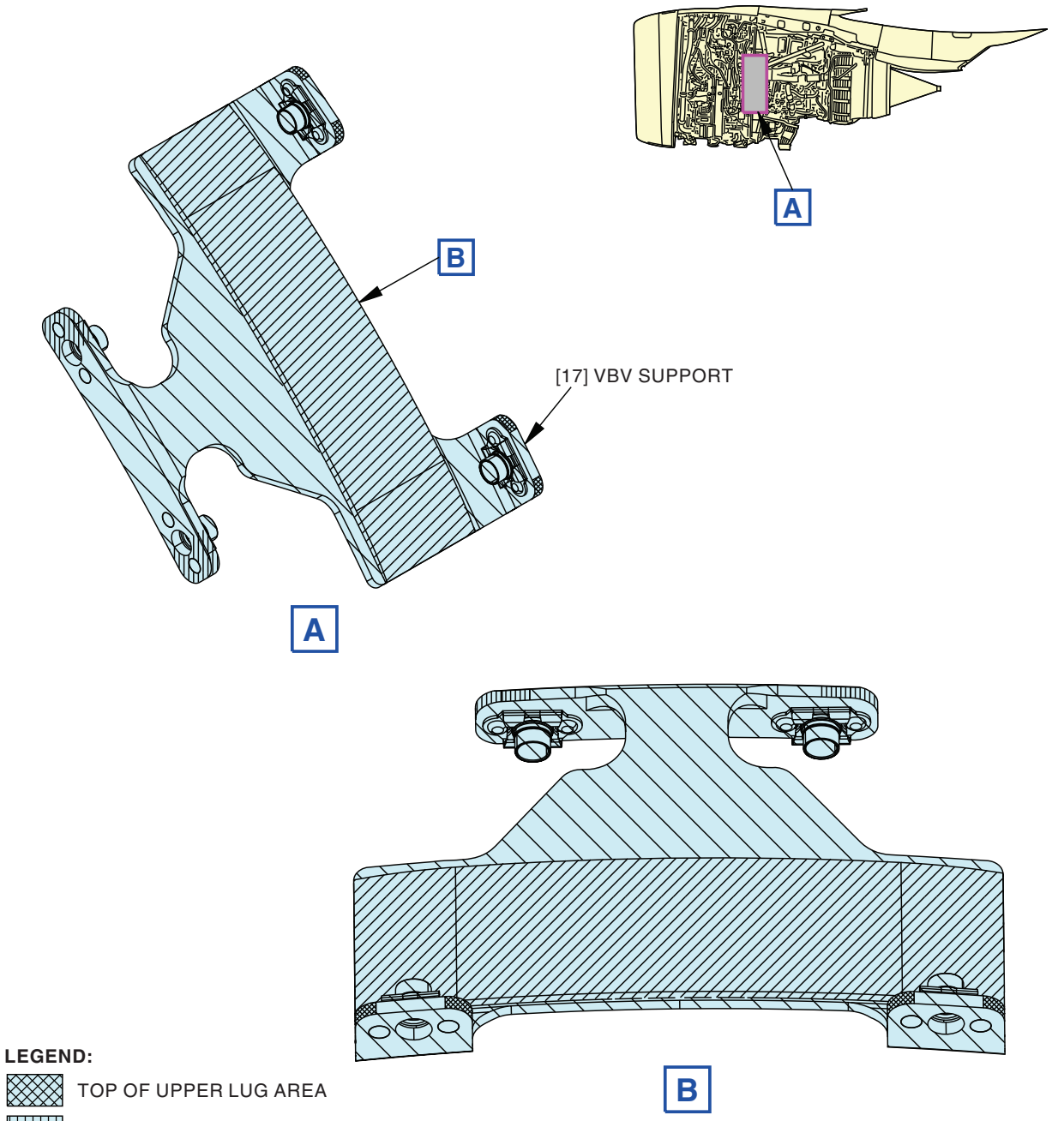
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800054A00101
2565724 S0000612702_V1

Gasket Inspection
Figure 617/72-23-00-990-817-G00

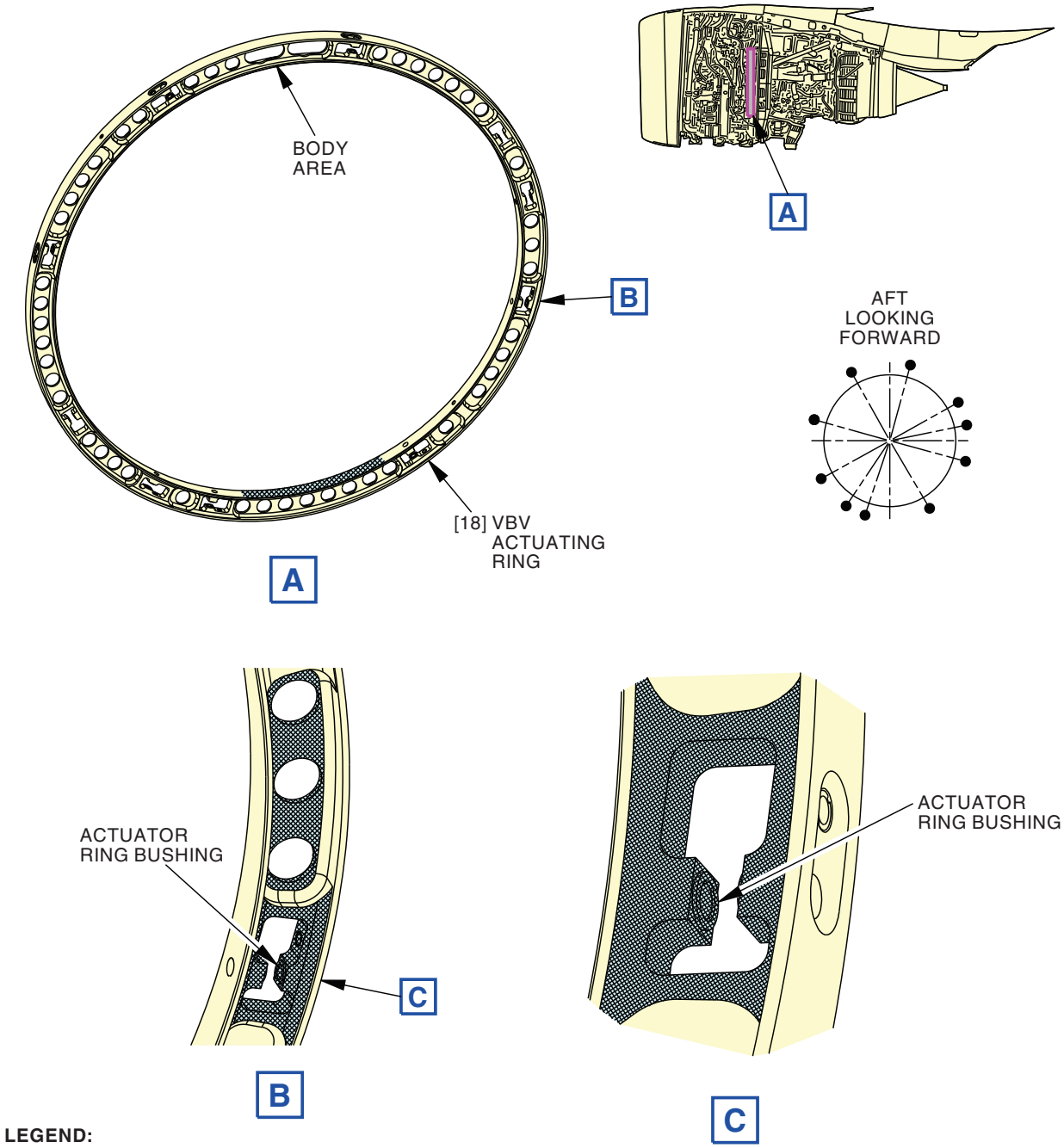
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800055A00101
2565718 S0000612703_V1

VBV Support Inspection
Figure 618/72-23-00-990-818-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

AREA A

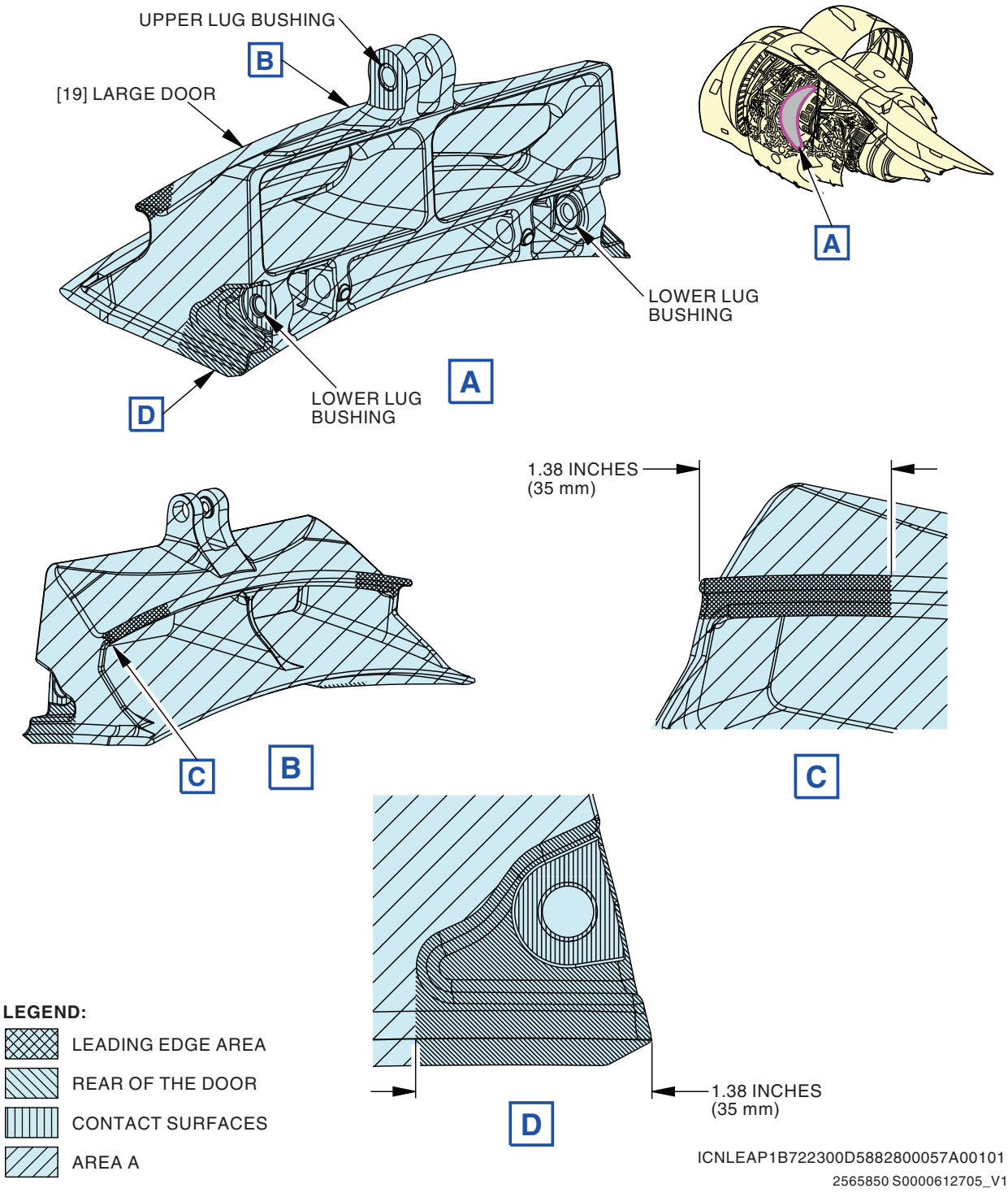
NOTE:

AREA A IS APPLICABLE TO THE TWO SIDES OF THE PART

ICNLEAP1B722300D5882800056A00101
2565723 S0000612704_V1

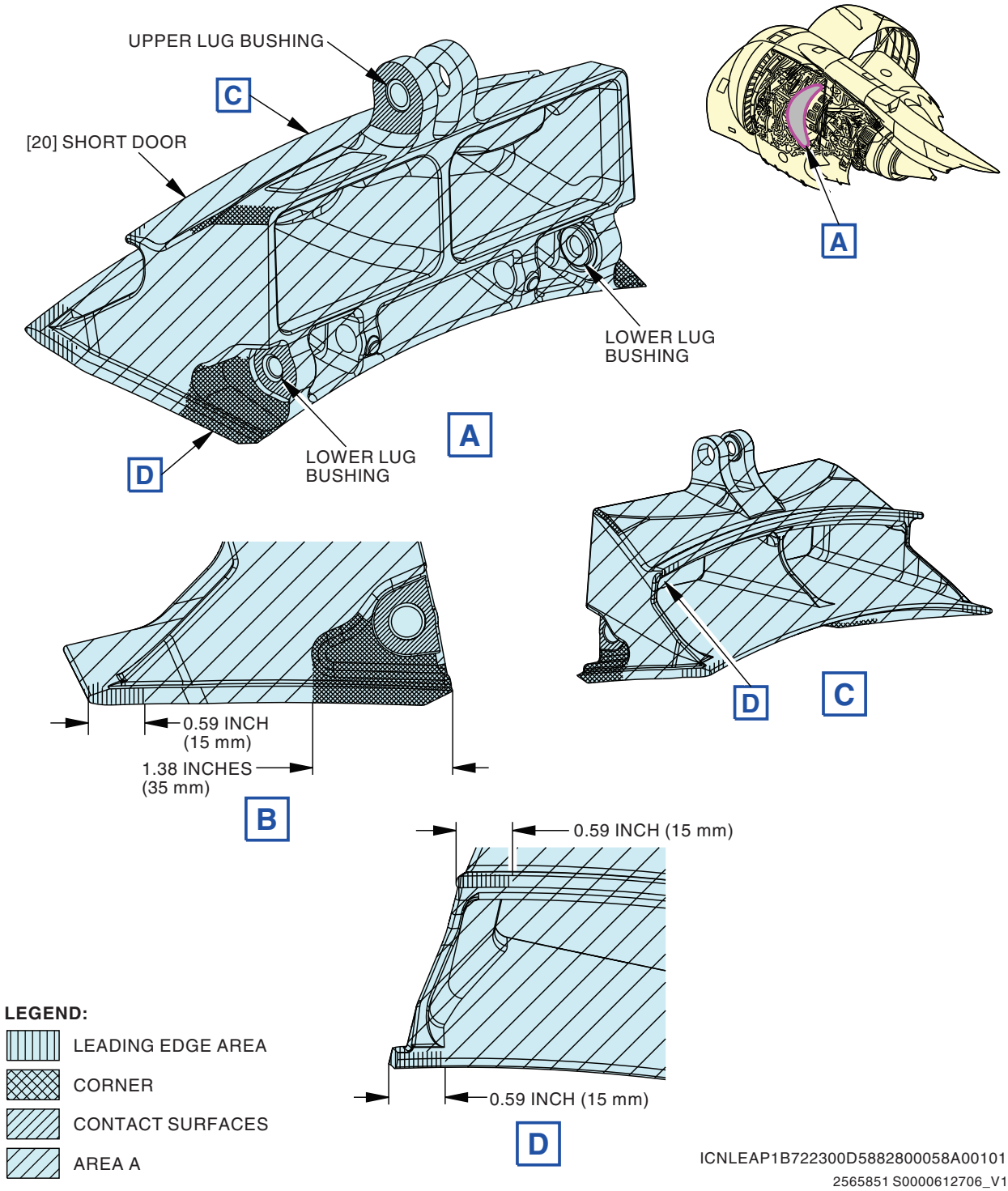
VBV Actuating Ring Inspection
Figure 619/72-23-00-990-819-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VBV Large Door Inspection
Figure 620/72-23-00-990-820-G00

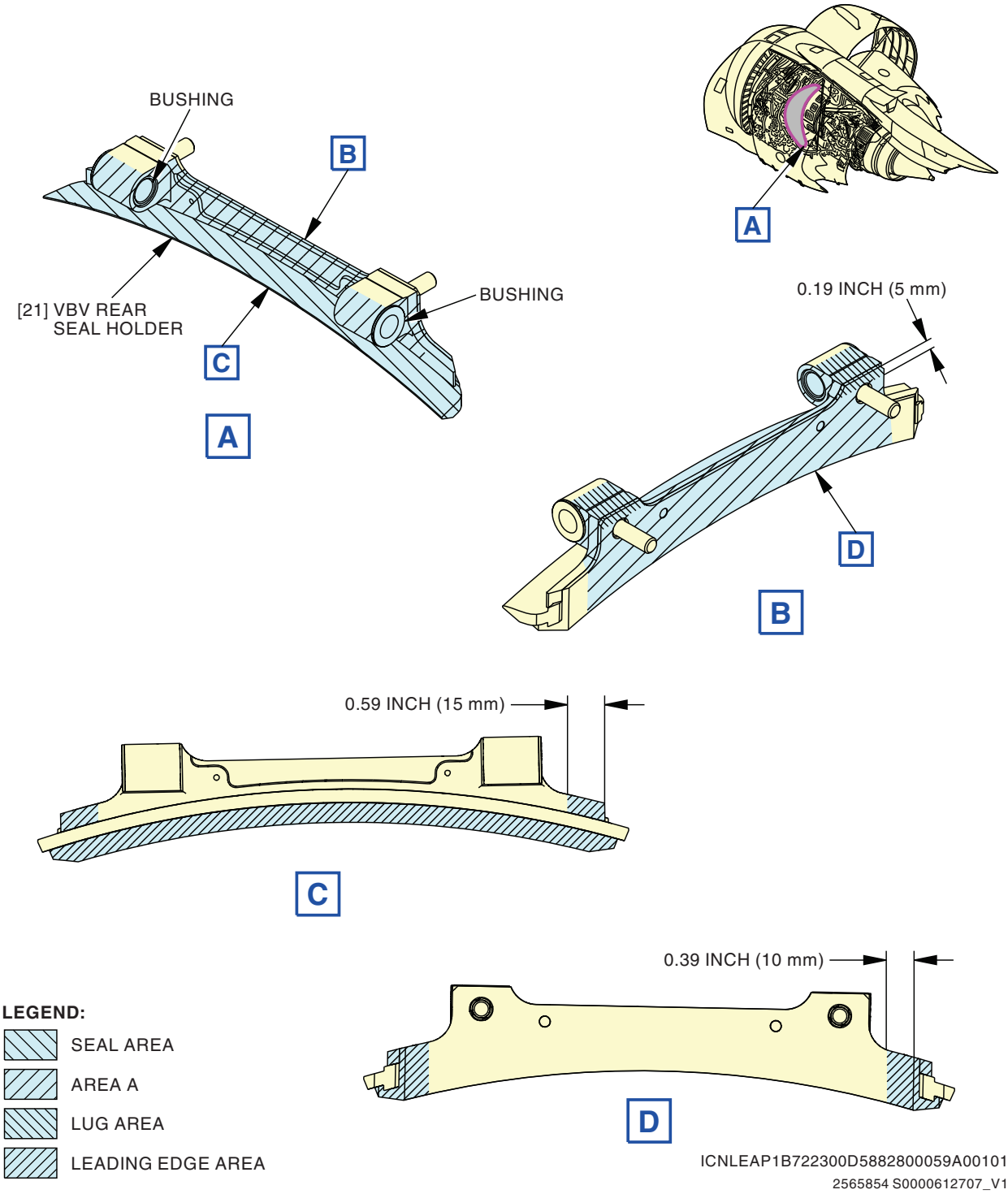
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800058A00101
2565851 S0000612706_V1

VBV Short Door Inspection
Figure 621/72-23-00-990-821-G00

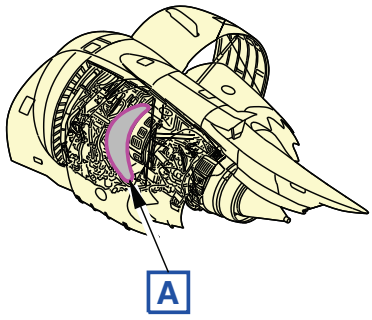
ECCN 9E991 BOEING PROPRIETARY - See title page for details

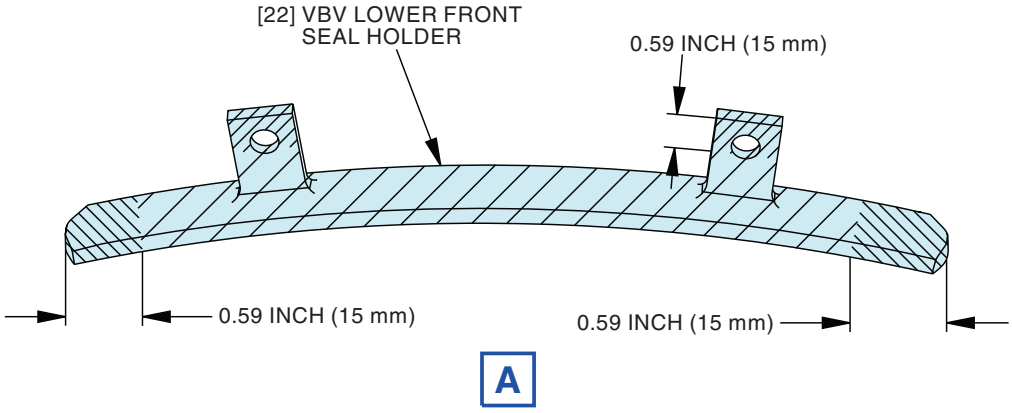


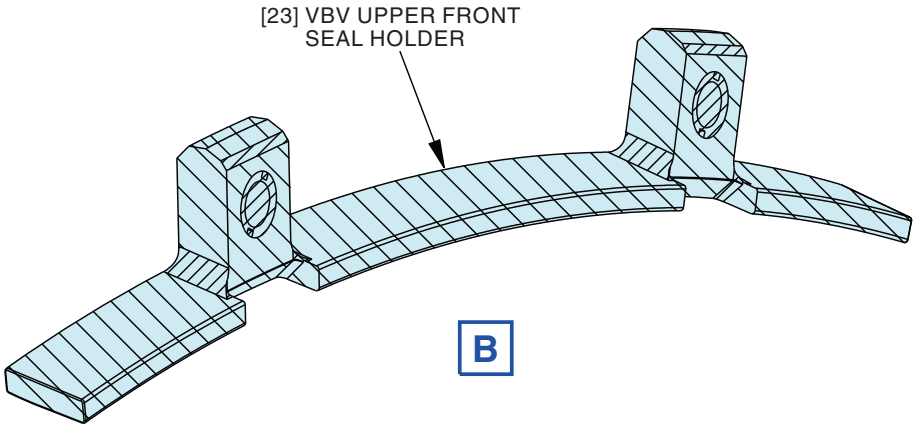
ICNLEAP1B722300D5882800059A00101
2565854 S0000612707_V1

VBV Rear Seal Holder Inspection
Figure 622/72-23-00-990-822-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details







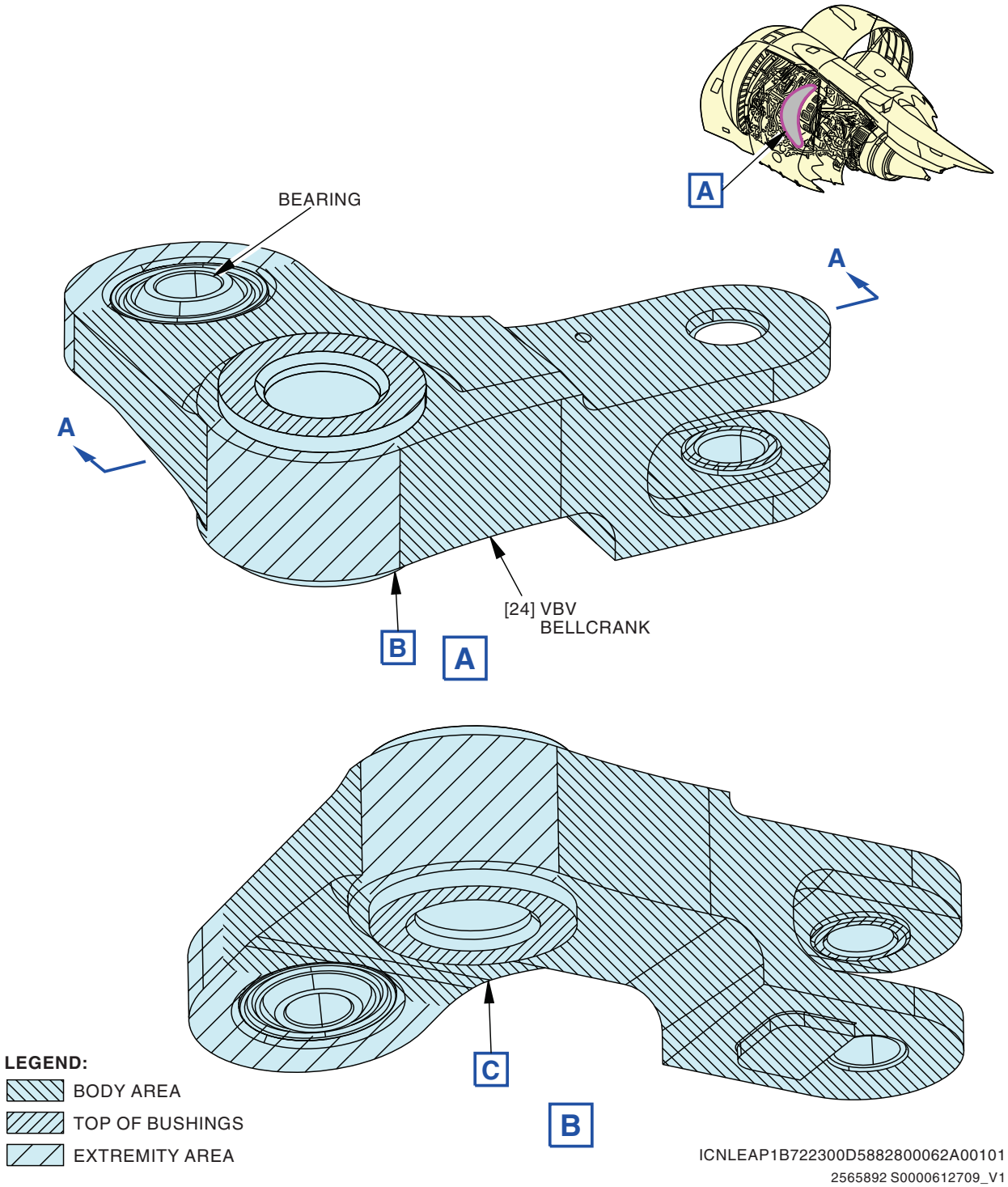
LEGEND:

- LUG AREA
- AREA A
- SIDE AREA
- BODY AREA

ICNLEAP1B722300D5882800060A00101
2565876 S0000612708_V1

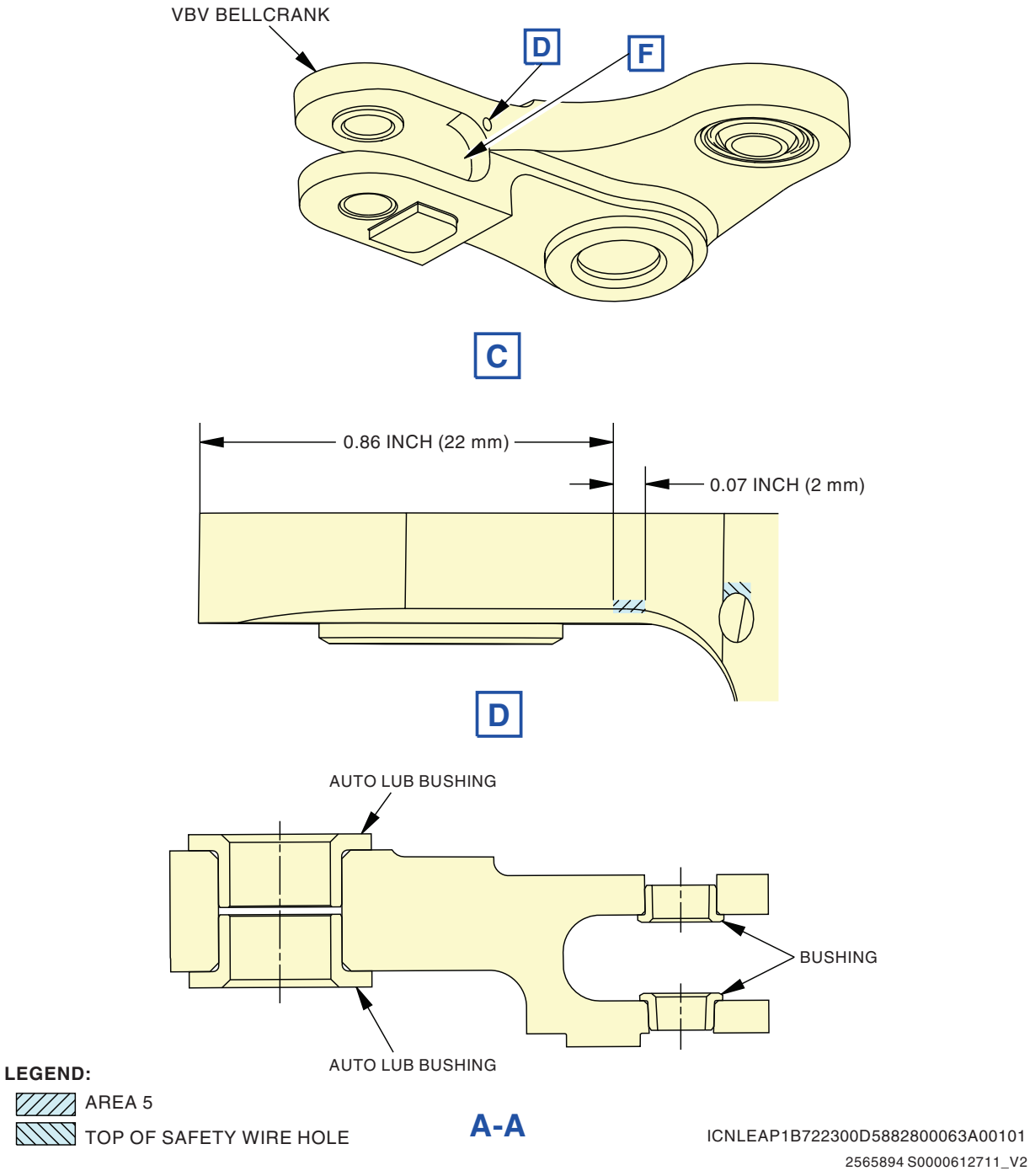
VBV Lower and Upper Front Seal Holder Inspection
Figure 623/72-23-00-990-823-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



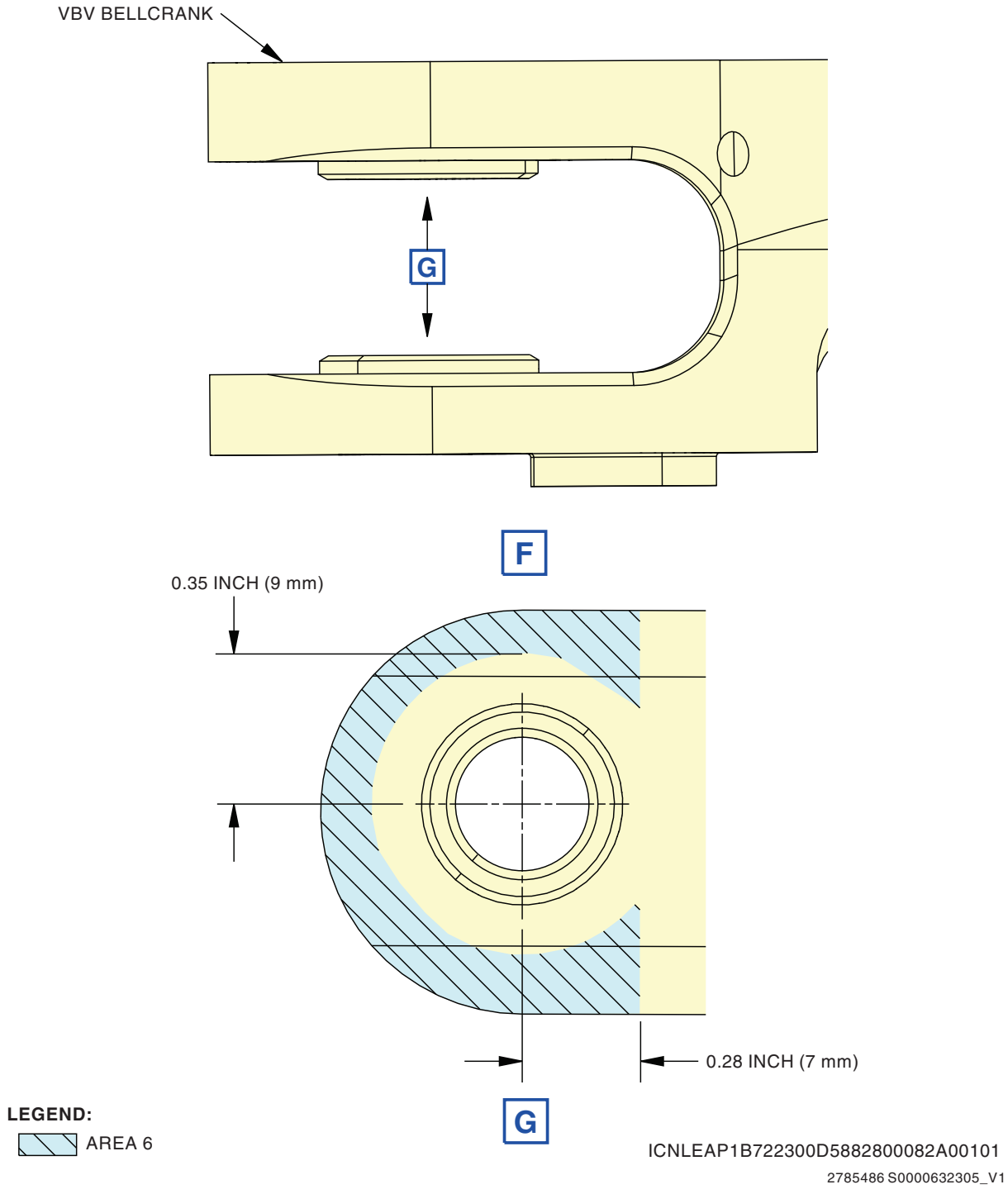
VBV Bellcrank Inspection
Figure 624/72-23-00-990-824-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



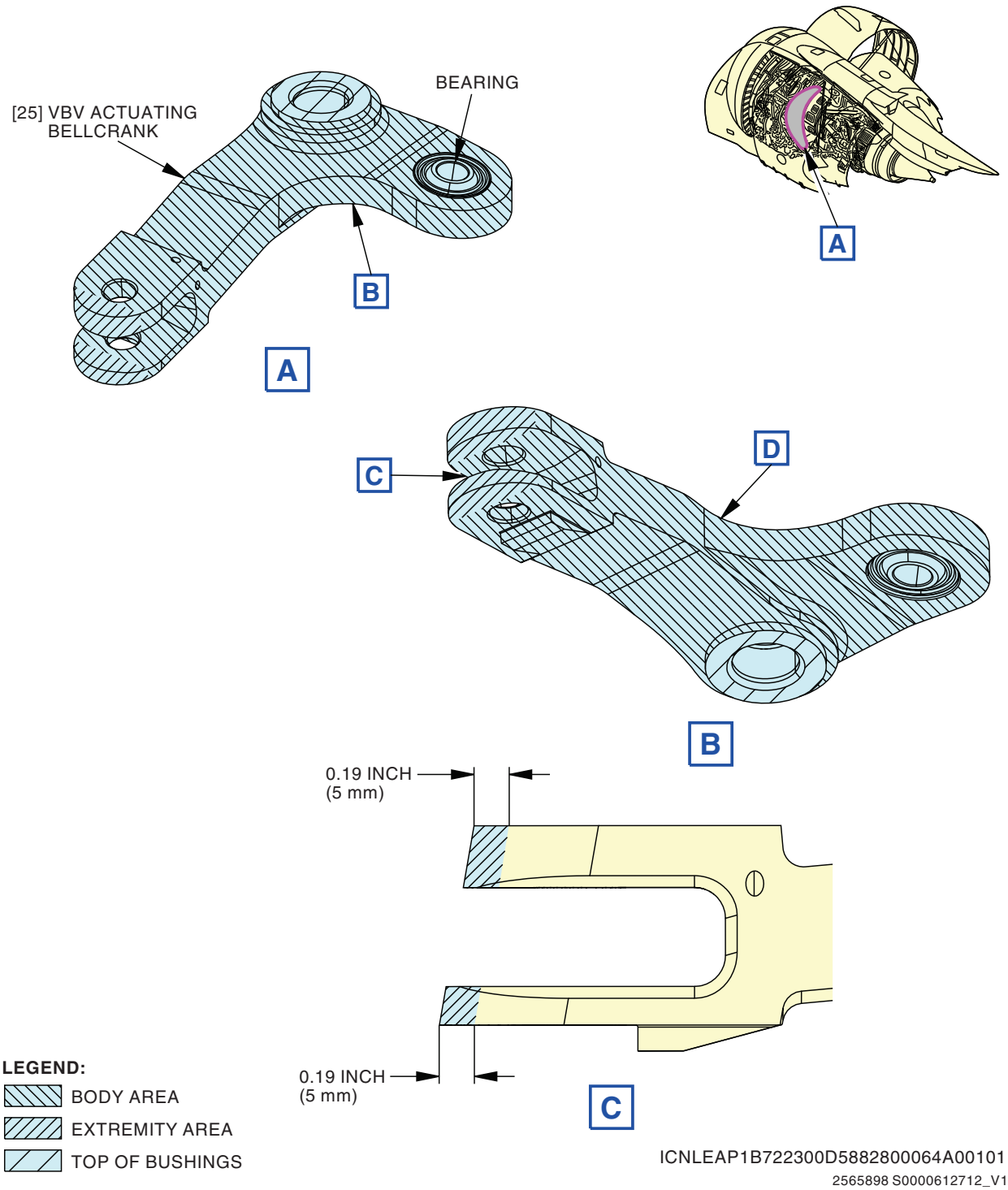
VBV Bellcrank Inspection
Figure 624/72-23-00-990-824-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VBV Bellcrank Inspection
Figure 624/72-23-00-990-824-G00 (Sheet 3 of 3)

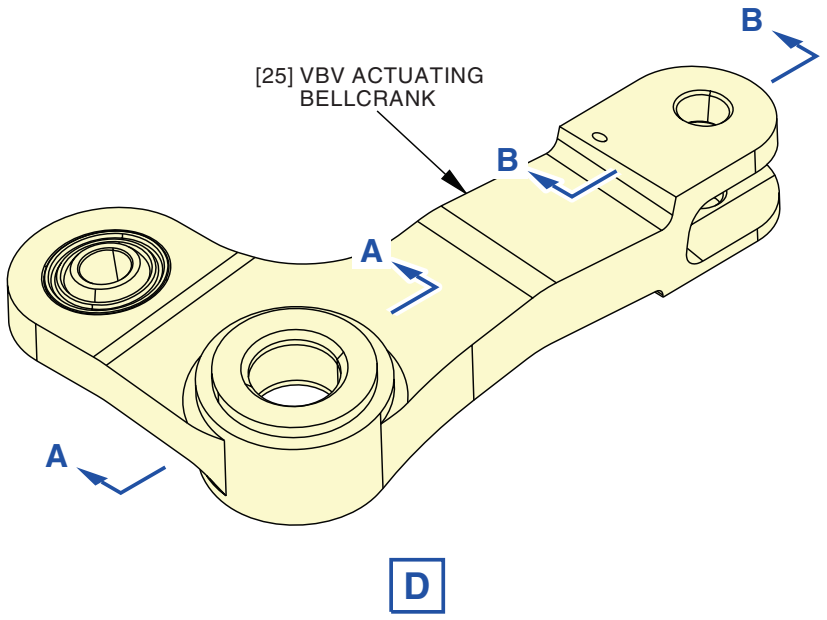
ECCN 9E991 BOEING PROPRIETARY - See title page for details

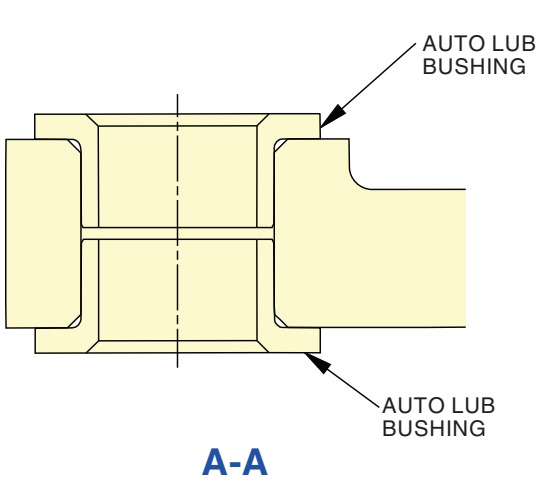


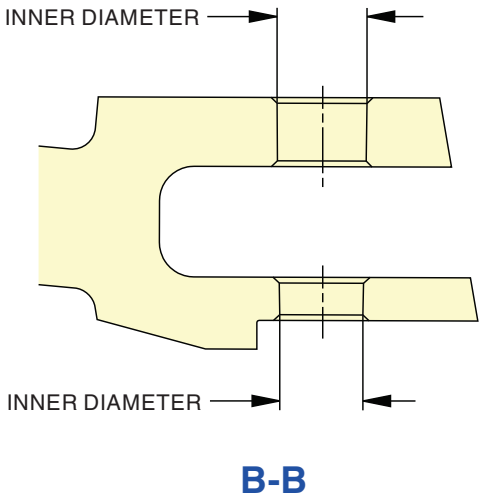
ICNLEAP1B722300D5882800064A00101
2565898 S0000612712_V1

VBV Actuating Bellcrank Inspection
Figure 625/72-23-00-990-825-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



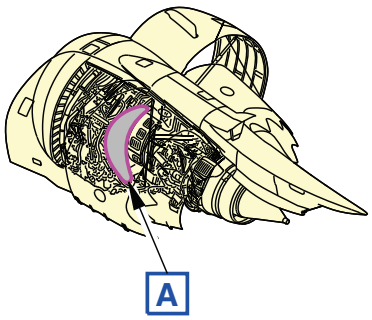


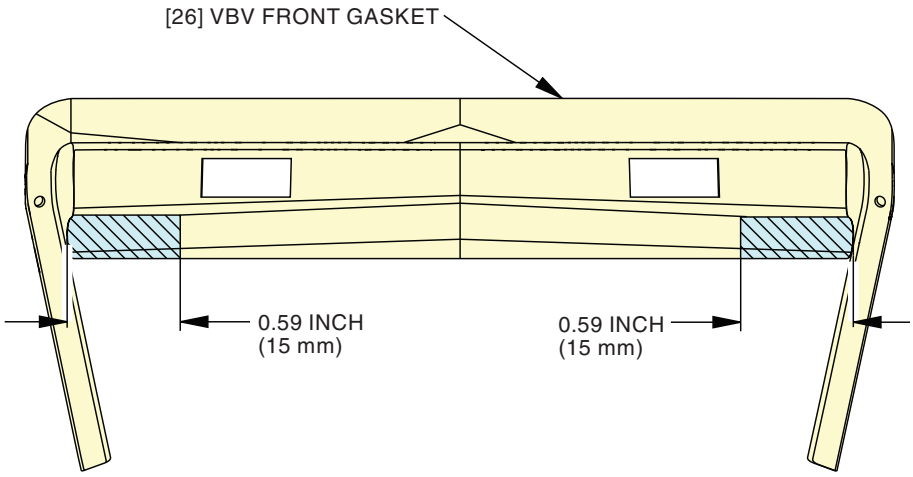


ICNLEAP1B722300D5882800065A00101
2565905 S0000612713_V1

VBV Actuating Bellcrank Inspection
Figure 625/72-23-00-990-825-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





TYPICAL VIEW OF VBV FRONT GASKETS

A

LEGEND:

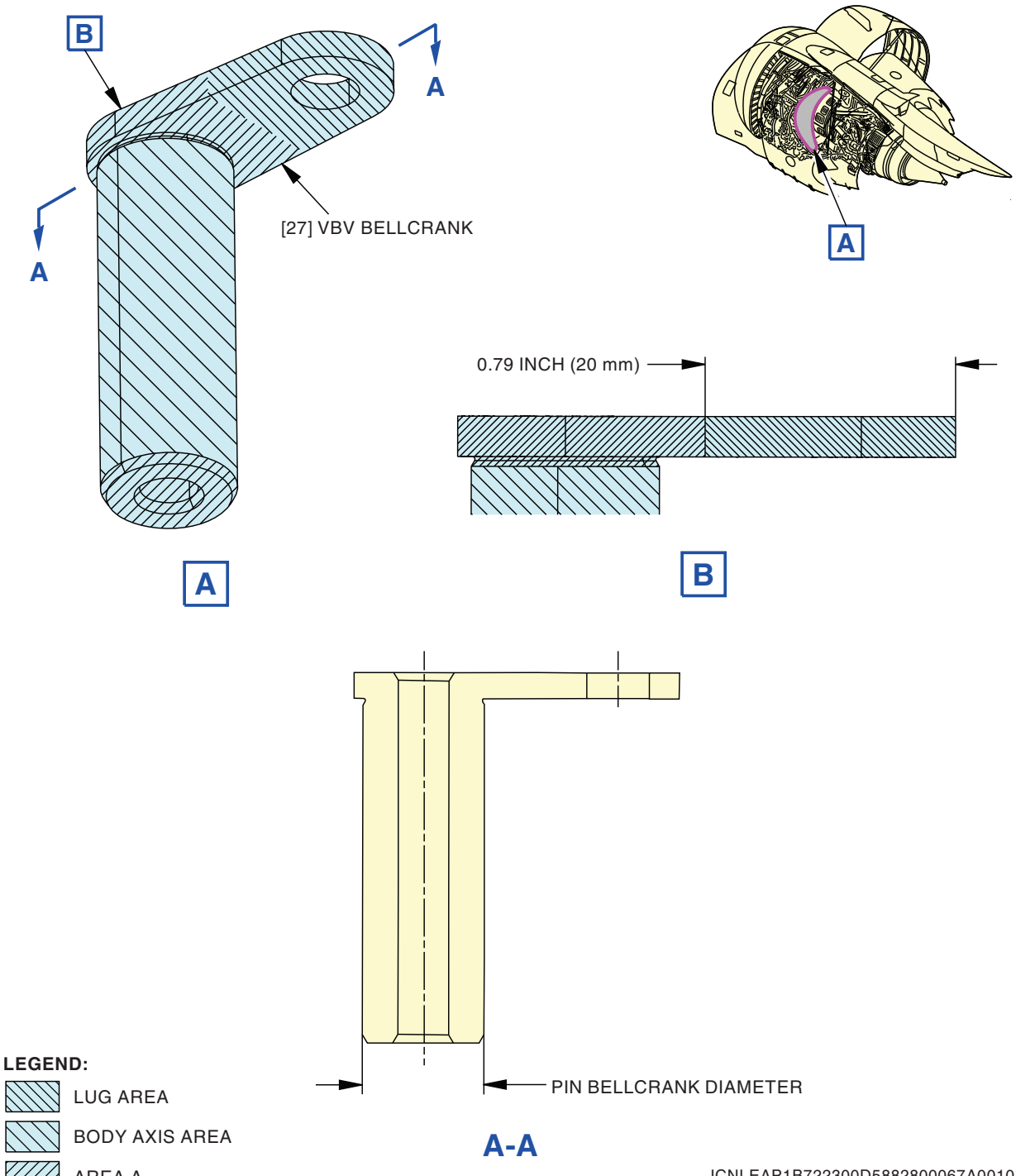
AREA 1

ICNLEAP1B722300D5882800066A00101

2565910 S0000612714_V1

VBV Front Gasket Inspection
Figure 626/72-23-00-990-826-G00

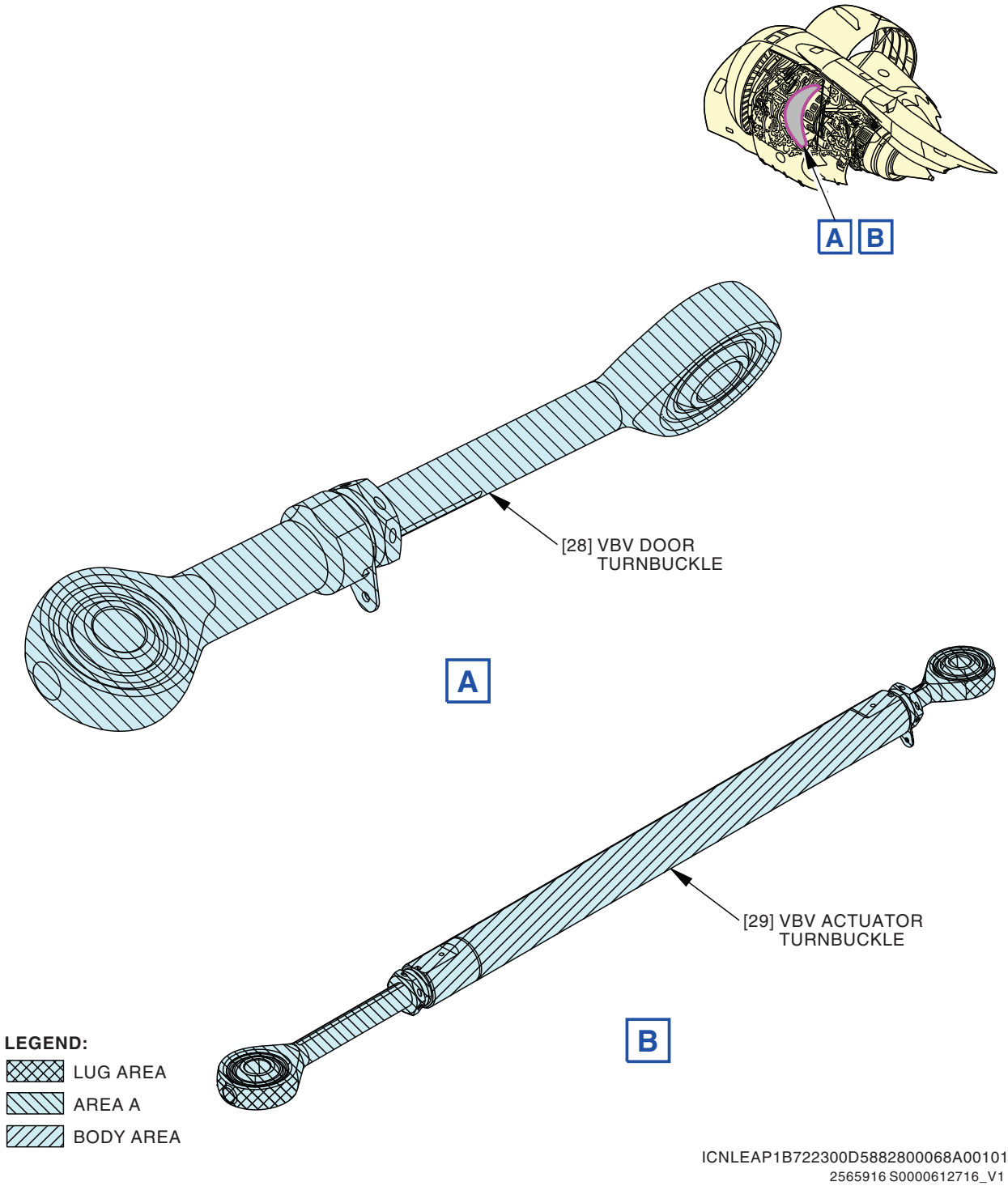
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800067A00101
2565914 S0000612715_V1

VBV Bellcrank Pin Inspection
Figure 627/72-23-00-990-827-G00

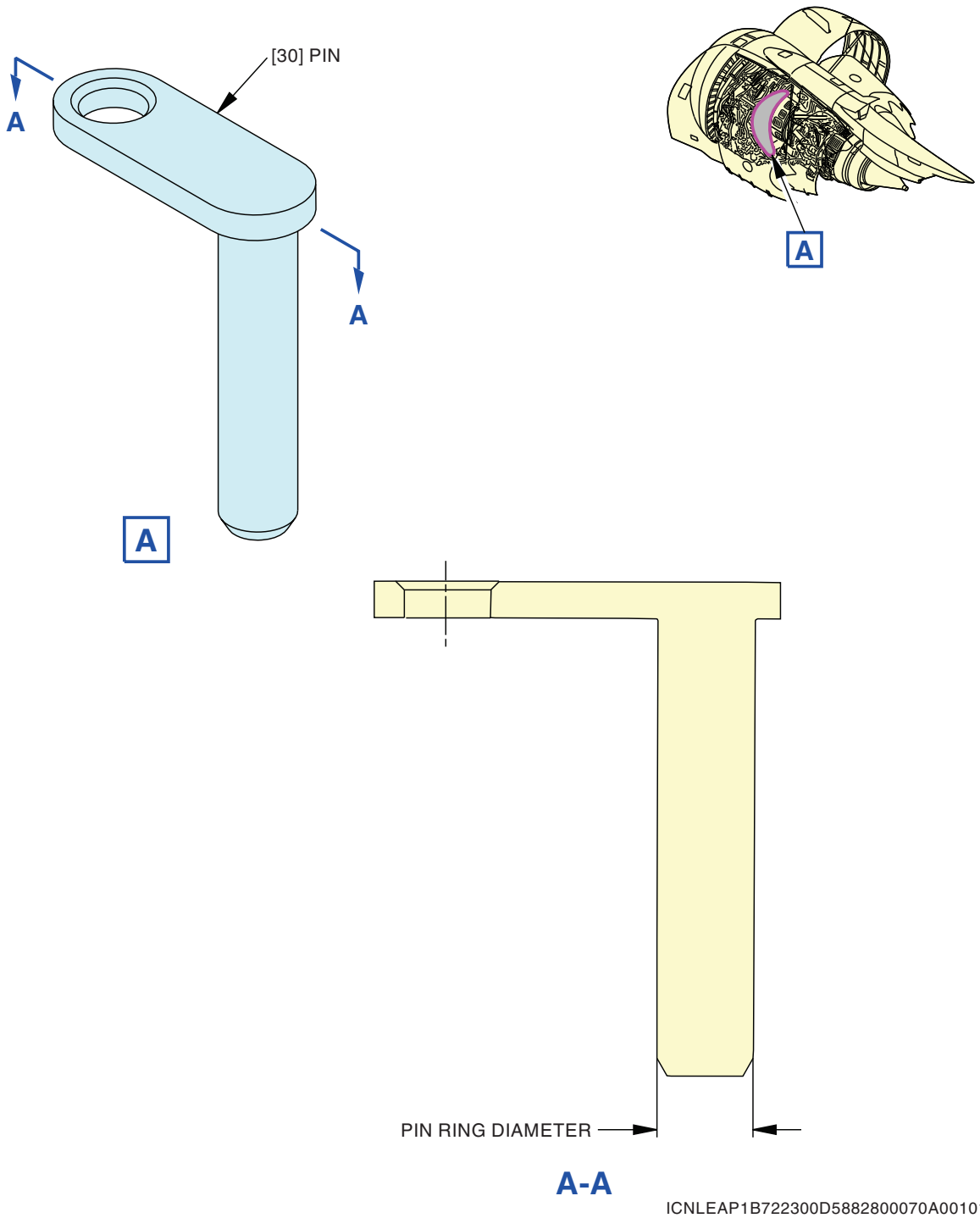
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800068A00101
2565916 S0000612716_V1

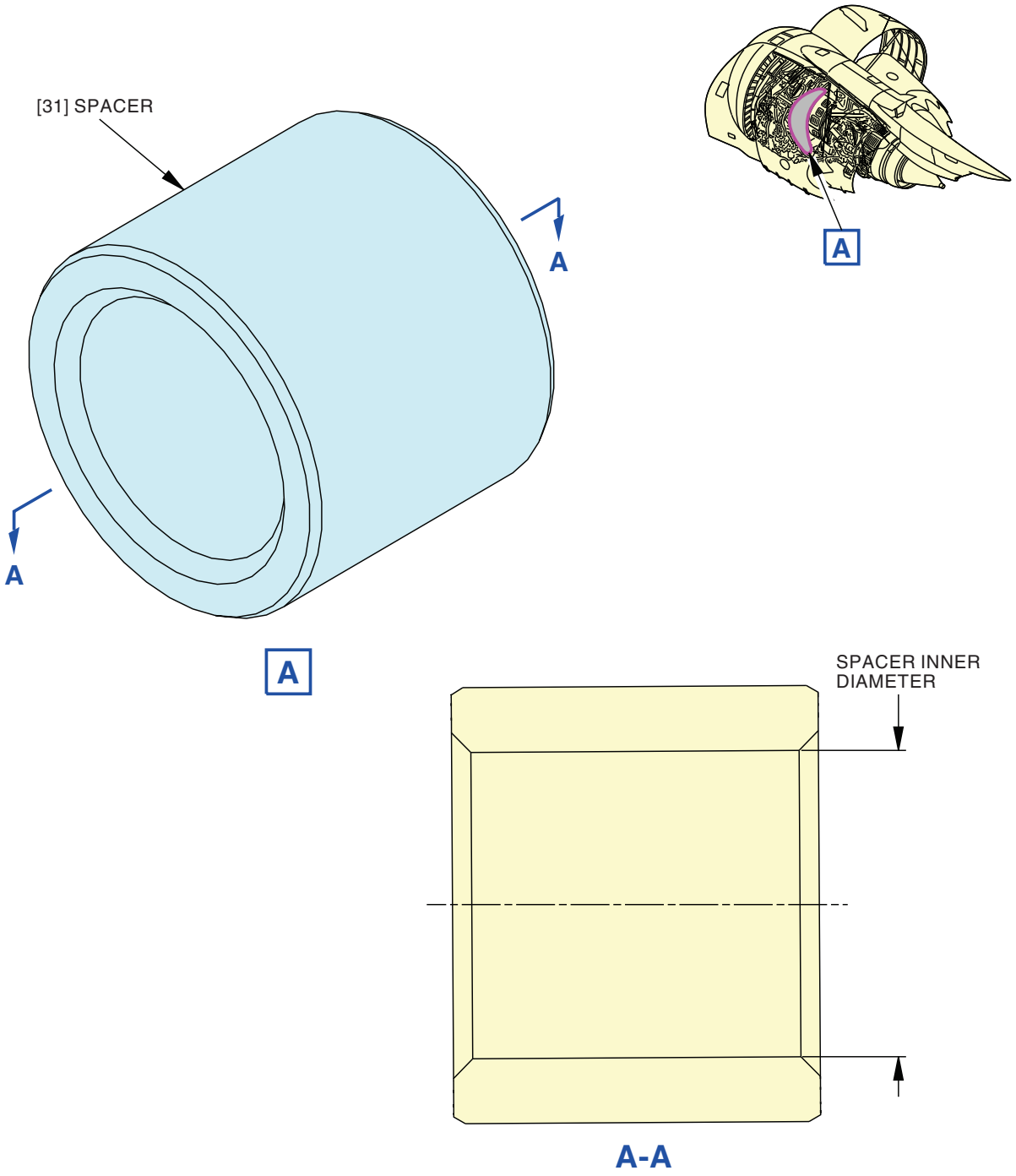
VBV Turnbuckle Inspection
Figure 628/72-23-00-990-828-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



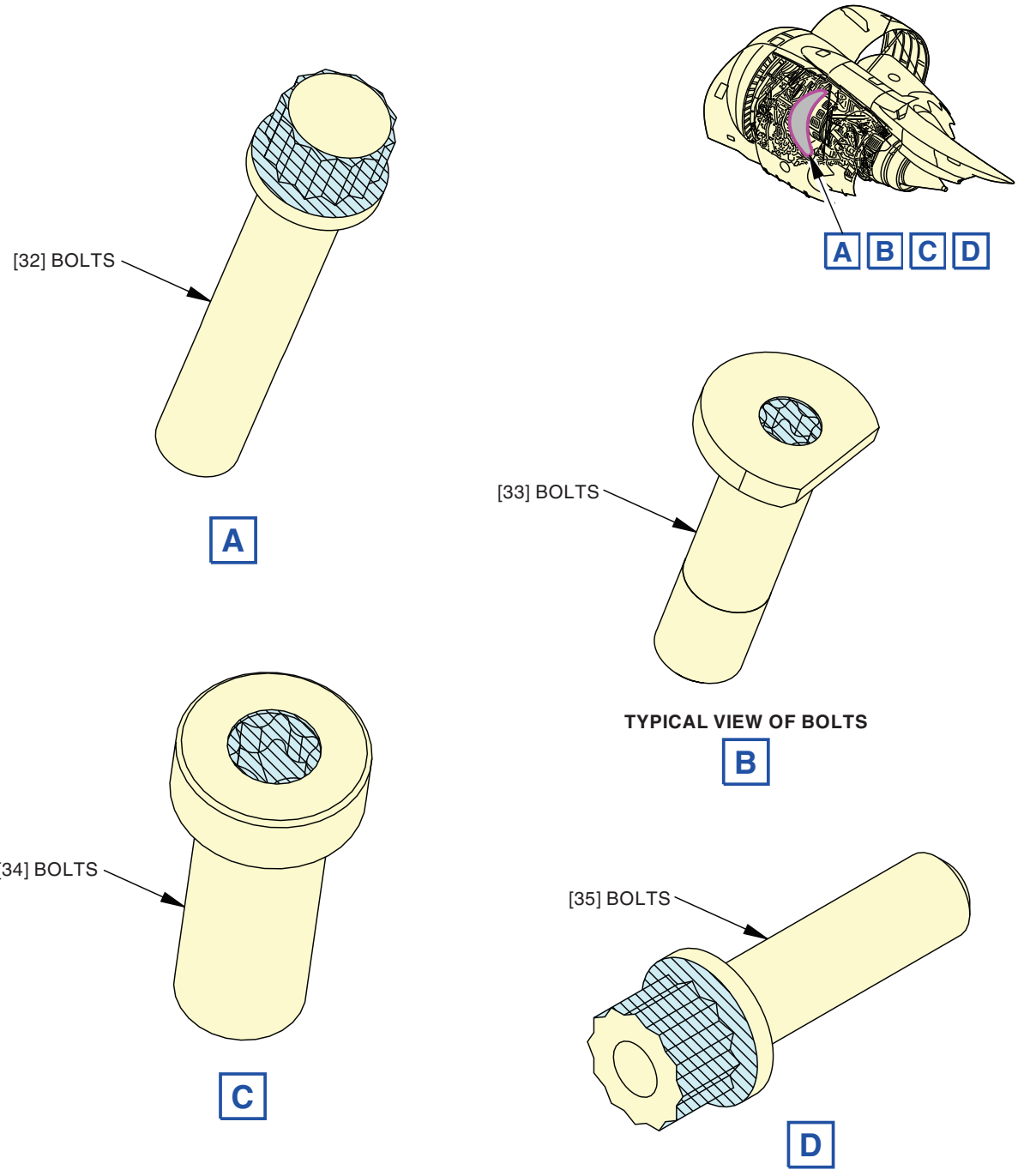
Pin Inspection
Figure 629/72-23-00-990-829-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



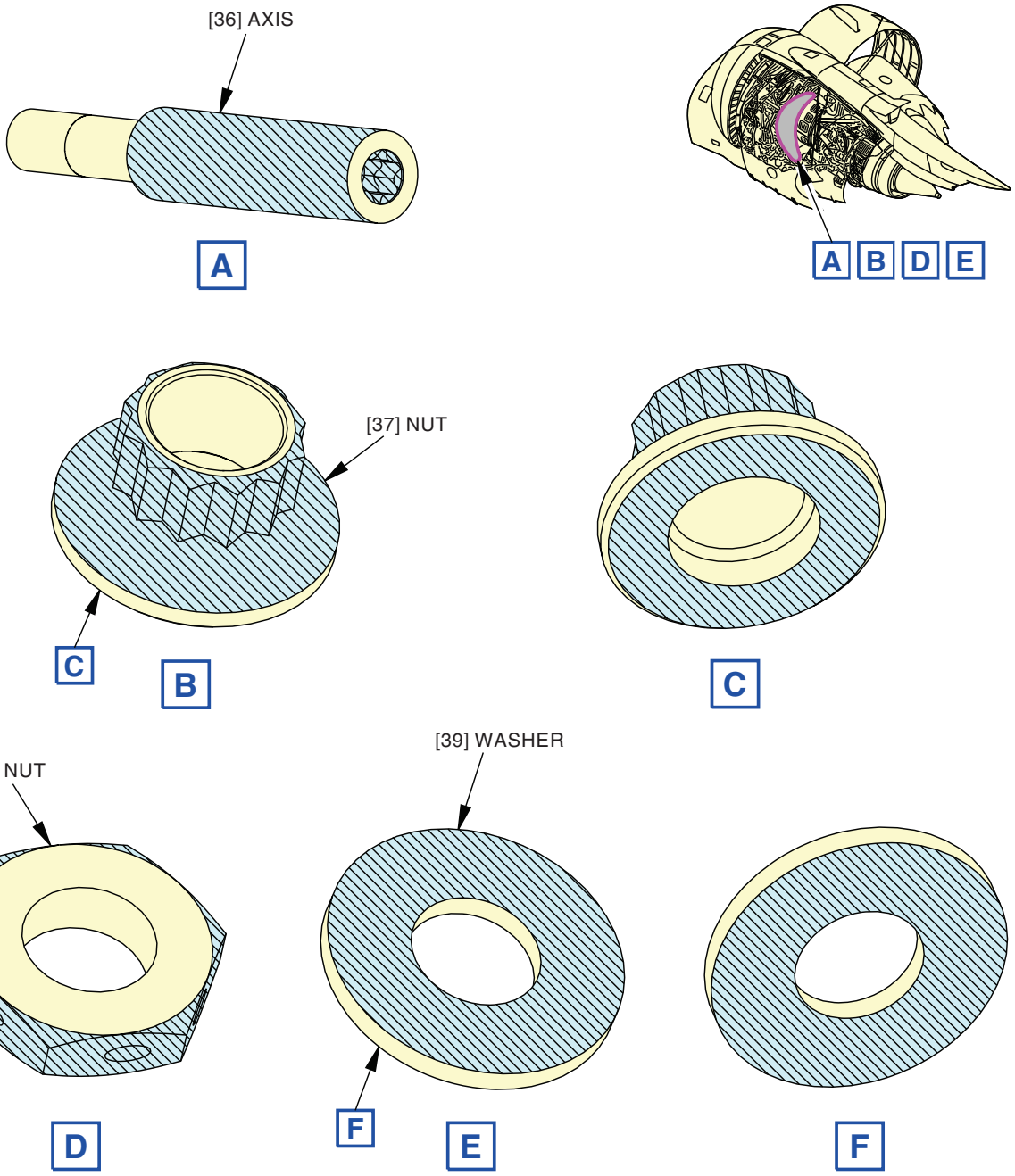
Spacer Inspection
Figure 630/72-23-00-990-830-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VBV Hardware Inspection
Figure 631/72-23-00-990-831-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



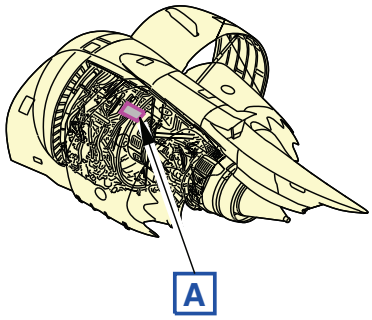
LEGEND:

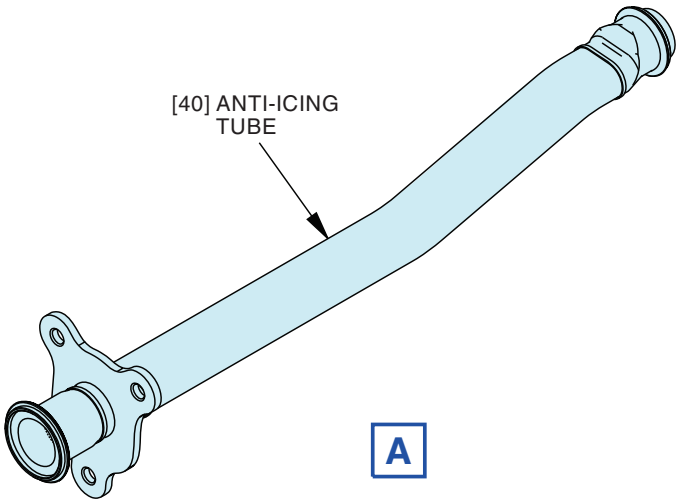
TOOLING CONTACT SURFACE

ICNLEAP1B722300D5882800076A00101
2565944 S0000612720_V1

VBV Hardware Inspection
Figure 631/72-23-00-990-831-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



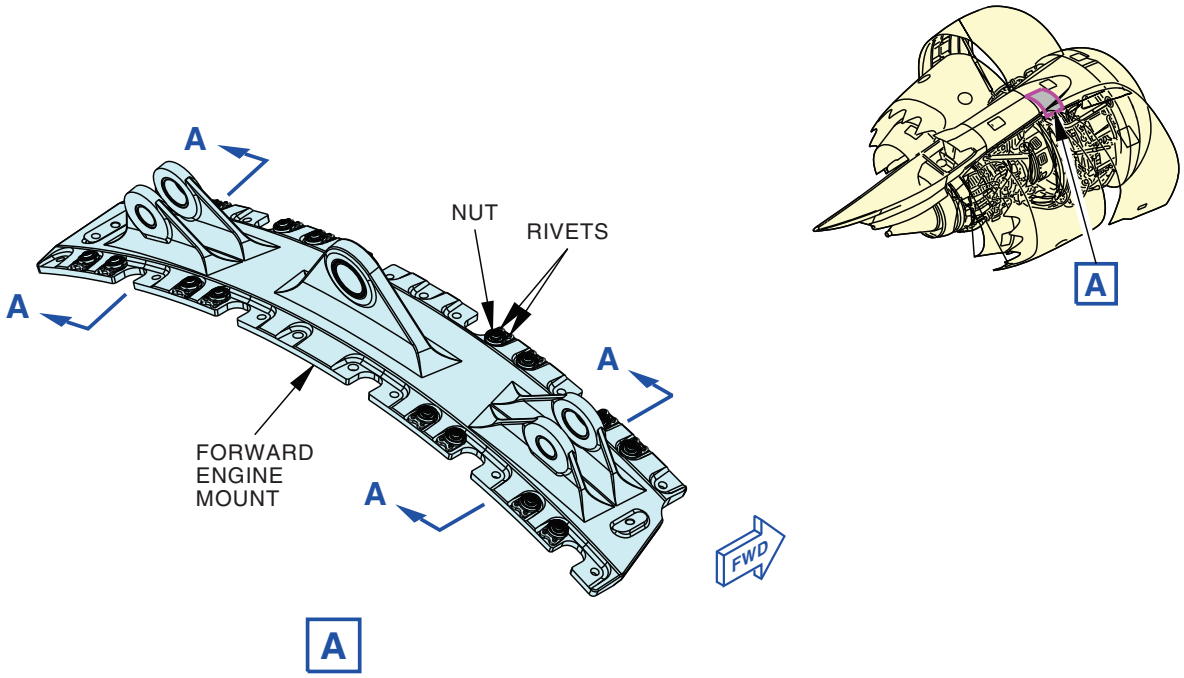


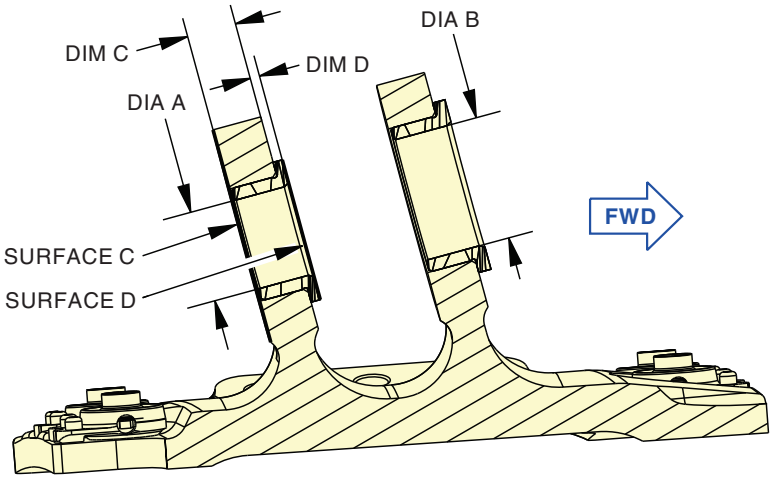
ICNLEAP1B722300D5882800080A00101

2565955 S0000612722_V1

Anti-Icing Tube Inspection
Figure 632/72-23-00-990-832-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



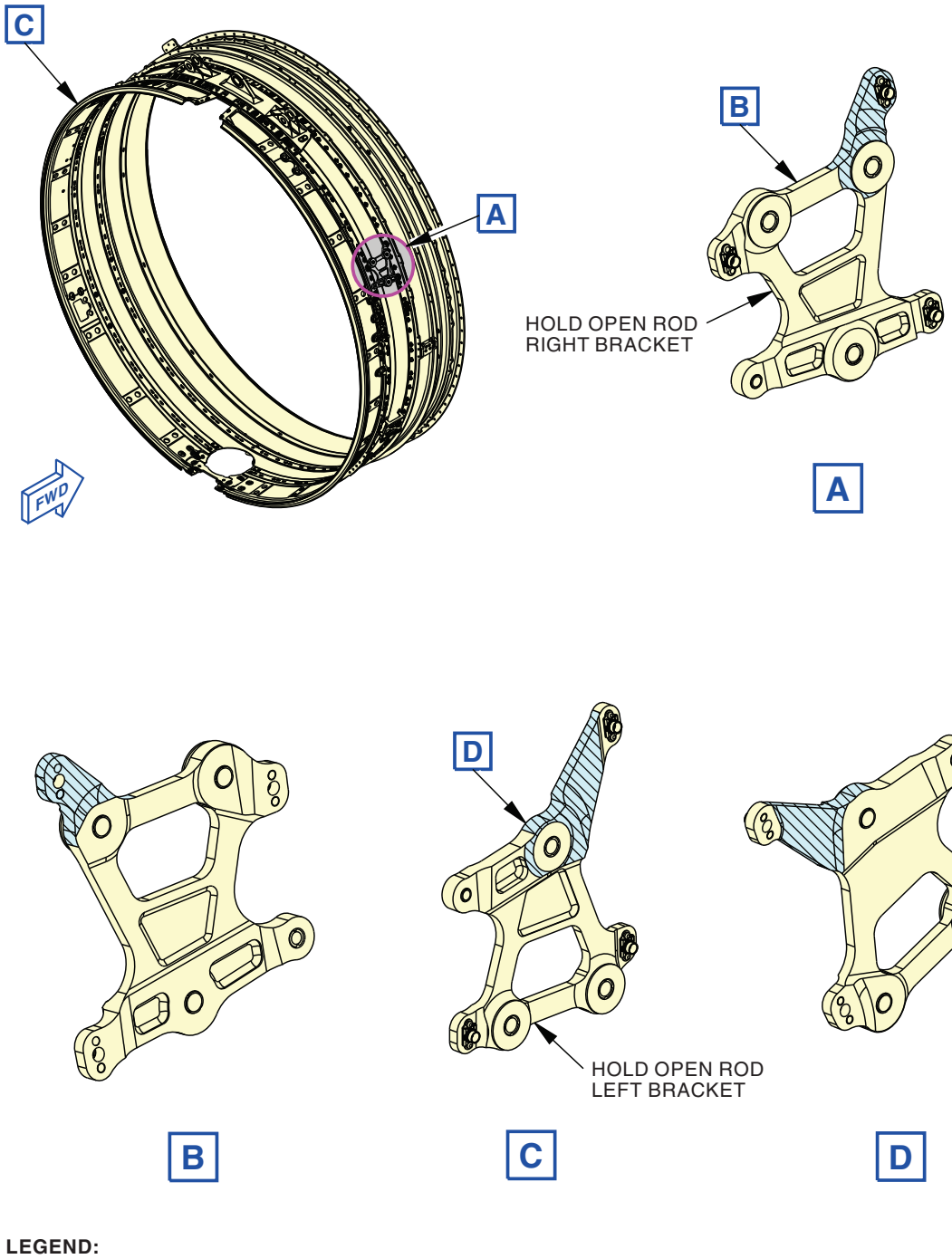


A-A

ICN LEAP 1B 722300 D 58828 00092 A001 01
2841416 S0000663022_V1

Forward Engine Mount Inspection
Figure 633/72-23-00-990-834-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

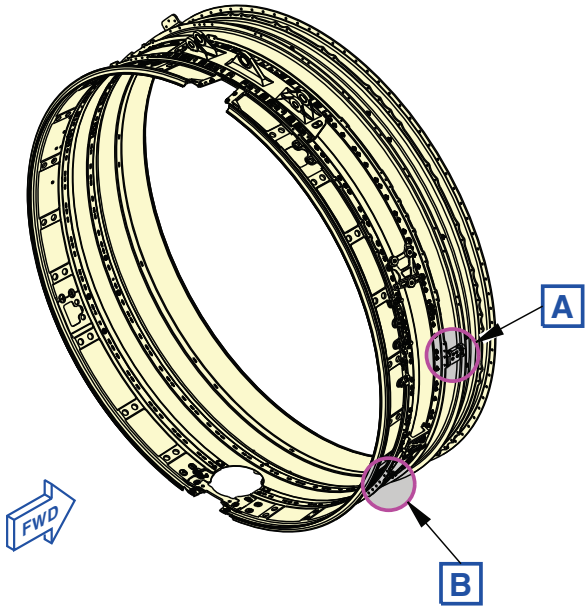
AREA A

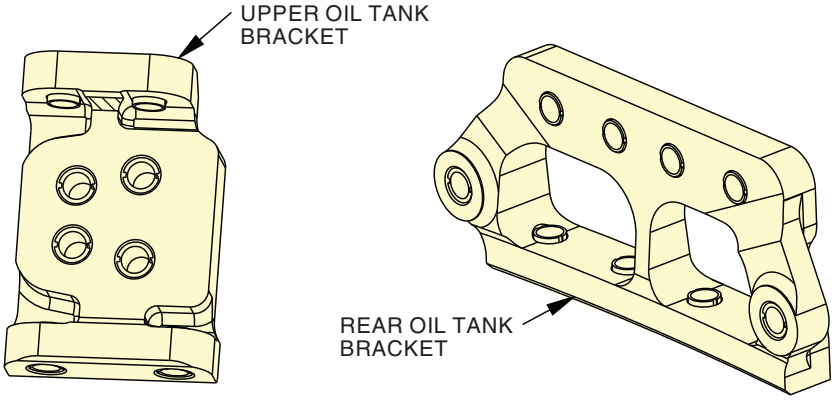
ICN LEAP 1B 722300 D 58828 00095 A001 01

2841471 S0000663025_V1

Hold Open Rod Brackets Inspection
Figure 634/72-23-00-990-837-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

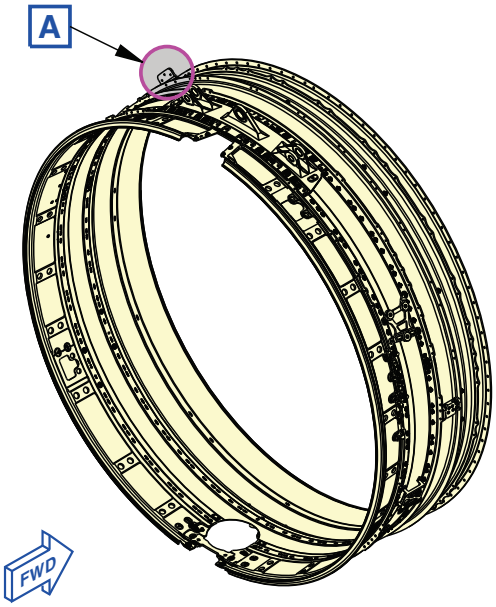
AREA A

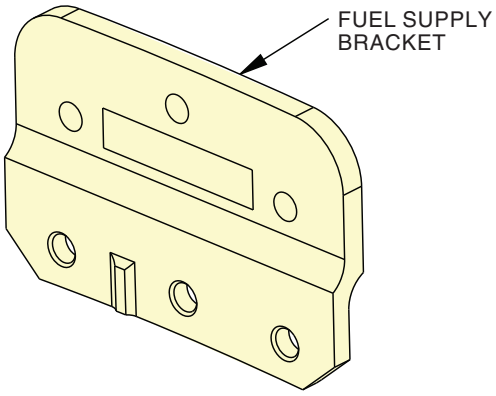
ICN LEAP 1B 722300 D 58828 00096 A001 01

2841648 S0000663026_V1

Oil Tank Brackets Inspection
Figure 635/72-23-00-990-838-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details





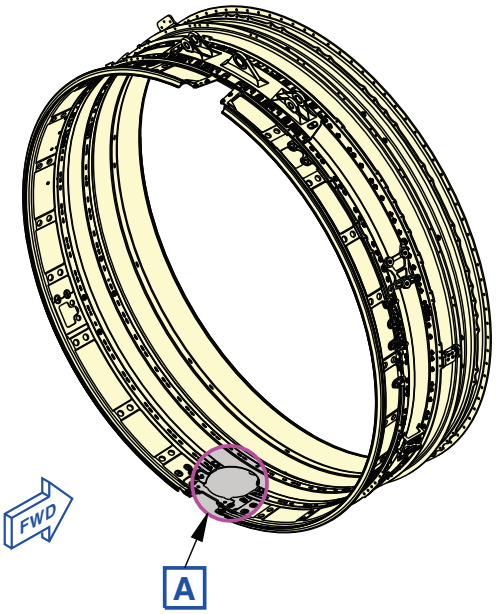
A

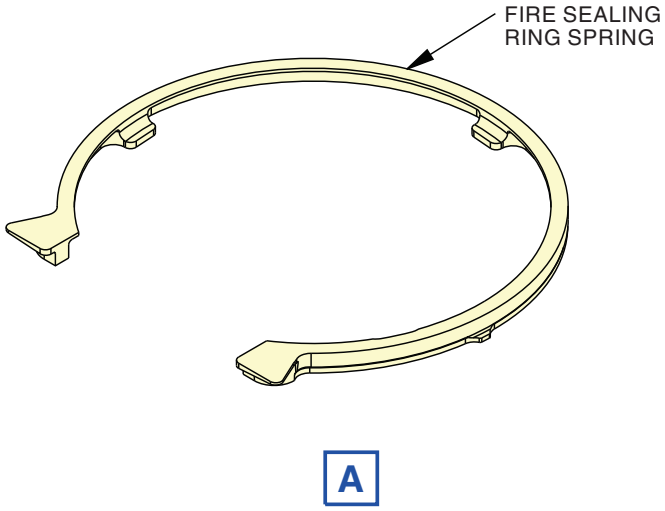
ICN LEAP 1B 722300 D 58828 00097 A001 01

2841701 S0000663027_V1

Fuel Supply Bracket Inspection
Figure 636/72-23-00-990-839-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



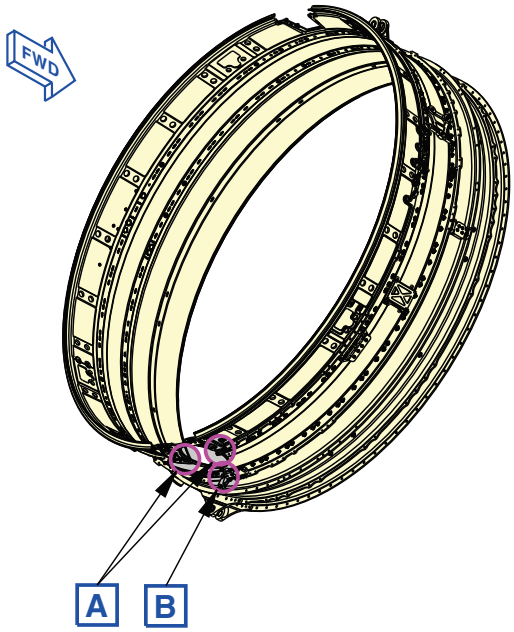


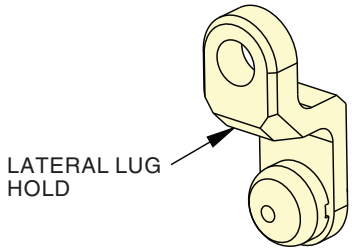
ICN LEAP 1B 722300 D 58828 00098 A001 01

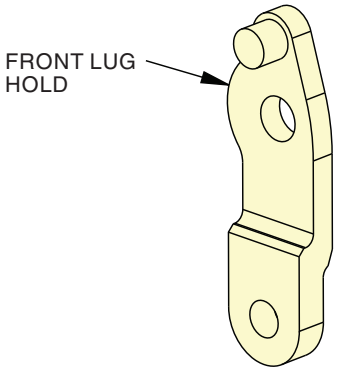
2841711 S0000663028_V1

Fire Sealing Ring Spring Inspection
Figure 637/72-23-00-990-840-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details





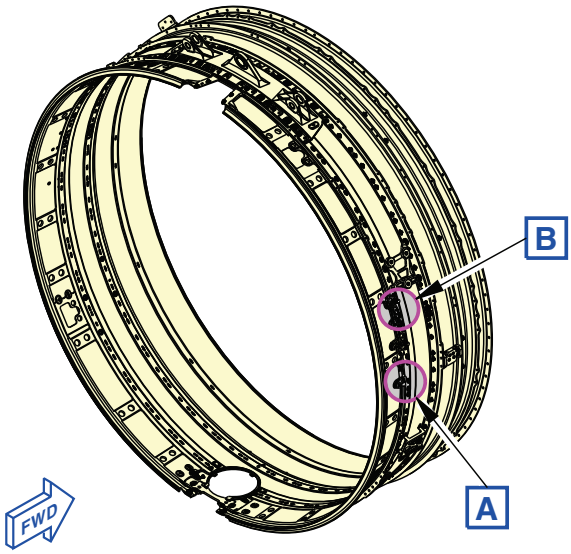


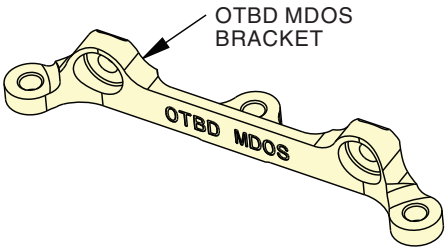
A B

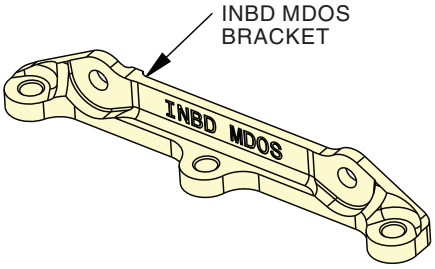
ICN LEAP 1B 722300 D 58828 00099 A001 01
2841724 S0000663029_V1

Front and Lateral Lug Hold Inspection
Figure 638/72-23-00-990-841-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details







A

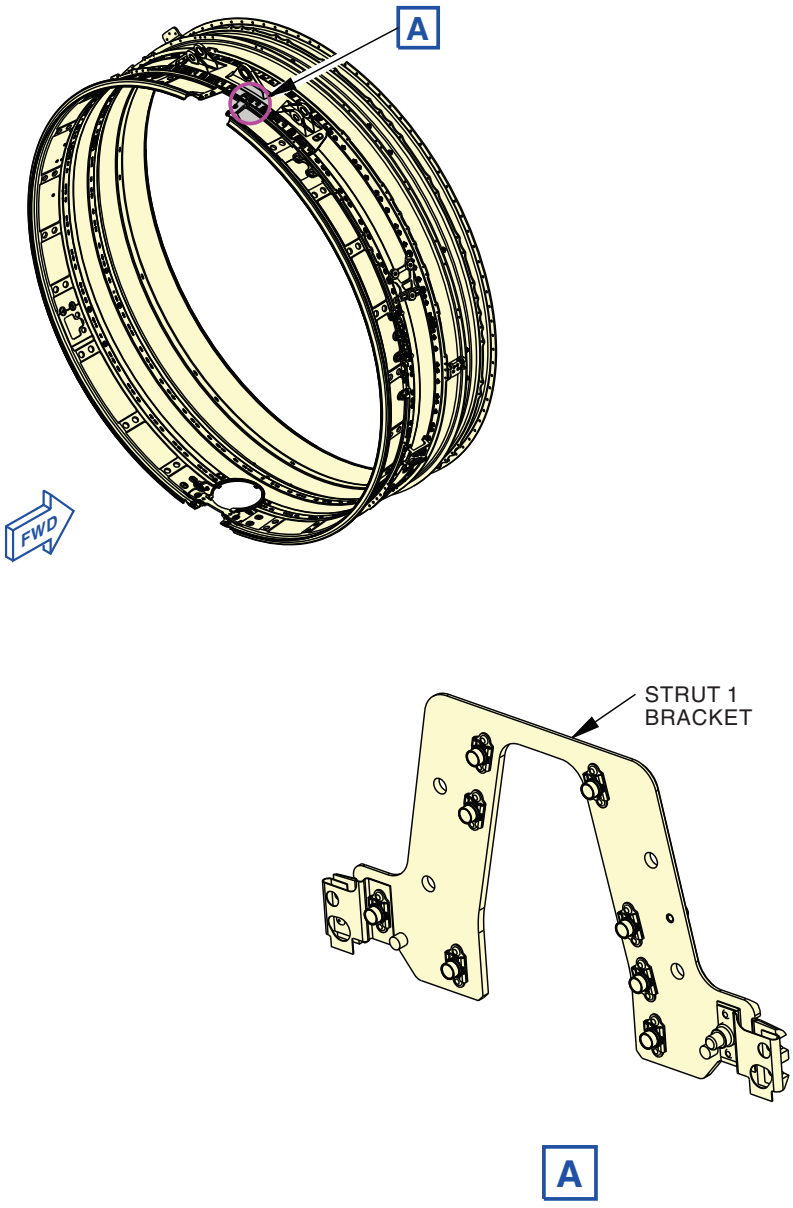
B

ICN LEAP 1B 722300 D 58828 00100 A001 01

2841758 S0000663030_V1

MDOS Brackets Inspection
Figure 639/72-23-00-990-842-G00

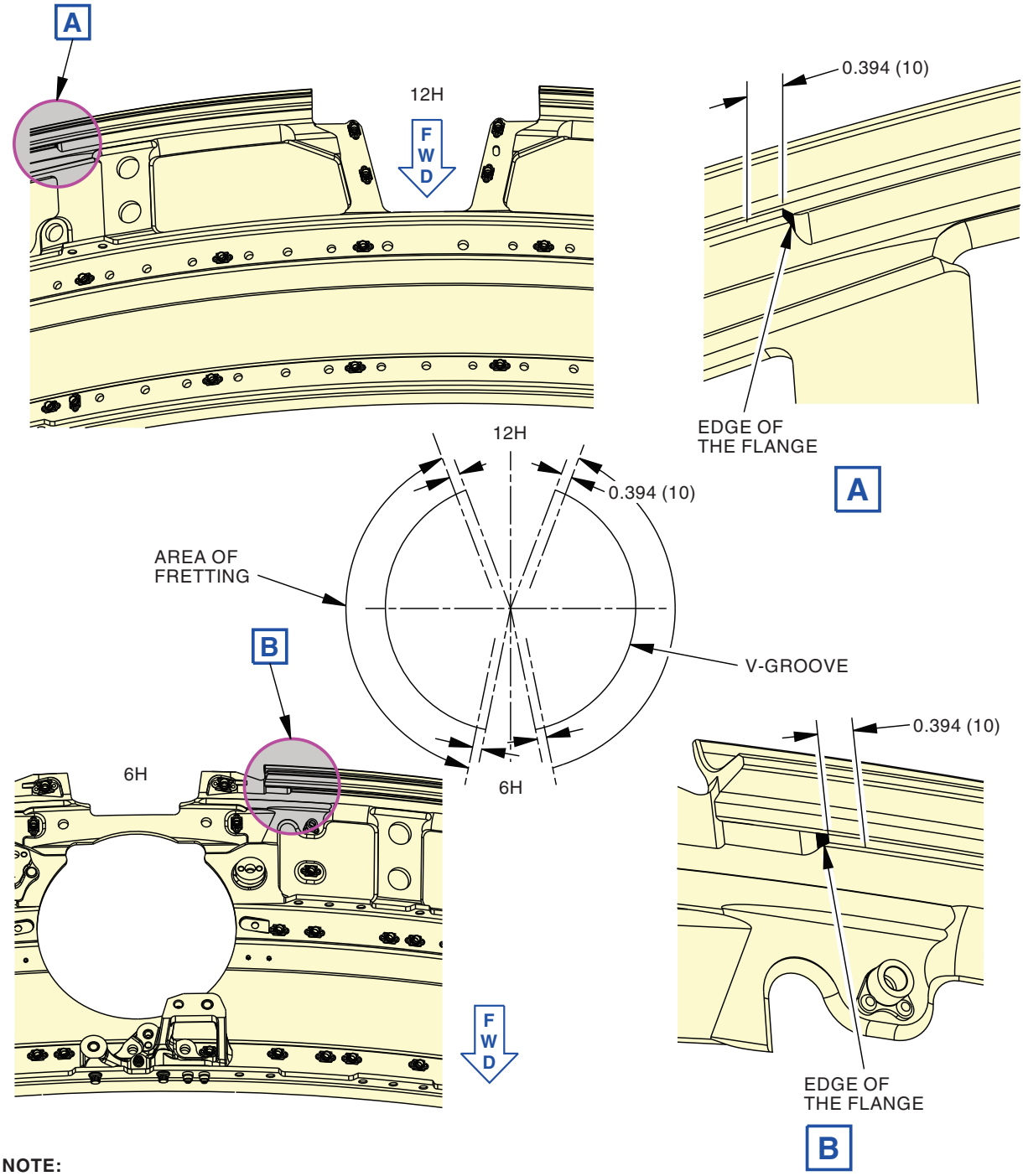
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN LEAP 1B 722300 D 58828 00101 A001 01
2841937 S0000663031_V1

Strut 1 Bracket Inspection
Figure 640/72-23-00-990-843-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

ICNLEAP1B722300D588280083A00101
2933119 S0000709894_V1

Fretting Area
Figure 641/72-23-00-990-844-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-00-211-805-G00

3. Fan Frame AGB Fittings and Axial Mount Inspection

(Figure 642)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do the inspection of the AGB fittings and the AGB axial mount.

NOTE: It is not necessary to remove the AGB for this inspection. If the AGB is installed on the engine, examine all areas of the AGB fittings and mounts that you can see. Some surfaces and dimensions are only visible when the AGB is removed from the engine. Proceed to measurements only when the AGB is removed from the engine.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-23-00-860-015-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-00-860-016-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Fan Frame AGB Fittings and Axial Mount Inspection

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-23-00-211-035-G00

(1) Examine the fan frame shroud [6].

(a) Do a visual examination of the AGB fittings (Figure 642 (Sheet 1) and Figure 642 (Sheet 2)).

1) Dents, nicks, scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth.

2) Tears/cracks

a) Not permitted.

3) Fretting/wear on mating surface DJ

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

a) Permitted up to 0.0007 in. (0.02 mm) in depth or if Dim DJ is between 0.0749 in. (1.90 mm) and 0.0826 in. (2.10 mm).

4) Fretting/wear on mating surface DO

a) Permitted up to 0.0007 in. (0.02 mm) in depth or if Dim DO is between 0.0749 in. (1.90 mm) and 0.0826 in. (2.10 mm).

5) Fretting/wear on mating surface DI

a) Permitted up to 0.0007 in. (0.02 mm) in depth or if Dim DI is between 0.0749 in. (1.90 mm) and 0.0826 in. (2.10 mm).

6) Fretting/wear on mating surface DN

a) Permitted up to 0.0007 in. (0.02 mm) in depth or if Dim DN is between 0.0749 in. (1.90 mm) and 0.0826 in. (2.10 mm).

7) Fretting/wear on diameter CE

a) Permitted if Dia CE is between 0.78740 in. (20.000 mm) and 0.78913 in. (20.044 mm).

8) Fretting/wear on diameter CG

a) Permitted if Dia CG is between 0.62992 in. (16.000 mm) and 0.63149 in. (16.040 mm).

9) Fretting/wear on diameter CH

a) Permitted if Dia CH is between 0.78740 in. (20.000 mm) and 0.78913 in. (20.044 mm).

10) Fretting/wear on diameter CI

a) Permitted if Dia CI is between 0.62992 in. (16.000 mm) and 0.63149 in. (16.040 mm).

(b) Do a visual examination of the AGB Axial Mount (Figure 642 (Sheet 1) and Figure 642 (Sheet 3)).

1) Dents, nicks, scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth.

2) Tears/cracks

a) Not permitted.

3) Fretting on mating surface DQ

a) Permitted if Dim DQ is between 0.0945 in. (2.40 mm) and 0.1023 in. (2.60 mm).

4) Fretting on mating surface DP

a) Permitted if Dim DP is between 0.5552 in. (14.10 mm) and 0.5708 in. (14.50 mm).

5) Fretting/wear on diameter AP

a) Permitted if Dia AP is between 0.96890 in. (24.610 mm) and 0.97598 in. (24.790 mm).

6) Fretting/wear on diameter AQ

a) Permitted if Dia AQ is between 0.62992 in. (16.000 mm) and 0.63149 in. (16.040 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

7) Fretting/wear on diameter AR

a) Permitted if Dia AR is between 0.78740 in. (20.000 mm) and 0.78913 in. (20.044 mm).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-23-00-211-050-G00

(2) Examine the fan frame shroud [6].

(a) Examine the AGB Axial Link Mount (Figure 642 (Sheet 4), Figure 642 (Sheet 5), and Figure 642 (Sheet 7)).

1) Do a visual examination of Dia F, Dia G, and Dia H.

a) Tears/cracks

<1> Not permitted.

b) Dents, nicks, scratches

<1> Not permitted.

c) Fretting on diameter F

<1> Permitted if Dia F is between 0.78740 in. (20.000 mm) and 0.78913 in. (20.044 mm).

d) Fretting on diameter G

<1> Permitted if Dia G is between 0.62992 in. (16.000 mm) and 0.63149 in. (16.040 mm).

e) Fretting on diameter H

<1> Permitted if Dia H is between 0.96851 in. (24.600 mm) and 0.97637 in. (24.800 mm).

2) Do a visual examination of Surface J and Surface I.

a) Tears/cracks

<1> Not permitted.

b) Dents, nicks, scratches

<1> Not permitted.

c) Fretting on surface J

<1> Permitted of Dim J is between 0.5552 in. (14.10 mm) and 0.5708 in. (14.50 mm).

d) Fretting on surface I

<1> Permitted of Dim I is between 0.0945 in. (2.400 mm) and 0.1023 in. (2.598 mm).

3) Do a visual examination of all other surfaces.

a) Tears/cracks

<1> Not permitted.

b) Dents, nicks, scratches

<1> Not permitted.

c) Fretting

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

<1> Not permitted.

4) Do a visual examination of bushings and inserts.

a) Bent, torn, worn, cut parts

<1> Not permitted.

b) Loose or missing parts

<1> Not permitted.

(b) Examine the AGB Mount Front Lower Link, Rear Lower Link, Front Upper Link, and Rear Upper Link (Figure 642 (Sheet 4), Figure 642 (Sheet 6), Figure 642 (Sheet 8)).

1) Do a visual examination of all surfaces.

a) Nicks and scratches

<1> Permitted up to 0.002 in. (0.05 mm) in depth.

b) Dents and pits

<1> Permitted up to 0.021 in. (0.53 mm) in depth.

c) Fretting

<1> Not permitted.

d) Tears/cracks

<1> Not permitted.

2) Do a visual examination of the bushings.

a) Bent, torn, worn bushings

<1> Not permitted.

b) Loose or missing bushings

<1> Not permitted.

PTF ALL

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-00-860-017-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-00-860-018-G00

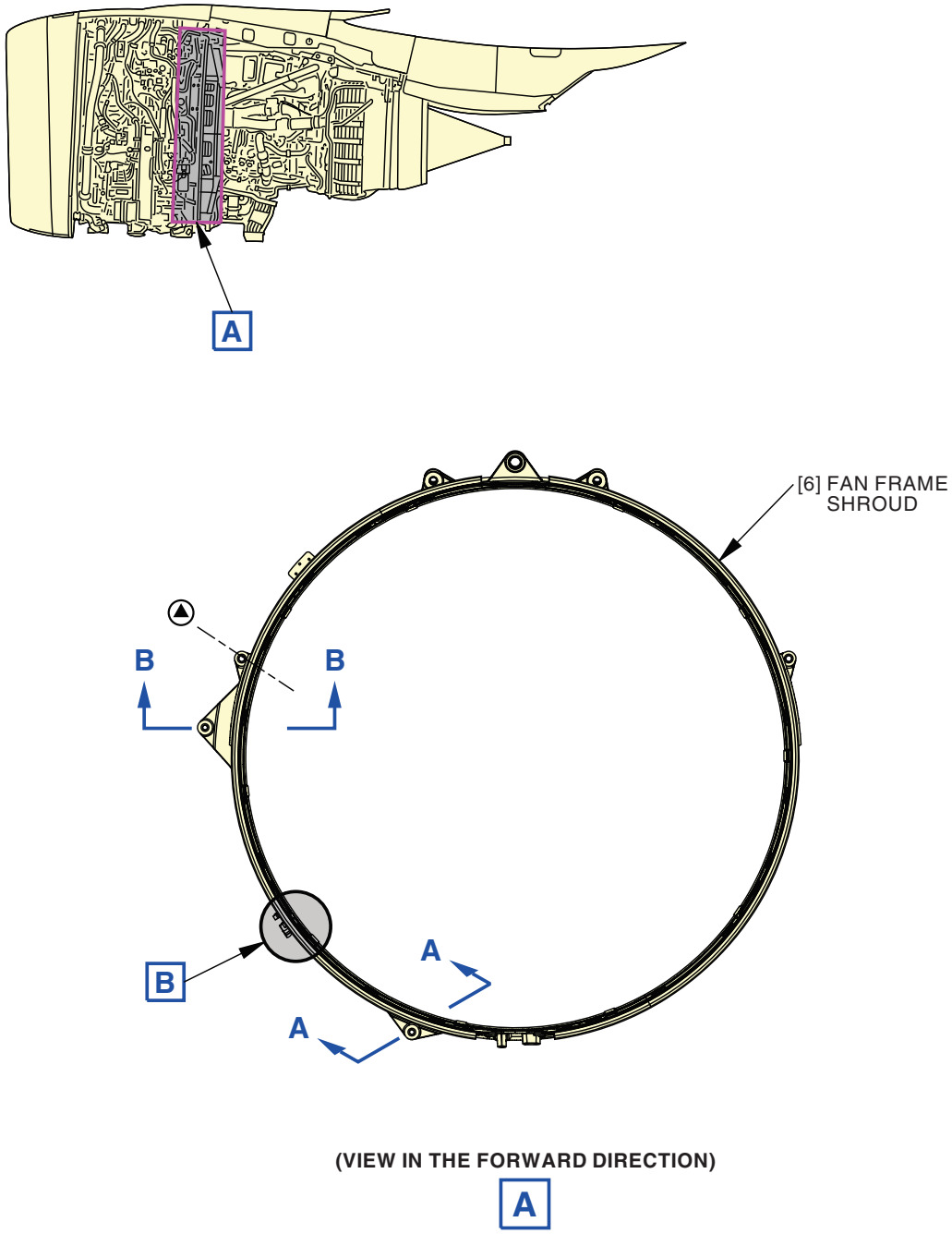
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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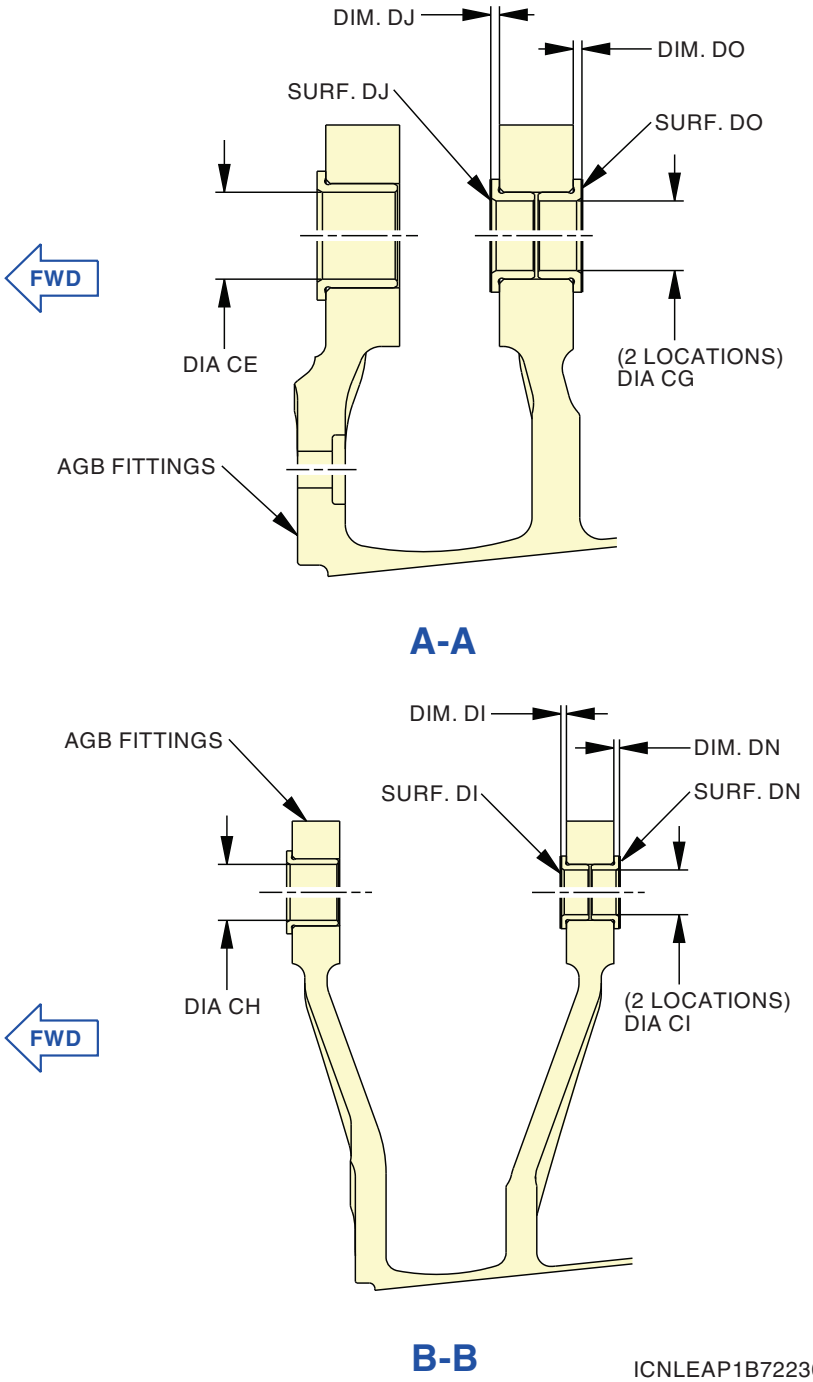
2752498 S0000628487_V1

Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 1 of 8)

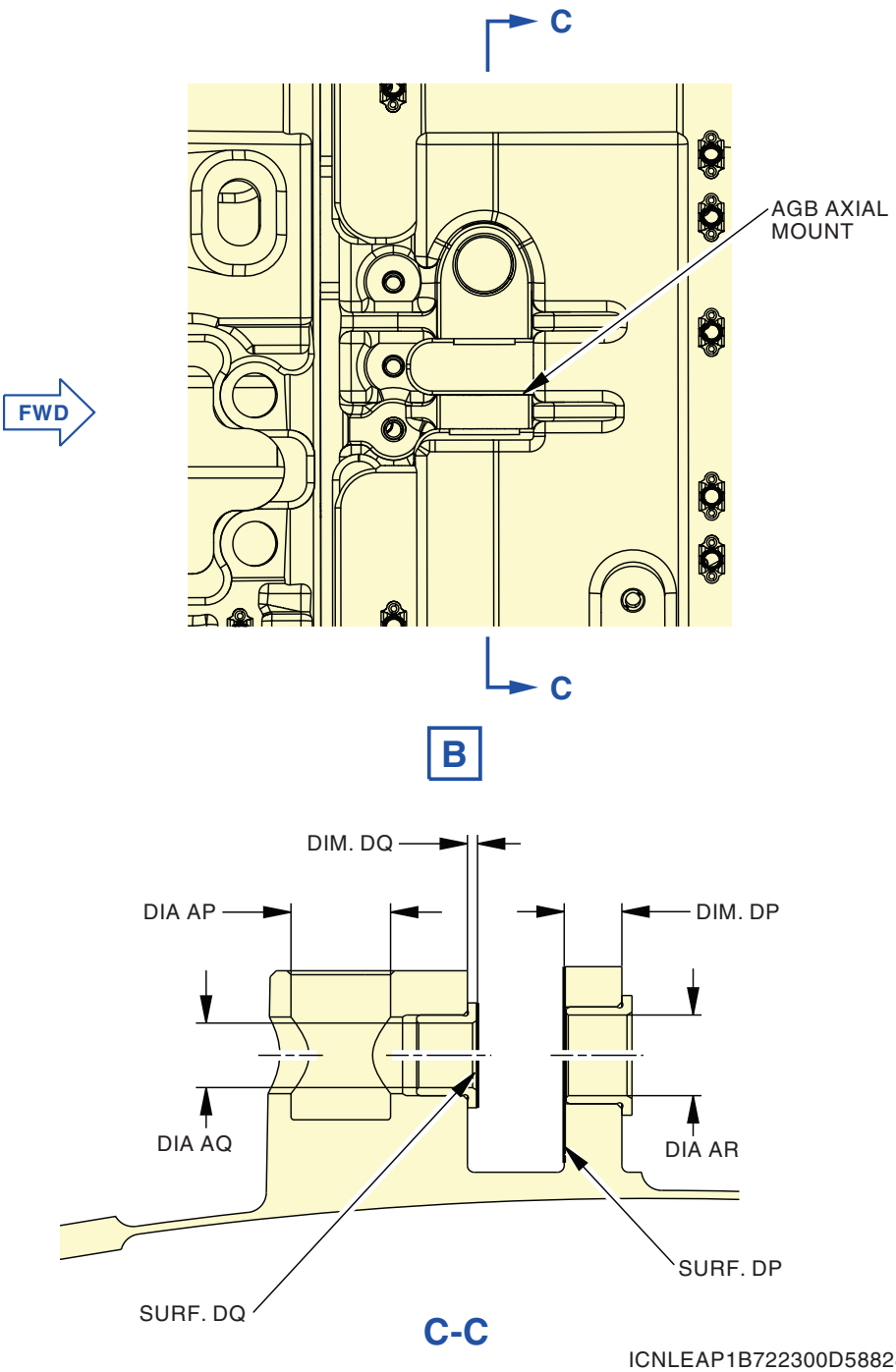
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722300D5882800021A00101
2752495 S0000628488_V1

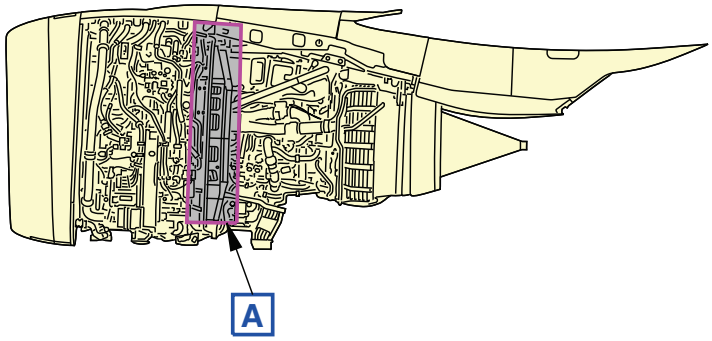
Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 2 of 8)

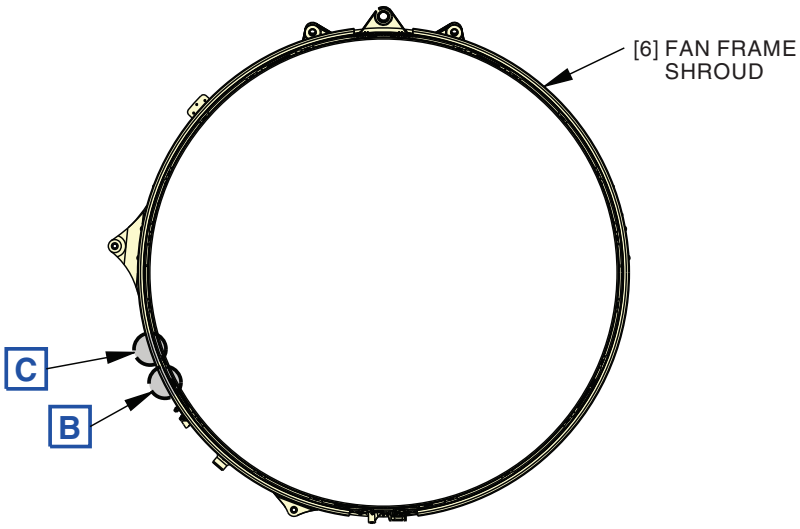
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





(VIEW IN THE FORWARD DIRECTION)

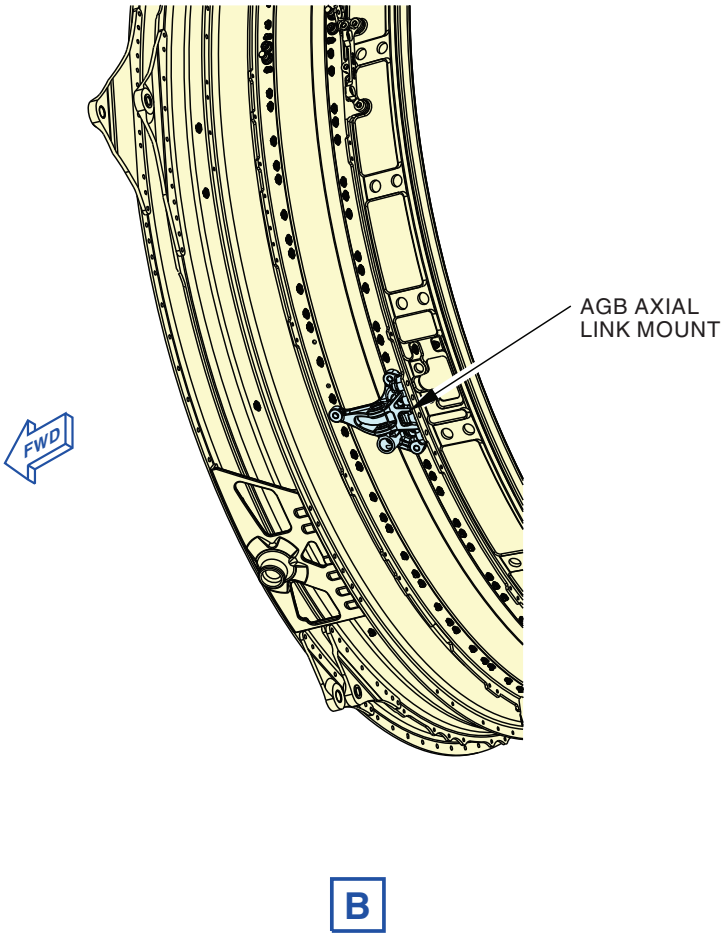
A

ICN LEAP 1B 722300 D 58828 00085 A001 01

2841941 S0000663118_V1

Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

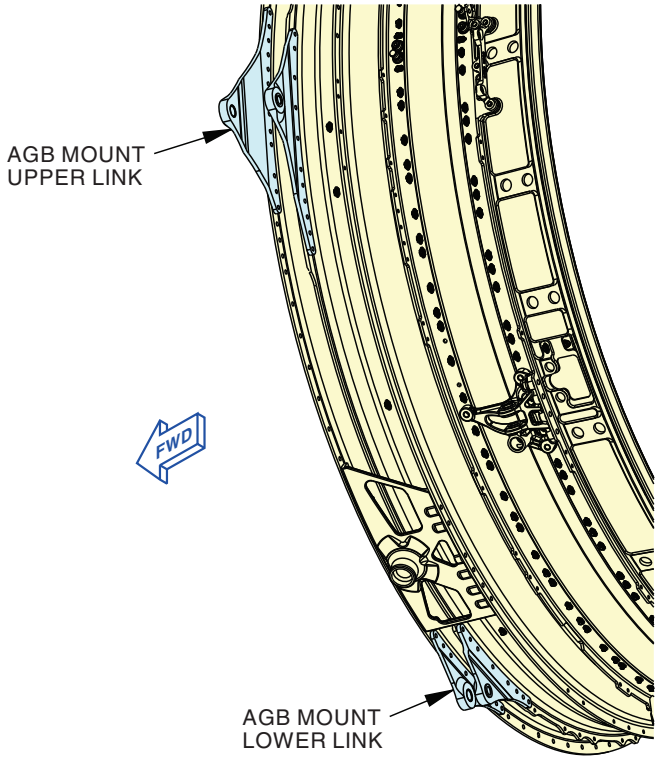


ICN LEAP 1B 722300 D 58828 00086 A001 01

2841961 S0000663119_V1

Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



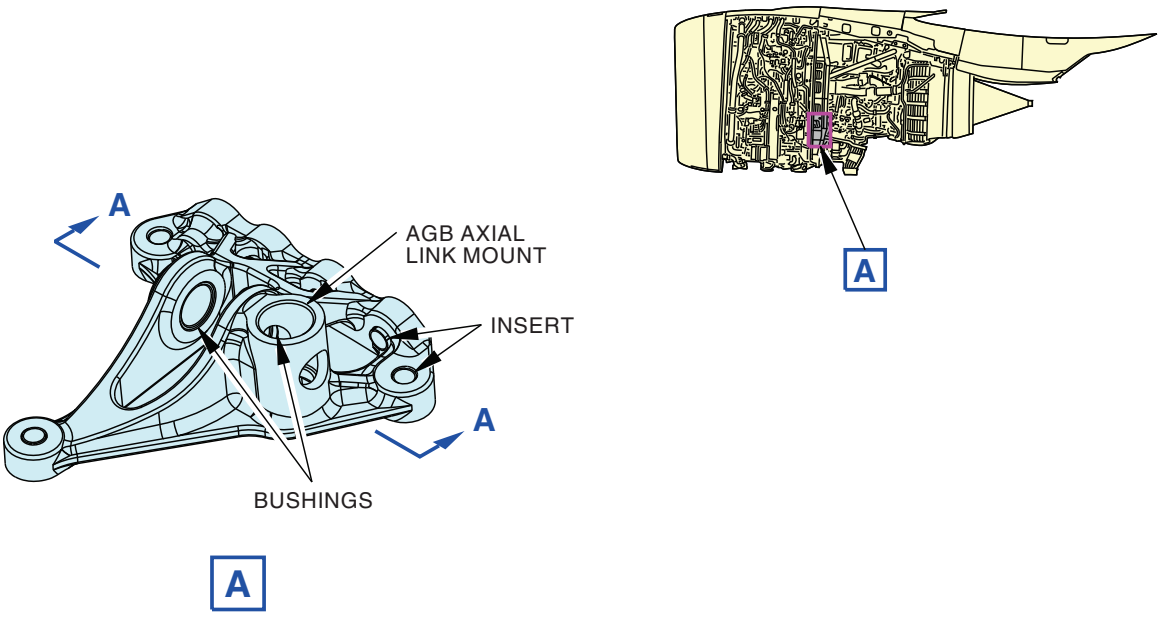
C

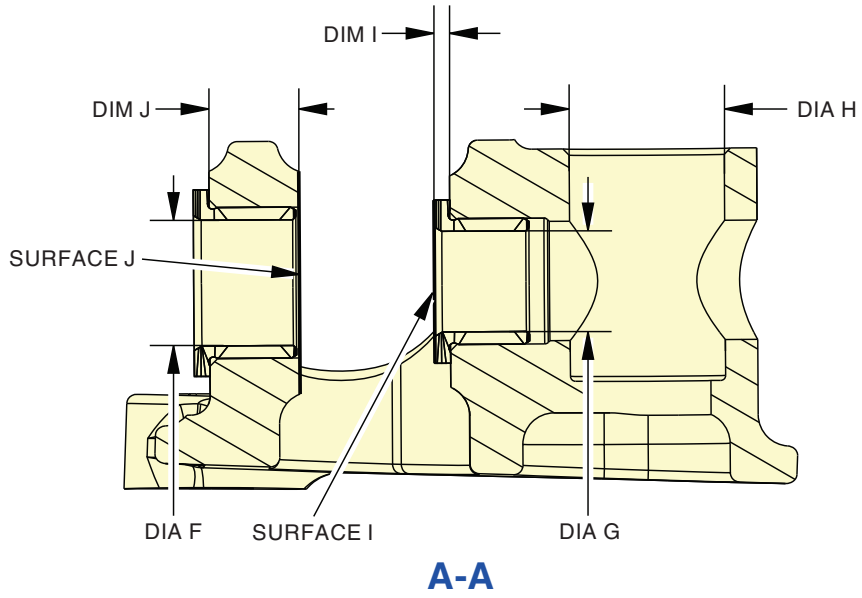
ICN LEAP 1B 722300 D 58828 00086 A001 01

2841965 S0000663120_V1

Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

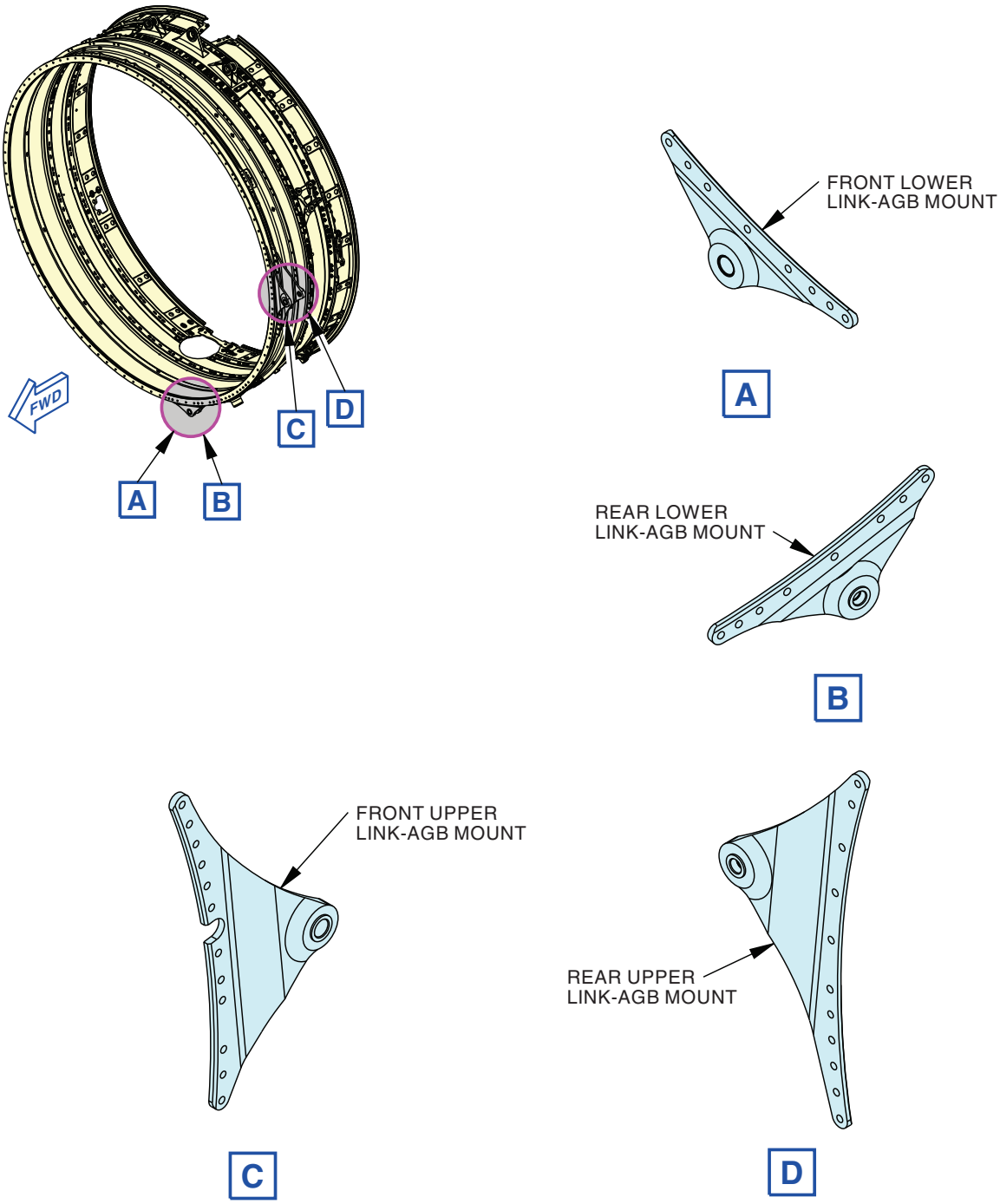




ICN LEAP 1B 722300 D 58828 00093 A001 01
2841436 S0000663023_V1

Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN LEAP 1B 722300 D 58828 00094 A001 01

2841449 S0000663024_V1

Fan Frame AGB Fittings and Axial Mount Inspection
Figure 642/72-23-00-990-833-G00 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME - REPAIRS

1. General

A. This procedure has one task:

(1) Blending of the strut 1 outer fitting

(2) Blending of the strut 5 outer fitting

(3) Replacement of the fan frame hub inserts.

TASK 72-23-00-300-801-G00

2. Strut 1 Outer Fitting - Blending

(Figure 801)

A. General

(1) This repair task gives instructions to do the blend repair on the right strut 1 outer fitting and the left strut 1 outer fitting.

B. References

Reference	Title
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-00-211-805-G00	Fan Frame AGB Fittings and Axial Mount Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51485 [S1093]	Etchant - Titanium Alloy	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Right strut 1 outer fitting	72-00-00-21-190	PTF ALL
2	Left strut 1 outer fitting	72-00-00-21-200	PTF ALL

F. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
414	Engine 1 - Fan Cowl, Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
422	Engine 2 - Nose Inlet Cowl
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right
425	Engine 2 - Thrust Reverser, Left

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Blending

SUBTASK 72-23-00-860-019-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-00-010-005-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

I. Strut 1 Outer Fitting - Blending

SUBTASK 72-23-00-211-052-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(1) Do a visual inspection of the part.

(a) Maximum of three blending are permitted.

(b) Make sure that the indication to be blended is on available surface and is not more than DIM. B (Figure 801).

1) Make sure that the depth DIM. B is less or equal to 0.023 in. (0.6 mm).

2) To locate the available areas, use the following dimensions - DIM. A is equal to 0.99 in. (25 mm).

SUBTASK 72-23-00-320-001-G00

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

(2) Do the blend of the right strut 1 outer fitting [1] or/and left strut 1 outer fitting [2].

NOTE: Remove minimum material more than the depth of deviation while blending.

NOTE: Blended areas must make a smooth contour into the adjacent areas to keep correct airflow.

(a) Method I

1) Remove the indication by hand blending (TASK 70-43-42-300-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Method II

1) Remove the indication by power blending (TASK 70-43-42-300-803-G00).

(c) If the minimum acceptable distance between two repaired areas is less than 40 times the DIM. B, mix them to get only one repaired area.

SUBTASK 72-23-00-110-001-G00

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(3) Clean the part using alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-00-350-001-G00

(4) Do a local swab etching of the repaired areas with titanium alloy etchant, G51485 [S1093] (TASK 70-43-24-100-802-G00).

SUBTASK 72-23-00-230-001-G00

(5) Do a fluorescent penetrant inspection of the repaired areas (TASK 70-11-06-230-803-G00).

(a) Use sensitivity level 2.

(b) Indications more than 0.0118 in. (0.30 mm) are not permitted.

(c) Linear indications are not permitted.

NOTE: Linear indications are indications that are at least four times longer than they are wide.

SUBTASK 72-23-00-211-053-G00

(6) Do a visual inspection of the parts (TASK 72-23-00-211-805-G00).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-00-410-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-00-860-020-G00

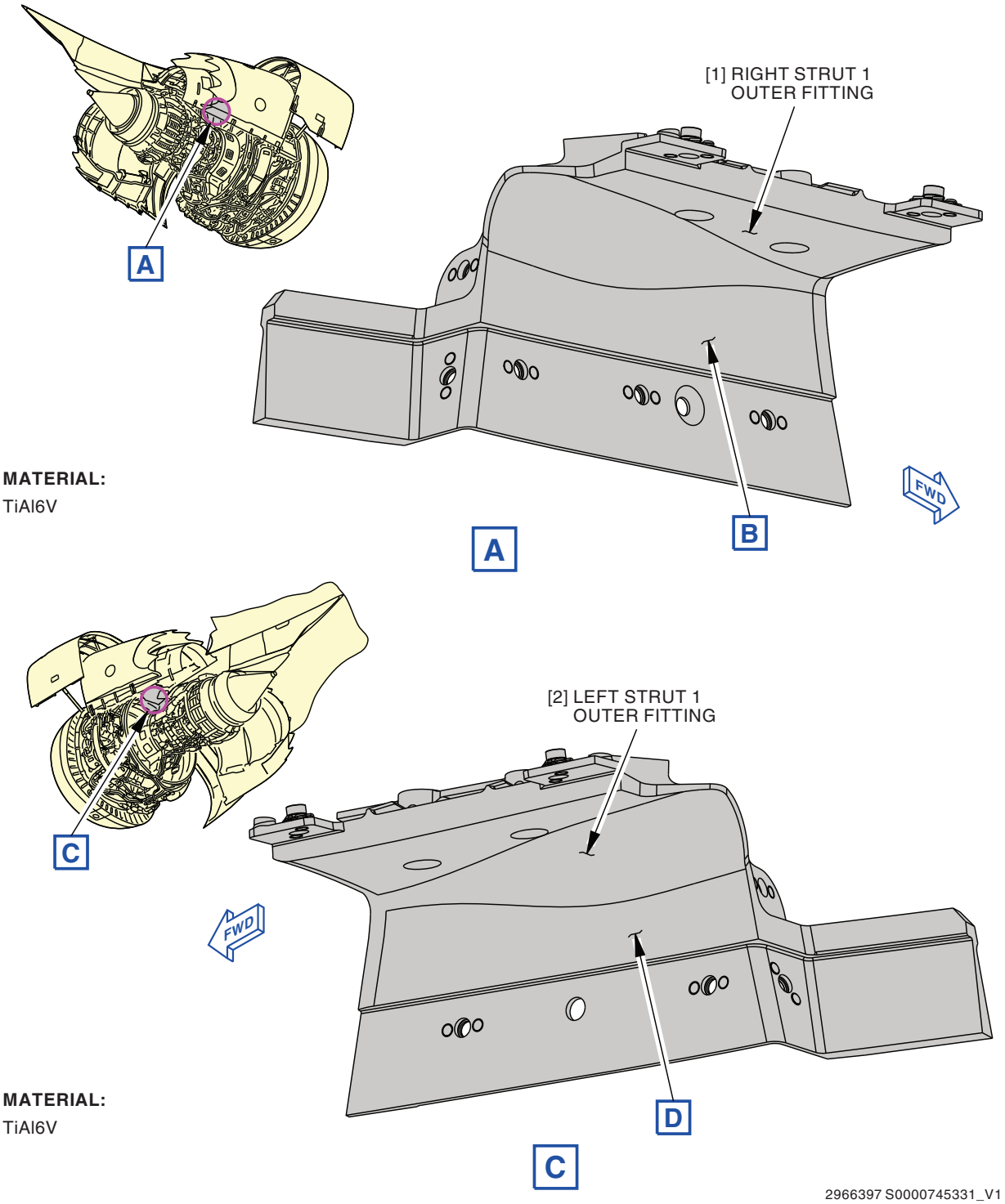
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

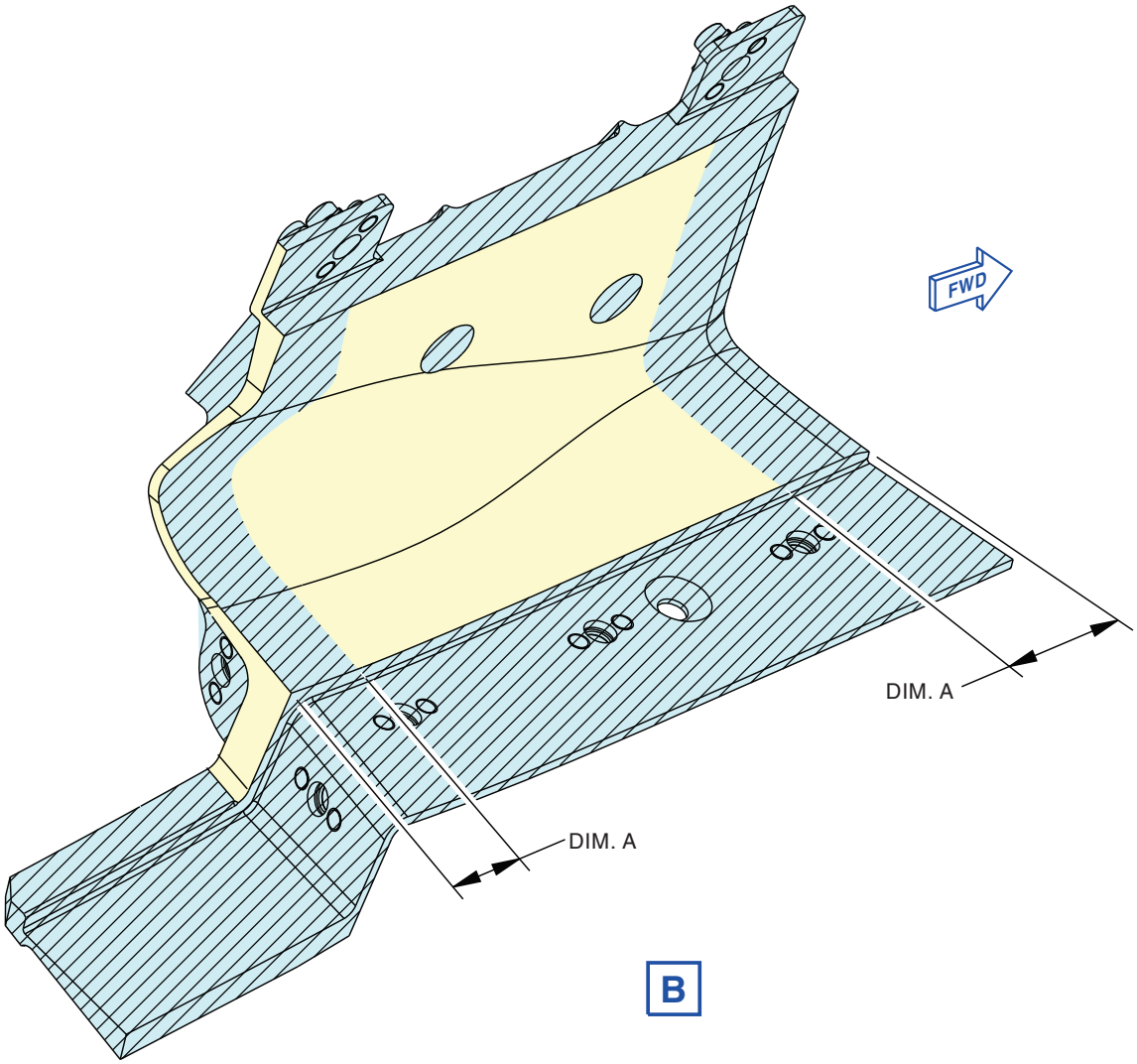
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Struts 1 Outer Fitting
Figure 801/72-23-00-990-849-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



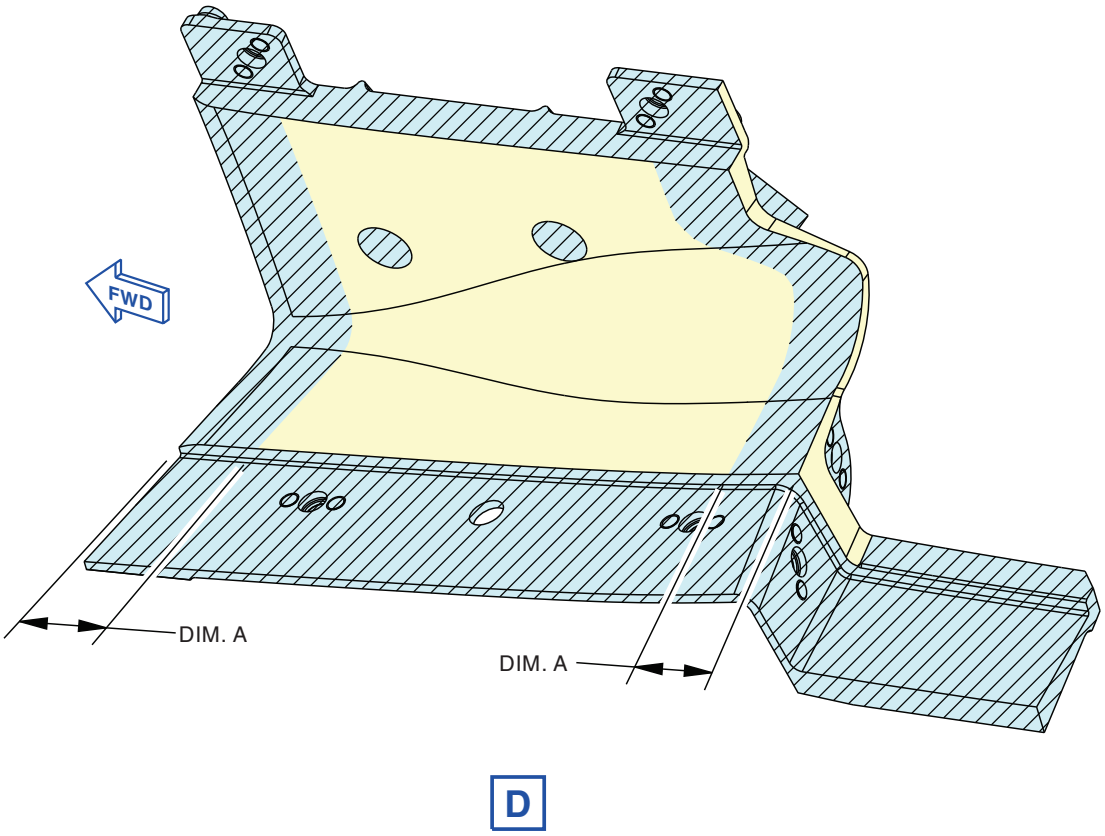
LEGEND:

NO BLENDING

2966399 S0000745332_V1

Struts 1 Outer Fitting
Figure 801/72-23-00-990-849-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



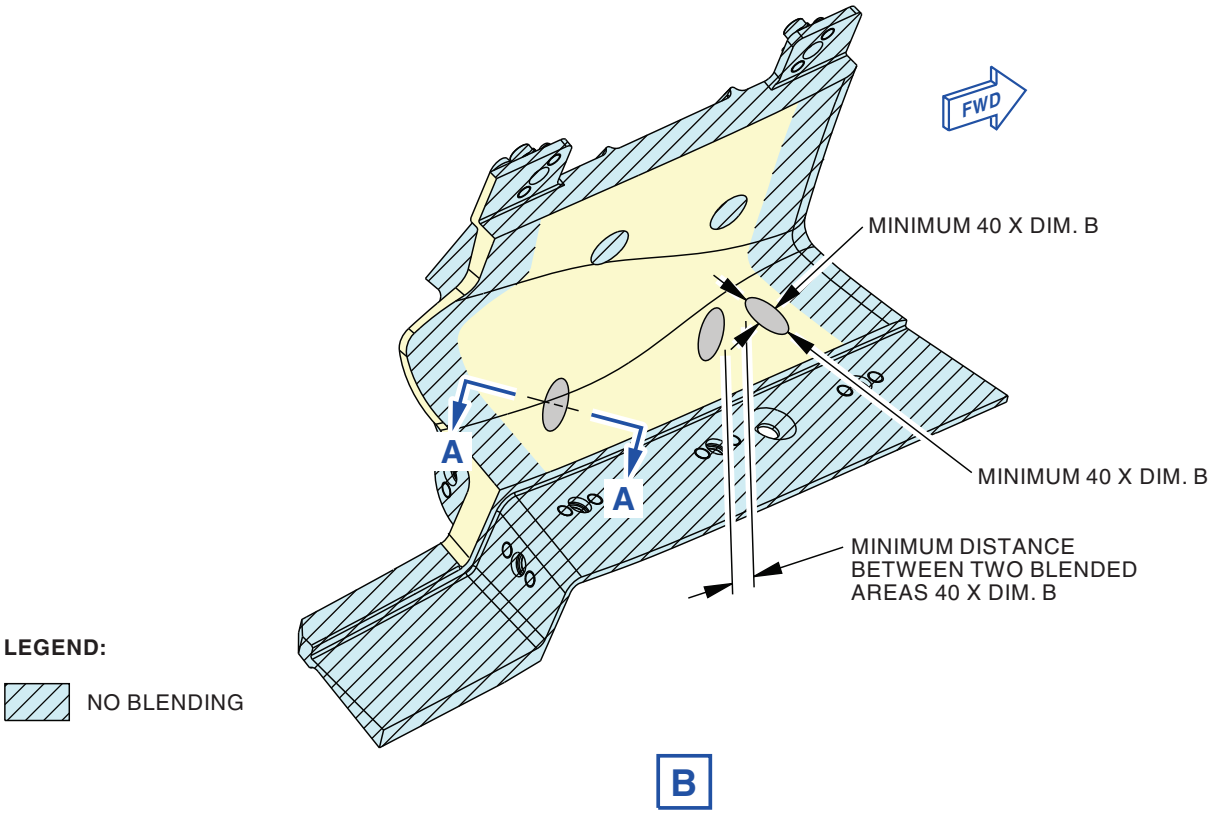
LEGEND:

NO BLENDING

2966488 S0000745333_V1

Struts 1 Outer Fitting
Figure 801/72-23-00-990-849-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ACCEPTABLE BLENDING PROFILE
A-A

2966491 S0000745334_V2

Struts 1 Outer Fitting
Figure 801/72-23-00-990-849-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-00-300-802-G00

3. Strut 5 Outer Fitting - Blending

(Figure 802)

A. General

(1) This repair task gives instructions to do the blend repair on the strut 5 outer fitting.

B. References

Reference	Title
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-00-211-804-G00	Fan Frame Module Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51485 [S1093]	Etchant - Titanium Alloy	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Strut 5 outer fitting	72-00-00-21-640	PTF ALL

F. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
422	Engine 2 - Nose Inlet Cowl
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right
425	Engine 2 - Thrust Reverser, Left

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Blending

SUBTASK 72-23-00-860-022-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-00-010-008-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

I. Strut 5 Outer Fitting Blending

SUBTASK 72-23-00-211-054-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(1) Do a visual inspection of the strut 5 outer fitting [3] (Figure 802).

(a) Maximum three blending are permitted.

(b) Make sure that the indication to be blended is on available surface and is not more than DIM. F.

1) Make sure that the depth DIM. F is less or equal to 0.023 in. (0.6 mm).

2) To locate the available areas, use the following dimensions:

a) DIM. A is equal to 0.79 in. (20 mm)

b) DIM. B is equal to 1.42 in. (36 mm)

c) DIM. C is equal to 1.74 in. (44 mm)

d) DIM. D is equal to 0.99 in. (25 mm)

e) DIM. E is equal to 1.23 in. (31 mm).

SUBTASK 72-23-00-320-002-G00

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

CAUTION REMOVE MINIMUM MATERIAL MORE THAN THE DEPTH OF DEVIATION WHEN YOU BLEND. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

CAUTION BLENDED AREAS MUST MAKE A SMOOTH CONTOUR INTO THE ADJACENT AREAS TO KEEP CORRECT AIRFLOW. IF YOU DO NOT OBEY, DAMAGE TO PARTS CAN OCCUR.

(2) Do the blend of the strut 5 outer fitting [3] (Figure 802).

(a) Do one of the following:

1) Remove the indication by hand blending (TASK 70-43-42-300-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Alternative procedure. Remove the indication by power blending (TASK 70-43-42-300-803-G00).

(b) If the minimum acceptable distance between two repaired areas is less than 40 times the DIM. F, mix them to get only one repaired area.

SUBTASK 72-23-00-110-002-G00

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(3) Clean the part with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-00-350-002-G00

(4) Do a local swab etching of the repaired areas with titanium alloy etchant, G51485 [S1093] (TASK 70-43-24-100-802-G00).

SUBTASK 72-23-00-230-002-G00

(5) Do a Fluorescent Penetrant Inspection (FPI) of the repaired areas (TASK 70-11-06-230-803-G00) .

(a) Use sensitivity level 2.

(b) Indications more than 0.0118 in. (0.30 mm) are not permitted.

(c) Linear indications are not permitted.

NOTE: Linear indications are indications that are at least four times longer than they are wide.

SUBTASK 72-23-00-211-055-G00

(6) Do a visual inspection of the strut 5 outer fitting [3] (TASK 72-23-00-211-804-G00).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-00-410-006-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-00-860-023-G00

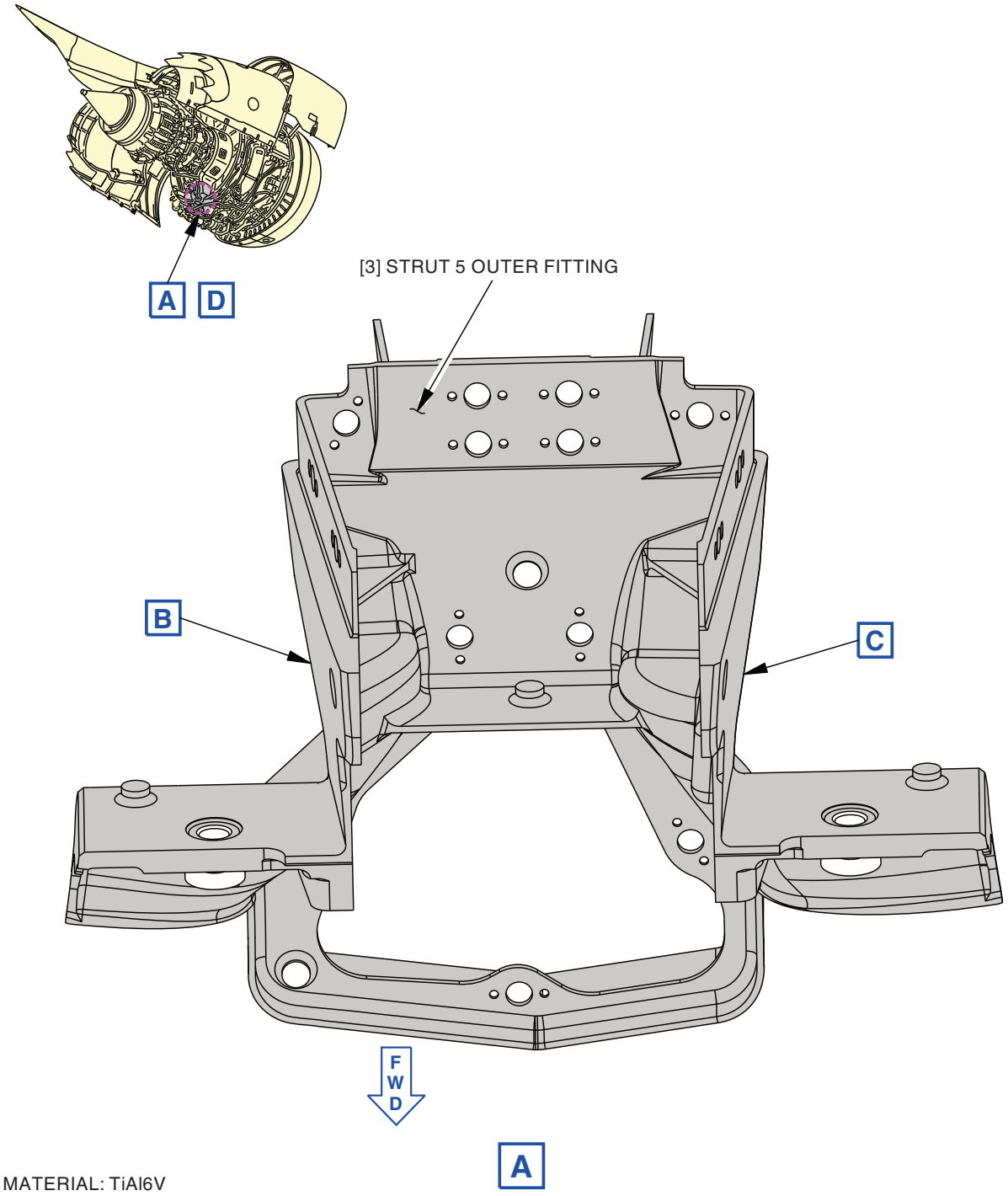
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

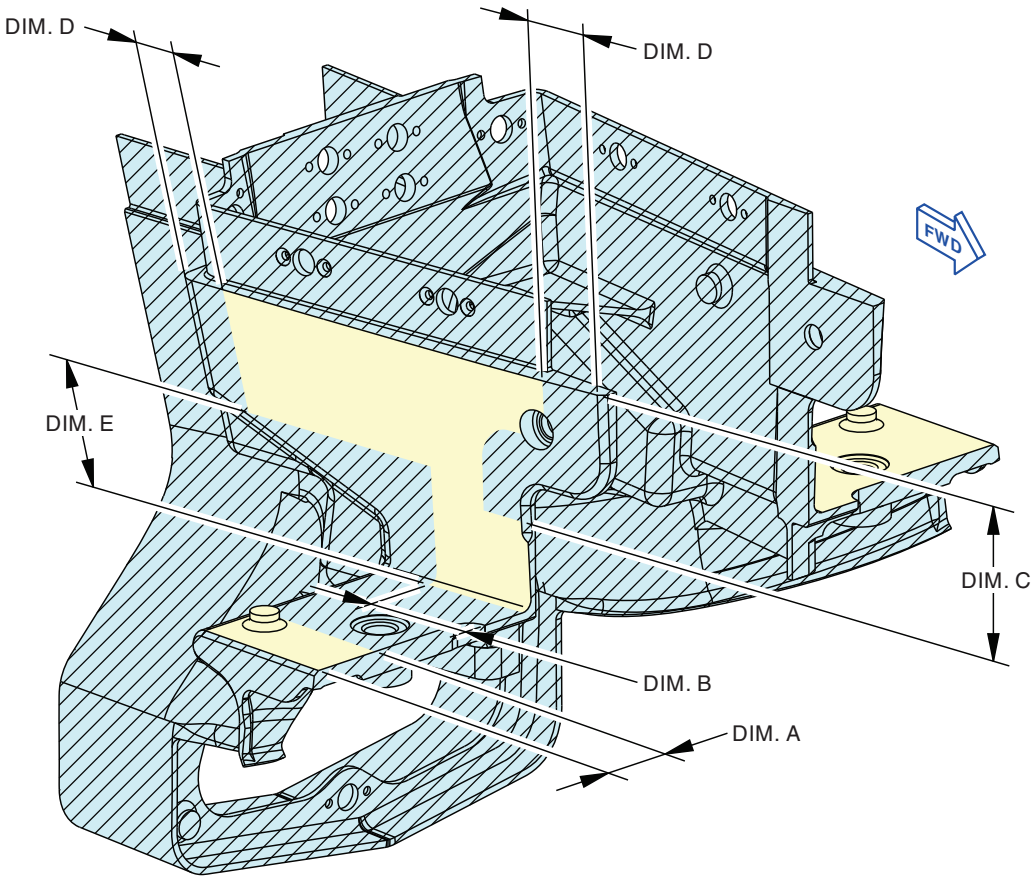
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



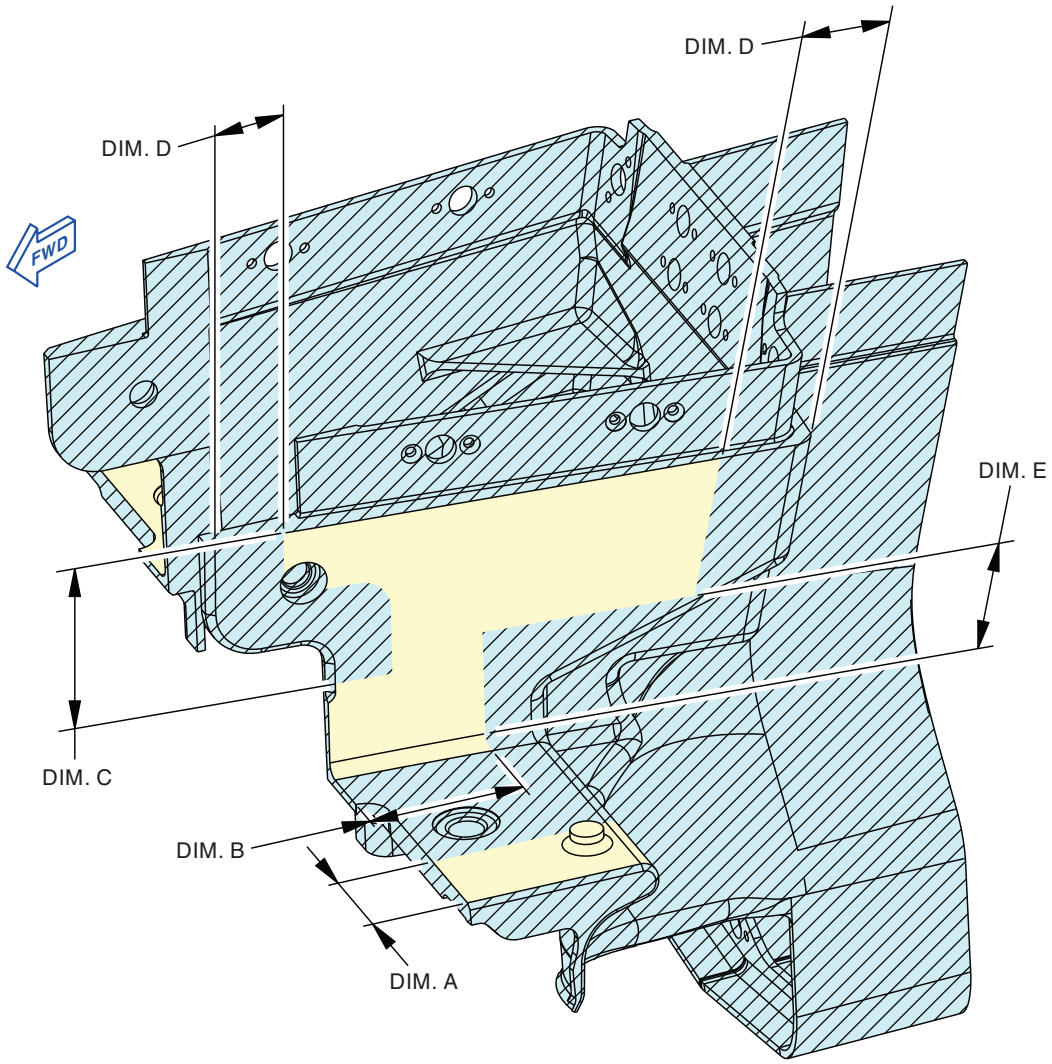
Strut 5 Outer Fitting Blending
Figure 802/72-23-00-990-853-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



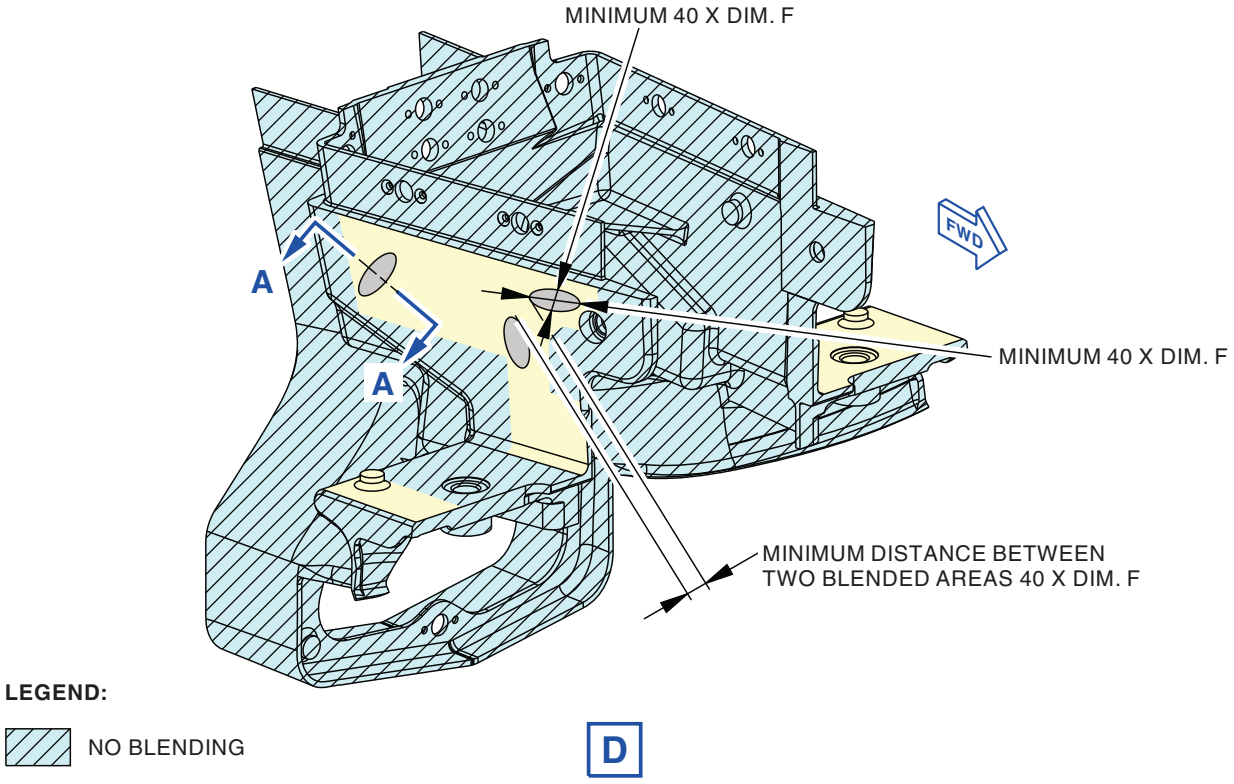
Strut 5 Outer Fitting Blending
Figure 802/72-23-00-990-853-G00 (Sheet 2 of 4)

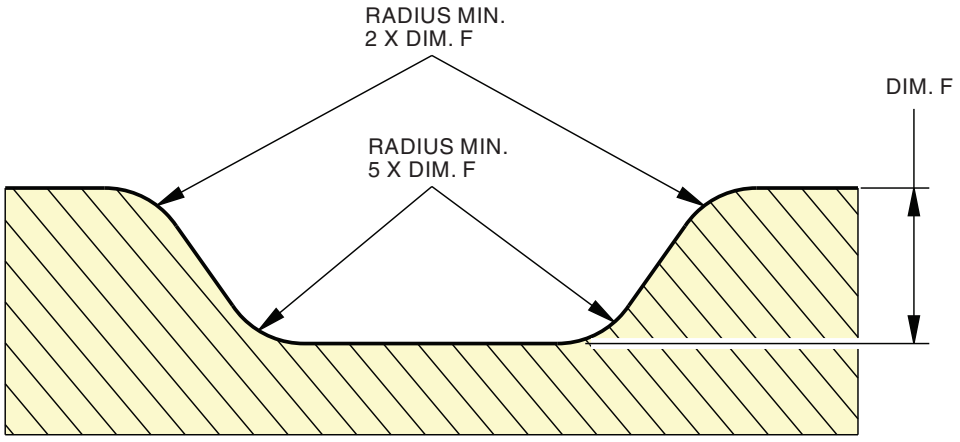
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut 5 Outer Fitting Blending
Figure 802/72-23-00-990-853-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





ACCEPTABLE BLENDING PROFILE
A-A

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Strut 5 Outer Fitting Blending
Figure 802/72-23-00-990-853-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-00-900-801-G00

4. Fan Frame Hub Inserts Replacement

(Figure 803, Figure 804, and Figure 805)

A. General

(1) This task gives instructions to replace the inserts on the fan frame hub.

B. References

Reference	Title
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
70-44-50-320-801-G00	Drilling (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-6559	Tool - Insert Broach

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
B50190 [CP1040]	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
B50344 [CP2726]	Solvent - Supersolve AS	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Replacement

SUBTASK 72-23-00-860-021-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-00-010-006-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Hub Inserts Replacement

SUBTASK 72-23-00-010-007-G00

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION BE CAREFUL WHEN YOU REMOVE THE KEY-LOCKED INSERT ON THE FAN FRAME HUB. IF YOU DO NOT OBEY, DAMAGE TO THE ADJACENT SURFACES CAN OCCUR.

(1) To remove the insert, do these steps (Figure 804):

(a) Do one of the following:

1) Drill the keys of the self-locking inserts [4] and inserts [5], do this task: Drilling, TASK 70-44-50-320-801-G00.

2) Alternative procedure. Grind the keys of the self-locking inserts [4] and inserts [5], do this task: Blending Procedures - Power Blending, TASK 70-43-42-300-803-G00.

(b) Push the self-broaching keys to the center.

(c) Put an extractor in place.

(d) Unscrew, remove, and discard the self-locking inserts [4] and inserts [5].

SUBTASK 72-23-00-100-001-G00

(2) Clean the insert recess with cloth moist with one of the following solvent:

(a) acetone solvent, B01058 [CP1039]

(b) solvent, B50190 [CP1040]

(c) alcohol, B00676 [CP1041]

(d) degreasing solvent, B50333 [CP2687]

(e) solvent, B50344 [CP2726].

SUBTASK 72-23-00-210-003-G00

(3) Do a visual inspection of the full area that is repaired (Figure 803):

(a) Burrs, arrachements, missing threads, and unwanted material are not permitted.

SUBTASK 72-23-00-420-001-G00

(4) Install new self-locking inserts [4] and inserts [5] as follows (Figure 805):

(a) Screw the self-locking inserts [4] and inserts [5].

(b) Align the self-broaching keys with the slots of the previous keys with a broach tool, STD-6559.

(c) Strike the top of the broach tool, STD-6559.

(d) Remove the broach tool, STD-6559.

SUBTASK 72-23-00-220-001-G00

(5) Do a dimensional inspection (Figure 805).

(a) Make sure that DIM. X is 0.0098 in. (0.25 mm) - 0.0295 in. (0.75 mm).

(b) Make sure that DIM. Y is 0.0051 in. (0.13 mm) - 0.0295 in. (0.75 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-00-410-007-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-00-860-024-G00

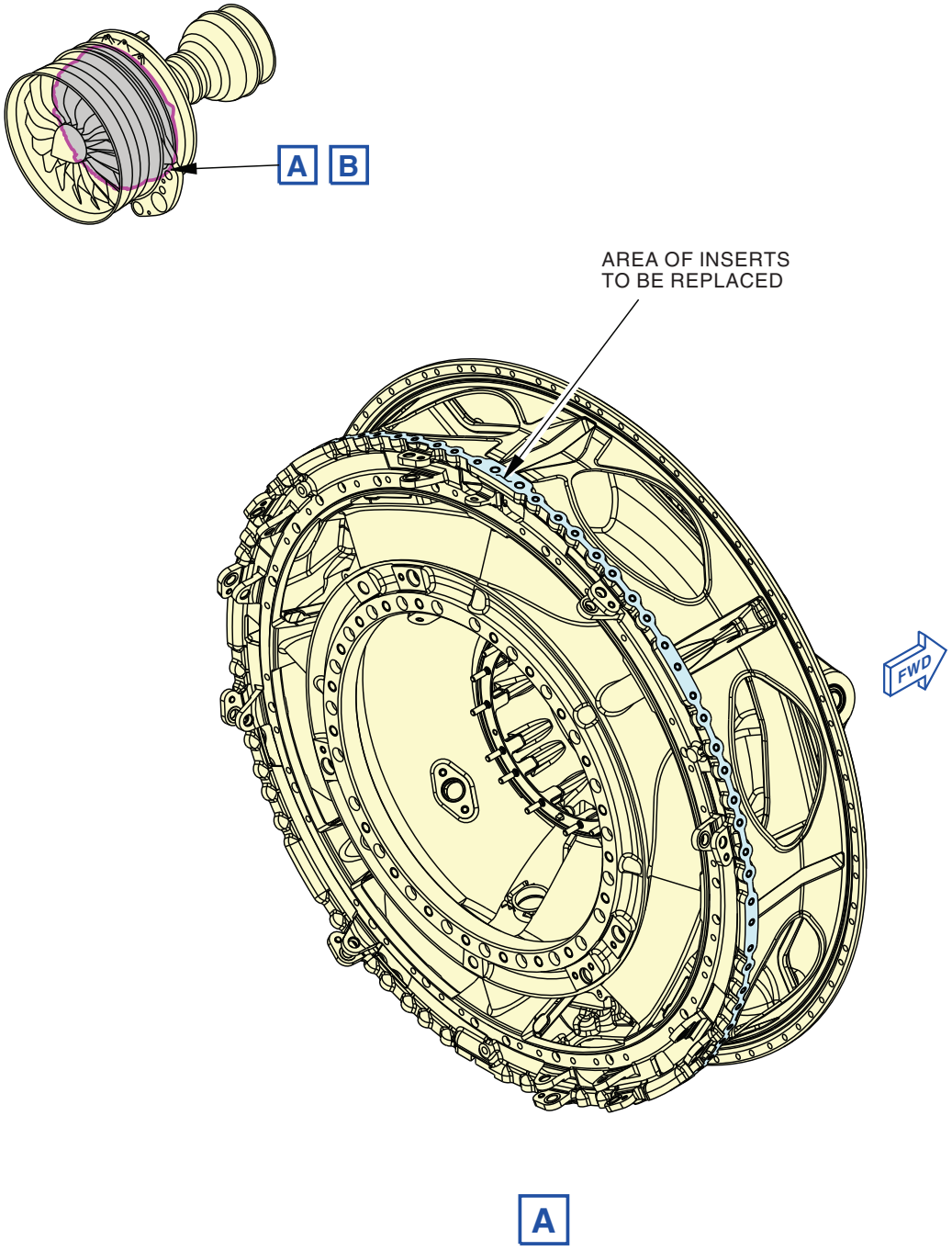
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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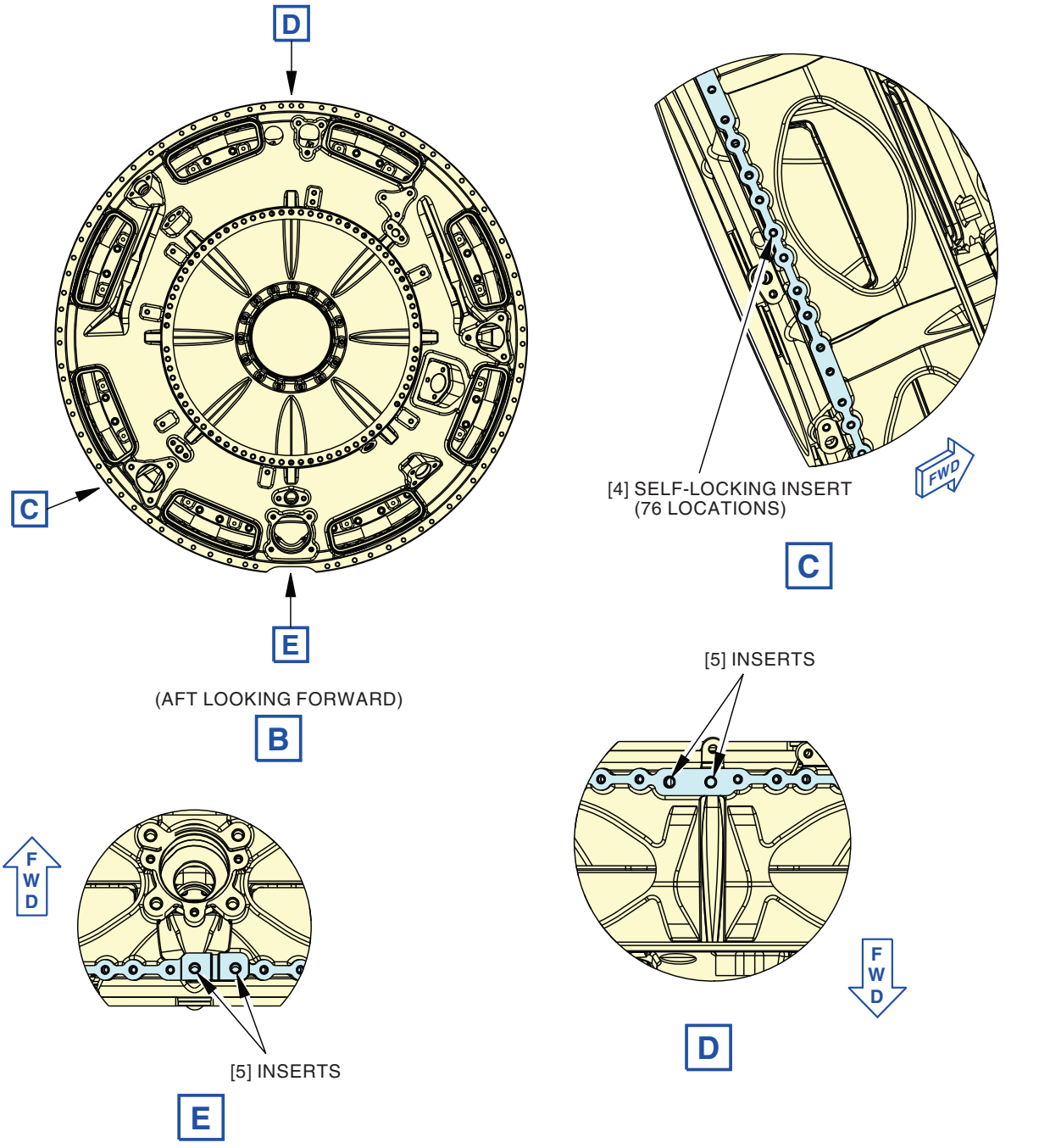
Fan Frame Hub Inserts Replacement
Figure 803/72-23-00-990-850-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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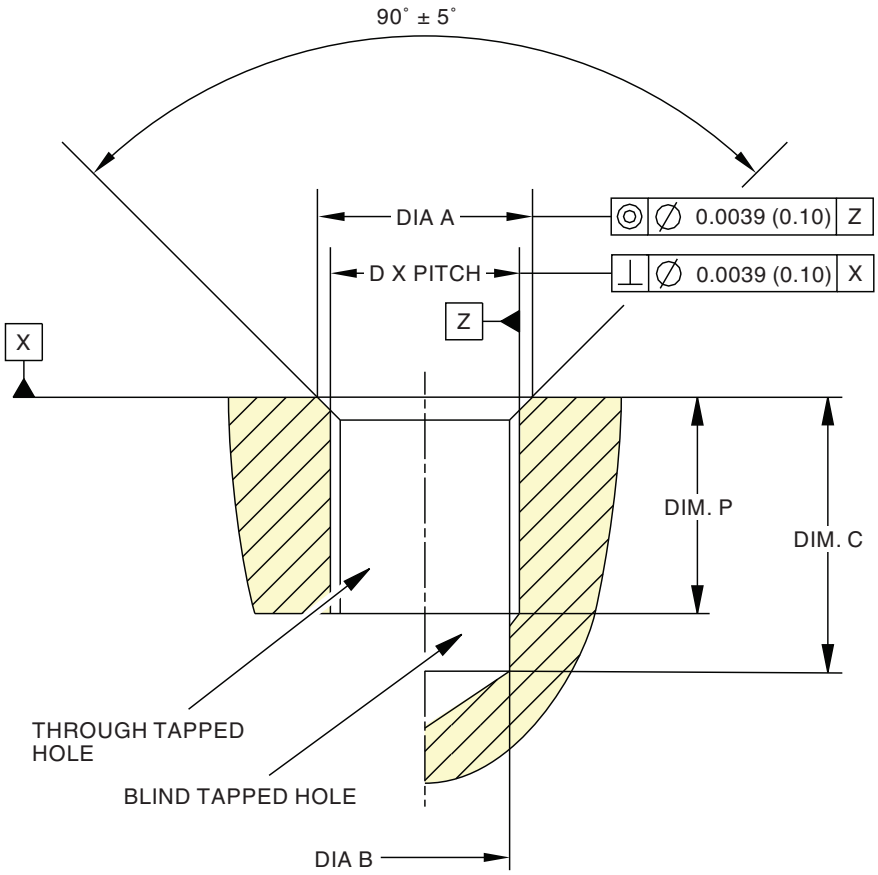
Fan Frame Hub Inserts Replacement
Figure 803/72-23-00-990-850-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



	KEY-LOCKED INSERT HOUSING DIMENSIONS									
ITEM	INSERT MINOR DIA	INSTALLATION THREAD (D X PITCH)	DIM. P		DIA A		DIA B		DIM. C	
			MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
[4] SELF-LOCKING INSERT	.2500-28UNJF-3B	.3750-16UNJS-3B	0.31 (8)	-	0.386 (9.8)	0.409 (10.4)	0.331 (8.4)	0.3366 (8.55)	0.650 (16.5)	0.689 (17.5)
[5] INSERT	.3750-24UNJF-3B	.5000-16UNJS-3B	0.47 (12)	-	0.5098 (12.95)	0.5334 (13.55)	0.4567 (11.60)	0.4626 (11.75)	0.7677 (19.50)	0.8070 (20.50)

NOTE:
DIMENSIONS ARE IN INCHES
WITH MILLIMETERS IN PARENTHESES.

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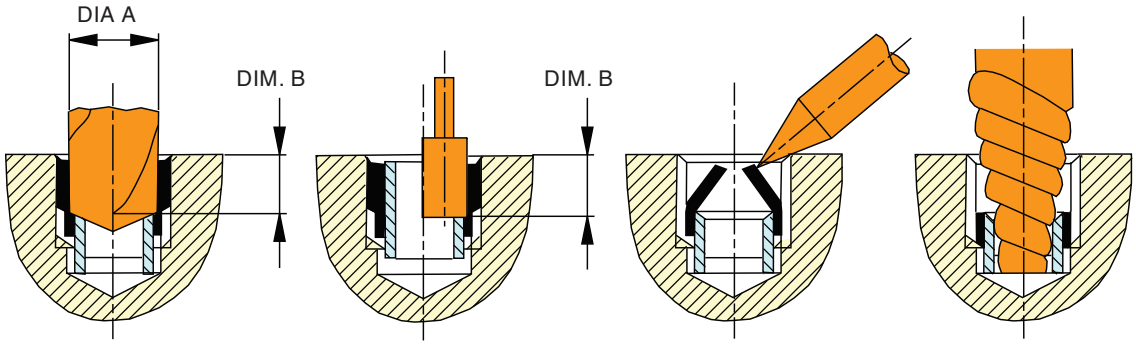
Fan Frame Hub Inserts Replacement
Figure 803/72-23-00-990-850-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ITEM	HOLE THREADING DIAMETER	DIA A	DIM. B
[4] SELF-LOCKING INSERT	.3750-16 (9.525)	0.283 (7.2)	0.189 (4.8)
[5] INSERT	.5000-16 (12.700)	0.406 (10.3)	

NOTE:
DIMENSIONS ARE IN INCHES
WITH MILLIMETERS IN PARENTHESES.

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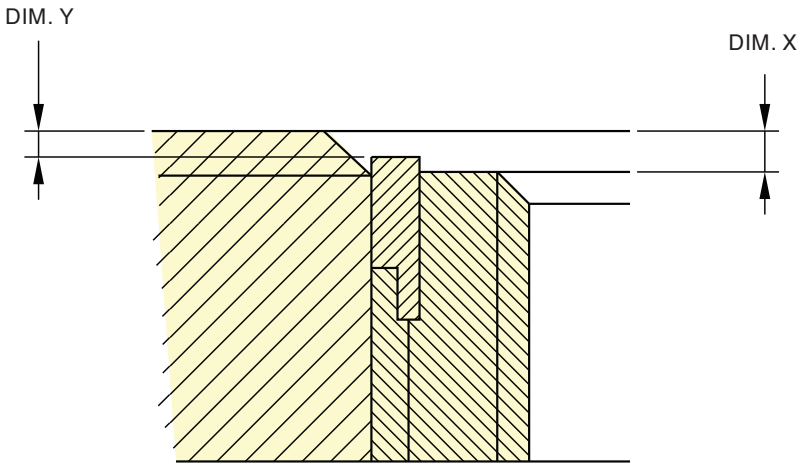
Insert Removal
Figure 804/72-23-00-990-851-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Dimensional Inspection
Figure 805/72-23-00-990-852-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN CASE - REPAIR

1. General

A. This procedure has these tasks:

(1) A repair of the fan case abradable coating

(2) A replacement of the fan case acoustic shroud bracket and/or attachment parts

(3) A replacement of the fan case counterplates

(4) A repair of the fan case impact on the trench filler

(5) A replacement of the fan case rivetless nut plate assembly

(6) A local repair of the fan case fireproof layer

(7) A local repair of the fan case abradable liner cracks.

TASK 72-23-01-300-801-G00

2. Fan Case Abradable Coating Repair

(Figure 801)

A. General

(1) This task gives the instructions to do a repair of the fan case abradable liner.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15597	Fixture, Local Repair - Abradable Liner Part #: 956A8082G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G02499 [CP2364]	Filler, Edge Sealing And Void Filling - Scotchweld EC 3524B/A	
G51335 [CP2674]	Lint-Free Cloth	
G51480 [CP2772]	Cloth - Abrasive, Alumina	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Repair

SUBTASK 72-23-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-01-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-23-01-100-001-G00

WARNING DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH YOU. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING ONLY USE CLEANING AGENTS WHEN THERE IS GOOD AIR FLOW THROUGH THE WORK AREA. THIS MATERIAL IS POISONOUS AND FLAMMABLE AND CAN CAUSE SKIN IRRITATION. OBEY THE MANUFACTURER'S INSTRUCTIONS. PUT ON PROTECTIVE CLOTHING. DO NOT GET THE CLEANING AGENTS IN YOUR MOUTH. DO NOT SMOKE. DO NOT BREATHE THE GAS. GET HELP IF YOUR SKIN OR EYES BECOME IRRITATED.

CAUTION DO NOT SAND OR CUT THE CARBON SKIN BELOW THE ABRADABLE MATERIAL. DAMAGE TO THE FAN CASE CAN OCCUR.

(3) Remove all deviations and/or unbonded abradable material from the liner.

(a) If it is necessary, use a plastic spatula or a cutter.

(b) Round the sharp edges.

(c) If it is necessary, remove the abradable material from the area to the repair until you see the carbon skin.

1) Do not damage the carbon skin.

NOTE: The remaining abradable must be clean and not disbonded.

SUBTASK 72-23-01-120-001-G00

(4) Smooth the repair area with an abrasive cloth, G51480 [CP2772].

SUBTASK 72-23-01-910-001-G00

WARNING USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

(5) Blow the repair area with dry and de-oiled air.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-01-100-002-G00

(6) Clean the area with a lint-free cloth, G51335 [CP2674], and alcohol, B00676 [CP1041].

H. Fan Case Abradable Coating Repair

SUBTASK 72-23-01-940-001-G00

CAUTION YOU MUST WEAR GLOVES THAT ARE FREE OF TALC OR SILICONE TO PREVENT CONTAMINATION OF ABRADABLE COMPOUND. IF NOT, YOU CAN CAUSE DAMAGE.

(1) Prepare the Scotchweld EC 3524B/A filler, G02499 [CP2364], as follows:

(a) Base Scotch-WeldTM EC-3524 Part B black: 100 parts by weight.

(b) Accelerator Scotch-WeldTM EC-3524 Part A: 94 parts by weight.

NOTE: If it is necessary, you can use Scotchweld 3524 B/A (BLUE) / EC 3524 B/A (BLUE) with the same proportions.

(c) Mix the base and hardener by hand until it becomes a homogeneous material.

NOTE: For Scotchweld EC 3524B/A BLACK, mixed compound color should be grey. For Scotchweld EC 3524B/A BLUE, mixed compound color should be pale blue.

(d) If it is necessary, heat the above mixture to 77°F (25°C) to make the application of product to the part easier. The most applicable work temperature is between 68°F (20°C) and 77°F (25°C).

SUBTASK 72-23-01-380-001-G00

(2) Apply the abradable material as follows:

NOTE: If you used the blue product, the final cosmetic result of the repaired area will be different from the original.

(a) If it is necessary, heat the area to repair or the working environment between 68°F (20°C) and 77°F (25°C) to make the application of product to the part easier.

(b) Make sure that the application surface is clean and not disbonded.

(c) Apply a very thin layer of abradable material on the area to repair and apply sufficient pressure with your thumb to obtain better results.

(d) Fill the remaining thickness with abradable material.

(e) Adjust the abradable liner with the repair fixture, SPL-15597, or a locally manufactured tool (Figure 801 (Sheet 3)).

1) Flush to the adjacent flow path area that is in good condition.

2) Adjust the radius A (Figure 801 (Sheet 2)).

SUBTASK 72-23-01-380-002-G00

CAUTION DO NOT APPLY HEAT TO THE FAN-CASE ADJACENT AREAS AND ADJACENT FAN BLADES TO THE REPAIR AREAS. A TEMPERATURE MORE THAN THE POLYMERISATION TEMPERATURE CAN CAUSE DAMAGE TO THE PARTS.

CAUTION DO NOT USE A HEAT GUN ON THE ABRADABLE MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE ABRADABLE MATERIAL CAN OCCUR.

(3) Do one of these procedures to polymerize the abradable material:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Polymerize at room temperature for 24 hours. Room temperature must be between 59°F (15°C) and 77°F (25°C). Use 77°F (25°C) to obtain better results.

(b) Polymerize at 122°F (50°C) - 140°F (60°C) for 5 hours.

(c) Polymerize at 158°F (70°C) - 194°F (90°C) for 2 hours.

SUBTASK 72-23-01-200-001-G00

(4) Do a hardness check of the abradable material.

(a) Make sure that the hardness is between 40 and 60 Shore D.

SUBTASK 72-23-01-220-001-G00

(5) Do an inspection of the repaired area.

(a) Do a visual inspection of the liner for bubbles or missing abradable material.

1) Air bubbles are not permitted.

2) Lack of abradable is not permitted. Do the repair again.

(b) Do a dimension check of the repaired area with the repair fixture, SPL-15597, or a locally manufactured tool (Figure 801 (Sheet 2), Figure 801 (Sheet 3)).

NOTE: The tool must be in contact with the adjacent unrepaired areas that are in good condition.

1) Bumped areas must not be more than 0.0098 in. (0.25 mm) thick.

2) Depressed areas must not be more than 0.0098 in. (0.25 mm) deep.

3) If necessary, blend the abradable to the final dimensions (Figure 801 (Sheet 3)).

(c) Do a check of the abradable material bonding with a wooden mallet.

NOTE: A dull sound shows that the abradable material is not correctly bonded.

(d) If the inspection shows an unserviceable condition, do the repair again.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-01-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-01-860-004-G00

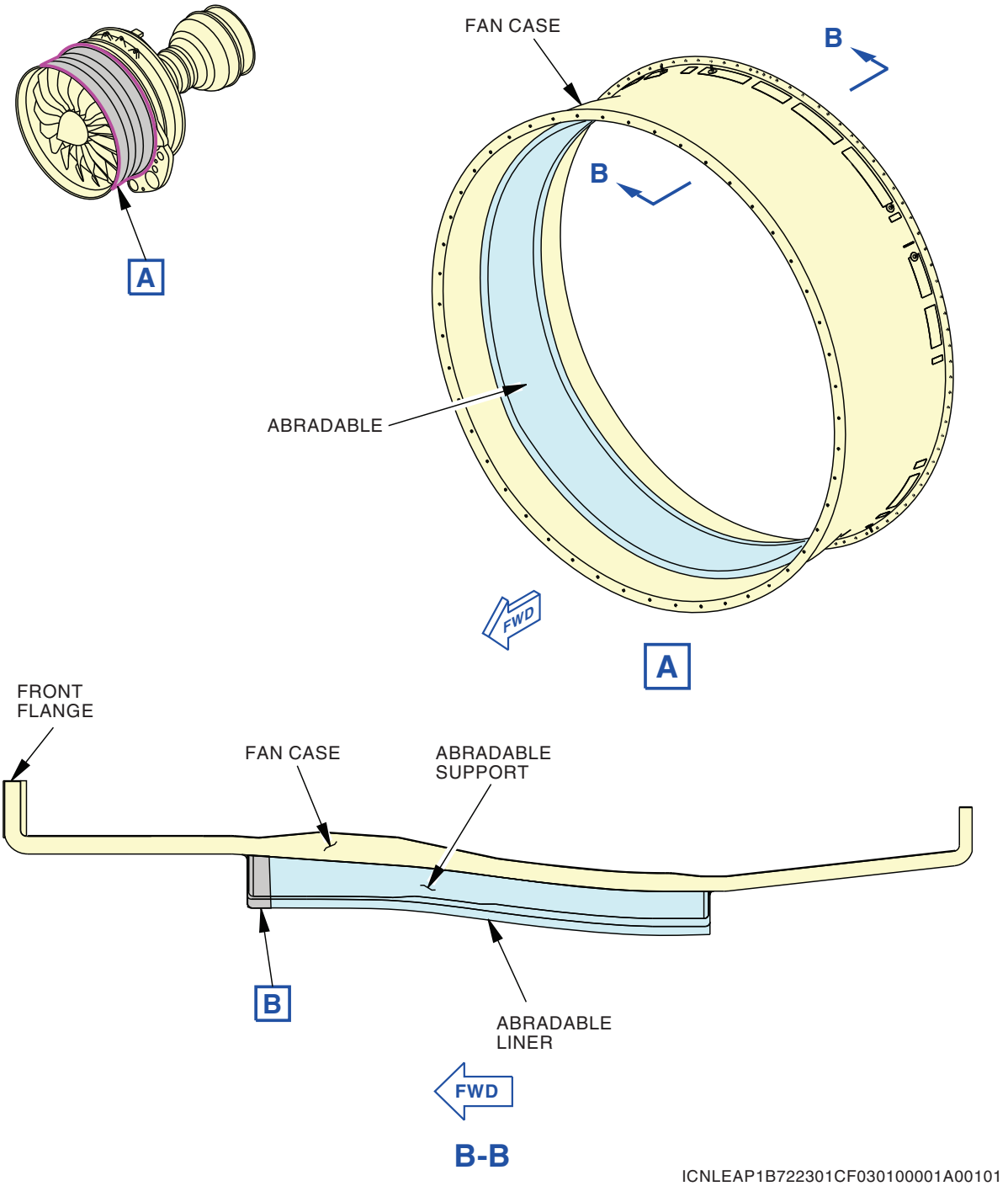
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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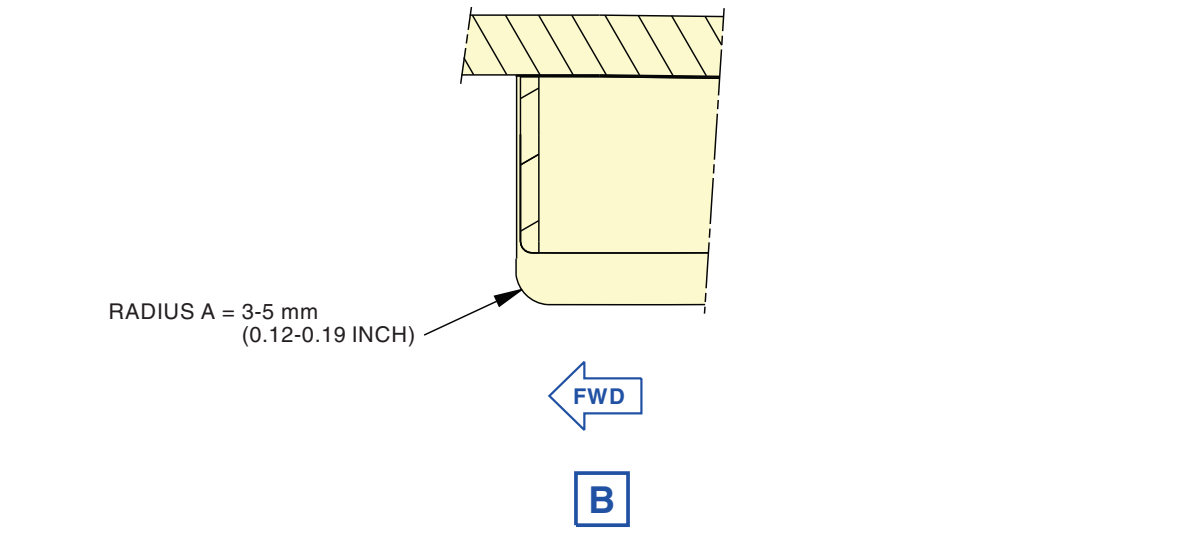
2799280 S0000637330_V1

Fan Case Abradable Coating Repair
Figure 801/72-23-01-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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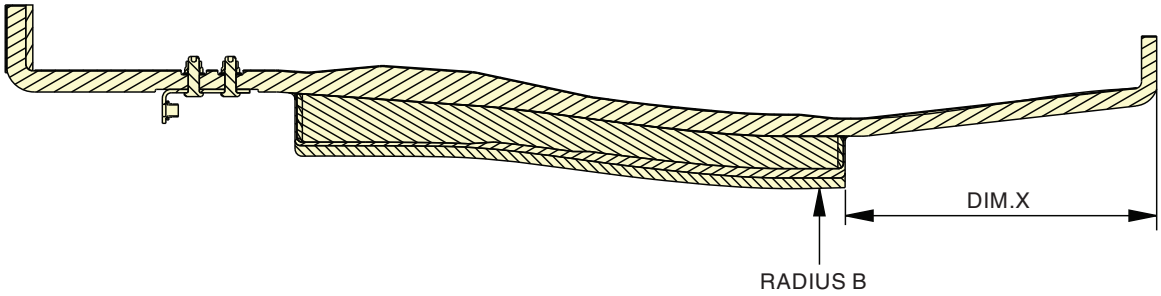
2799295 S0000640677_V1

Fan Case Abradable Coating Repair
Figure 801/72-23-01-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ABRADABLE PROFILE

DIM. X INCH	DIM. X (mm)	RADIUS B INCH	RADIUS B (mm)
7.17	(182)	33.99673	(863.517)
7.80	(198)	33.9831	(863.17)
8.43	(214)	33.98077	(863.112)
9.06	(230)	33.99494	(863.472)
9.69	(246)	34.03085	(864.384)
10.32	(262)	34.08589	(865.782)
10.95	(278)	34.14734	(867.343)
11.56	(294)	34.21242	(868.996)
12.21	(310)	34.28793	(870.914)
12.84	(326)	34.37161	(873.039)
13.47	(342)	34.45764	(875.224)
14.09	(358)	34.53927	(877.298)
14.72	(374)	34.60969	(879.086)
15.35	(390)	34.66443	(880.477)
15.98	(406)	34.69228	(881.184)
16.61	(422)	34.70272	(881.449)
17.24	(438)	34.70791	(881.581)
17.87	(454)	34.71075	(881.653)
18.50	(470)	34.7118	(881.68)
19.13	(486)	34.71146	(881.671)

NOTE:

DIM. X : DISTANCE BETWEEN THE AFT FLANGE AND THE MEASUREMENT POSITION.

RADIUS B : RADIUS BETWEEN THE FLOWPATH AND THE ENGINE ROTATIONAL AXIS.

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Fan Case Abradable Coating Repair
Figure 801/72-23-01-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-01-960-801-G00

3. Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement

(Figure 802)

A. General

(1) This task gives the instructions to replace the acoustic shroud bracket (fan case lug) and/or its attachment parts on the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
D50197	Lubricant - Cetyl Alcohol	SAE AS87132 (Supersedes MIL-L-87132)
G51335 [CP2674]	Lint-Free Cloth	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-01-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-01-860-006-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement

SUBTASK 72-23-01-020-001-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION

DO NOT TOUCH THE FAN CASE SKIN DURING THE REMOVAL AND INSTALLATION. DAMAGE TO THE FAN CASE CAN OCCUR.

(1) Remove the parts to replace.

(a) Discard the nut.

SUBTASK 72-23-01-100-003-G00

(2) Clean the area with a moist lint-free cloth, G51335 [CP2674] and alcohol, B00676 [CP1041].

SUBTASK 72-23-01-220-002-G00

(3) Do a visual inspection of the fan case area where you will install the new parts, including the fan case lug holes. All indications are permitted as follows:

(a) If not more than 0.028 in. (0.7 mm) each.

(b) If the distance between two indications is more than 0.98 in. (25 mm).

SUBTASK 72-23-01-220-003-G00

(4) Do a dimensional check of the fan case lug holes.

(a) DIA A must be between 0.2496 in. (6.34 mm) and 0.2516 in. (6.39 mm).

SUBTASK 72-23-01-420-001-G00

(5) Install the parts.

(a) Do a visual inspection of the bolts and washers.

1) Make sure that the bolt is still lubricated, and if it is necessary apply lubricant, D50197 to the thread.

2) If they are damaged, replace the bolts and washers.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-01-860-007-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-01-860-008-G00

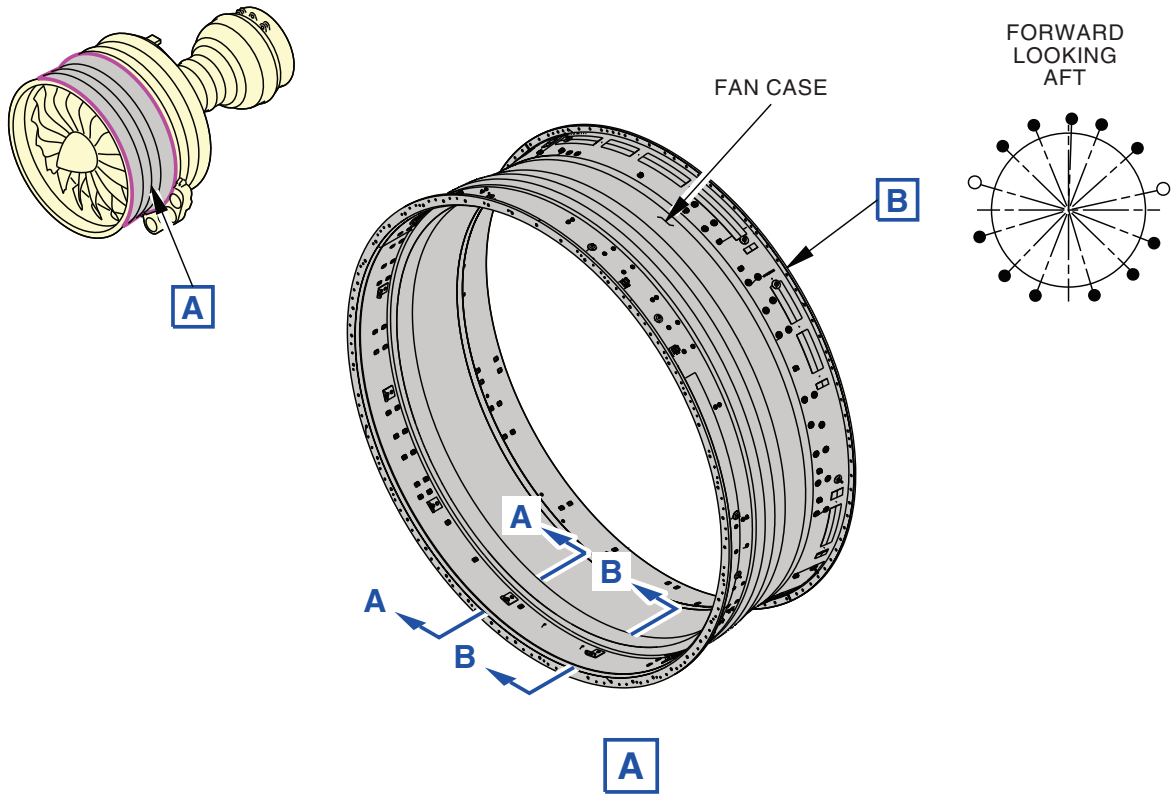
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

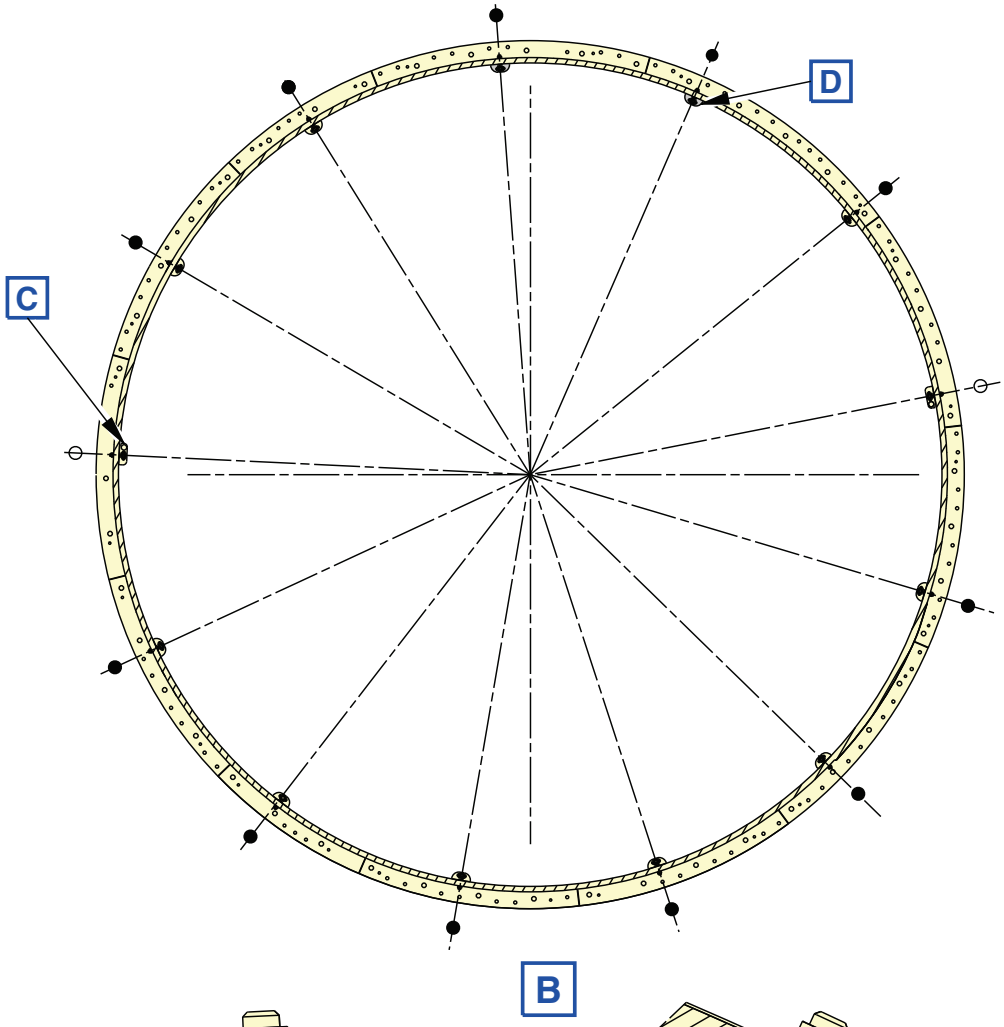
ECCN 9E991 BOEING PROPRIETARY - See title page for details

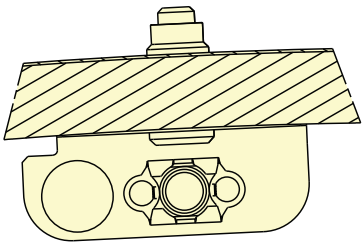


2799842 S0000640687_V1

Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement
Figure 802/72-23-01-990-802-G00 (Sheet 1 of 3)

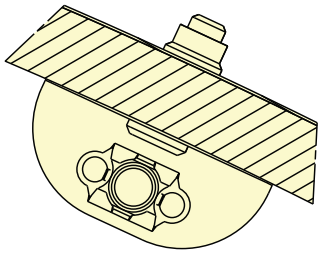
ECCN 9E991 BOEING PROPRIETARY - See title page for details





BRACKET TYPE 2
2 LOCATIONS (○)

C



BRACKET TYPE 1
11 LOCATIONS (●)

D

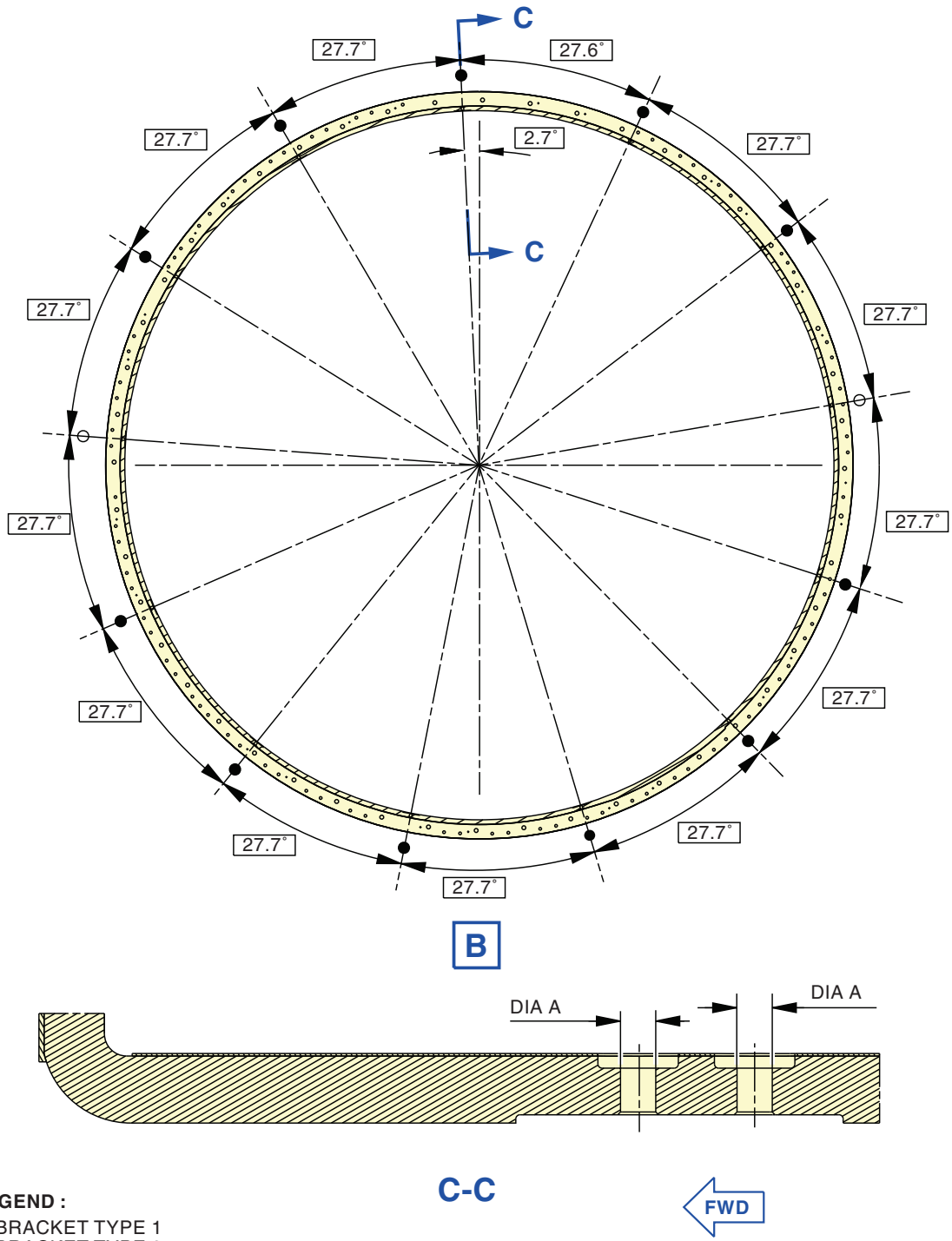
LEGEND :

● BRACKET TYPE 1
○ BRACKET TYPE 2

2799847 S0000640688_V1

Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement
Figure 802/72-23-01-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement
Figure 802/72-23-01-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-01-960-802-G00

4. Fan Case Counterplates Replacement

(Figure 803)

A. General

(1) This task gives instructions to replace the counterplates (plate stiffener) on the fan case.

(2) The Table 801 and Table 802 give a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 801/72-23-01-993-802-G00

Part Number	Designation	Configuration
364-050-013-0	FAN CASE	1
364-050-014-0	FAN CASE	2
364-050-016-0	FAN CASE	2
364-050-017-0	FAN CASE	2
364-050-099-0	FAN CASE	2

Table 802/72-23-01-993-801-G00

Part Number	Description	Quantity	Configuration	Legend
364-854-001-0	PLATE STIFFENER FWD	AR	1	[5]
364-854-111-0	PLATE STIFFENER FWD	AR	1	[3]
364-854-201-0	PLATE STIFFENER FWD	AR	1	[4]
364-854-301-0	PLATE STIFFENER FWD	AR	1	[6]
364-854-401-0	PLATE STIFFENER FWD	AR	1	[7]
364-854-503-0	PLATE STIFFENER FWD	AR	1	[8]
364-854-611-0	PLATE STIFFENER FWD	AR	1	[1]
364-854-710-0	PLATE STIFFENER FWD	AR	1	[2]
364-854-002-0	PLATE STIFFENER FWD	AR	2	[13]
364-854-112-0	PLATE STIFFENER FWD	AR	2	[11]
364-854-202-0	PLATE STIFFENER FWD	AR	2	[12]
364-854-302-0	PLATE STIFFENER FWD	AR	2	[14]
364-854-402-0	PLATE STIFFENER FWD	AR	2	[15]
364-854-504-0	PLATE STIFFENER FWD	AR	2	[16]
364-854-612-0	PLATE STIFFENER FWD	AR	2	[9]
364-854-711-0	PLATE STIFFENER FWD	AR	2	[10]
AN123552	COUNTERSUNK HEAD RIVET	AR	1 and 2	-

B. References

Reference	Title
70-43-11-300-801-G00	Formed Rivet Assemblies Repair (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-01-860-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-01-860-010-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Case Counterplates Replacement

SUBTASK 72-23-01-960-001-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

(1) Replace the plate stiffener as follows:

CAUTION DO NOT TOUCH THE ADJACENT PARTS WHEN YOU DRILL OR GRIND THE RIVET HEAD. DAMAGE TO THE PARTS CAN OCCUR.

(a) Remove and discard the rivets (TASK 70-43-11-300-801-G00).

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

(b) Remove the particles from the part.

(c) Clean the repaired area with alcohol, B00676 [CP1041].

(d) Do a visual inspection of the holes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Deviation to a correct condition is not permitted.

(e) Install and make sure that the plate stiffener is installed correctly (TASK 70-43-11-300-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-01-860-011-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-01-860-012-G00

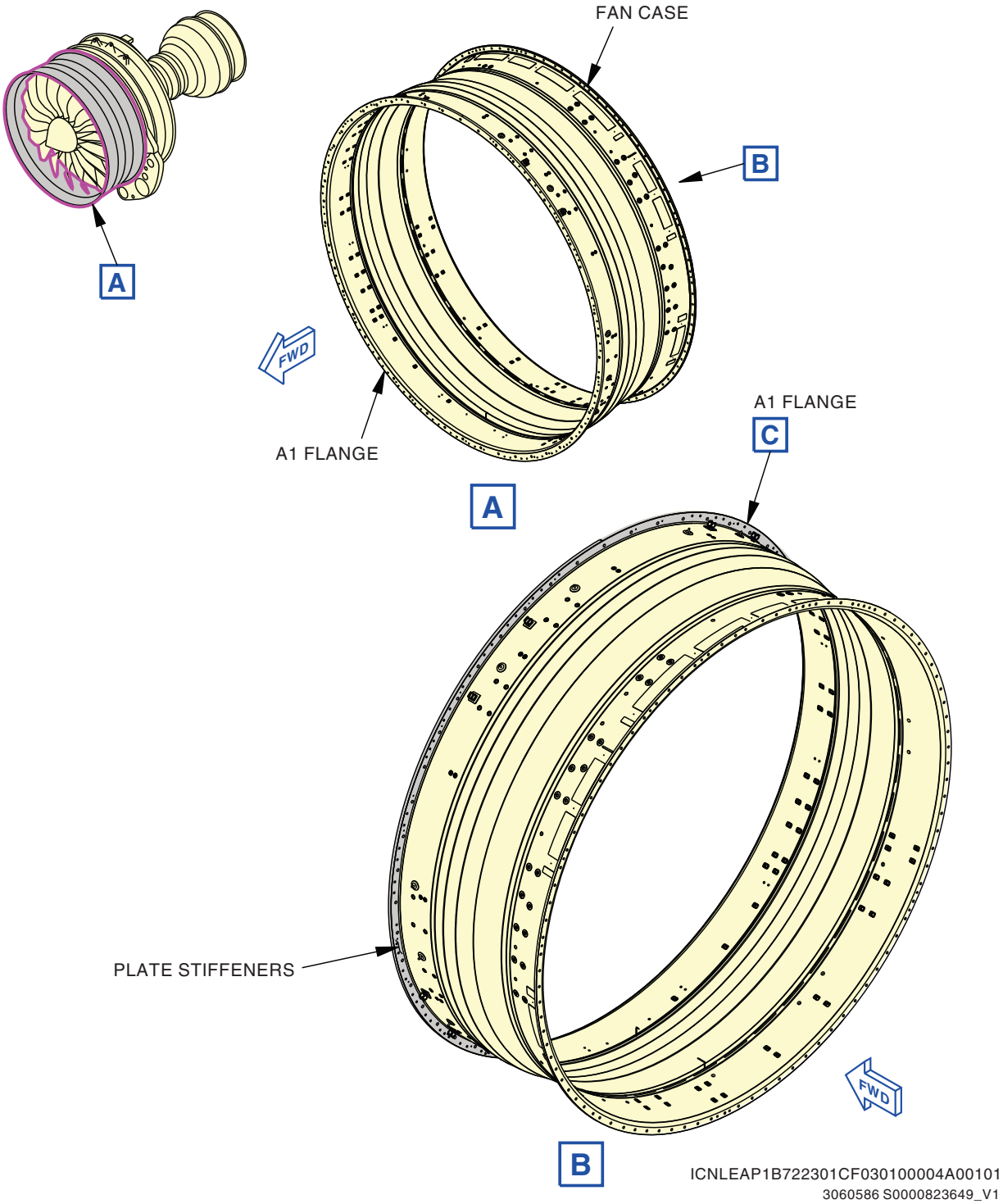
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

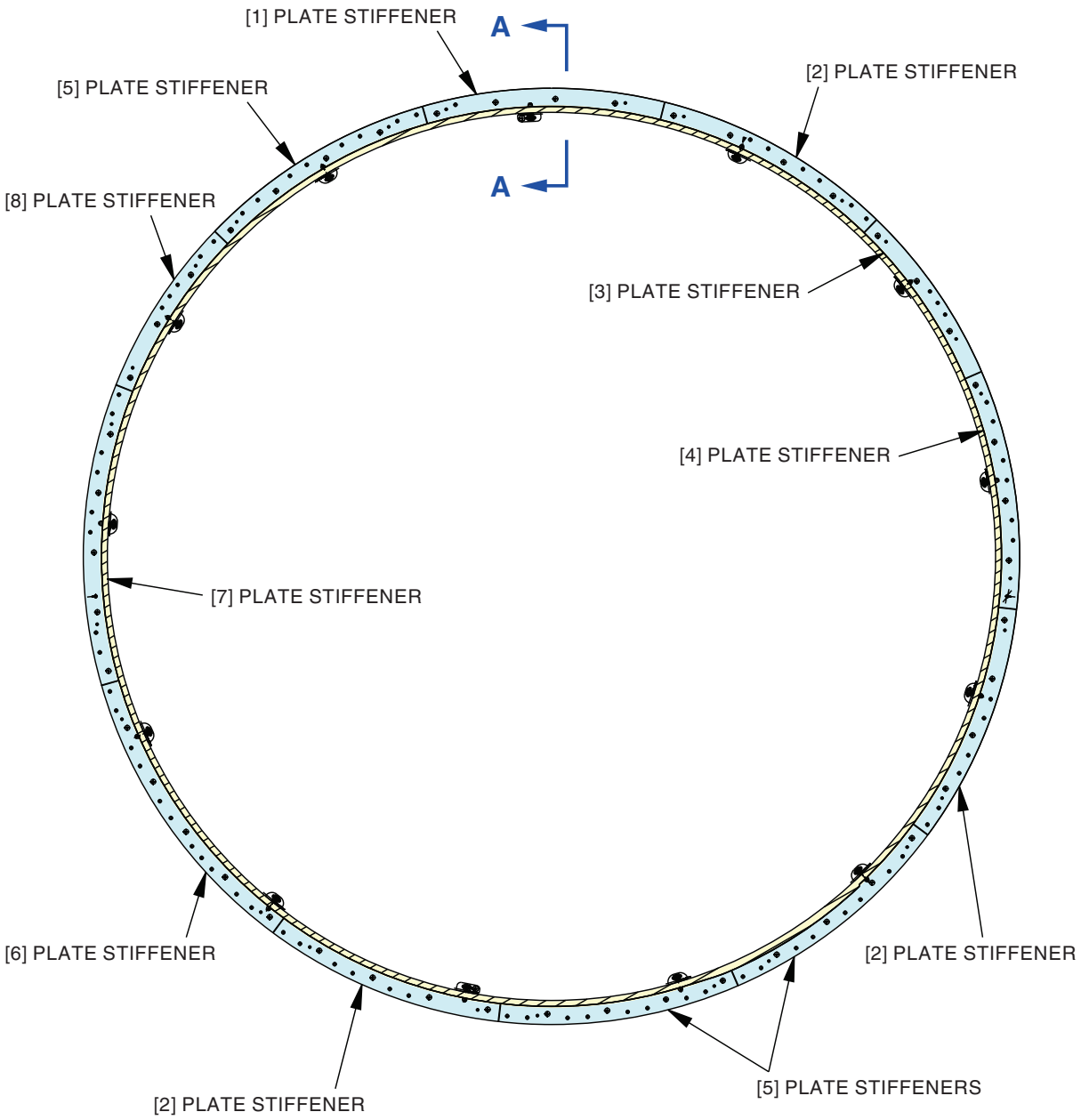
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Case Counterplates Replacement
Figure 803/72-23-01-990-803-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A1 FLANGE
(FAN CASE CONFIGURATION 1)
(VIEW IN THE FORWARD DIRECTION)

C

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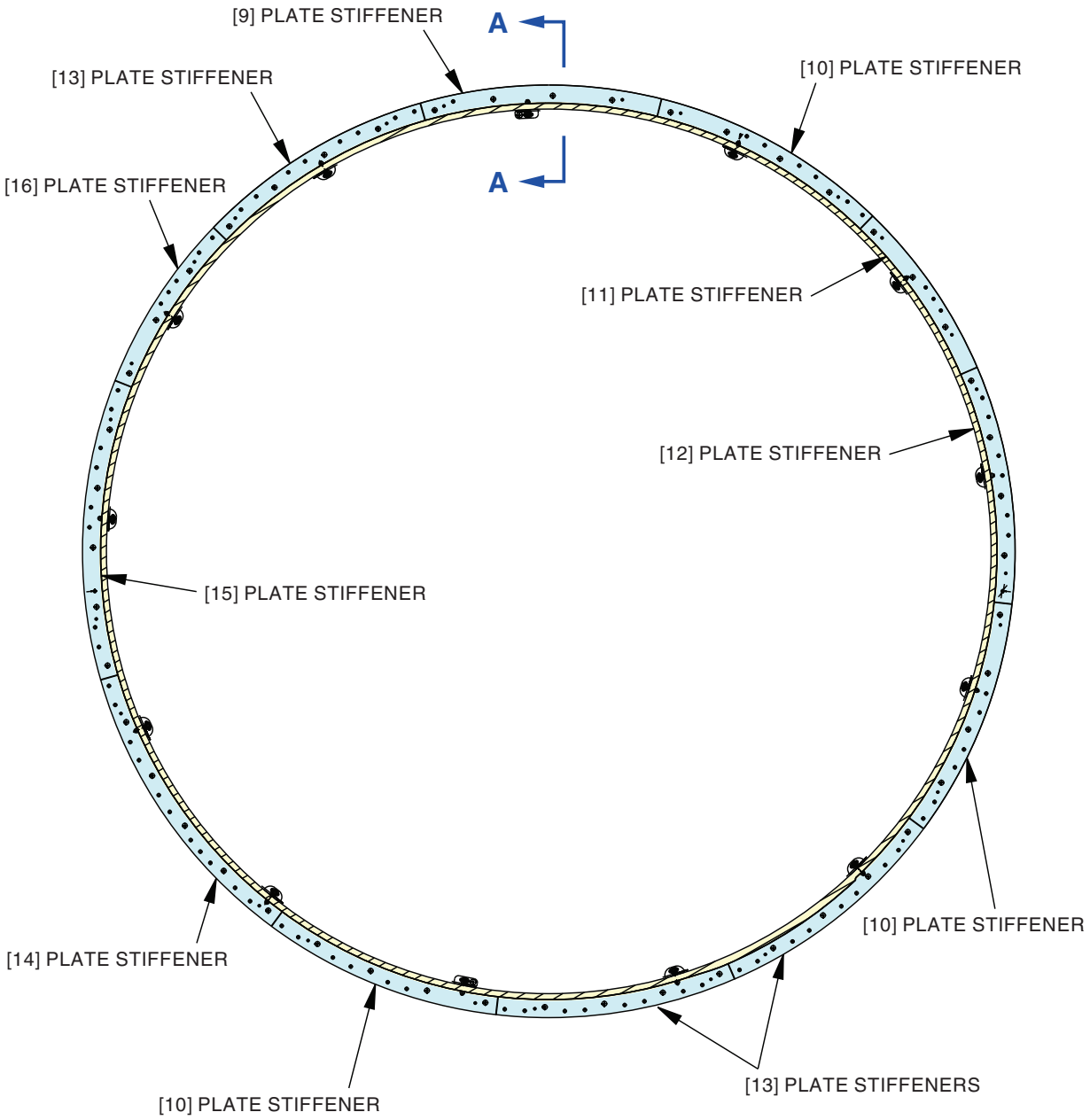
3060587 S0000823650_V1

Fan Case Counterplates Replacement
Figure 803/72-23-01-990-803-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A1 FLANGE
(FAN CASE CONFIGURATION 2)
(VIEW IN THE FORWARD DIRECTION)

C

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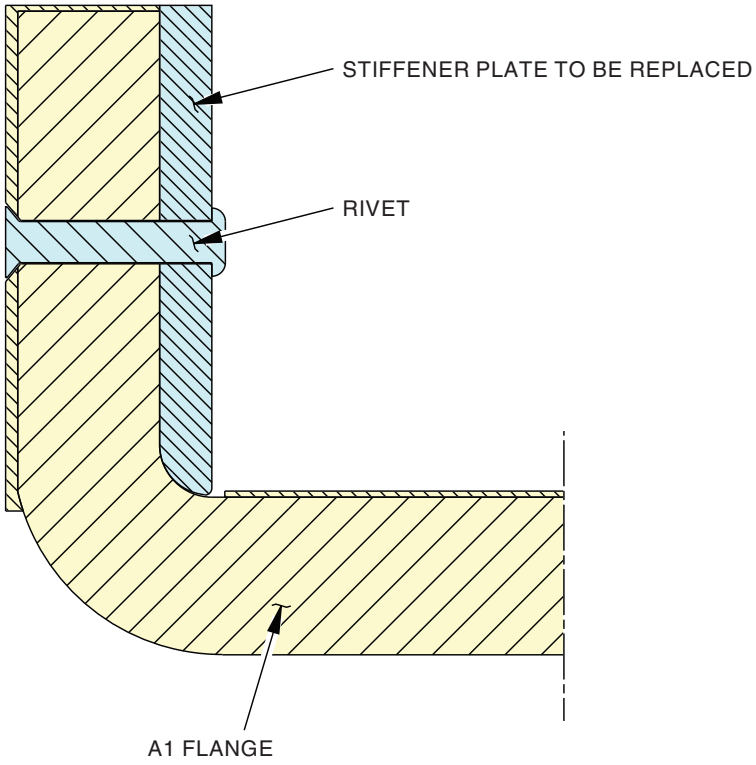
3060588 S0000823651_V1

Fan Case Counterplates Replacement
Figure 803/72-23-01-990-803-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A

ICNLEAP1B722301CF030100007A00101
3060590 S0000823652_V1

Fan Case Counterplates Replacement
Figure 803/72-23-01-990-803-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-01-960-803-G00

5. Fan Case Impact Repair on the Trench Filler

(Figure 804, Figure 805, Figure 806, and Figure 807)

A. General

(1) This task gives instructions to repair the fan case trench filler with abradable material.

(2) The table below gives a list of part numbers that are applicable to this repair.

NOTE: All part numbers are applicable to all paragraphs unless specified differently.

Table 803/72-23-01-993-808-G00 Applicable Part Numbers

Part Number	Designation
364-050-013-0	CASE, FAN
364-050-014-0	CASE, FAN
364-050-016-0	CASE, FAN
364-050-017-0*[1]	CASE, FAN
364-050-099-0	CASE, FAN

*[1] This repair is not applicable for part number 364-050-017-0 if it marked with rectangular or stepped block (Figure 807).

(3) The table below gives dimensional information for the repair application.

Table 804/72-23-01-993-803-G00 Dimensional Information

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
DIM. A	-	-	-	1.6098 in. (40.89 mm)	-
DIM. B	-	-	-	0.8677 in. (22.04 mm)	-
DIM. C	-	-	-	1.2094 in. (30.72 mm)	-
DIM. D	-	-	-	0.9342 in. (23.73 mm)	-
DIM. E	-	-	-	1.1968 in. (30.40 mm)	-
DIM. F	-	-	-	0.338 in. (8.6 mm)	-
DIM. G	-	-	-	0.6409 in. (16.28 mm)	-
DIM. H	-	-	-	0.338 in. (8.6 mm)	-
DIM. I	-	-	-	0.23 in. (6 mm)	-
DIM. J	-	-	-	8.8169 in. (223.95 mm)	-
DIM. K	-	-	-	6.4913 in. (164.88 mm)	-

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B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
C50435 [CP8019]	Varnish Base - A1500-UVR	
G51480 [CP2772]	Cloth - Abrasive, Alumina	
G51697 [CP2929]	Tape - Flashbreaker 1	
G51699 [CP8024]	Label - Paper, Cotton	
G51894 [CP2923]	Compound - Epoxy, Void Filling and Edge Sealing, Scotch-Weld EC-3524 B/A Black	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Repair

SUBTASK 72-23-01-860-013-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

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(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-01-860-014-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Case Impact Repair on the Trench Filler

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH YOU. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

SUBTASK 72-23-01-350-001-G00

(1) Find the location of the bonded repair labels on the fan case (Figure 804).

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SUBTASK 72-23-01-350-002-G00

(2) Find if it is possible to repair the part again.

(a) Several indications can be repaired in one repair area if the total diameter of the repaired area does not exceed 2.000 in. (50.8 mm).

(b) The total area of all repaired areas must not exceed 12.6 in^2 (8129 mm^2).

(c) The distance between the contours of two repaired areas must be more than four times the largest repaired diameter of the two.

(d) The maximum permitted number of bonded repair labels is 10 (Figure 805).

SUBTASK 72-23-01-300-003-G00

(3) Prepare the area to repair as follows (Figure 806):

(a) Do not go through the bottom skin of the trench filler or the part cannot be repaired.

(b) Remove all abradable and trench filler material that you need to repair as follows:

1) Cut a circular shape around the indication.

a) Make sure that the diameter of the circle is 0.252 in. (6.4 mm) larger that the indication, but does not exceed 2.000 in. (50.8 mm).

b) Make sure that the depth of the cut does not exceed 0.255 in. (6.5 mm) to prevent cutting through the bottom skin of the trench filler.

2) Sand the area to repair with abrasive cloth, G51480 [CP2772], or aluminium oxide abrasive sheets (120 grit or finer) to get a clean surface for material filling and bonding (TASK 70-43-42-300-802-G00).

3) Remove all unwanted material from the area to repair.

4) Clean manually with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-01-300-004-G00

CAUTION

YOU MUST WEAR GLOVES THAT ARE FREE OF TALC OR SILICONE TO PREVENT CONTAMINATION OF ABRADABLE COMPOUND. IF NOT, YOU CAN CAUSE DAMAGE.

(4) Fill with abradable material as follows (View C, Figure 806):

(a) Prepare the abradable material:

1) Prepare epoxy compound, G51894 [CP2923], as follows:

a) Base Scotch-Weld TM EC-3524 Part B black or Base Scotch-Weld TM EC-3515 Part B black: 100 parts by weight.

b) Accelerator Scotch-Weld TM EC-3524 Part A or Accelerator Scotch-Weld TM EC-3515 Part A: 94 parts by weight.

2) Stir the above mixture manually until you get a homogenous material with a dark grey color.

a) Mix both parts slowly to prevent air bubbles in the mixture.

3) Increase the temperature of the above mixture to 68°F (20°C) - 77°F (25°C) to make the application of the product to the part easier.

(b) Fill-up the area to repair with the abradable material until the carbon plies (View C, Figure 806).

1) Increase the temperature of the area to repair or the working environment to 68°F (20°C) - 77°F (25°C) to make the application of the product to the part easier.

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2) Apply a very thin layer of abradable material on the area to repair.

a) Apply sufficient pressure to get better results.

3) Fill the remaining thickness with abradable material.

NOTE: Fill the abradable with an overthickness compared to the flush thickness of the trench filler round. Avoid overflowing the surface of the hole when applying the abradable material.

SUBTASK 72-23-01-300-006-G00

CAUTION

DO NOT INCREASE THE TEMPERATURE OF THE FAN CASE ADJACENT AREAS AND ADJACENT FAN BLADES TO THE REPAIR AREAS. A TEMPERATURE MORE THAN THE POLYMERIZATION TEMPERATURE WILL OCCUR AND CAN CAUSE DAMAGE THE PARTS.

CAUTION

DO NOT USE A HEAT GUN ON THE ABRADABLE MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE ABRADABLE MATERIAL CAN OCCUR.

(5) Do the polymerization of the abradable material as follows:

(a) Alternative procedure available. Polymerize at room temperature for a minimum of 24 hours.

NOTE: Room temperature must be between 59°F (15°C) - 77°F (25°C). Use 77°F (25°C) to get better results.

(b) Alternative procedure. Polymerize at a temperature between 122°F (50°C) - 140°F (60°C) for a minimum of 5 hours.

(c) Alternative procedure. Polymerize at a temperature between 158°F (70°C) - 194°F (90°C) for a minimum of 3 hours.

SUBTASK 72-23-01-700-001-G00

(6) Do a hardness check.

(a) Make sure that the hardness is between 40 - 60 Shore D.

SUBTASK 72-23-01-120-002-G00

(7) Sand the abradable surface flush to the surrounding healthy surface with abrasive cloth, G51480 [CP2772] (View C, Figure 806, TASK 70-43-42-300-802-G00).

SUBTASK 72-23-01-210-001-G00

(8) Do an inspection of the repaired area as follows:

(a) Do a visual inspection of the liner for bubbles or missed areas:

1) Air bubbles are not permitted.

a) If it necessary, refer to the steps above to do the repair again.

2) Lack of abradable material is not permitted.

a) If it necessary, refer to the steps above to do the repair again.

(b) If the inspection shows an unserviceable condition, refer to the steps above to do the repair again.

SUBTASK 72-23-01-800-001-G00

(9) Record the trench filler repair on the fan case as follows:

(a) Record the repair information (Figure 805).

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(b) Write the repair information in inches and millimeters on cotton paper label, G51699 [CP8024], with an Optical Character Recognition (OCR) scan legible print font as follows (Figure 805):

1) Put the repair labels as follows (Figure 804):
   a) On the outer skin of the fan case.
   b) Between 101 degrees and 111 degrees counterclockwise from 12:00 o'clock position.
   c) At a minimum of 6.10 in. (155 mm) from aft face of the A1 fan case flange and a minimum of 3.94 in. (100 mm) from the left tapped hole.
2) Do the display and content of information for the repair label as shown on Figure 805.
3) The character height must be between 0.1158 in. (2.94 mm) - 0.1362 in. (3.46 mm).
4) The space between two characters must be between 0.1158 in. (2.94 mm) - 0.1362 in. (3.46 mm).
5) The space between two lines must be at least 1.4 times the height.
6) The margin between the information written and the sides of the repair label must be at least 1.5 times the space between two characters.
7) Make sure that the ink is dry.
8) Make sure that the repair label is legible.

(c) Bond the repair label on the fan case as follows:

1) Remove grease from the area of the repair label with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).
2) Do a visual examination of the area of the repair label.
   a) Make sure that the surface is mat.

      NOTE: A bright area is not permitted.
   b) Make sure that there is no residual varnish.
   c) Make sure that there is no frayed fibers or cracks.
3) Apply mask around the area of the repair label with Flashbreaker 1 tape, G51697 [CP2929].
4) Bond the repair label as follows:
   a) Apply A1500-UVR varnish base, C50435 [CP8019].
   b) Apply the repair label on the varnished area.
   c) Apply A1500-UVR varnish base, C50435 [CP8019].
      <1> Make sure that you apply A1500-UVR varnish base, C50435 [CP8019], to the full surface of the repair label.
5) Do the polymerization of the A1500-UVR varnish base, C50435 [CP8019], as follows:
   a) Alternative procedure available. Polymerize at room temperature for a minimum of 7 to 9 hours.

      NOTE: Room temperature must be between 59°F (15°C) - 77°F (25°C).
   b) Alternative procedure. Polymerize at 122°F (50°C) for a minimum of 3 hours.
6) Do a visual examination:

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a) Foreign objects more than 0.03 in. (1 mm) in diameter are not permitted.

b) Air bubbles more than 0.03 in. (1 mm) in diameter are not permitted.

c) Non-impregnated areas more than 0.0062 in^2 (4 mm^2) are not permitted.

d) The repair label must be legible.

SUBTASK 72-23-01-340-001-G00

(10) Do this task: Fan Case Abradable Liner Cracks Local Repair, TASK 72-23-01-300-803-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-01-860-015-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-01-860-016-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

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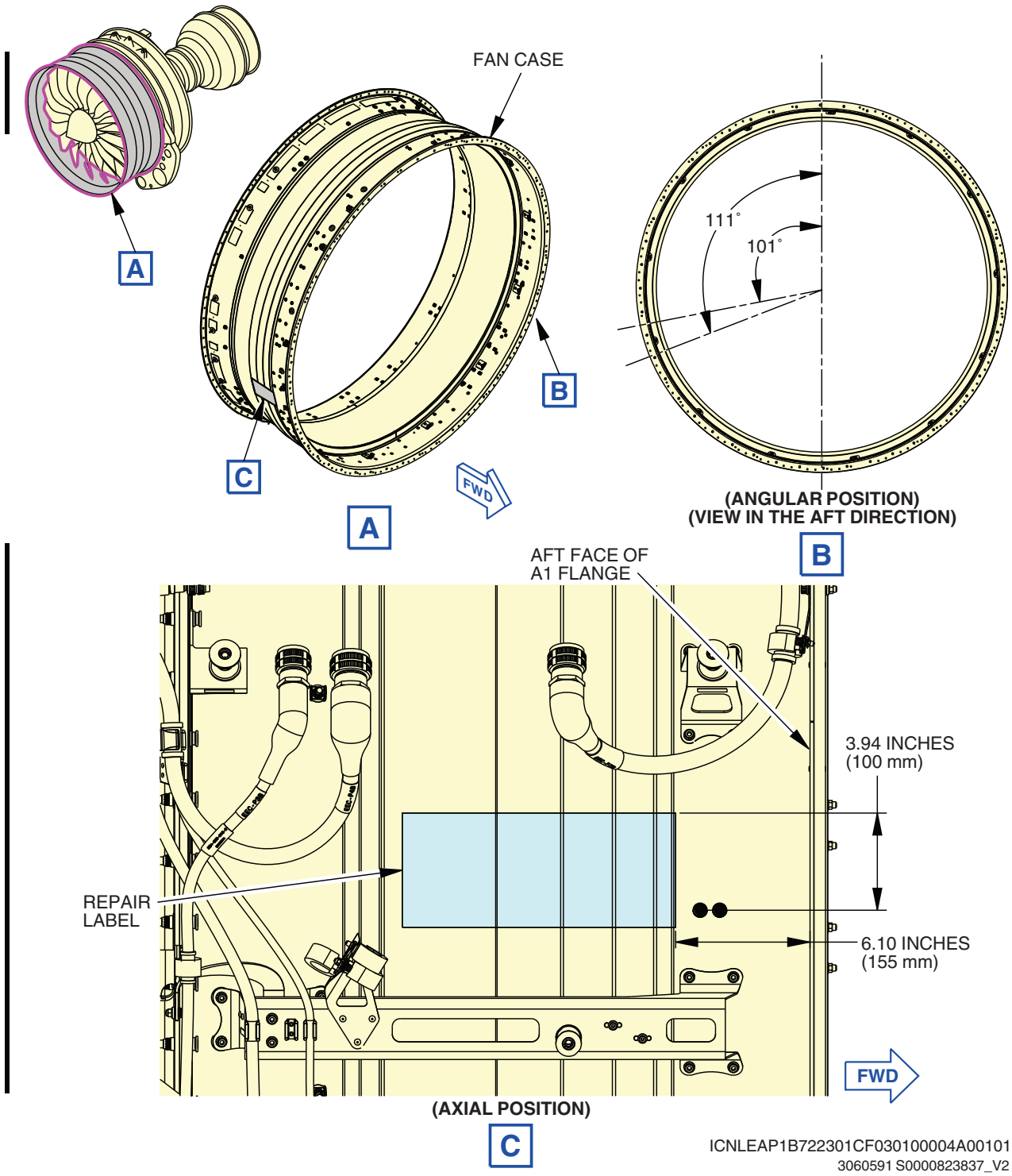
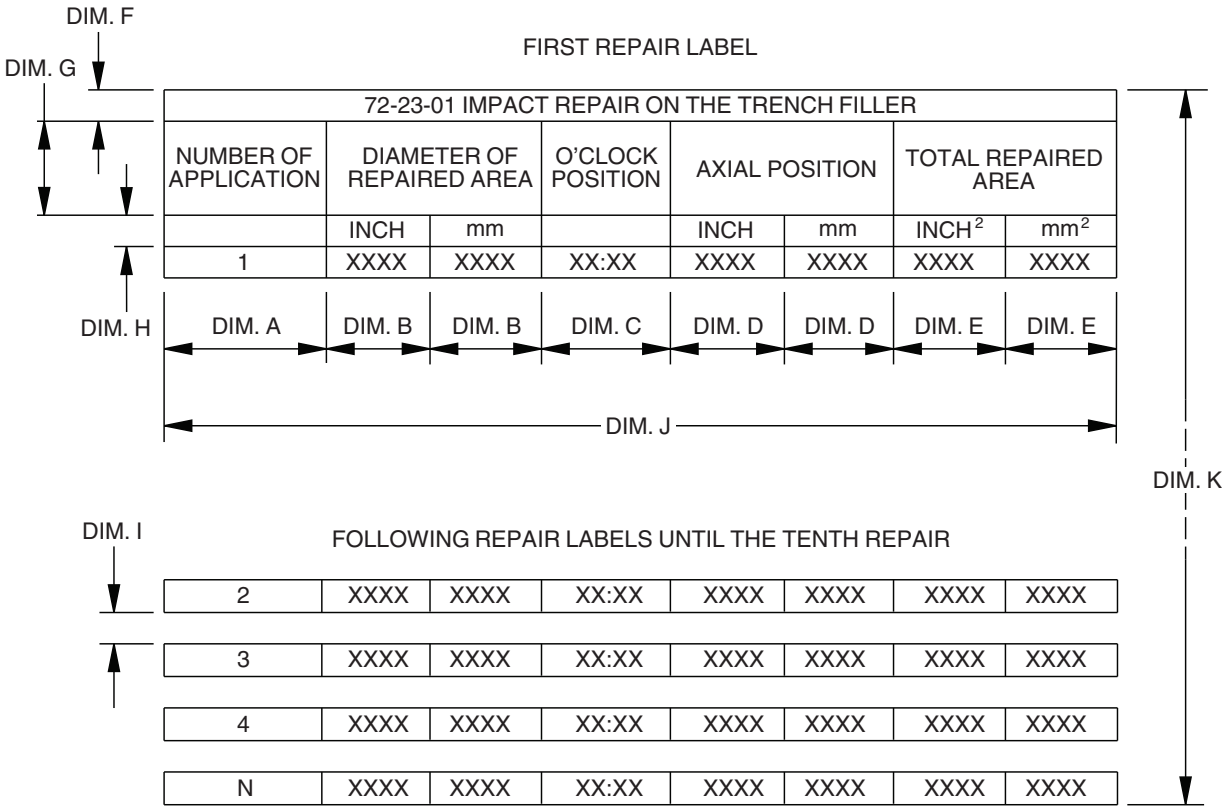


Repair Label Position
Figure 804/72-23-01-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



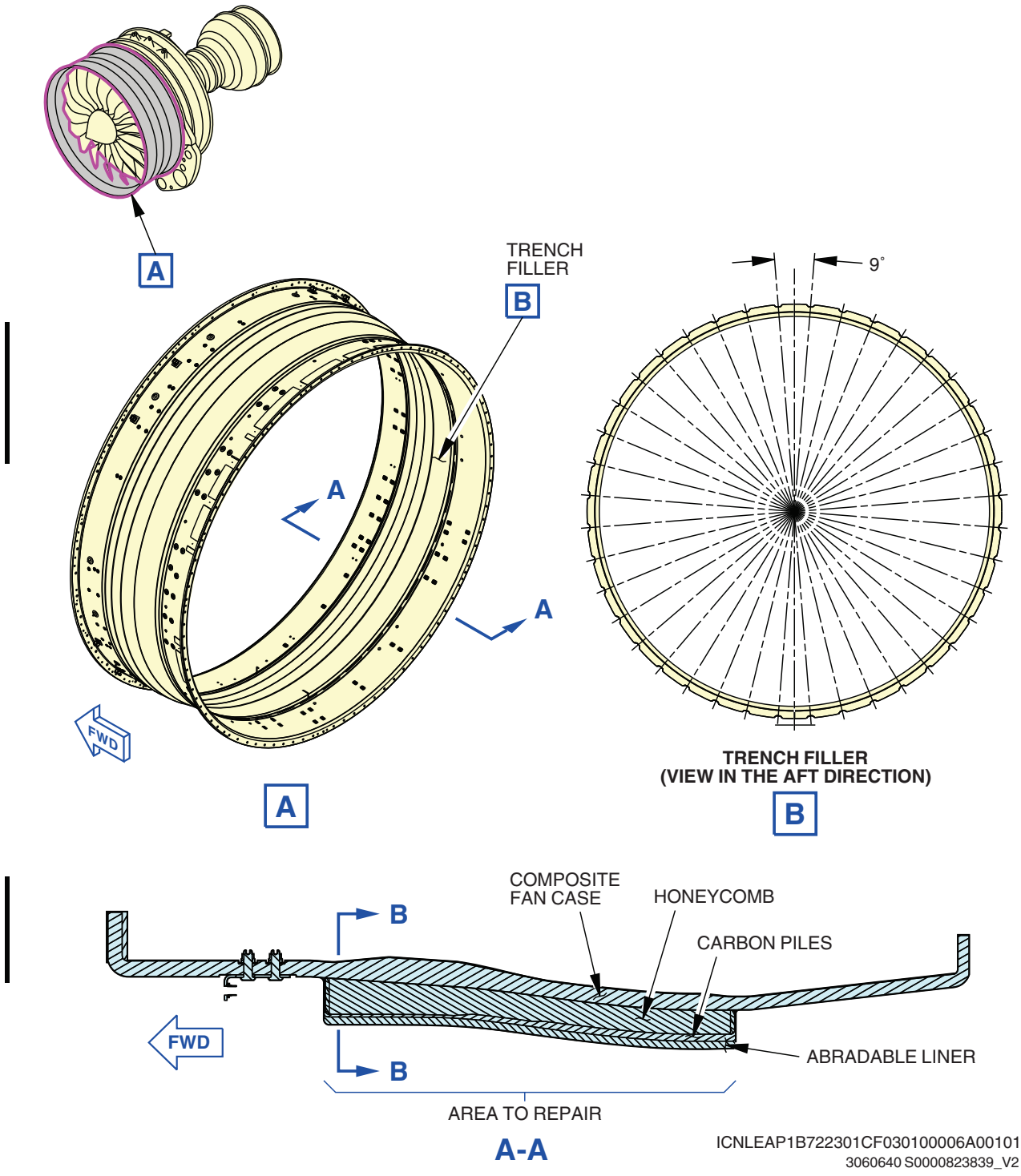
NOTE:

AXIAL POSITION: DISTANCE BETWEEN THE CENTER OF THE REPAIRED AREA AND AFT FACE OF THE A1 FAN CASE FLANGE. O'CLOCK POSITION: ANGULAR POSITION OF THE CENTER OF THE REPAIRED DIAMETER TOTAL REPAIRED AREA: SUM OF THE CURRENT REPAIRED SURFACE AND THE SURFACES REPAIRED BEFORE

ICNLEAP1B722301CF030100005A00101
3060635 S0000823838_V2

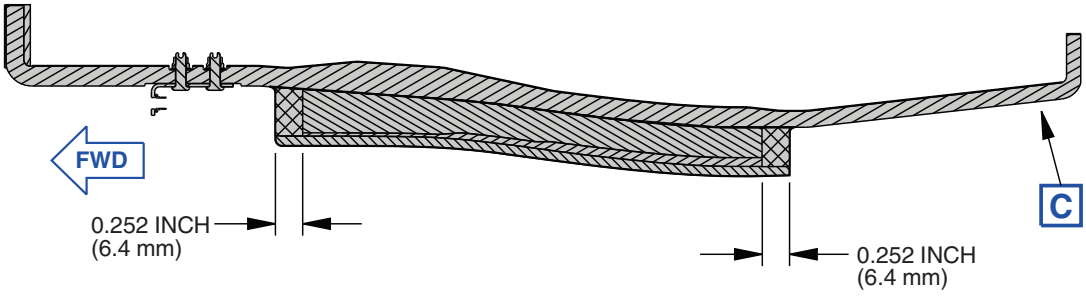
Display, Dimensions, and Content of the Repair Label
Figure 805/72-23-01-990-805-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



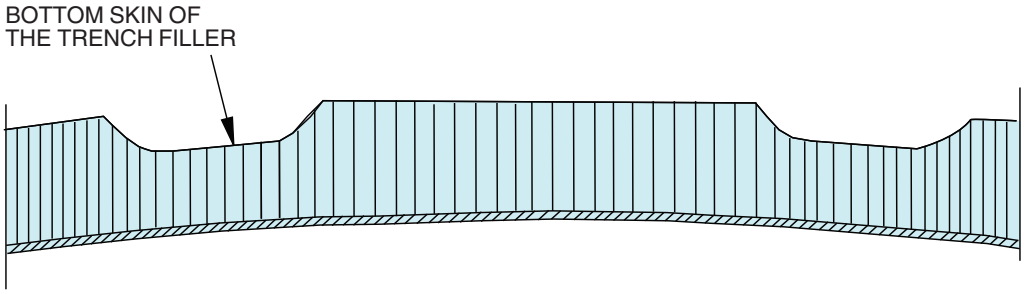
Fan Case Impact Repair on the Trench Filler
Figure 806/72-23-01-990-806-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(DIMENSIONAL INFORMATION)

A-A



B-B

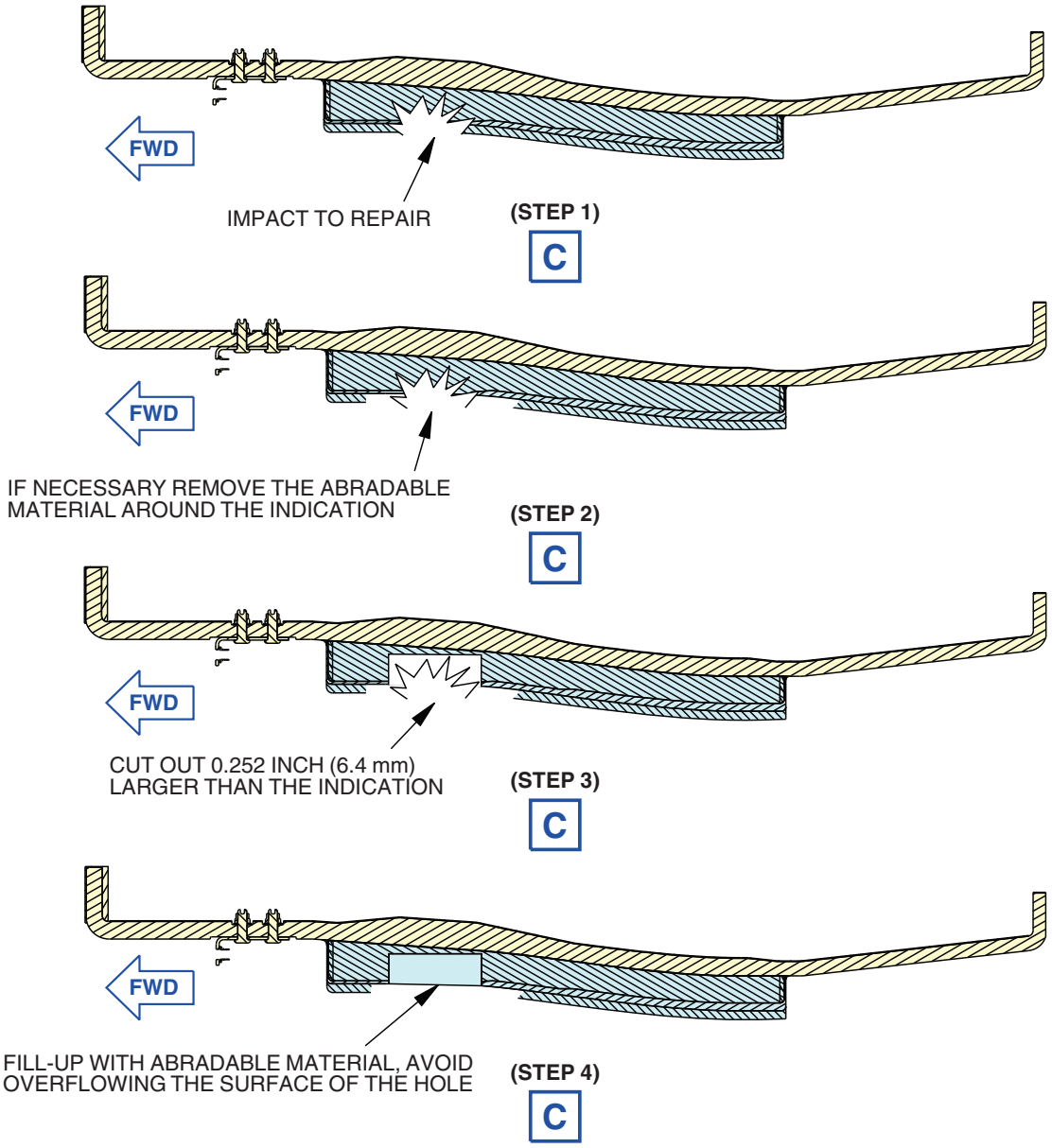
LEGEND:

NOT REPAIRABLE AREA

ICNLEAP1B722301CF030100007A00101
3060641 S0000823840_V2

Fan Case Impact Repair on the Trench Filler
Figure 806/72-23-01-990-806-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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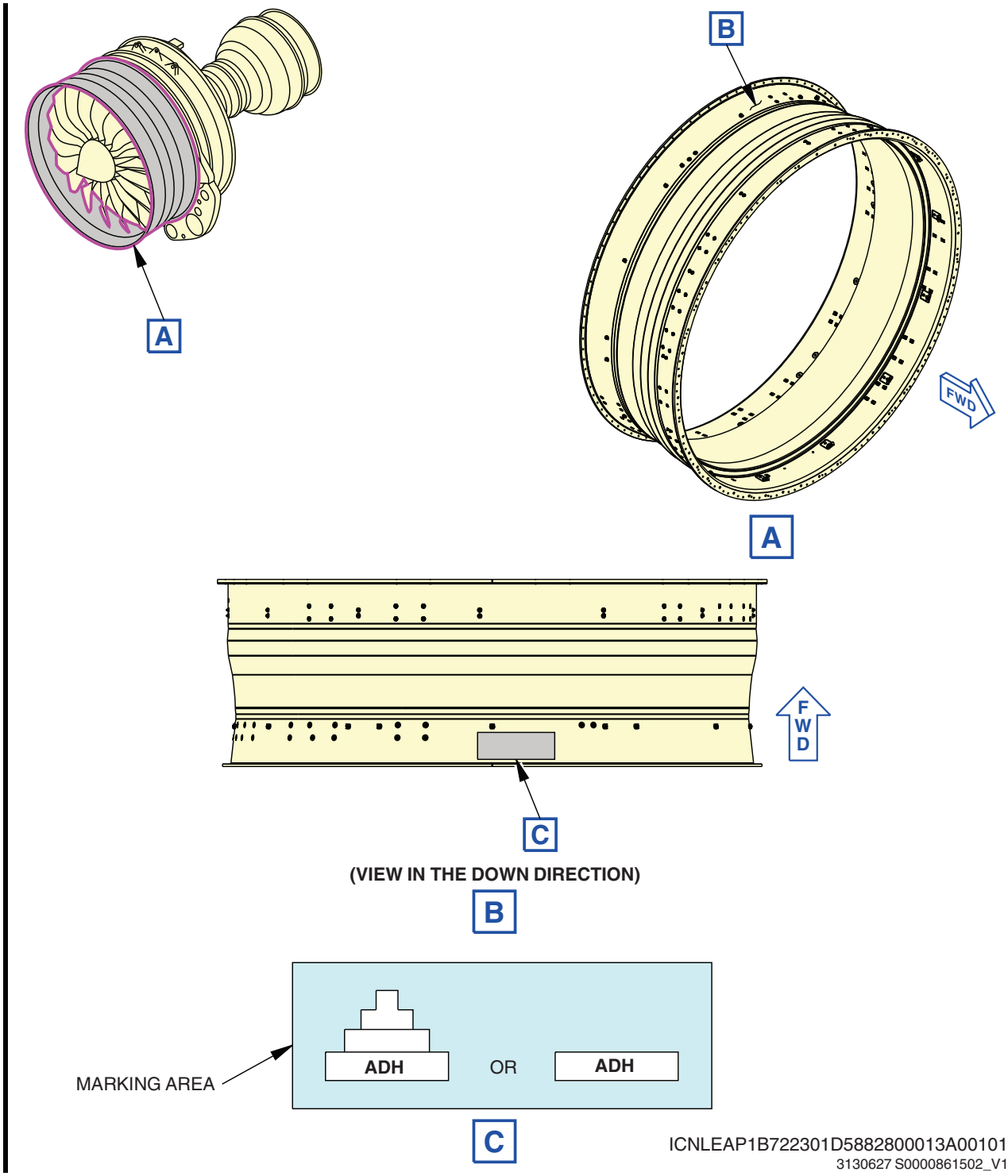
Fan Case Impact Repair on the Trench Filler
Figure 806/72-23-01-990-806-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Special Character for Marking
Figure 807/72-23-01-990-811-G00

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TASK 72-23-01-960-804-G00

6. Fan Case Rivetless Nut Plate Assembly Replacement

(Figure 808)

A. General

(1) This task gives instructions to replace the rivetless nut plate assembly of the fan case.

(2) The Table 805 and Table 806 give a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 805/72-23-01-993-804-G00

Part Number	Description	Quantity
649-313-102-0	Rivetless Nut Plate	AR

Table 806/72-23-01-993-805-G00

Part Number	Service Bulletin	Designation
364-050-013-0	-	FAN CASE
364-050-014-0	72-0061	FAN CASE
364-050-016-0	72-0181	FAN CASE
364-050-099-0	72-0181	FAN CASE

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-52-300-801-G00	Banded Self-Locking Rivetless Nut Plate Assemblies Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-01-860-017-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-01-860-018-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Case Rivetless Nut Plate Assembly Replacement

SUBTASK 72-23-01-960-002-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

(1) Replace the rivetless nut plate assembly as follows:

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

CAUTION DO NOT TOUCH THE ADJACENT PARTS WHEN YOU DRILL OR GRIND THE RIVET HEAD. DAMAGE TO THE PARTS CAN OCCUR.

(a) Remove and discard the rivetless nut plate assembly (TASK 70-43-52-300-801-G00).

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(b) Clean the area where you will install a new rivetless nut plate assembly with a cloth moistened with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(c) Do a visual inspection of the hole of the fan case.

1) Deviation to a correct condition is not permitted.

(d) Install and make sure that the new rivetless nut plate assembly is installed correctly (TASK 70-43-52-300-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-01-860-019-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

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1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-01-860-020-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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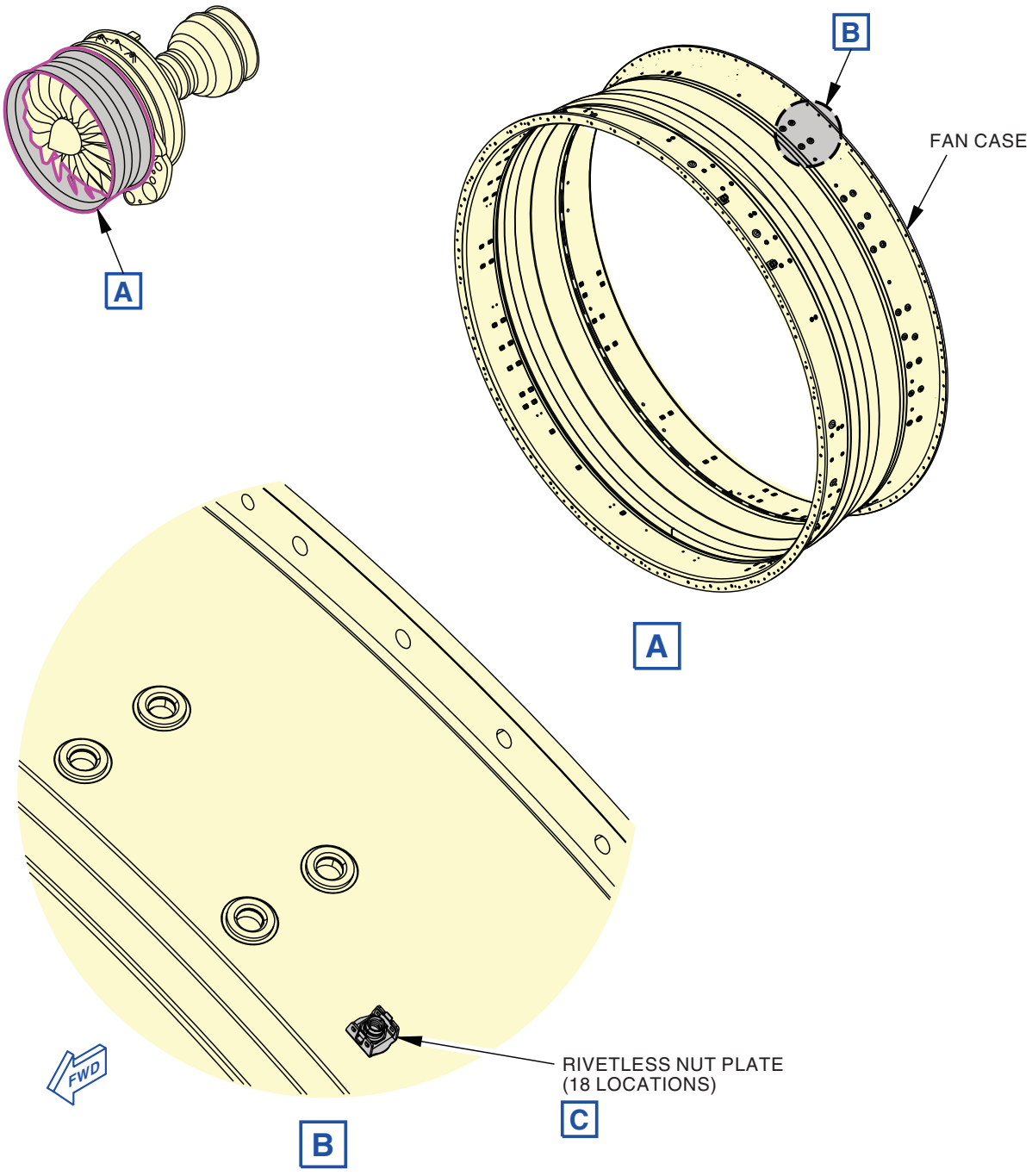


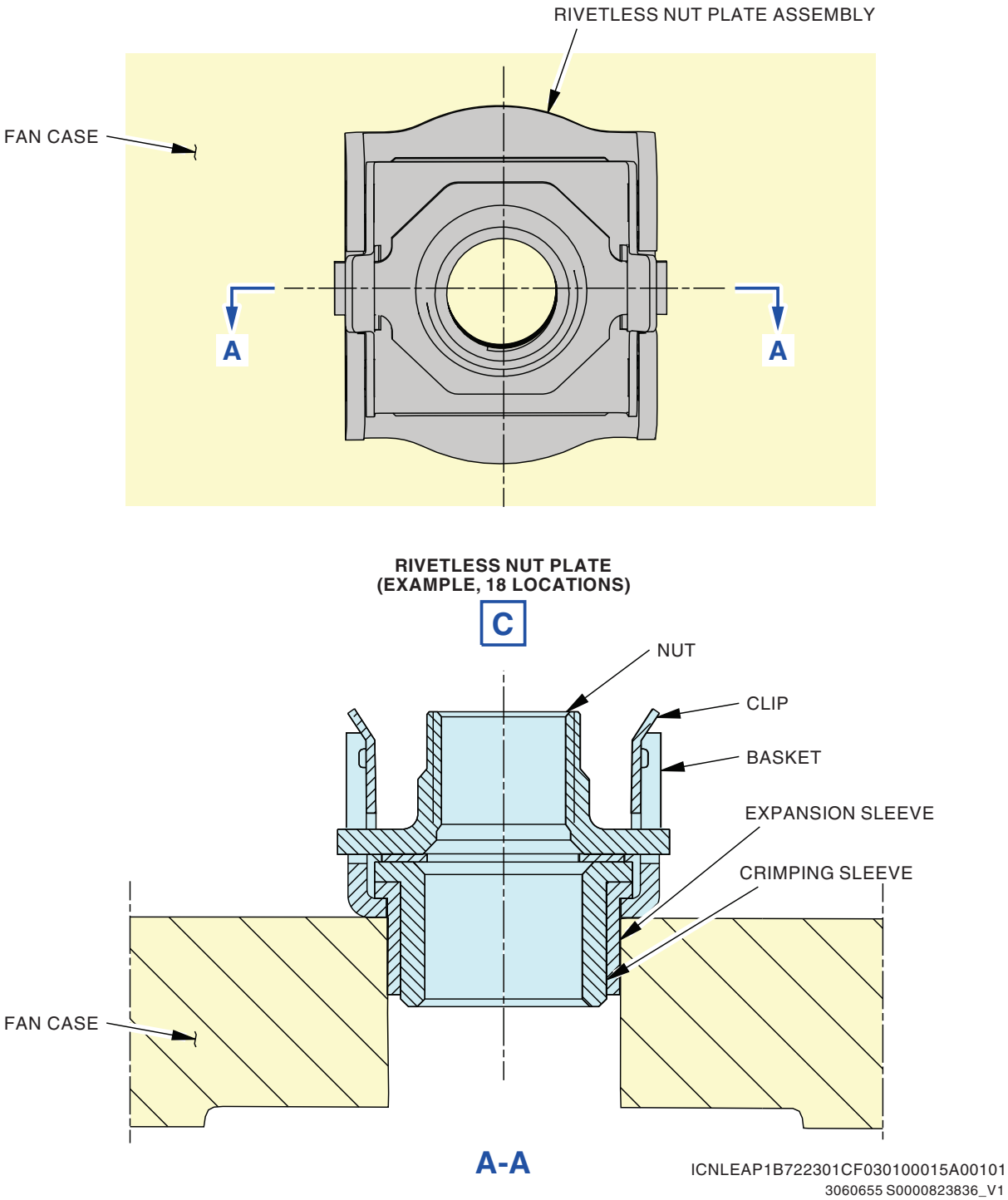
NOTE:

FAN CASE MATERIAL: COMPOUND MATERIAL
RIVETLESS NUT PLATE 18 LOCATIONS MATERIAL: Z6NCT25 AND Z6CNT18

ICNLEAP1B722301CF030100008A00101
3060654 S0000823835_V1

Fan Case Rivetless Nut Plate Assembly Replacement
Figure 808/72-23-01-990-807-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Case Rivetless Nut Plate Assembly Replacement
Figure 808/72-23-01-990-807-G00 (Sheet 2 of 2)

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TASK 72-23-01-300-802-G00

7. Fan Case Fireproof Layer Local Repair

(Figure 809, Figure 810, and Figure 811)

A. General

(1) This task gives instructions to repair the fireproof layer of the fan case.

(2) The table below gives a list of part numbers that are applicable to this repair.

NOTE: All part numbers are applicable to all paragraphs unless specified differently.

Table 807/72-23-01-993-809-G00 Applicable Part Numbers

Part Number	Designation
364-050-013-0	FAN CASE
364-050-014-0	FAN CASE
364-050-016-0	FAN CASE
364-050-017-0	FAN CASE
364-050-099-0	FAN CASE

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-196	Container - Metal
STD-197	Container - Plastic, Polyethylene or Polypropylene
STD-809	Spatula
STD-858	Tag - DO NOT OPERATE
STD-11587	Syringe (Commercial available)

D. Consumable Materials

Reference	Description	Specification
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
A50633 [CP2466]	Adhesive - Epoxy Resin (Hysol EA 9396)	
B00676 [CP1041]	Alcohol - Isopropyl	
G50381	Abrasive - Aluminum Oxide Paper, 180 Grit	

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(Continued)

Reference	Description	Specification
G51448	Abrasive - Aluminum Oxide Paper, 220 grit or finer	
G51693 [CP2967]	Tape - Teflease MG2	
G51697 [CP2929]	Tape - Flashbreaker 1	
G51826 [CP2508]	Tape - Flashbreaker No. 5	
G51996 [CP2958]	Tape - Pressure Sensitive - Flashbreaker 2	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Repair

SUBTASK 72-23-01-860-021-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-01-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Case Fireproof Layer Local Repair

SUBTASK 72-23-01-220-004-G00

(1) Identify and calculate the surfaces of the fireproof layer to be repaired on the external surface of the fan case (Figure 809 and Figure 810).

NOTE: If the carbon under the fireproof layer is damaged, it is not permitted to do the repair.

(a) Calculate the surfaces and add areas repaired before with Dapco 2100 sealant, A50483 [CP2927], and/or Hysol EA 9396 adhesive, A50633 [CP2466].

(b) Make sure that the fan case is repairable with the criteria that follow:

1) For the forward face of A1 flange area (repaired with Dapco 2100 sealant, A50483 [CP2927], or Hysol EA 9396 adhesive, A50633 [CP2466]):

a) The total cumulated repaired surface must be 3.72 in^2 (2400 mm^2) maximum for the areas repaired with Dapco 2100 sealant, A50483 [CP2927].

b) The total cumulated repaired surface must be 4.805 in^2 (3100 mm^2) maximum for the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466].

2) For all other repairable areas on the fan case (barrel and outer diameter of the flanges repaired only with Dapco 2100 sealant, A50483 [CP2927]):

a) The total cumulated repaired surface must be 31.00 in^2 (20,000 mm^2) maximum.

b) The distance between the area to repair and areas repaired before must be more than 7.88 in. (200 mm).

(c) If it is necessary to repair the area repaired before with Hysol EA 9396 adhesive, A50633 [CP2466], or Dapco 2100 sealant, A50483 [CP2927], without removing additional glass plies, it is not necessary to remove the plies and clean the area to repair with dry and de-oiled compressed air.

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SUBTASK 72-23-01-300-007-G00

WARNING

DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH. USE LOCAL MECHANICAL EXHAUST AIRFLOW OR AN APPROVED RESPIRATOR. USE PROTECTION EQUIPMENT. IF YOU DO NOT OBEY, THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION.

CAUTION

DO NOT SAND OR CUT THE CARBON SKIN BELOW THE GLASS PLIES. IF YOU DO NOT OBEY, DAMAGE TO THE FAN CASE CAN OCCUR.

(2) If the fan case is repairable, remove the plies to repair.

NOTE: You can cut the plies one by one or all at the same time if other criteria are followed and the visual inspection of the glass ply or carbon skin under is performed just after to make sure that the layer under is not damaged.

(a) Apply masking tape Flashbreaker 1 tape, G51697 [CP2929], Flashbreaker 5 tape, G51826 [CP2508], or Flashbreaker 2 tape, G51996 [CP2958], around the area to repair.

(b) Cut the plies around the indication.

1) If the indication is on the barrel, cut larger than the maximum length of the indication with these criteria:

- Sharp edges are not permitted
- The surface of the repaired area must be 3.10 in^2 (2000 mm^2) maximum
- The repaired area must be outside a diameter of 2.29 in. (58 mm) around holes on the barrel.

2) If the indication is on the outer diameter of the flanges, cut larger than the maximum length of the indication on each side with this criteria:

- The length of the repaired area must be 19.68 in. (500 mm) maximum.

3) If the indication is on forward face of the A1 flange area, cut larger than the maximum length to cover the indication with these criteria:

- The surface of one repaired area with Dapco 2100 sealant, A50483 [CP2927], must be 0.93 in^2 (600 mm^2) maximum
- The surface of one repaired area with Hysol EA 9396 adhesive, A50633 [CP2466], must be 0.07 in^2 (45 mm^2) maximum
- The distance between the area repaired with Dapco 2100 sealant, A50483 [CP2927], and edges of the modular holes must be 0.40 in. (10 mm) minimum (Figure 811)
- The distance between the edge of the modular holes and the end of the area repaired with Hysol EA 9396 adhesive, A50633 [CP2466], must be 0.60 in. (15 mm) maximum (Figure 811).

(c) Do a visual inspection to make sure that the glass layer or carbon skin layer below is not damaged.

1) Nicks, dents, scratches, cracks, delaminations, and tears are not permitted.

(d) Recalculate the cumulated repaired surfaces with the real cut-out areas to make sure that the fan case is repairable with these criteria:

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1) For the forward face of A1 flange area (repaired with Dapco 2100 sealant, A50483 [CP2927], or Hysol EA 9396 adhesive, A50633 [CP2466]):
   - The total cumulated repaired surface must be 3.72 in^2 (2400 mm^2) maximum for the areas repaired with Dapco 2100 sealant, A50483 [CP2927]
   - The total cumulated repaired surface must be 4.805 in^2 (3100 mm^2) maximum for the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466].
2) For all other repairable areas on the fan case (barrel and outer diameter of the flanges repaired only with Dapco 2100 sealant, A50483 [CP2927]):
   - The total cumulated repaired surface must be 31.00 in^2 (20,000 mm^2) maximum
   - The distance between the area to repair and areas repaired before must be more than 7.88 in. (200 mm).

(e) Remove the masking.

SUBTASK 72-23-01-100-004-G00

WARNING USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

(3) Clean the area to repair with dry and de-oiled compressed air.

SUBTASK 72-23-01-300-008-G00

CAUTION DO NOT SAND OR CUT THE CARBON SKIN BELOW THE GLASS PLIES. IF YOU DO NOT OBEY, DAMAGE TO THE FAN CASE CAN OCCUR.

(4) Prepare the surface of the area to repair.

(a) Apply Flashbreaker 1 tape, G51697 [CP2929], Flashbreaker 5 tape, G51826 [CP2508], or Flashbreaker 2 tape, G51996 [CP2958], around the area to repair.

(b) If it is necessary, remove the disbonded, worn, or polluted Dapco 2100 sealant, A50483 [CP2927], or Hysol EA 9396 adhesive, A50633 [CP2466], from the previous repaired area.

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(c) Manually sand the area to repair with 220 grit or finer abrasive paper, G51448 (TASK 70-43-42-300-802-G00).

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(d) Clean the area to repair with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(e) Remove the masking.

(f) The maximal time between the surface preparation and application of Dapco 2100 sealant, A50483 [CP2927], is 14 days.

1) If the 14 days time limit is exceeded, do these steps:

WARNING USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

a) Clean the area to repair with dry and de-oiled compressed air.

b) Apply Flashbreaker 1 tape, G51697 [CP2929], Flashbreaker 5 tape, G51826 [CP2508], or Flashbreaker 2 tape, G51996 [CP2958], around the area to repair.

c) If it is necessary, remove the disbonded, worn, or polluted Dapco 2100 sealant, A50483 [CP2927], or Hysol EA 9396 adhesive, A50633 [CP2466], from the previous repaired area.

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

d) Manually sand the area to repair with 220 grit or finer abrasive paper, G51448 (TASK 70-43-42-300-802-G00).

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

e) Clean the area to repair with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

f) Remove the masking.

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SUBTASK 72-23-01-300-009-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING USE PERSONAL PROTECTION EQUIPMENT AND MECHANICAL EXHAUST VENTILATION. THE EPOXY RESIN AND THE CATALYST IS AN IRRITANT, ALLERGEN, AND HAS POISONOUS DETERIORATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS CAN OCCUR.

(5) Apply Dapco 2100 sealant, A50483 [CP2927], or Hysol EA 9396 adhesive, A50633 [CP2466], on the area to repair.

(a) If the Hysol EA 9396 adhesive, A50633 [CP2466], will be used, prepare the Hysol EA 9396 adhesive, A50633 [CP2466], with a Part A viscosity between 400 and 1200 poises.

1) Use a clean and disposable metal container, STD-196, or plastic container, STD-197, and a clean tool to mix the epoxy resin.

2) Make an estimate of the quantity of Hysol EA 9396 adhesive, A50633 [CP2466], that is necessary to repair the glass fiber.

a) Prepare 0.45 oz (13 g) - 3.52 oz (100 g) of Hysol EA 9396 adhesive, A50633 [CP2466].

3) Prepare a mixture in the following weight proportions.

a) Hysol EA 9396 adhesive, A50633 [CP2466] Part A: 100 parts

b) Hysol EA 9396 adhesive, A50633 [CP2466] Part B: 30 parts ± 10%.

4) Mix the 2 parts to have a homogeneous green structure.

(b) If it is necessary to apply Dapco 2100 sealant, A50483 [CP2927], and Hysol EA 9396 adhesive, A50633 [CP2466], on the same area to repair, apply the products one after another.

1) The polymerization of the first product must be fully complete before you apply the next one.

(c) If it is necessary to repair aftward flange radius area, do not overflow the glass cloth on the barrel when applying Dapco 2100 sealant, A50483 [CP2927] (Figure 809).

(d) Apply one layer of Flashbreaker 1 tape, G51697 [CP2929], or Teflease MG2 tape, G51693 [CP2967], around the area to repair.

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(e) Clean the area to repair with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

1) It is necessary to wait 10 minutes minimum but not more than 120 minutes before Dapco 2100 sealant, A50483 [CP2927], and Hysol EA 9396 adhesive, A50633 [CP2466], application.

a) Make sure that the alcohol is evaporated.

2) If more than 120 minutes pass, clean the area to repair with alcohol, B00676 [CP1041] again.

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(f) Apply Dapco 2100 sealant, A50483 [CP2927], or Hysol EA 9396 adhesive, A50633 [CP2466], with a spatula, STD-809, or syringe, STD-11587, on the area to repair.

1) For the areas repaired with Dapco 2100 sealant, A50483 [CP2927], do these steps:

a) Make sure that the repaired area thickness is equal to the thickness of the glass plies removed before in addition with the thickness of the masking tape on the adjacent areas.

b) Make sure that the repaired surface is flush with the masking tape on the adjacent areas.

2) For the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466], do these steps:

NOTE: It is recommended to limit air bubbles during application of Hysol EA 9396 adhesive, A50633 [CP2466], to have a homogeneous surface.

NOTE: Hysol EA 9396 adhesive, A50633 [CP2466], is a low viscosity product. It is recommended to control the position of the part before curing to avoid the product to flow.

a) Wait at least 5 minutes for adhesive to spread.

b) Add an excess thickness of adhesive if the finding is not completely filled.

<1> No cavity permitted.

c) Examine the repaired area.

<1> The thickness must be homogeneous.

<a> No cavity and no pollution are permitted.

SUBTASK 72-23-01-300-010-G00

(6) If the area repaired with Dapco 2100 sealant, A50483 [CP2927], remove the masking.

SUBTASK 72-23-01-300-011-G00

(7) Cure the repaired area.

(a) If the areas repaired with Dapco 2100 sealant, A50483 [CP2927], do the polymerization (Table 808, Table 809).

Table 808/72-23-01-993-806-G00 Polymerization Cycles for DAPCO on the Barrel or on the Outer Diameter of the Flanges

Temperature °F (°C)	Relative Humidity %	Polymerization time
54 ≤ T < 59 (12 ≤ T < 15)	10 ≤ RH ≤ 80	24 h minimum
59 ≤ T < 68 (15 ≤ T < 20)	10 ≤ RH ≤ 80	8 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	10 ≤ RH < 20	3 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	20 ≤ RH < 40	2 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	40 ≤ RH ≤ 80	1 h minimum
86 ≤ T < 104 (30 ≤ T < 40)	10 ≤ RH < 30	2 h minimum
86 ≤ T < 104 (30 ≤ T < 40)	30 ≤ RH ≤ 80	1 h minimum
104 ≤ T ≤ 131 (40 ≤ T ≤ 55)	10 ≤ RH ≤ 80	1 h minimum

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Table 809/72-23-01-993-807-G00 Polymerization Cycles for DAPCO on the Forward Face of A1 Flange

Temperature °F (°C)	Relative Humidity %	Polymerization time
54 ≤ T < 59 (12 ≤ T < 15)	10 ≤ RH ≤ 80	48 h minimum
59 ≤ T < 68 (15 ≤ T < 20)	10 ≤ RH ≤ 80	20 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	10 ≤ RH < 20	12 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	20 ≤ RH < 30	8 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	30 ≤ RH < 40	6 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	40 ≤ RH < 60	4 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	60 ≤ RH < 70	3 h minimum
68 ≤ T < 86 (20 ≤ T < 30)	70 ≤ RH ≤ 80	2 h minimum
86 ≤ T < 104 (30 ≤ T < 40)	10 ≤ RH < 20	8 h minimum
86 ≤ T < 104 (30 ≤ T < 40)	20 ≤ RH < 30	5 h minimum
86 ≤ T < 104 (30 ≤ T < 40)	30 ≤ RH < 40	4 h minimum
86 ≤ T < 104 (30 ≤ T < 40)	40 ≤ RH < 50	3 h minimum
86 ≤ T < 104 (30 ≤ T < 40)	50 ≤ RH ≤ 80	2 h minimum
104 ≤ T < 122 (40 ≤ T < 50)	10 ≤ RH < 20	5 h minimum
104 ≤ T < 122 (40 ≤ T < 50)	20 ≤ RH < 40	3 h minimum
104 ≤ T < 122 (40 ≤ T < 50)	40 ≤ RH < 60	2 h minimum
104 ≤ T < 122 (40 ≤ T < 50)	60 ≤ RH ≤ 80	1 h minimum
122 ≤ T ≤ 131 (50 ≤ T ≤ 55)	10 ≤ RH < 20	3 h minimum
122 ≤ T ≤ 131 (50 ≤ T ≤ 55)	20 ≤ RH < 40	2 h minimum
122 ≤ T ≤ 131 (50 ≤ T ≤ 55)	40 ≤ RH ≤ 80	1 h minimum

(b) If the repaired area is located both on the outer diameter of the flanges and on the forward face of A1 flange and you repair all the area with Dapco 2100 sealant, A50483 [CP2927], do the longest polymerization cycle proposed (forward face of A1 flange cycles) for the associated humidity and temperature parameters.

(c) Alternative procedures available. If the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466], do the polymerization at 140°F (60°C) - 158°F (70°C) for one hour minimum.

(d) Alternative procedure. If the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466], do the polymerization at 167°F (75°C) - 185°F (85°C) for 30 minutes minimum.

(e) Remove the masking after curing the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466].

SUBTASK 72-23-01-300-012-G00

(8) Remove the unwanted material.

(a) If the areas repaired with Dapco 2100 sealant, A50483 [CP2927], remove the unwanted Dapco 2100 sealant, A50483 [CP2927], with a sharp tool.

1) Make sure that you do not remove the parent material.

(b) If the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466], remove the unwanted material.

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1) Apply Flashbreaker 1 tape, G51697 [CP2929], Flashbreaker 5 tape, G51826 [CP2508], or Flashbreaker 2 tape, G51996 [CP2958], around the unwanted excess of Hysol EA 9396 adhesive, A50633 [CP2466].

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

2) Remove the unwanted Hysol EA 9396 adhesive, A50633 [CP2466], with 180 grit abrasive paper, G50381, or finer (TASK 70-43-42-300-802-G00).
   a) Make sure that you do not remove the parent material.
3) Remove the masking.

SUBTASK 72-23-01-100-005-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(9) If it is necessary, clean the repaired area with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-01-211-001-G00

(10) Do the final inspection of the repaired area.
   (a) If the areas repaired with Dapco 2100 sealant, A50483 [CP2927], do these steps:
      1) Make sure that the surface is smooth, elastic, and not tacky.
      2) Make sure that the color is uniform.
      3) Make sure that there are no bubbles, wrinckles, and foreign objects on the surface.
      4) Make sure that the maximum excess thickness of the repaired area with adjacent areas is 0 in. (0 mm) - 0.011 in. (0.3 mm) on the barrel or on the outer diameter of the flanges.
         a) Underthickness is not permitted.
      5) Make sure that the maximum excess thickness of the repaired area with adjacent areas is 0 in. (0 mm) - 0.003 in. (0.1 mm) on the forward face of A1 flange.
         a) Underthickness is not permitted.
   (b) If the areas repaired with Hysol EA 9396 adhesive, A50633 [CP2466], do these steps:
      1) Make sure that the surface appearance is homogeneous along the entire surface.
      2) Make sure that there is no offset or cavity on the surface.
      3) Make sure that there is no pollution on the surface.
      4) Make sure that the surface of the repaired area is flush with the adjacent areas.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-01-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-01-860-022-G00

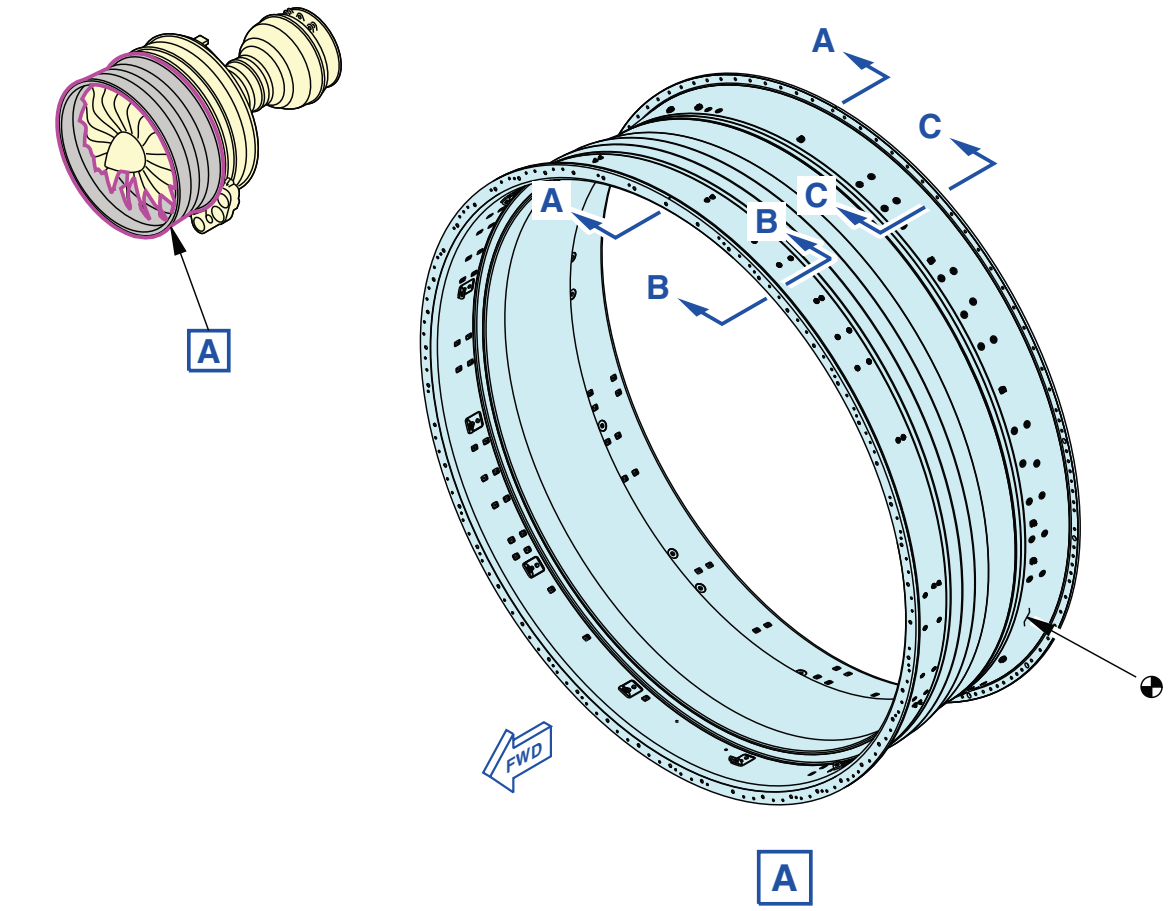
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

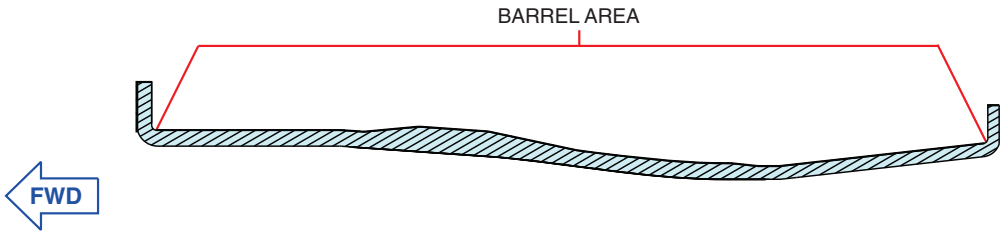
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





A-A

LEGEND:

- COMPOUND MATERIAL (EPOXY RESIN AND CARBON FIBER + GLASS FIBER
- REPAIRABLE AREA (TWO GLASS PLIES)

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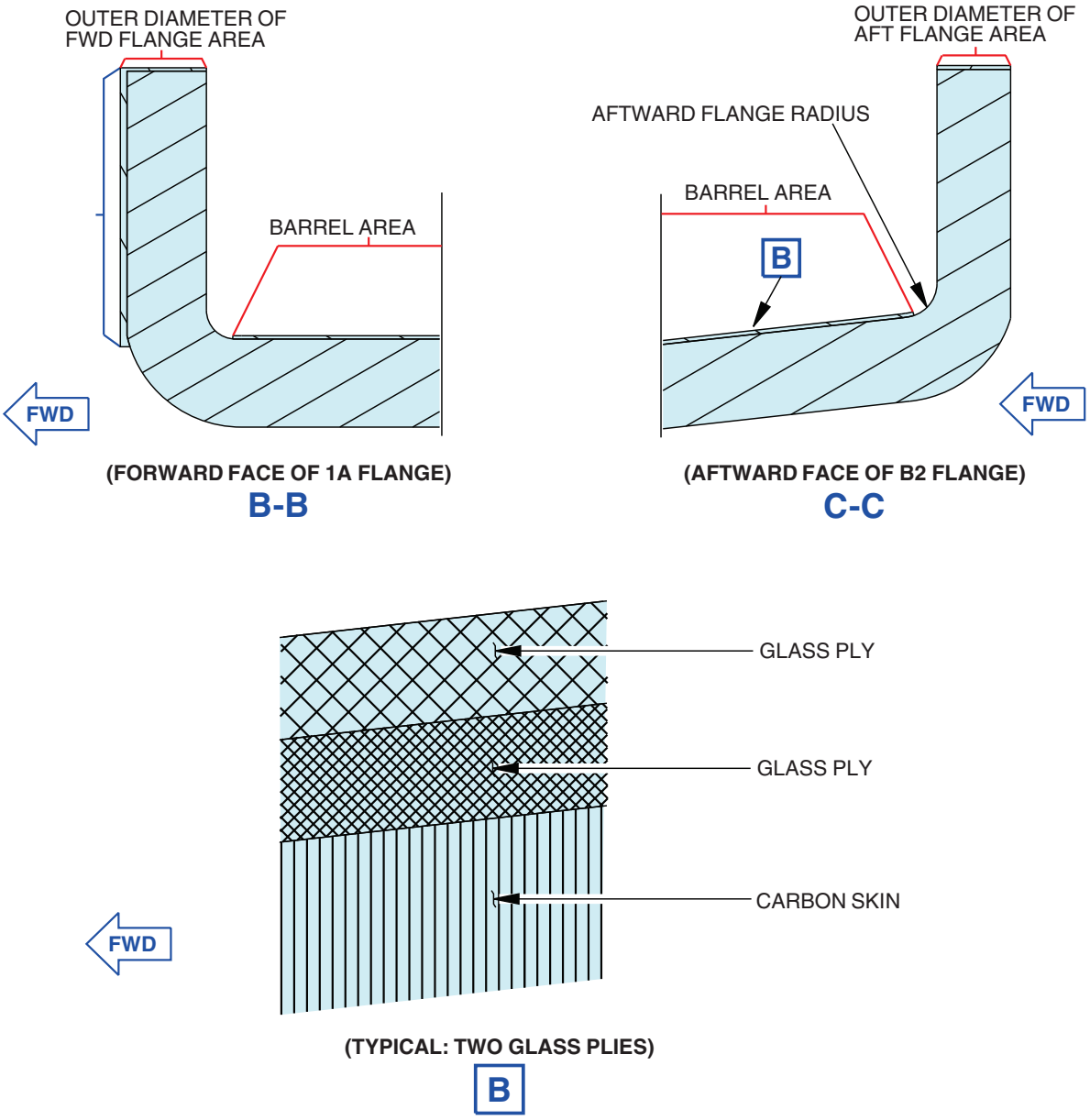
Repairable Areas
Figure 809/72-23-01-990-808-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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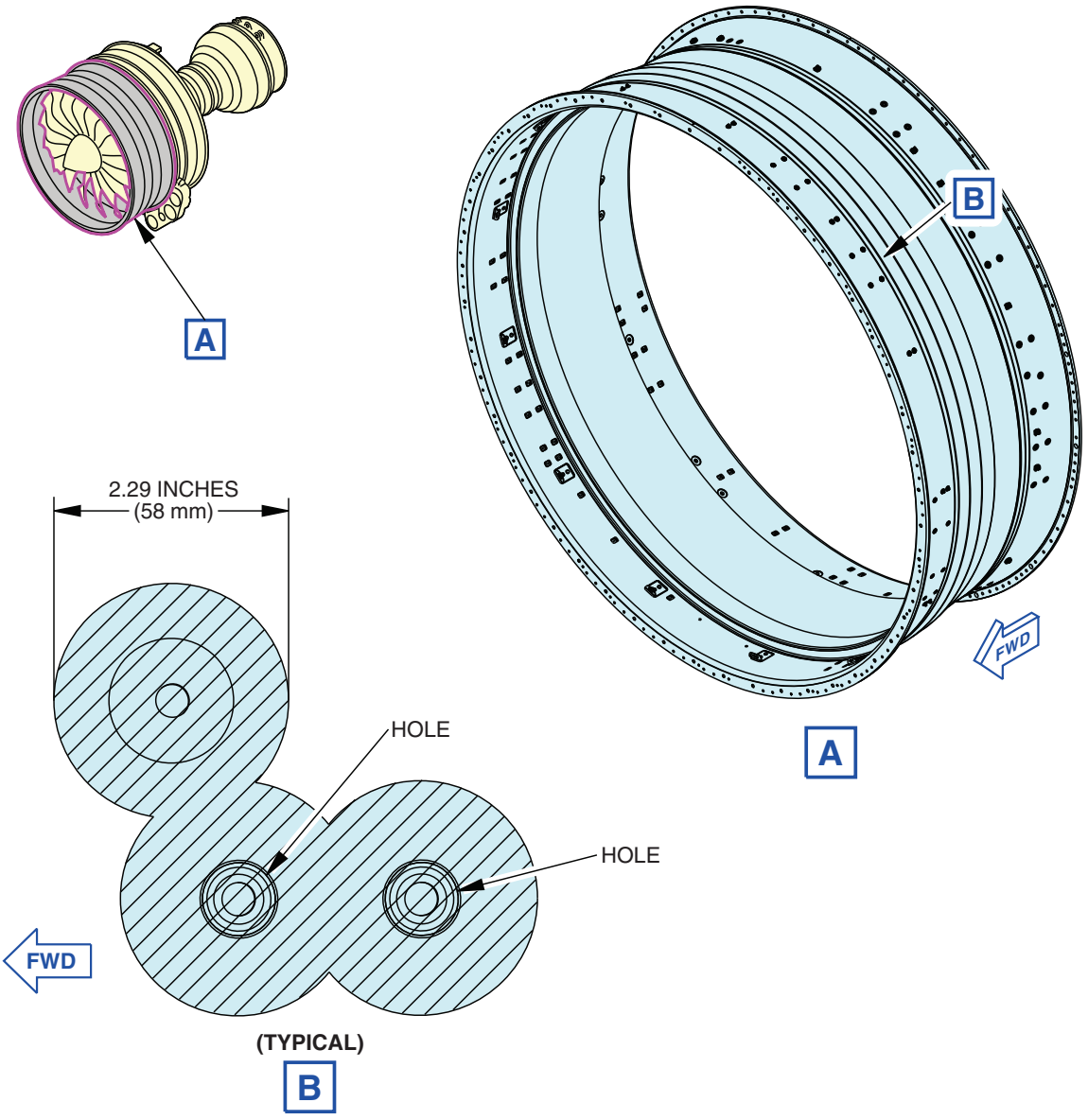
Repairable Areas
Figure 809/72-23-01-990-808-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

NON-REPAIRABLE AREAS

NOTE:

THIS FIGURE NOT REPRESENT ALL TYPE OF HOLES OF THE BARREL ON THE FAN CASE

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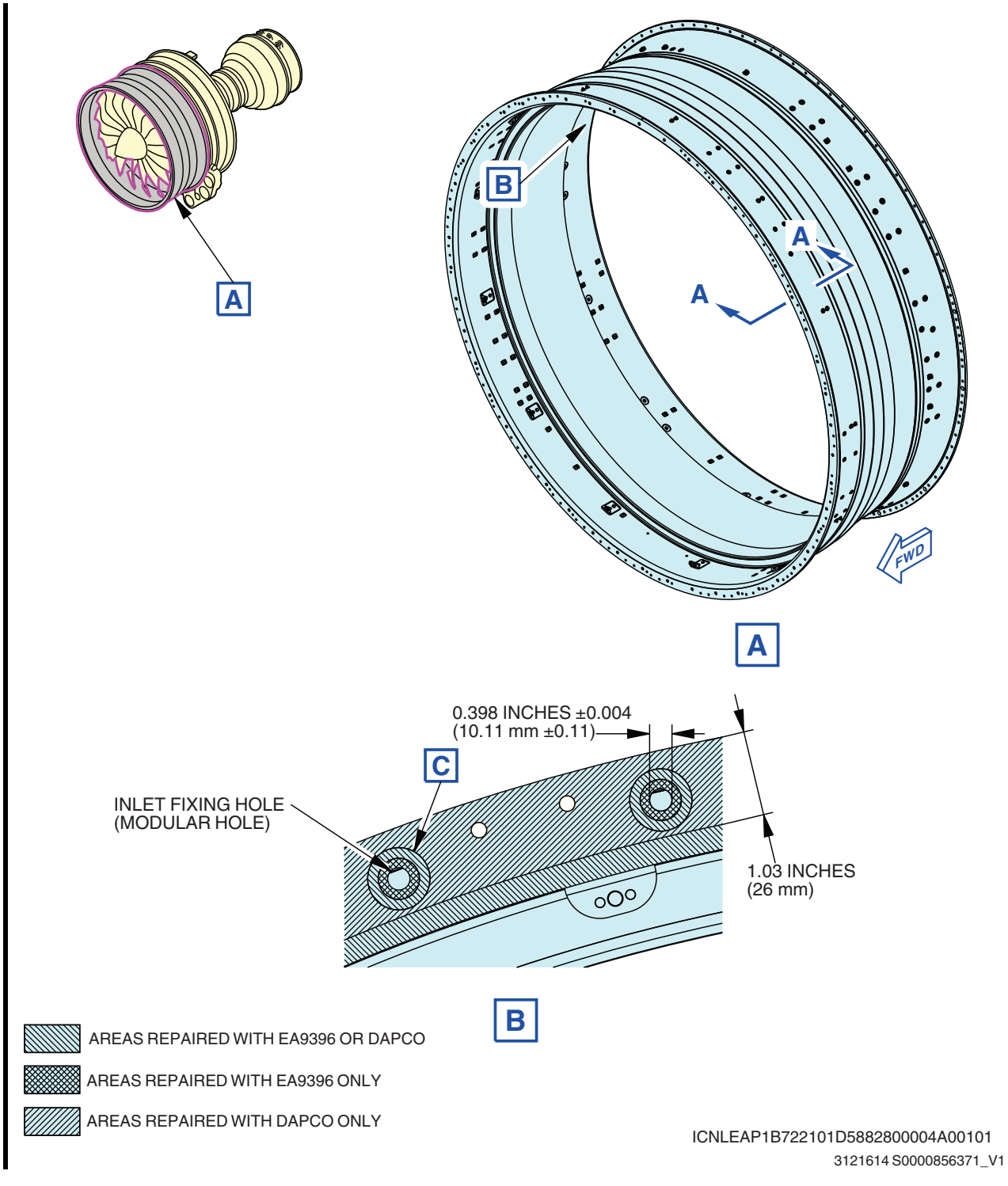
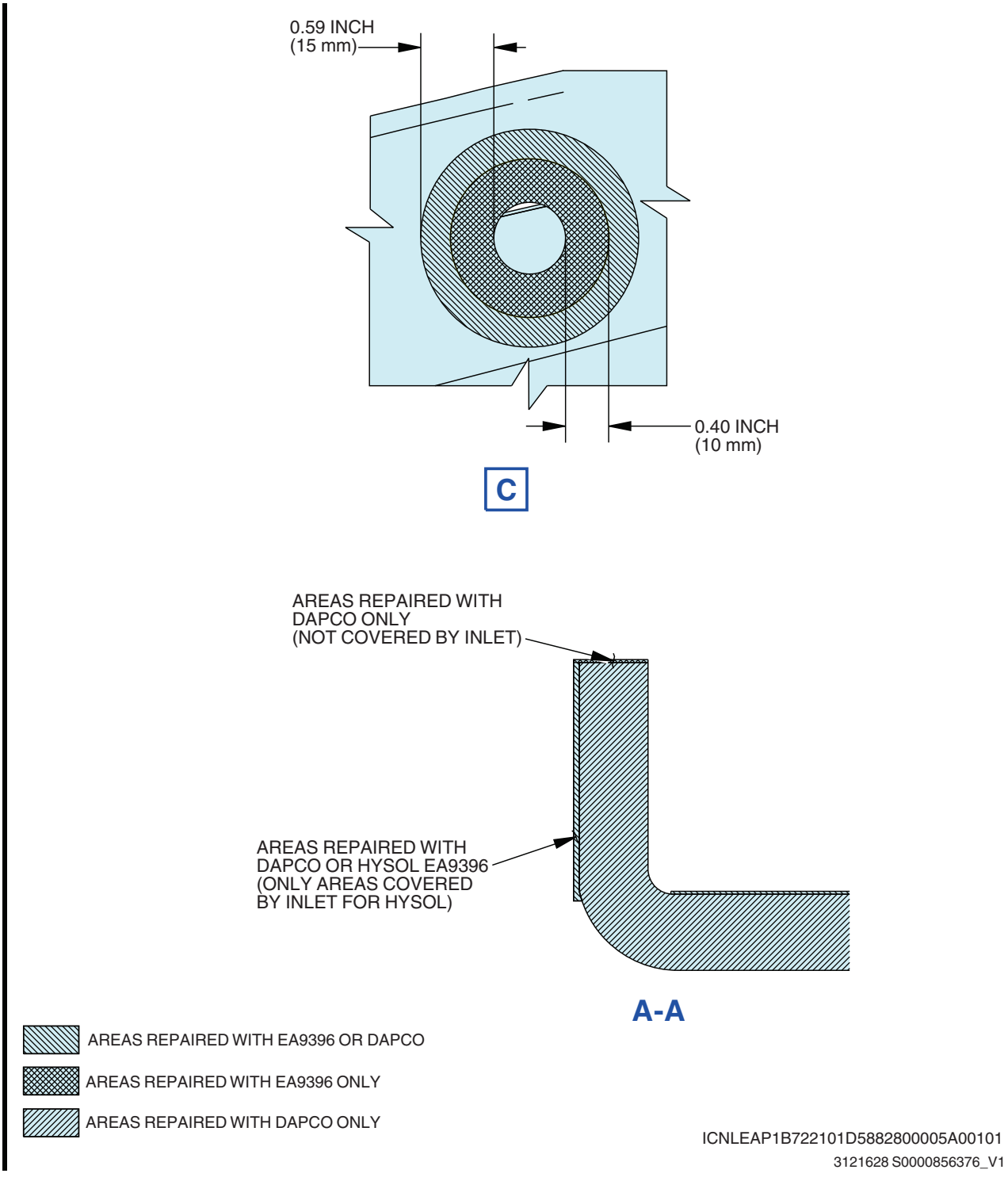
Non-Repairable Areas
Figure 810/72-23-01-990-809-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Face of A1 Flange
Figure 811/72-23-01-990-810-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Face of A1 Flange
Figure 811/72-23-01-990-810-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-01-300-803-G00

8. Fan Case Abradable Liner Cracks Local Repair

(Figure 812 and Figure 813)

A. General

(1) This task gives the instructions to do a local repair for the liner cracks of the fan case abradable.

(2) The table below gives a list of part numbers that are applicable to this repair.

NOTE: All part numbers are applicable to all paragraphs unless specified differently.

Table 810/72-23-01-993-810-G00 Applicable Part Numbers

Part Number	Designation
364-050-013-0	FAN CASE
364-050-014-0	FAN CASE
364-050-016-0	FAN CASE
364-050-017-0	FAN CASE
364-050-099-0	FAN CASE

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15597	Fixture, Local Repair - Abradable Liner Part #: 956A8082G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51335 [CP2674]	Lint-Free Cloth	
G51480 [CP2772]	Cloth - Abrasive, Alumina	
G51894 [CP2923]	Compound - Epoxy, Void Filling and Edge Sealing, Scotch-Weld EC-3524 B/A Black	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Repair

SUBTASK 72-23-01-860-023-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-01-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-23-01-100-006-G00

WARNING DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH YOU. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING ONLY USE CLEANING AGENTS WHEN THERE IS GOOD AIR FLOW THROUGH THE WORK AREA. THIS MATERIAL IS POISONOUS AND FLAMMABLE AND CAN CAUSE SKIN IRRITATION. OBEY THE MANUFACTURER'S INSTRUCTIONS. PUT ON PROTECTIVE CLOTHING. DO NOT GET THE CLEANING AGENTS IN YOUR MOUTH. DO NOT SMOKE. DO NOT BREATHE THE GAS. GET HELP IF YOUR SKIN OR EYES BECOME IRRITATED.

CAUTION DO NOT SAND OR CUT THE CARBON SKIN BELOW THE ABRADABLE MATERIAL. DAMAGE TO THE FAN CASE CAN OCCUR.

(3) Remove all deviations and/or unbonded abradable material from the liner.

(a) If it is necessary, use a plastic spatula or a cutter.

(b) Round the sharp edges.

(c) If it is necessary, remove the abradable material from the area to repair until you see the carbon skin.

1) Do not damage the carbon skin.

(d) Do not open the crack more than 0.295 in. (7.5 mm) in each directions around the original crack profile.

(e) Make sure that the remaining abradable is clean and not disbonded.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-01-120-003-G00

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(4) Smooth the area to repair with abrasive cloth, G51480 [CP2772] (TASK 70-43-42-300-802-G00).

SUBTASK 72-23-01-100-007-G00

WARNING USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

(5) Blow the area to repair with dry and de-oiled air.

SUBTASK 72-23-01-110-001-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(6) Clean the area to repair with a lint-free cloth, G51335 [CP2674], and alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-01-940-002-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

CAUTION YOU MUST WEAR GLOVES THAT ARE FREE OF TALC OR SILICONE TO PREVENT CONTAMINATION OF ABRADABLE COMPOUND. IF NOT, YOU CAN CAUSE DAMAGE.

(7) Prepare the epoxy compound, G51894 [CP2923].

(a) Base Scotch-WeldTM EC-3524 Part B black: 100 parts by weight.

(b) Accelerator Scotch-WeldTM EC-3524 Part A: 94 parts by weight.

(c) Mix the base and hardener by hand until it becomes a homogeneous material with a dark grey color.

1) Mix both parts slowly to prevent air bubbles in the mixture.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If it is necessary, heat the above mixture to 77°F (25°C) to make the application of product to the part easier.

NOTE: The recommended work temperature is 68°F (20°C) - 77°F (25°C).

H. Fan Case Abradable Liner Cracks Local Repair

SUBTASK 72-23-01-380-003-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

CAUTION YOU MUST WEAR GLOVES THAT ARE FREE OF TALC OR SILICONE TO PREVENT CONTAMINATION OF ABRADABLE COMPOUND. IF NOT, YOU CAN CAUSE DAMAGE.

(1) Apply the abradable material as follows:

(a) If it is necessary, heat the area to repair or the working environment to 68°F (20°C) - 77°F (25°C) to make the application of product to the part easier.

(b) Make sure that the application surface is clean and not disbonded.

(c) Apply a very thin layer of abradable material epoxy compound, G51894 [CP2923], on the area to repair and apply sufficient pressure with your thumb to obtain better results.

(d) Fill the remaining thickness with abradable material epoxy compound, G51894 [CP2923].

(e) Adjust the abradable liner with the repair fixture, SPL-15597, or a locally manufactured tool.

1) Flush to the adjacent flow path area that is in good condition.

SUBTASK 72-23-01-380-004-G00

CAUTION DO NOT APPLY HEAT TO THE FAN-CASE ADJACENT AREAS AND ADJACENT FAN BLADES TO THE REPAIR AREAS. A TEMPERATURE MORE THAN THE POLYMERISATION TEMPERATURE CAN CAUSE DAMAGE TO THE PARTS.

(2) Do one of these steps to polymerize the abradable material:

(a) Polymerize at room temperature for 24 hours.

1) Room temperature must be between 59°F (15°C) and 77°F (25°C).

a) Use 77°F (25°C) to get better results.

(b) Polymerize at 122°F (50°C) - 140°F (60°C) for 5 hours.

(c) Polymerize at 158°F (70°C) - 194°F (90°C) for 2 hours.

SUBTASK 72-23-01-200-002-G00

(3) Do a hardness check of the abradable material.

(a) Make sure that the hardness is between 40 and 60 Shore D.

SUBTASK 72-23-01-220-005-G00

(4) Do an inspection of the repaired area.

(a) Do a visual inspection of the liner for bubbles or missing abradable material.

1) Air bubbles are not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Lack of abradable is not permitted.

(b) Do a dimension check of the repaired area with the repair fixture, SPL-15597, or a locally manufactured tool (View A-A, Figure 812 and Figure 813).

NOTE: The tool must be in contact with the adjacent unrepaired areas that are in good condition.

1) Bumped and depressed areas are not permitted.

2) The repaired area must have a smooth transition with the surrounding abradable material.

WARNING

DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

3) If it is necessary, blend the abradable to the final dimensions with P220 or thinner abrasive cloth, G51480 [CP2772] (View A-A, Figure 812 and Figure 813) (TASK 70-43-42-300-802-G00).

(c) Do a check of the abradable material bonding with a wooden mallet.

NOTE: A dull sound shows that the abradable material is not correctly bonded.

(d) If the inspection shows an unserviceable condition, do the repair again.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-01-410-002-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-01-860-024-G00

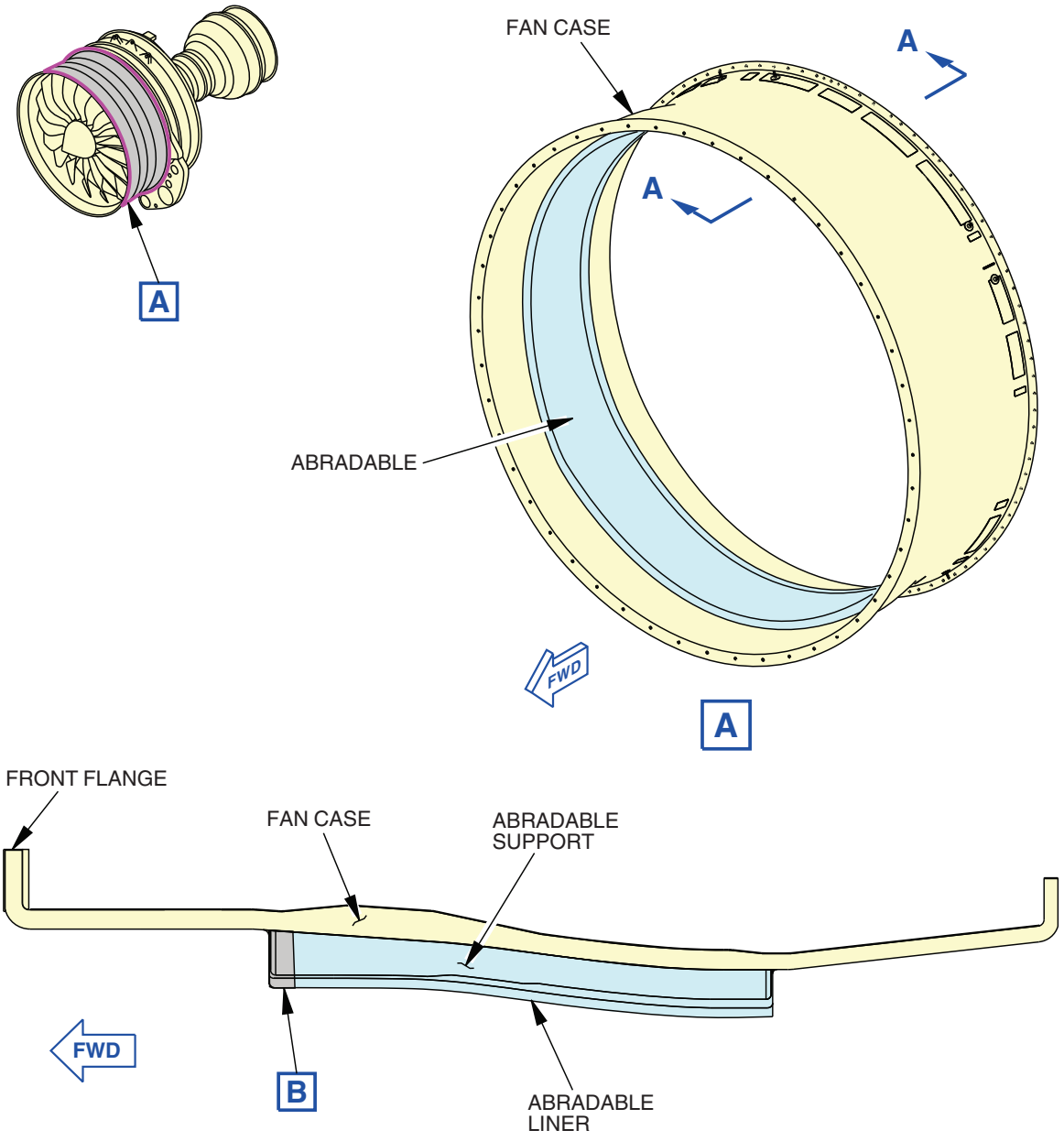
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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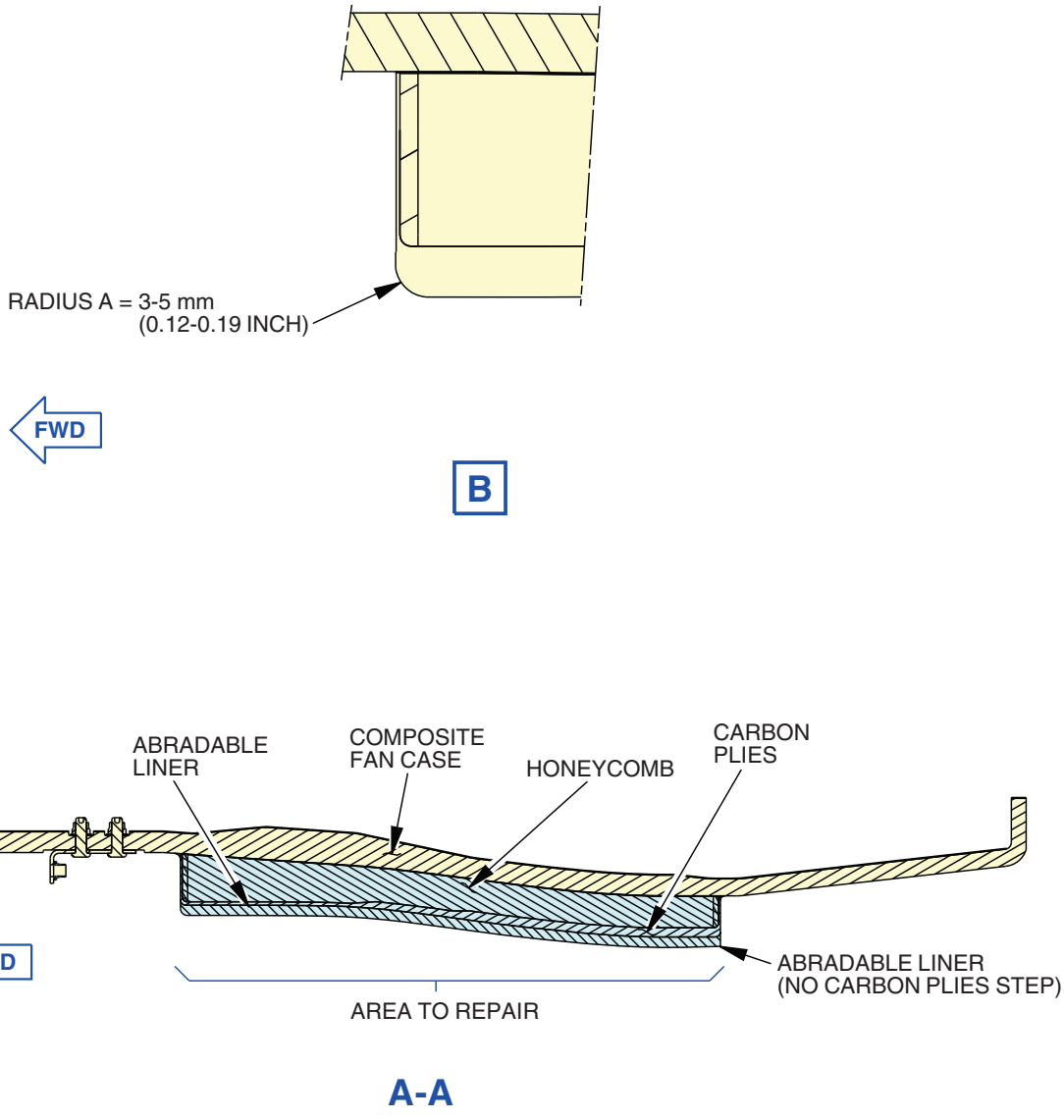
Abradable Material cracks repair
Figure 812/72-23-01-990-812-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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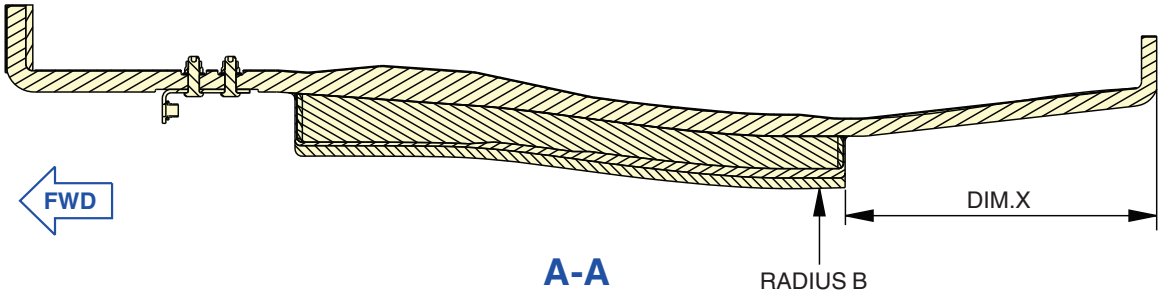
Abradable Material cracks repair
Figure 812/72-23-01-990-812-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ABRADABLE PROFILE

DIM. X INCH	DIM. X (mm)	RADIUS B INCH	RADIUS B (mm)
7.17	(182)	33.99673	(863.517)
7.80	(198)	33.9831	(863.17)
8.43	(214)	33.98077	(863.112)
9.06	(230)	33.99494	(863.472)
9.69	(246)	34.03085	(864.384)
10.32	(262)	34.08589	(865.782)
10.95	(278)	34.14734	(867.343)
11.56	(294)	34.21242	(868.996)
12.21	(310)	34.28793	(870.914)
12.84	(326)	34.37161	(873.039)
13.47	(342)	34.45764	(875.224)
14.09	(358)	34.53927	(877.298)
14.72	(374)	34.60969	(879.086)
15.35	(390)	34.66443	(880.477)
15.98	(406)	34.69228	(881.184)
16.61	(422)	34.70272	(881.449)
17.24	(438)	34.70791	(881.581)
17.87	(454)	34.71075	(881.653)
18.50	(470)	34.7118	(881.68)
19.13	(486)	34.71146	(881.671)

NOTE:

DIM. X : DISTANCE BETWEEN THE AFT FLANGE AND THE MEASUREMENT POSITION.

RADIUS B : RADIUS BETWEEN THE FLOWPATH AND THE ENGINE ROTATIONAL AXIS.

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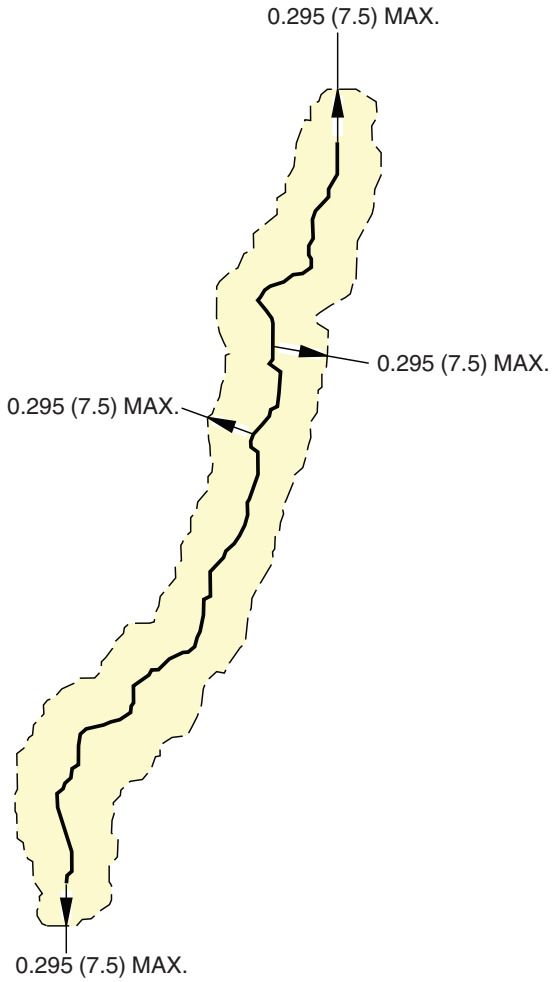
Abradable Material cracks repair
Figure 812/72-23-01-990-812-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

ORIGINAL CRACK PROFILE

OPENED CRACK PROFILE

NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES)

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Abradable Material Cracks Opening Profile Limits
Figure 813/72-23-01-990-813-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME MODULE SEGMENT SHROUDS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fan Frame Module Segment Shrouds Removal

(2) Fan Frame Module Segment Shrouds Installation.

TASK 72-23-04-000-801-G00

2. Fan Frame Module Segment Shrouds Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the segment shrouds for the fan frame module.

NOTE: The VBV grid seals located under the VBV grids can be red or blue.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
434	Engine 1 - Aft Strut Fairing
444	Engine 2 - Aft Strut Fairing

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 72-23-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Fan Frame Module Segment Shrouds Removal

SUBTASK 72-23-04-020-001-G00

CAUTION

IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(1) Remove the segment shroud 5 [1] as follows:

(a) Install the protector, SPL-14947.

(b) Install the protective covers on the grids of segment shroud 5 [1].

(c) Remove the two bolts [4] from the two saddle clamps [3] that attach the duct WYE [2] to the inlet duct [13].

(d) Loosen the screw [85] of the clamp [86].

(e) Loosen the screw [87] of the clamp [88].

(f) Remove the duct WYE [2] from the inlet duct [13] and from the HPTACC valve [89] and the LPTACC valve [90].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(g) Remove the two bolts [92] that attach the diffuser plate [93] to the VBV duct 7 [14].

PTF ALL

(h) Remove the screw [8] that attaches the segment shroud 5 [1] to the strut 1 pad (left) [7].

(i) If the segment shroud 4 [18] is still in place, remove the screw [5] that attaches the segment shroud 5 [1] to the junction shroud fitting [21].

(j) If the segment shroud 4 [18] is still in place, remove the three screws [6] that attach the segment shroud 5 [1] to the junction shroud fitting [21].

(k) Remove the five bolts [11] that attach the segment shroud 5 [1] to the retainer ring [10].

(l) Loosen the three screws [9] that attach the segment shroud 5 [1] to the strut 1 pad (left) [7].

(m) Loosen but do not remove the eight screws [77] from the VBV duct 7 [14] and VBV duct 8 [16].

CAUTION

MAKE SURE THAT YOU HOLD THE SEGMENT SHROUD WHILE YOU REMOVE THE LAST SCREWS. THE SHROUD CAN FALL AND DAMAGE CAN OCCUR.

(n) Remove the three screws [9].

(o) Remove the segment shroud 5 [1] with the VBV duct 7 [14], VBV 7 seal [15], VBV duct 8 [16] and VBV 8 seal [17] from the fan frame hub [12].

NOTE: The VBV 7 seal [15] and VBV 8 seal [17] can be orange or gray.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(p) Install the protective covers on the two VBV doors of the fan frame hub [12].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(q) Install the protective covers on the two VBV doors of the fan frame hub [12] and on the diffuser plate [93].

PTF ALL

SUBTASK 72-23-04-020-002-G00

CAUTION IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(2) Remove the segment shroud 4 [18] as follows:

(a) Install the protector, SPL-14947.

(b) Install the protective covers on the grids of segment shroud 4 [18].

(c) Unclamp the harness [82] from the bracket [19].

(d) Unclamp the harness [84] from the bracket [83].

(e) If the segment shroud 5 [1] is still in place, remove the screw [22] that attaches the segment shroud 4 [18] to the junction shroud fitting [21].

(f) Remove the screw [25] that attaches the segment shroud 4 [18] to the strut 5 pad (left) [24].

(g) If the segment shroud 5 [1] is still in place, loosen the three screws [23] that attach the segment shroud 4 [18] to the junction shroud fitting [21].

(h) Remove the five bolts [27] that attach the segment shroud 4 [18] to the retainer ring [10].

(i) Loosen the three screws [26] that attach the segment shroud 4 [18] to the strut 5 pad (left) [24].

(j) Loosen but do not remove the eight screws [78] from the VBV duct 5 [28] and VBV duct 6 [30].

(k) If the segment shroud 5 [1] is still in place, remove the three screws [23].

CAUTION MAKE SURE THAT YOU HOLD THE SEGMENT SHROUD WHILE YOU REMOVE THE LAST SCREWS. THE SHROUD CAN FALL AND DAMAGE CAN OCCUR.

(l) Remove the three screws [26].

(m) Remove the segment shroud 4 [18] with the VBV duct 5 [28], VBV 5 seal [29], VBV duct 6 [30] and VBV 6 seal [31] from the fan frame hub [12].

NOTE: The VBV 5 seal [29] and VBV 6 seal [31] can be orange or gray.

(n) Install the protective covers on the two VBV doors of the fan frame hub [12].

SUBTASK 72-23-04-020-003-G00

CAUTION IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(3) Remove the segment shroud 2 [32] as follows:

(a) Install the protector, SPL-14947.

(b) Install protective covers on the grids of segment shroud 2 [32].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(c) Remove the bolt [35] that attaches the clamp [34] to the bracket [33].

PTF ALL

(d) Remove the two screws [36] that attach the bracket [33] to the segment shroud 2 [32].

(e) Remove the two screws [38] that attach the bracket [37] to the segment shroud 2 [32].

(f) Remove the two screws [40] that attach the bracket [39] to the segment shroud 2 [32].

(g) If the segment shroud 1 [50] is still in place, remove the screw [41] that attaches the segment shroud 2 [32] to the junction shroud fitting [53].

(h) If the segment shroud 1 [50] is still in place, loosen the three screws [42] that attach the segment shroud 2 [32] to the junction shroud fitting [53].

(i) If the segment shroud 3 [62] is still in place, remove the screw [43] that attaches the segment shroud 2 [32] to the junction shroud fitting [68].

CAUTION

MAKE SURE THAT YOU HOLD THE SEGMENT SHROUD WHILE YOU REMOVE THE LAST SCREWS. THE SHROUD CAN FALL AND DAMAGE CAN OCCUR.

(j) If the segment shroud 3 [62] is still in place, remove the three screws [44] that attach the segment shroud 2 [32] to the junction shroud fitting [68].

(k) Remove the five bolts [45] that attach the segment shroud 2 [32] to the retainer ring [10].

(l) Loosen but do not remove the eight screws [79] from the VBV duct 2 [46] and VBV duct 3 [48].

(m) If the segment shroud 1 [50] is still in place, remove the three screws [42] that attach the segment shroud 2 [32] to the junction shroud fitting [53].

(n) Remove the segment shroud 2 [32] with the VBV duct 2 [46], VBV 2 seal [47], VBV duct 3 [48] and VBV 3 seal [49] from the fan frame hub [12].

NOTE: The VBV 2 seal [47] and VBV 3 seal [49] can be orange or gray.

(o) Install the protective covers on the two VBV doors of the fan frame hub [12].

SUBTASK 72-23-04-020-004-G00

CAUTION

IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(4) Remove the segment shroud 1 [50] as follows:

(a) Install the protector, SPL-14947.

(b) Install protective covers on the grids of segment shroud 1 [50].

(c) Remove the two screws [52] that attach the bracket [51] to the segment shroud 1 [50].

(d) If the segment shroud 2 [32] is still in place, remove the screw [54] that attaches the segment shroud 1 [50] to the junction shroud fitting [53].

(e) If the segment shroud 2 [32] is still in place, remove the three screws [55] that attach the segment shroud 1 [50] to the junction shroud fitting [53].

(f) Remove the screw [57] that attaches the segment shroud 1 [50] to the strut 1 pad (right) [56].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Loosen the three screws [58] that attach the segment shroud 1 [50] to the strut 1 pad (right) [56].

(h) Remove the three bolts [59] that attach the segment shroud 1 [50] to the retainer ring [10].

(i) Loosen but do not remove the four screws [80] from the VBV duct 1 [60].

CAUTION MAKE SURE THAT YOU HOLD THE SEGMENT SHROUD WHILE YOU REMOVE THE LAST SCREWS. THE SHROUD CAN FALL AND DAMAGE CAN OCCUR.

(j) Remove the three screws [58] that attache the segment shroud 1 [50] to the strut 1 pad (right) [56].

(k) Remove the segment shroud 1 [50] with the VBV duct 1 [60] and VBV 1 seal [61] from the fan frame hub [12].

NOTE: The VBV 1 seal [61] can be orange or gray.

(l) Install the protective covers on the VBV door of the fan frame hub [12].

SUBTASK 72-23-04-020-005-G00

CAUTION IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(5) Remove the segment shroud 3 [62] as follows:

(a) Install the protector, SPL-14947.

(b) Loosen the screw [65] from the clamp [64] that attaches the flex duct [63] to the air duct [91].

(c) Disconnect the flex duct [63] from the air duct [91].

(d) Remove the two screws [67] that attach the bracket [66] to the segment shroud 3 [62].

(e) If the segment shroud 2 [32] is still in place, remove the screw [69] that attaches the segment shroud 3 [62] to the junction shroud fitting [68].

(f) If the segment shroud 2 [32] is still in place, remove the three screws [70] that attach the segment shroud 3 [62] to the junction shroud fitting [68].

(g) Remove the screw [72] that attaches the segment shroud 3 [62] to the strut 5 pad (right) [71].

CAUTION MAKE SURE THAT YOU HOLD THE SEGMENT SHROUD WHILE YOU REMOVE THE LAST SCREWS. THE SHROUD CAN FALL AND DAMAGE CAN OCCUR.

(h) Remove the three screws [73] that attach the segment shroud 3 [62] to the strut 5 pad (right) [71].

(i) Remove the four bolts [74] that attach the segment shroud 3 [62] to the retainer ring [10].

(j) Loosen but do not remove the four screws [81] from the VBV duct 4 [75].

(k) Remove the segment shroud 3 [62] with the VBV duct 4 [75], VBV 4 seal [76] and flex duct [63] from the fan frame hub [12].

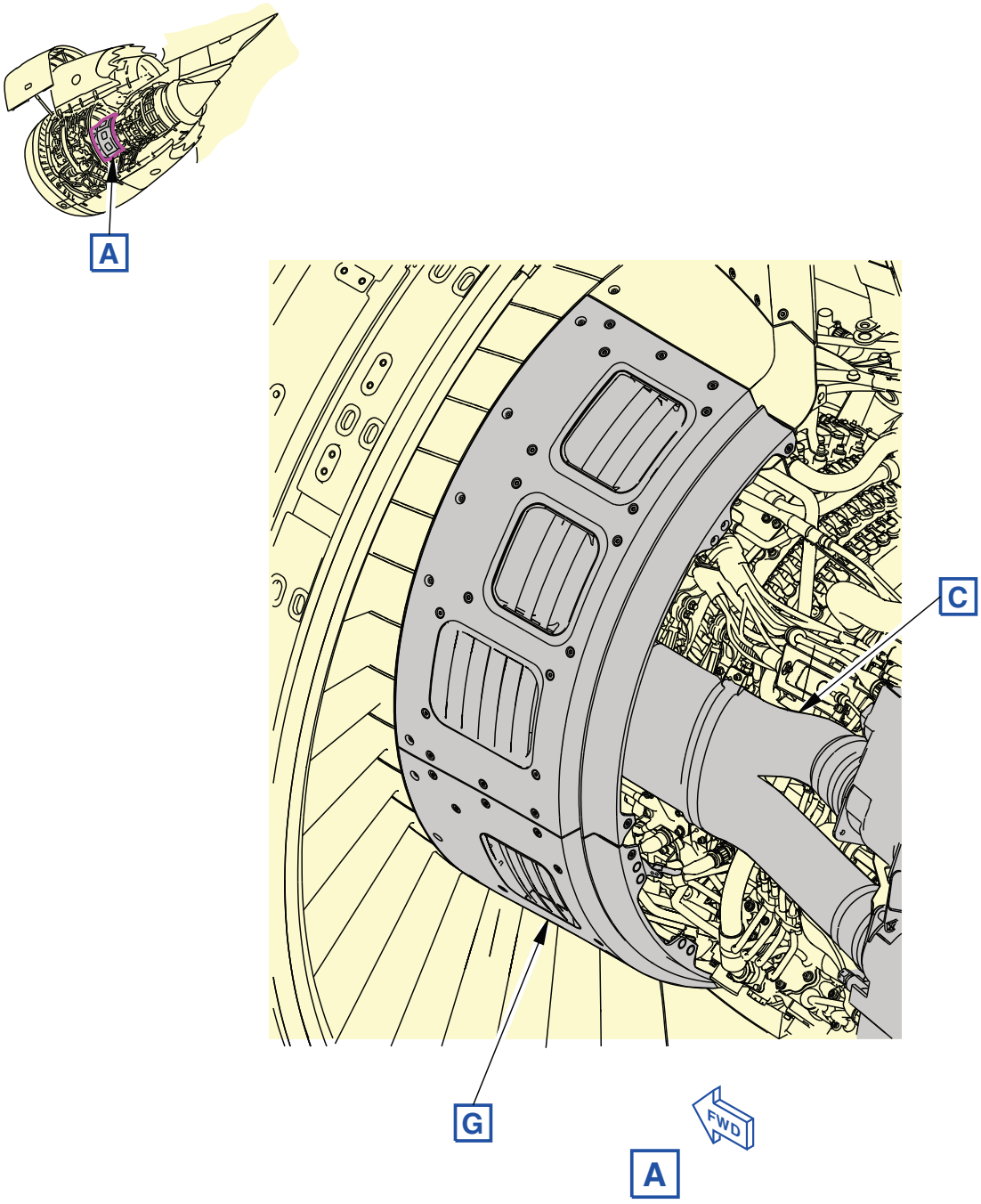
NOTE: The VBV 4 seal [76] can be orange or gray.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Install the protective covers on the VBV door of the fan frame hub [12] and the air duct [91].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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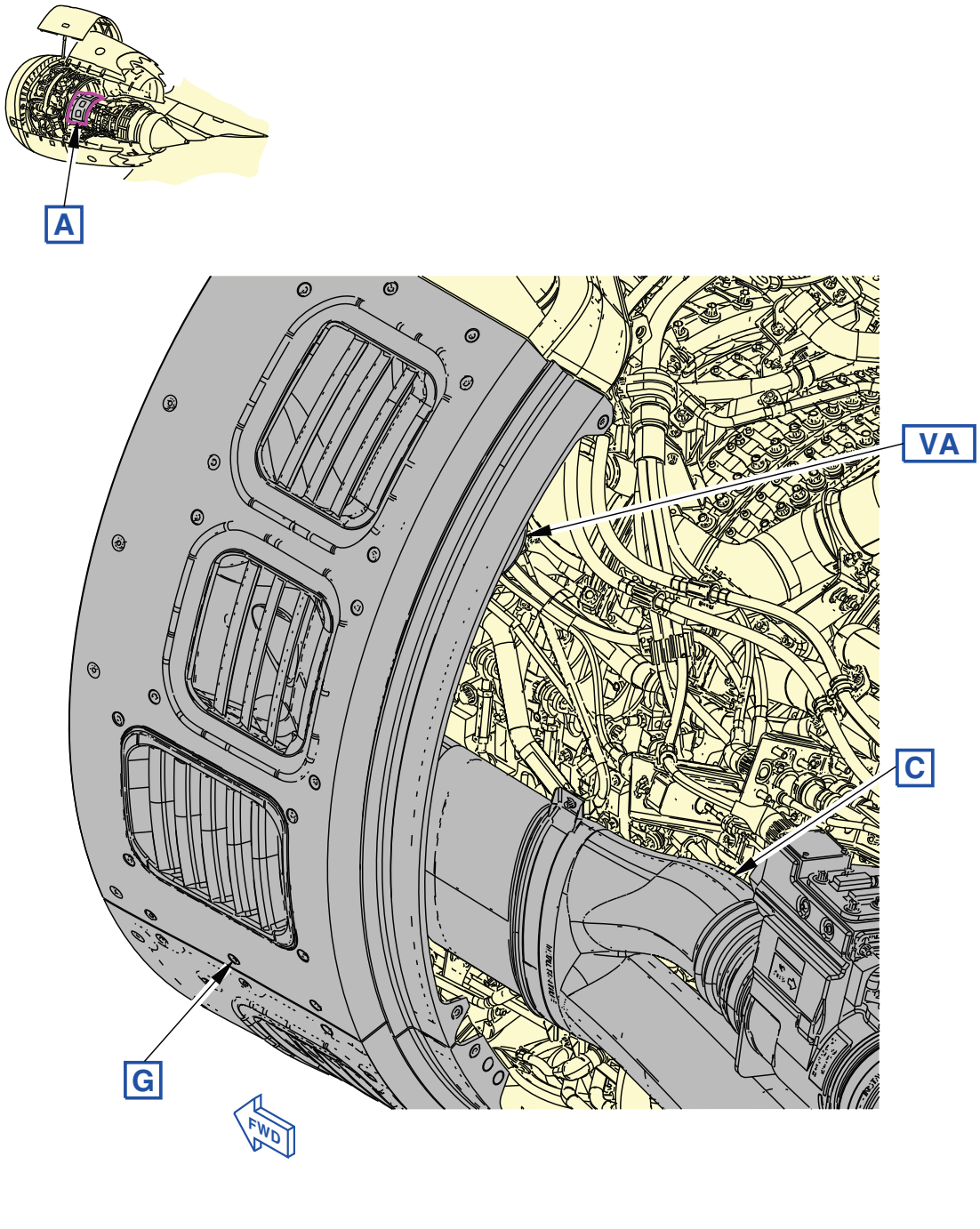
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 1 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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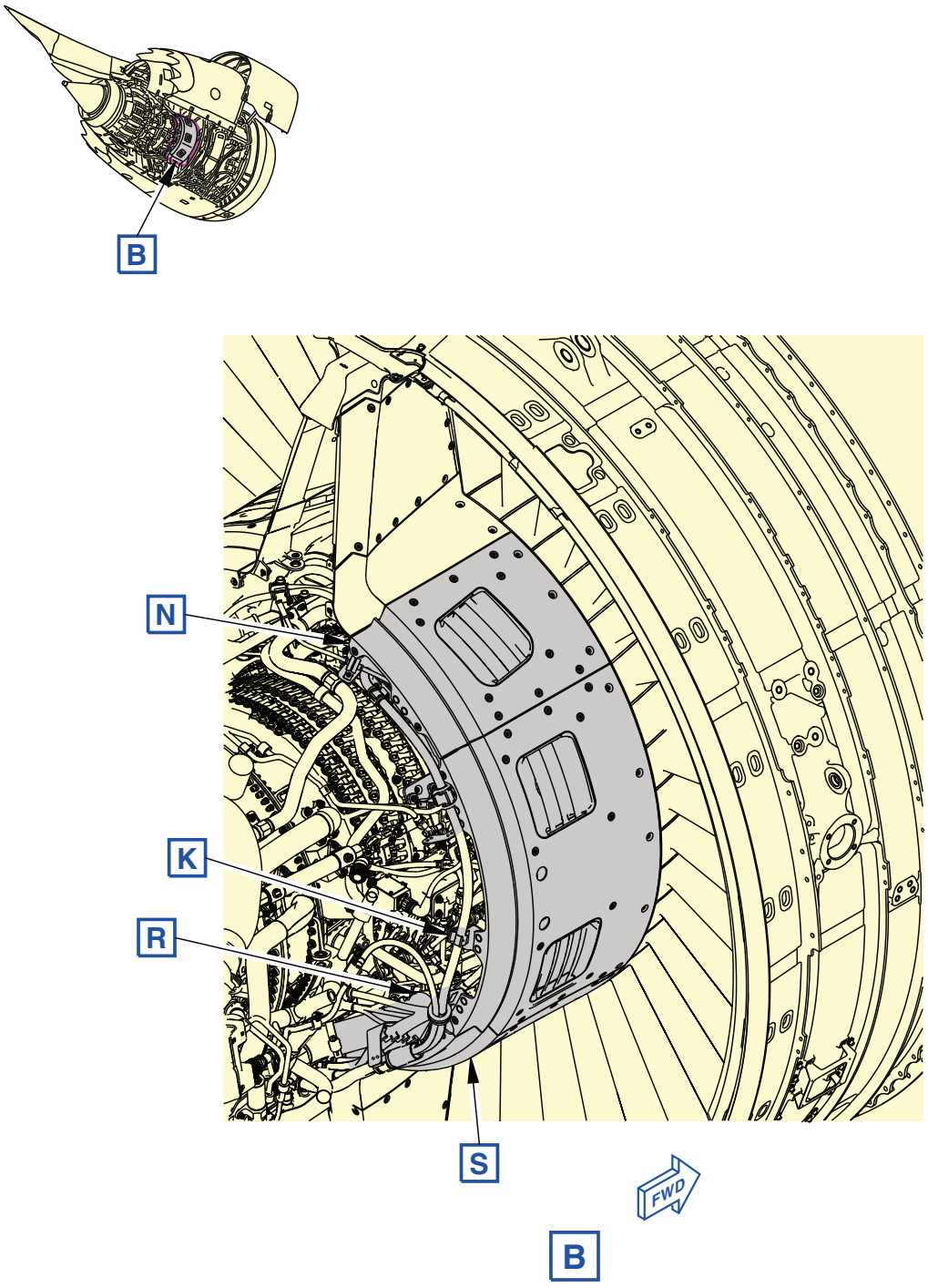
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 2 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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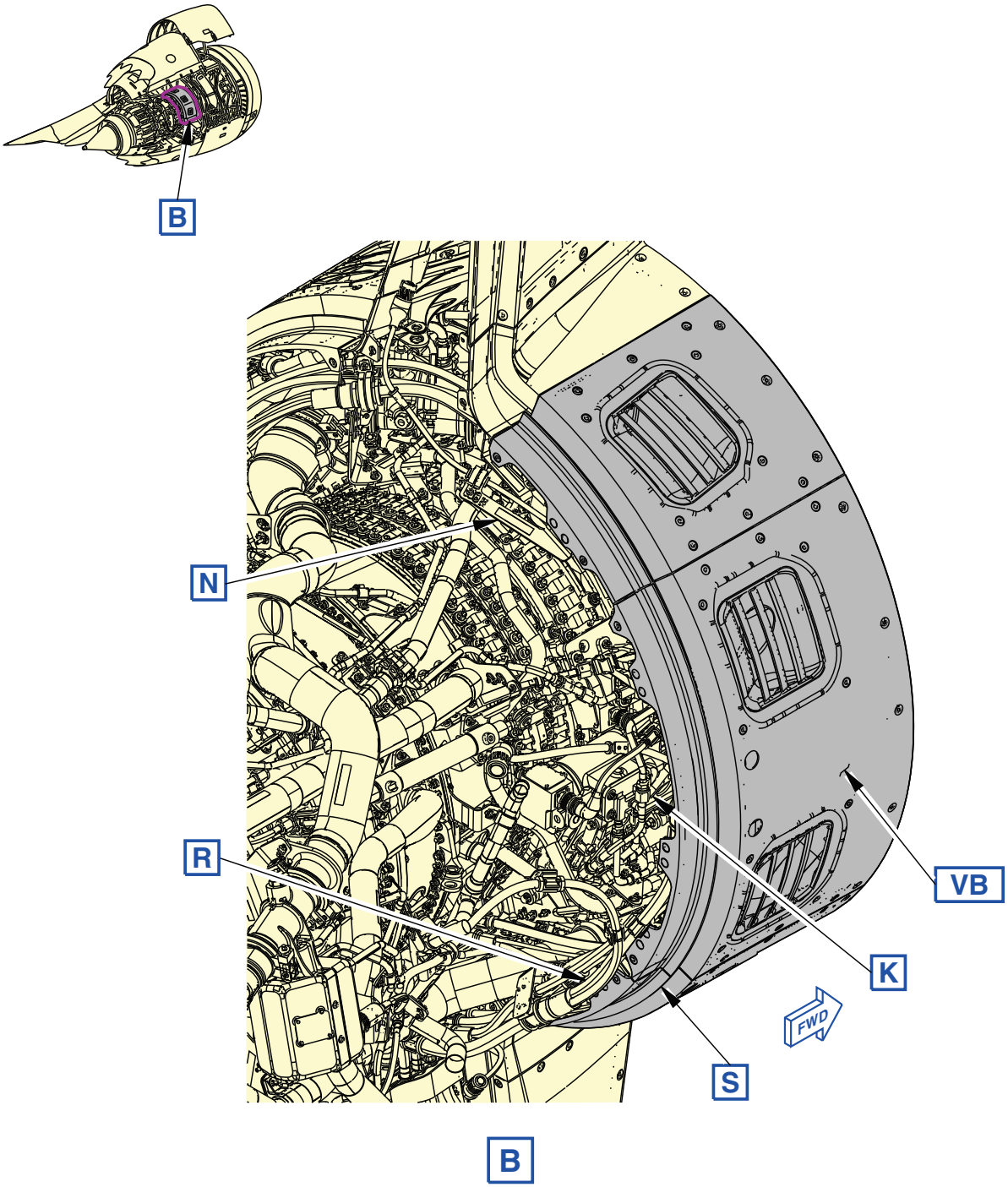
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 3 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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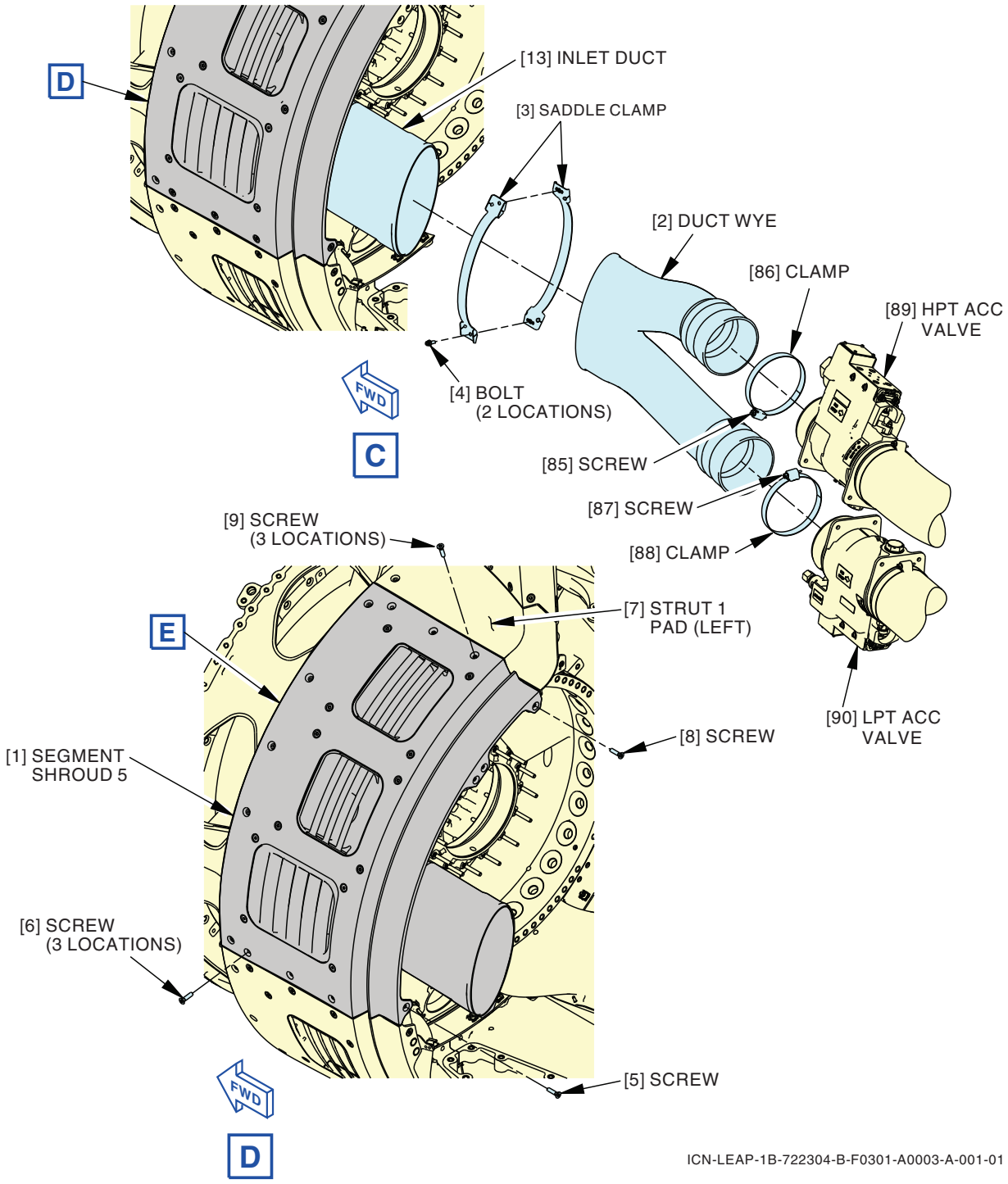
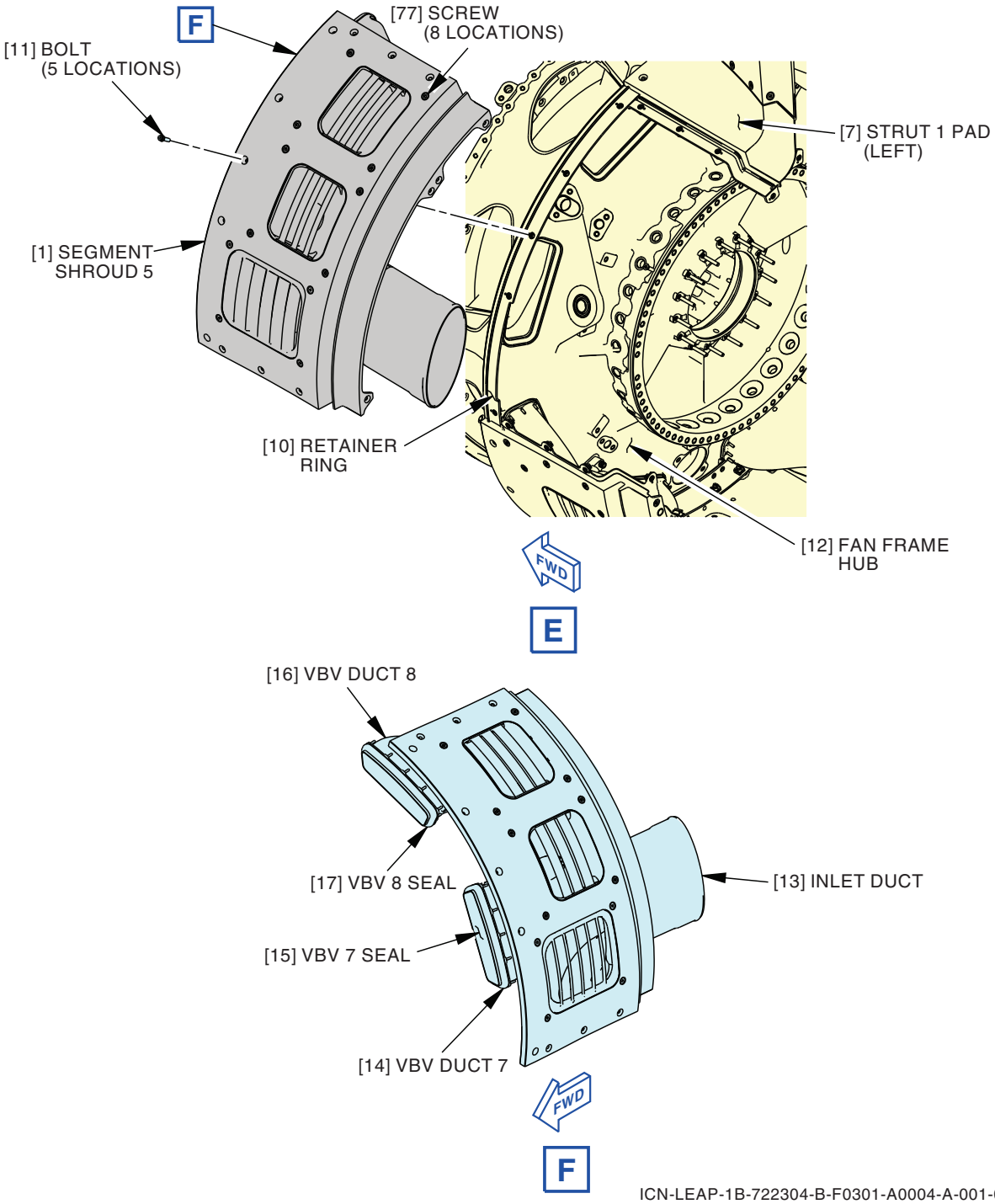
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 4 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



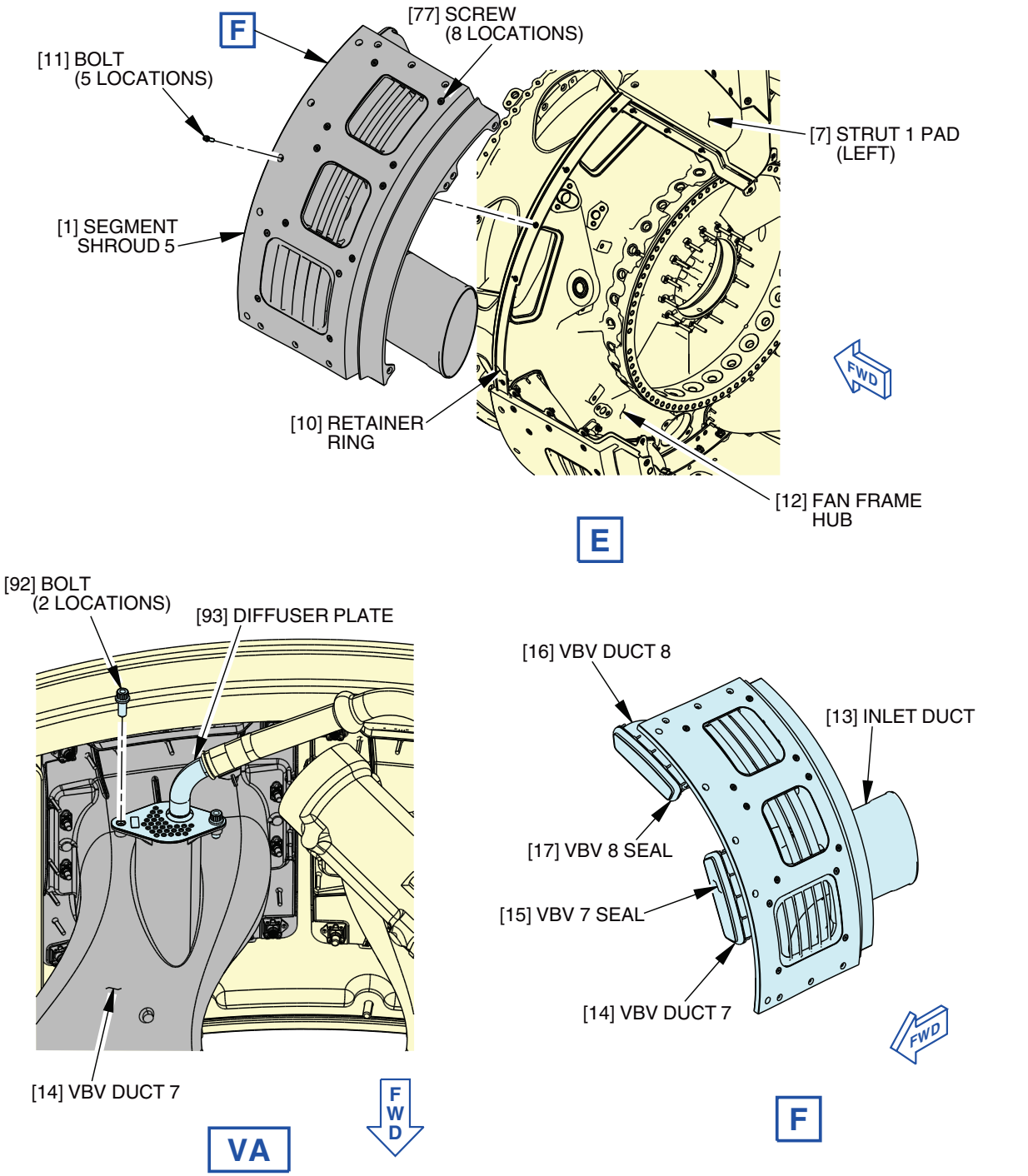
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 5 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 6 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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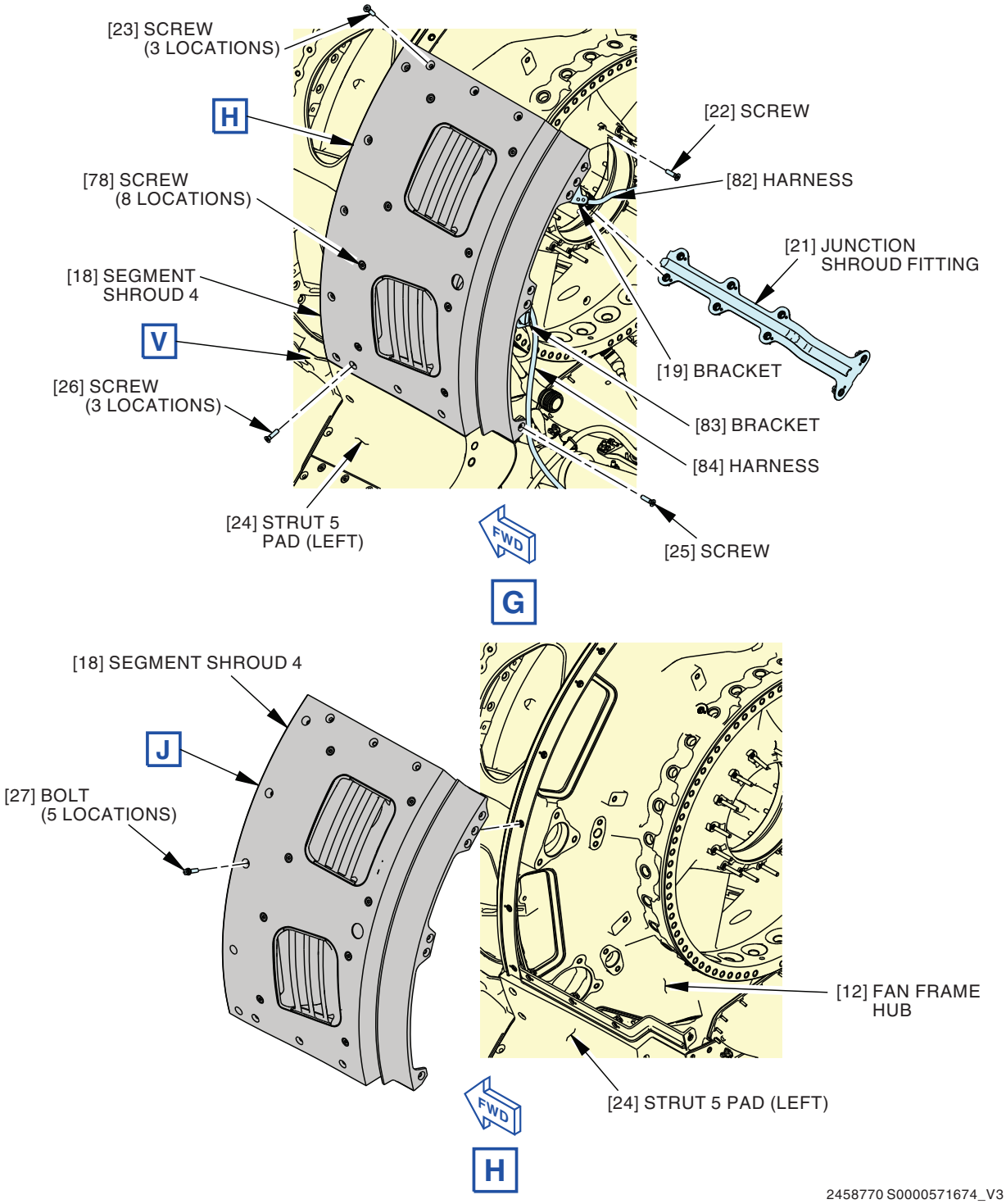
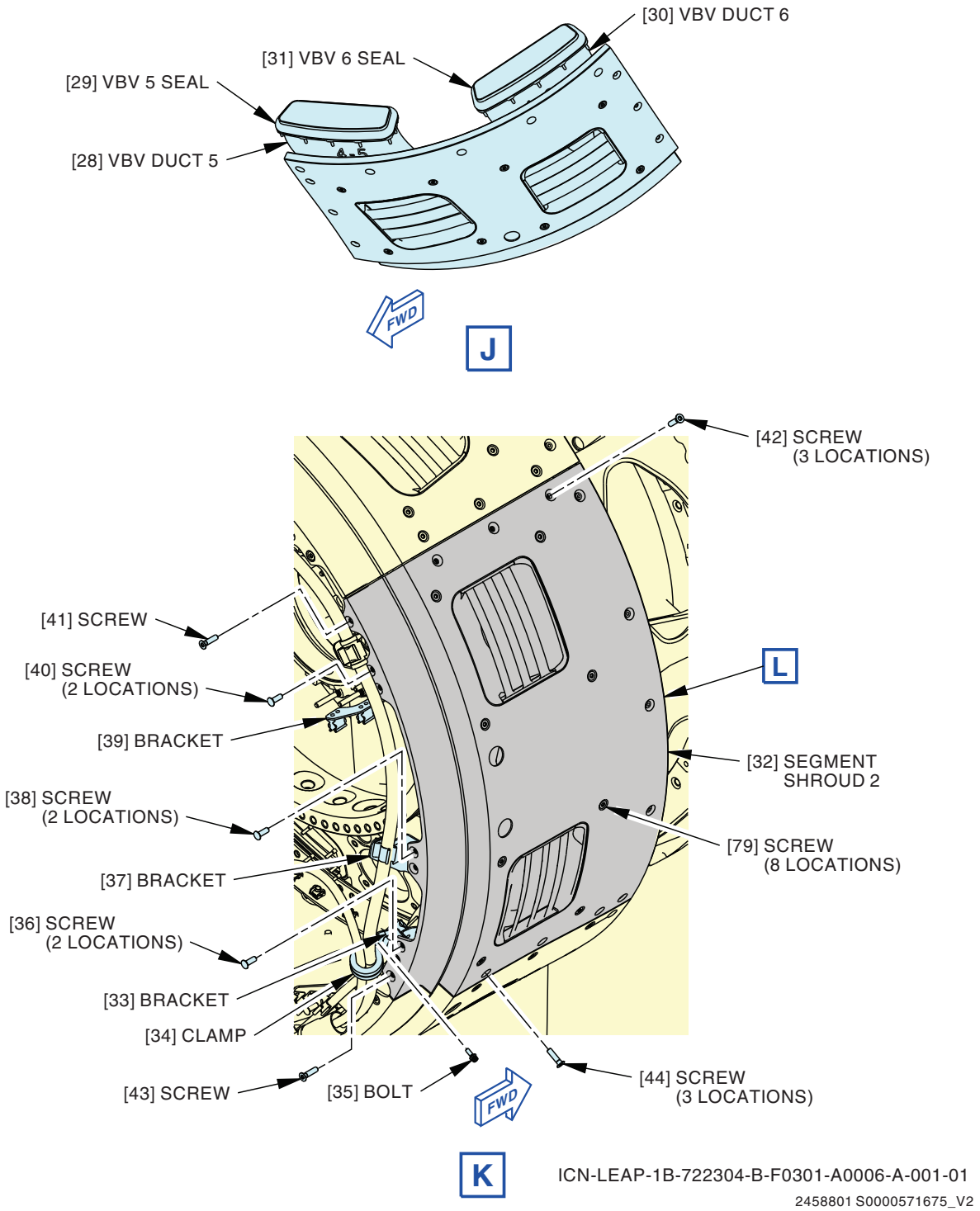
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 7 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



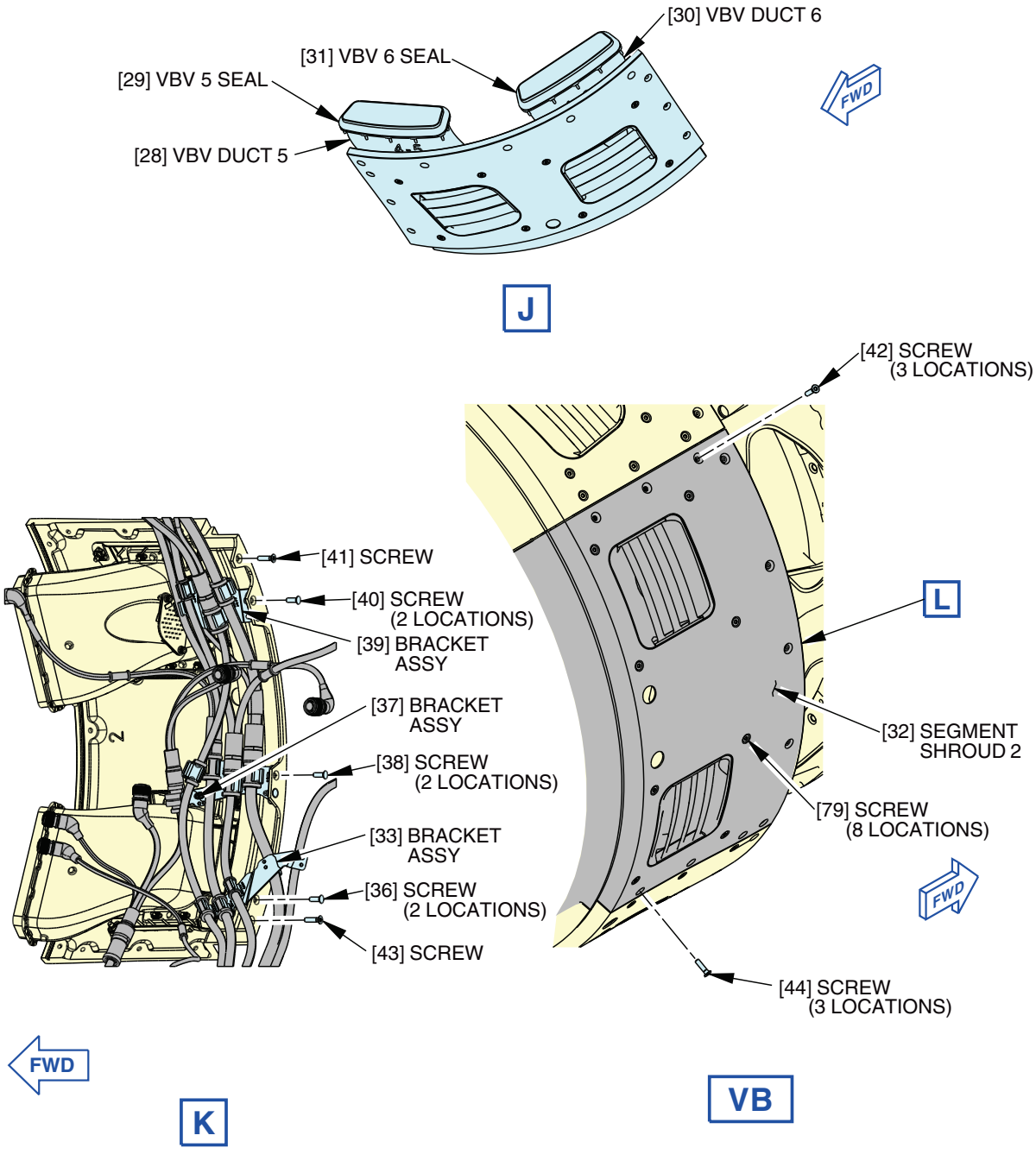
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 8 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 9 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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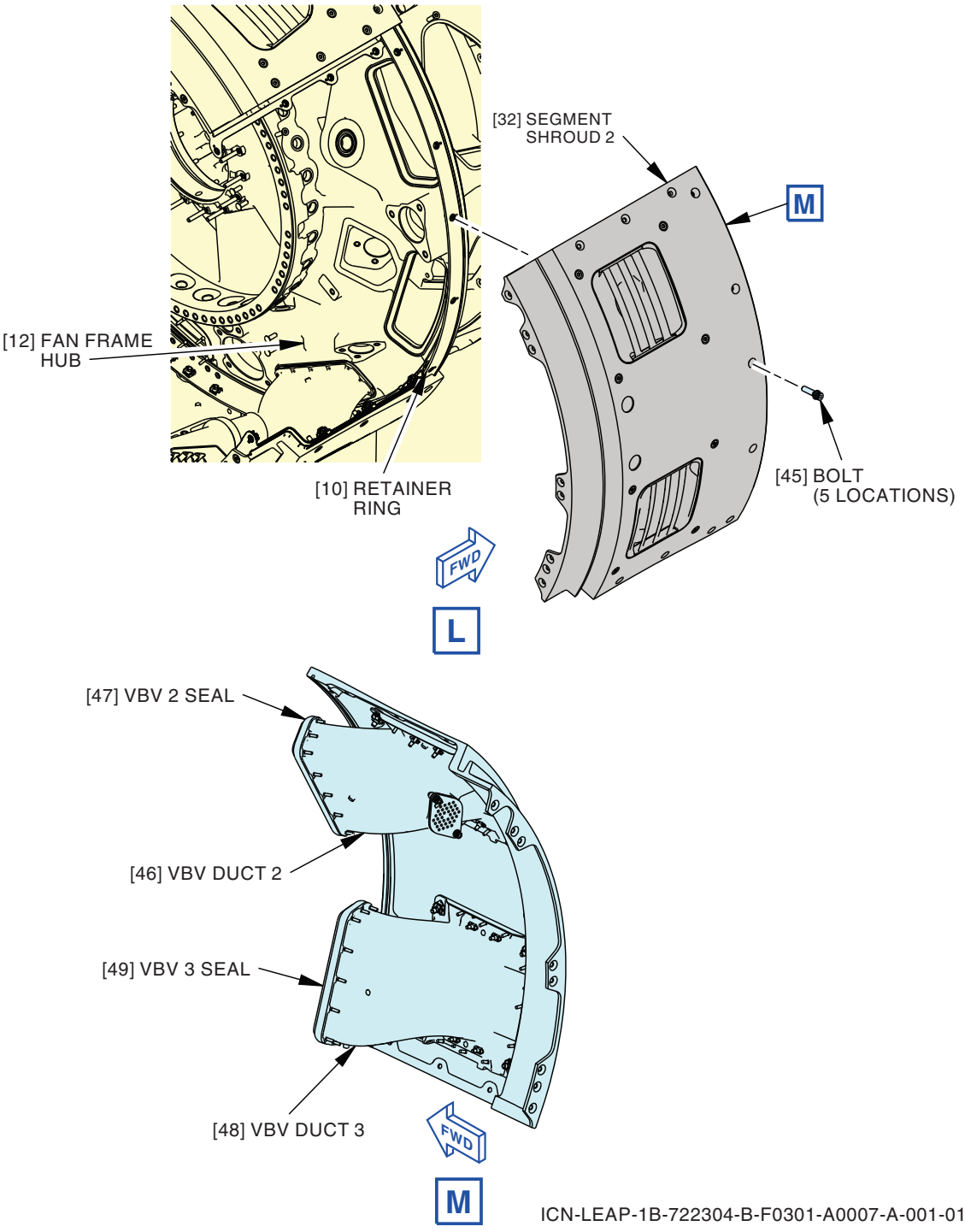
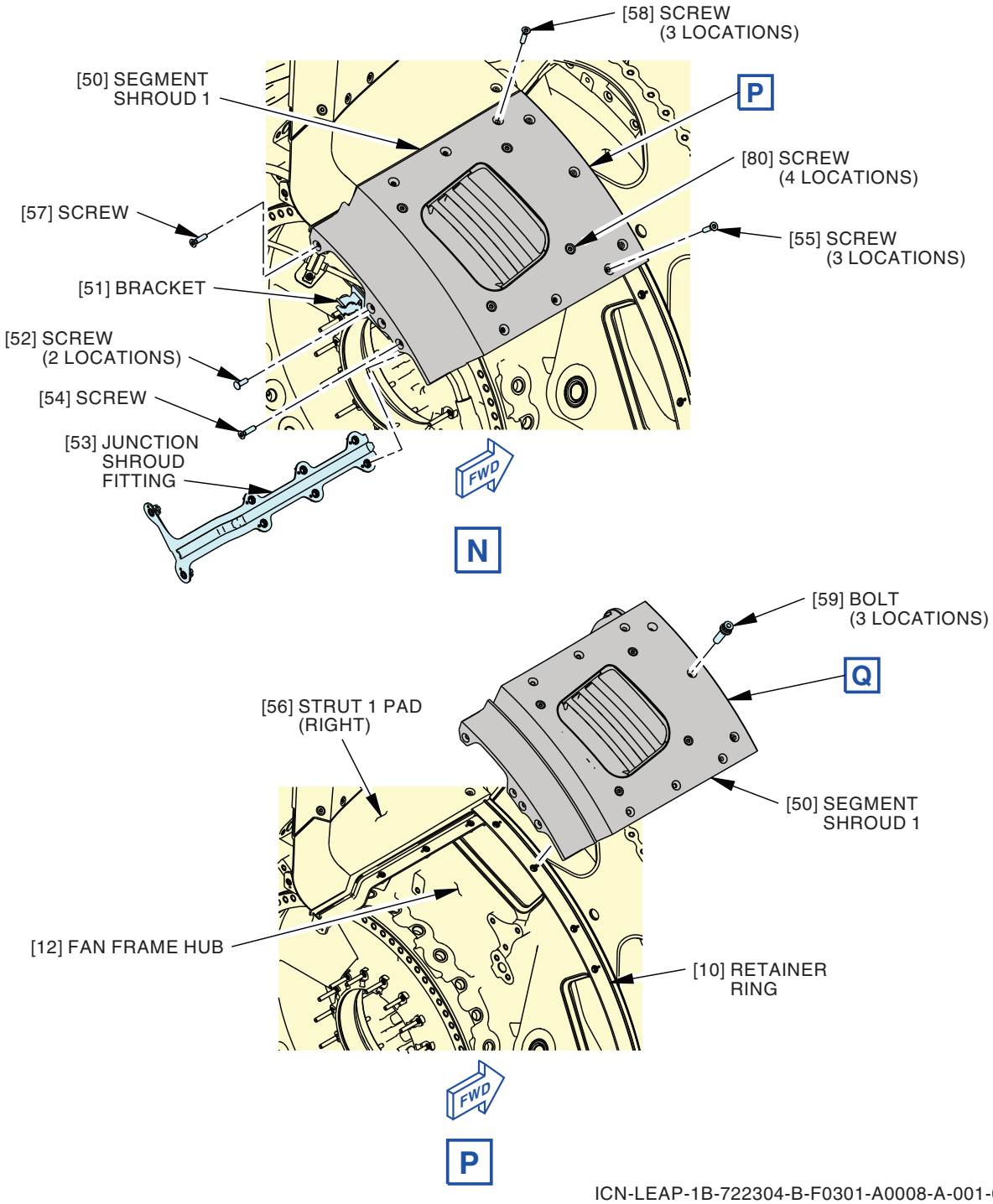
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 10 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 11 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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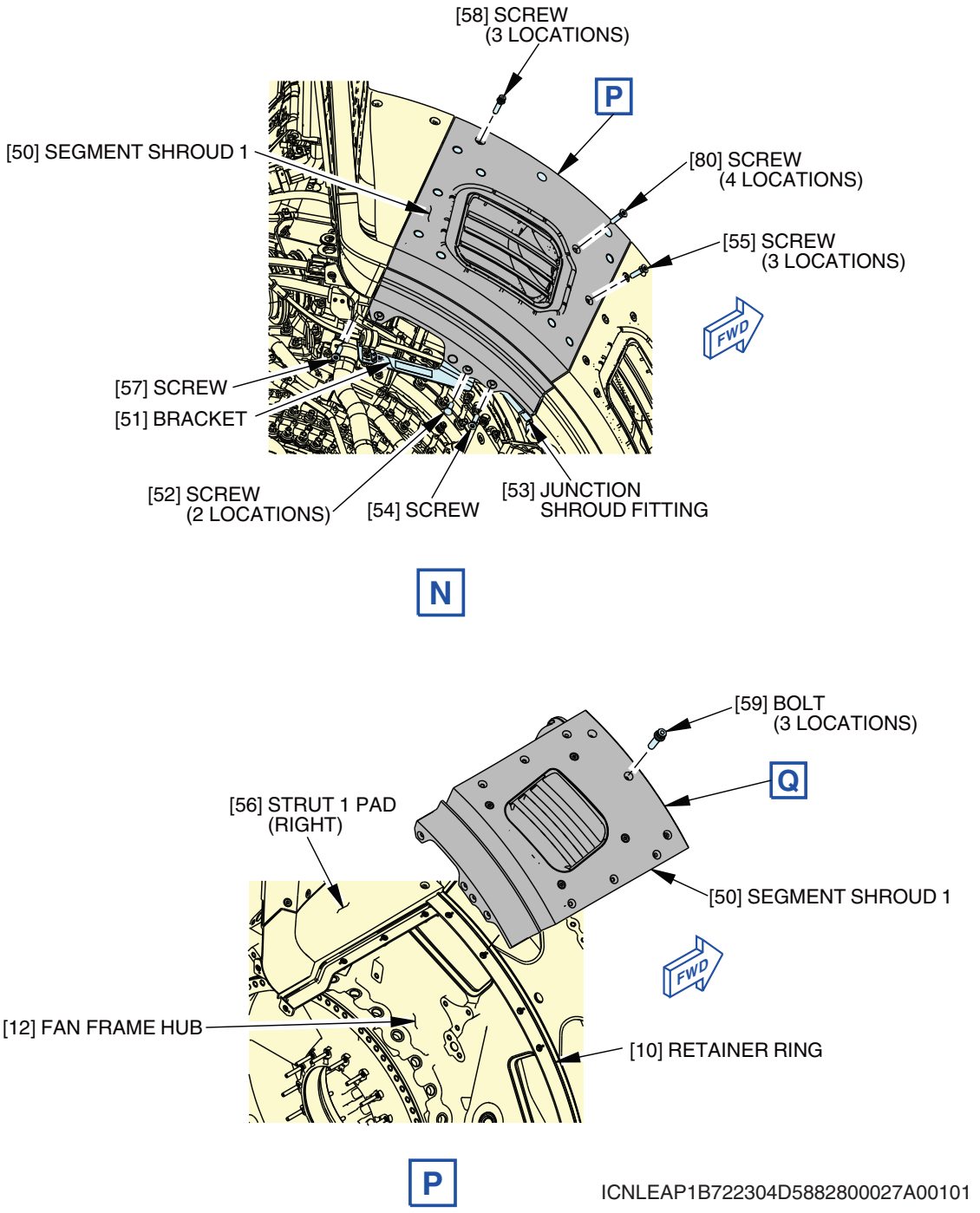
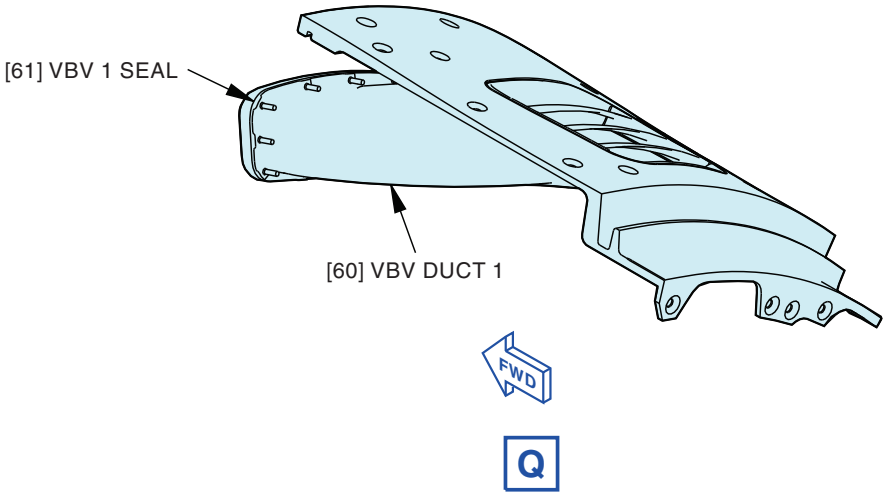
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 12 of 18)

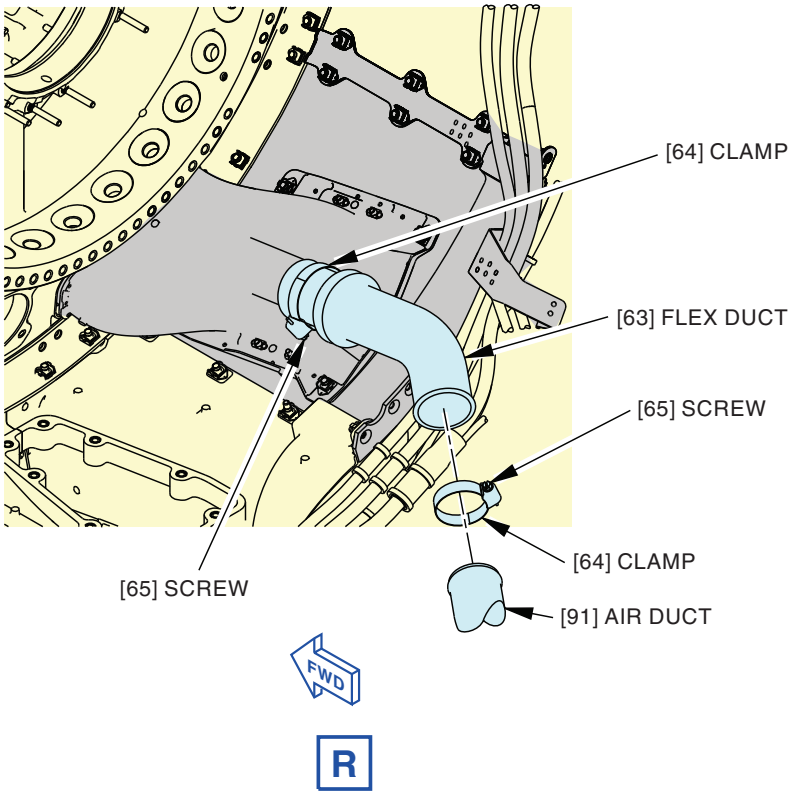
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 13 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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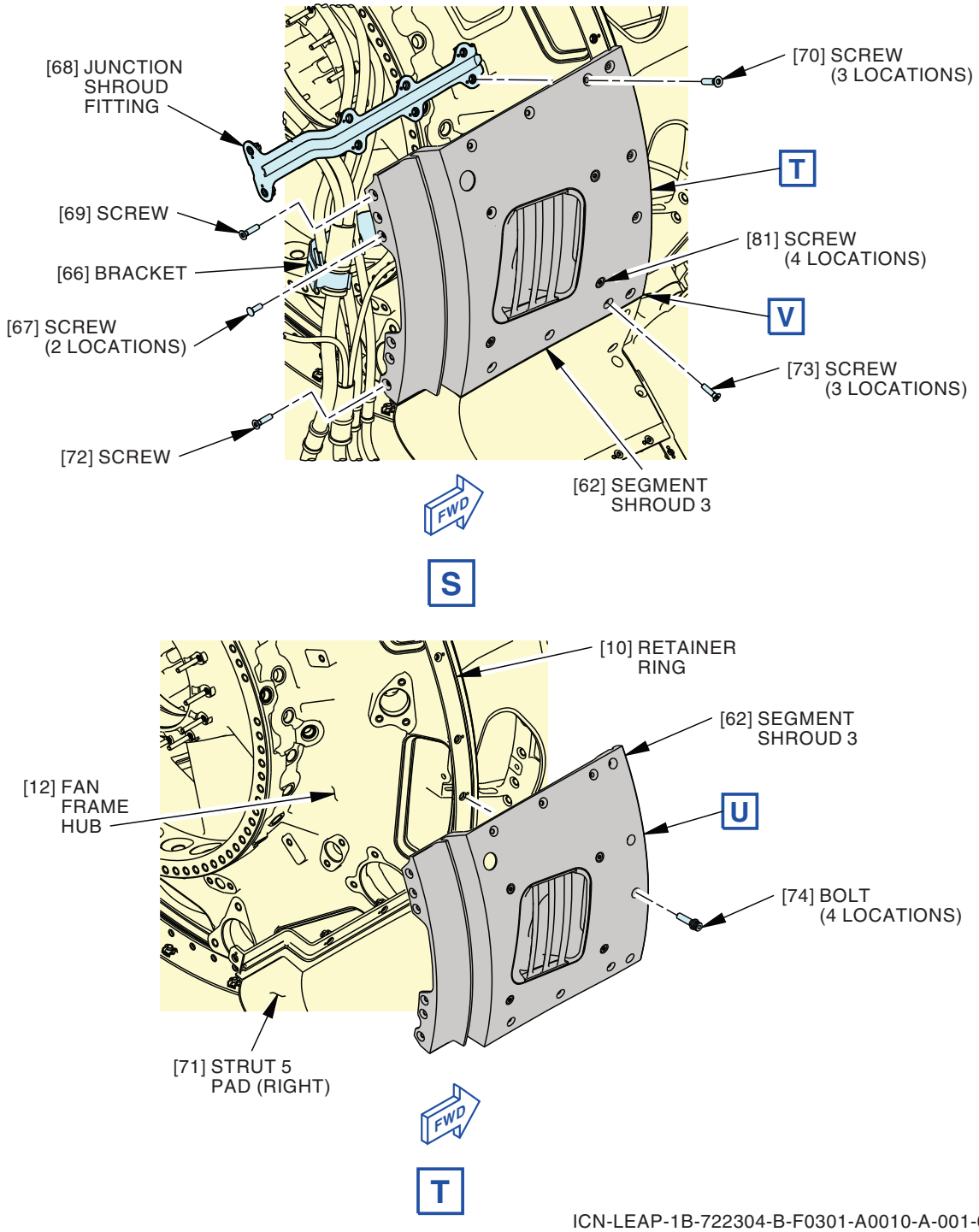
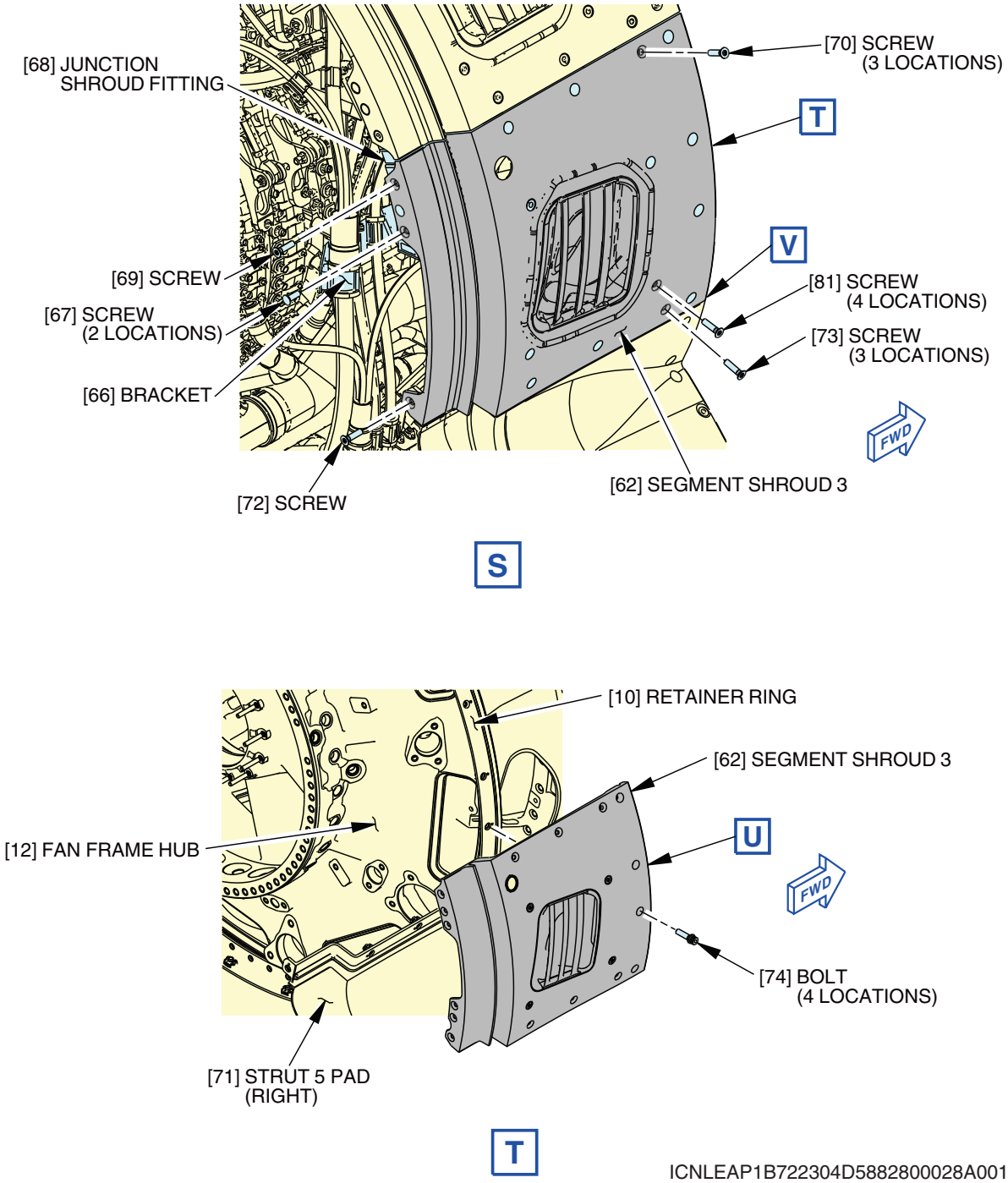
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 14 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



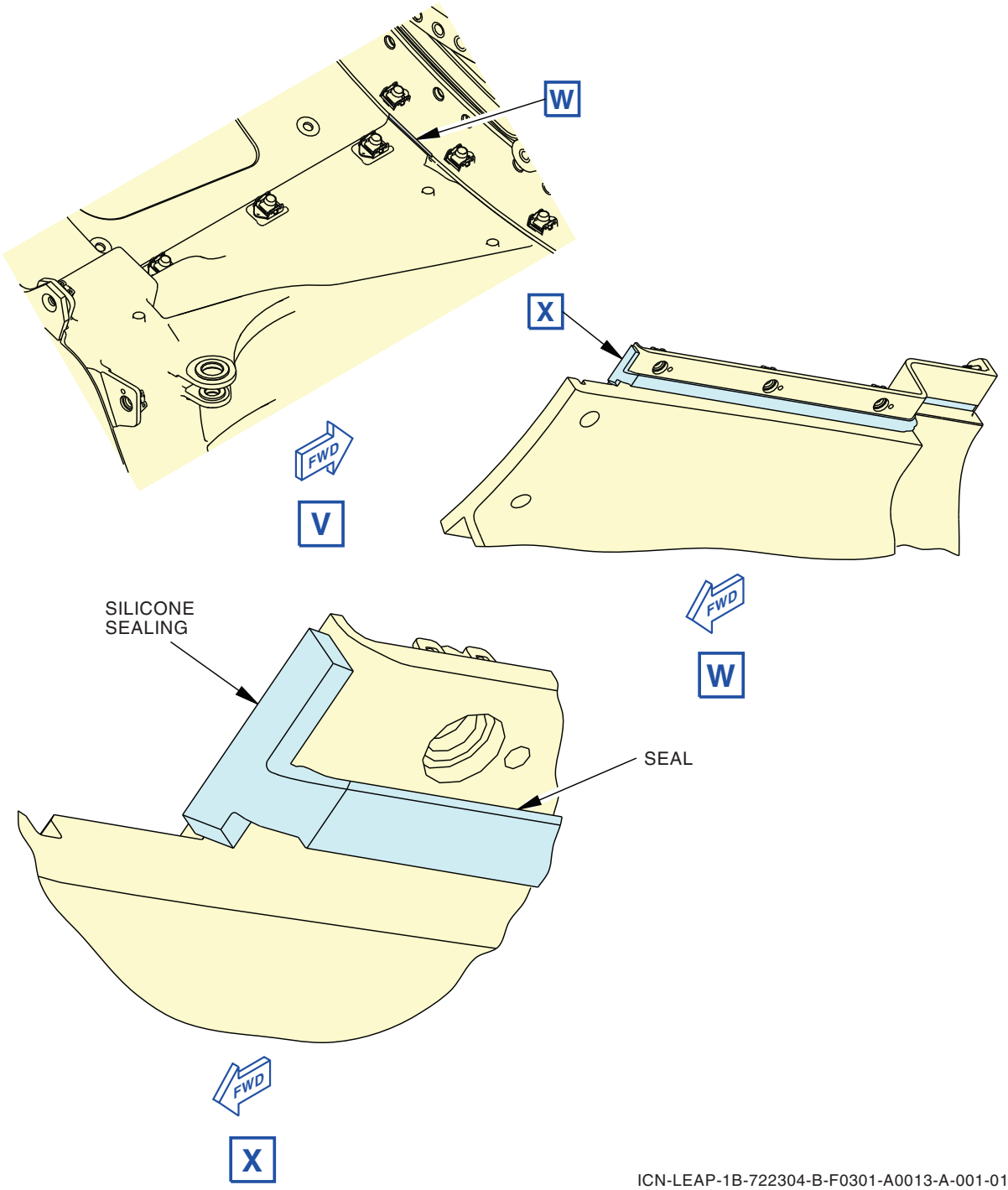
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 15 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 16 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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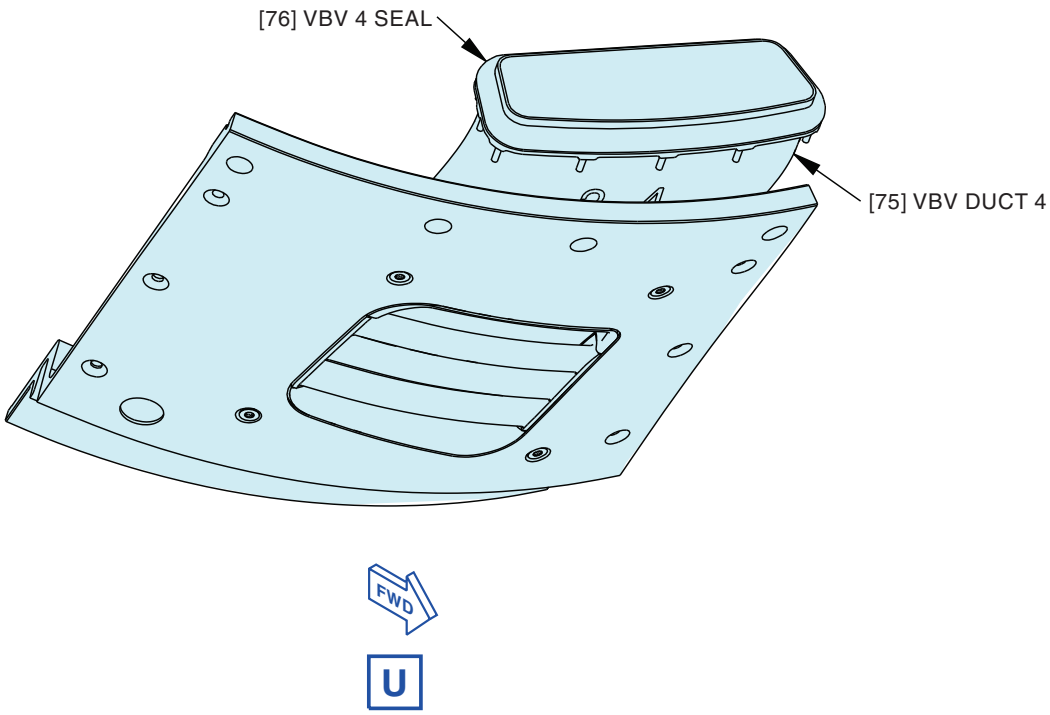
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 17 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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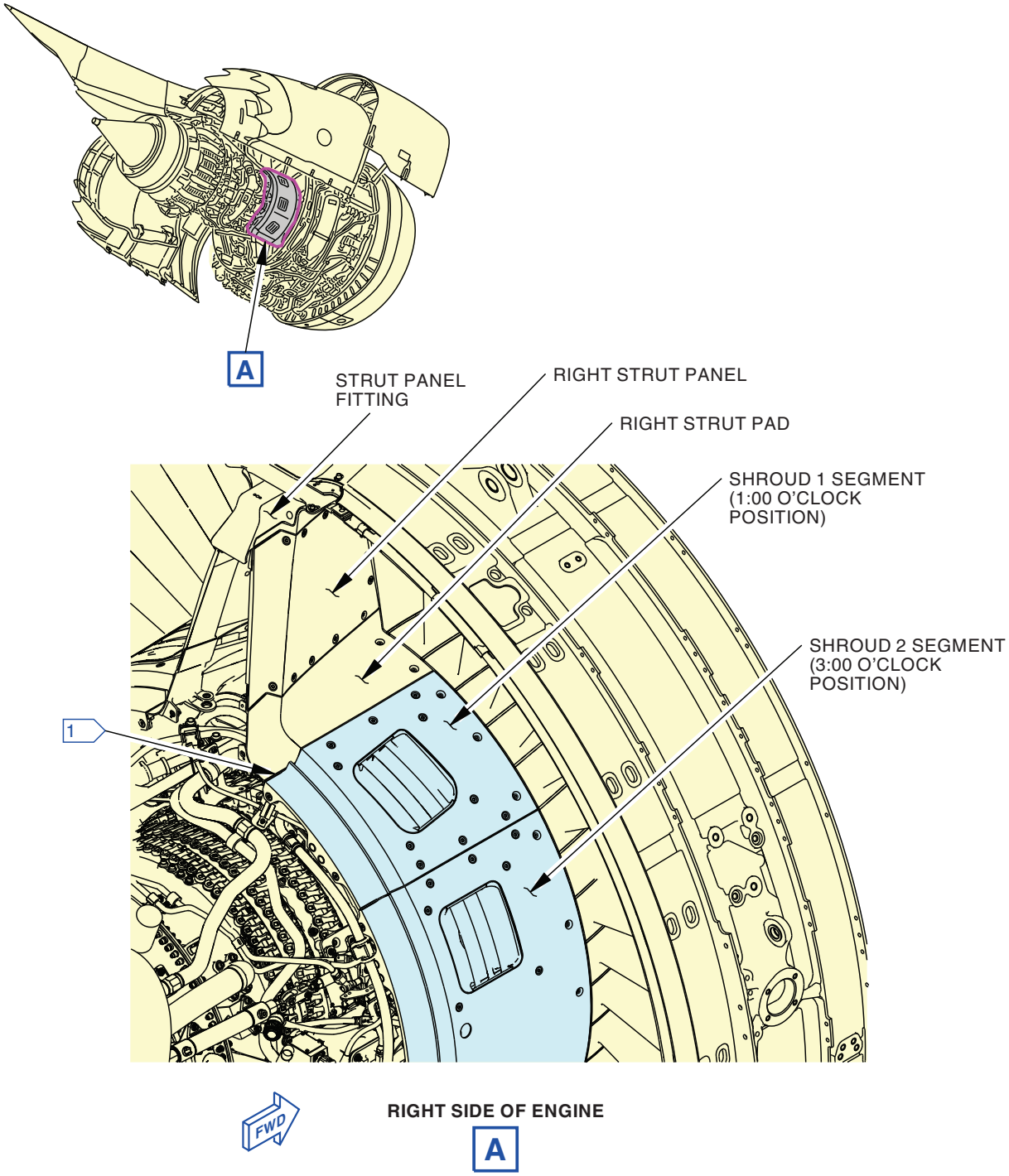
Fan Frame Module Segment Shrouds Installation
Figure 401/72-23-04-990-801-G00 (Sheet 18 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Panels and Shroud Segments Firewall Sealant Installation
Figure 402/72-23-04-990-802-G00 (Sheet 1 of 2)

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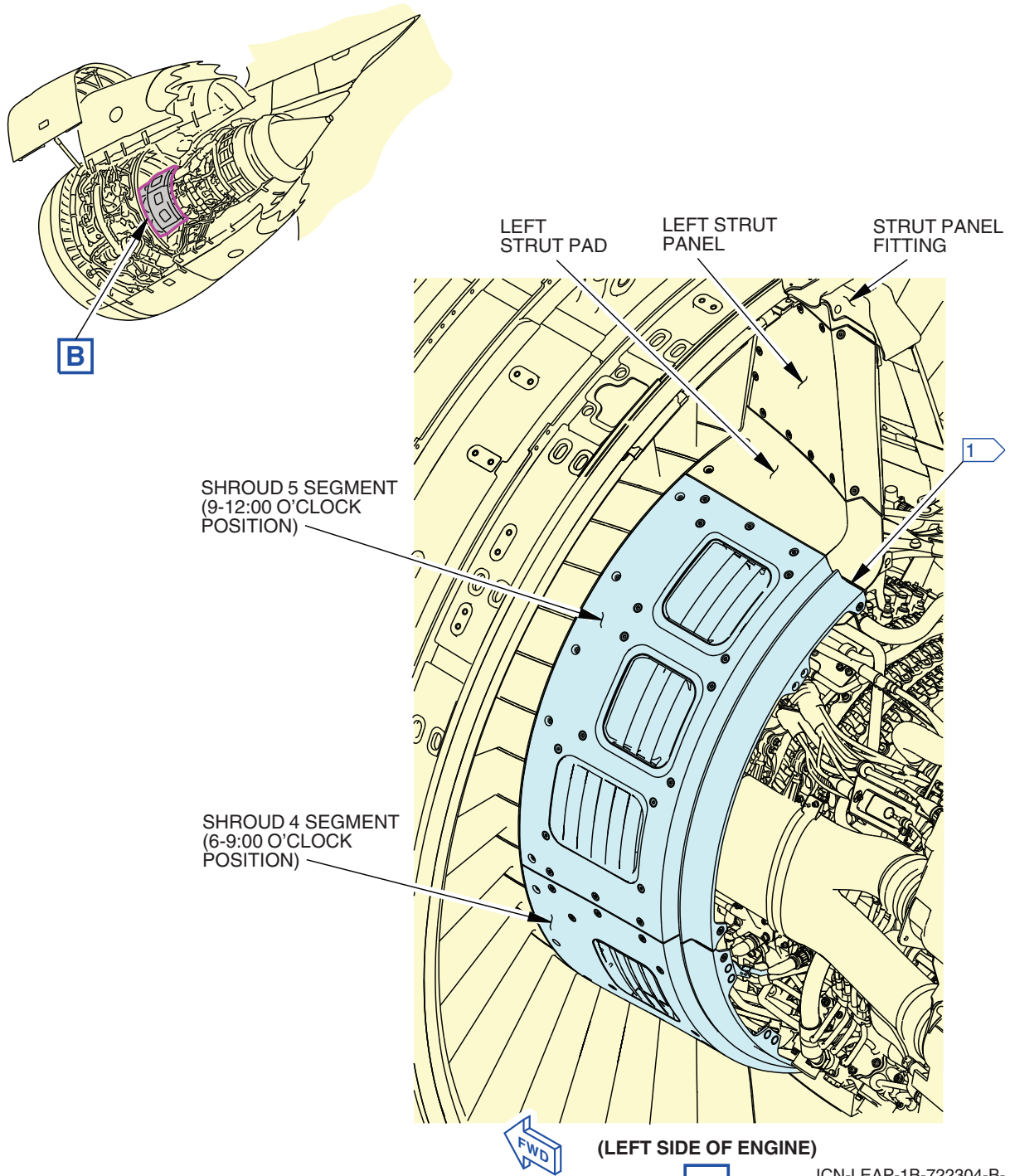

Strut Panels and Shroud Segments Firewall Sealant Installation
Figure 402/72-23-04-990-802-G00 (Sheet 2 of 2)

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TASK 72-23-04-400-801-G00

3. Fan Frame Module Segment Shrouds Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the segment shrouds for the fan frame module.

NOTE: The VBV grid seals located under the VBV grids can be red or blue.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G51328 [CP2674]	Cloth - Lint-Free	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Segment shroud 5	72-00-00-08-340	PTF ALL
15	VBV 7 seal	72-00-00-08-220	PTF ALL
17	VBV 8 seal	72-00-00-08-150	PTF ALL
18	Segment shroud 4	72-00-00-08-650	PTF ALL
29	VBV 5 seal	72-00-00-08-580	PTF ALL
31	VBV 6 seal	72-00-00-08-150	PTF ALL
		72-00-00-08-510	PTF ALL
32	Segment shroud 2	72-00-00-07-380	PTF ALL
47	VBV 2 seal	72-00-00-07-270	PTF ALL
49	VBV 3 seal	72-00-00-07-340	PTF ALL
50	Segment shroud 1	72-00-00-07-860	PTF ALL
61	VBV 1 seal	72-00-00-07-790	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
62	Segment shroud 3	72-00-00-07-570	PTF ALL
76	VBV 4 seal	72-00-00-07-630	PTF ALL

F. Location Zones

Zone	Area
434	Engine 1 - Aft Strut Fairing
444	Engine 2 - Aft Strut Fairing

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Module Segment Shrouds Installation

SUBTASK 72-23-04-210-001-G00

CAUTION IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

CAUTION TIGHTEN THE SEGMENT SHROUDS OF THE ENGINE FAN FRAME MODULE AT THE SAME TIME IF YOU REMOVED THE ADJACENT SEGMENT SHROUDS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Make sure that the mating faces are clean and clear of unwanted materials.

SUBTASK 72-23-04-420-001-G00

CAUTION IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(2) Install the segment shroud 3 [62].

(a) Remove the protective covers from the air duct [91] and fan frame hub [12].

(b) Lubricate the threads of the screws [67] with engine oil, D00599 [CP2442].

(c) If the segment shroud 2 [32] is still in place, lubricate the threads and pressure faces of the screw [69] and screws [70] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the threads and pressure faces of the screw [72], screws [73], and bolts [74] with graphite mineral vaseline, D00601 [CP2101].

(e) If it is necessary, lubricate the threads and pressure faces of the screws [81] with graphite mineral vaseline, D00601 [CP2101].

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(f) If the sealant is partially or totally missing, apply Dapco 2100 sealant, A50483 [CP2927], to these areas (Views X and W, Figure 401):

1) The forward edge of the strut 5 pad (right).

a) Apply firewall sealant at the strut panel.

2) The segment shroud 3 [62].

a) Apply firewall sealant at the shroud segment edge.

(g) Install the segment shroud 3 [62] with the VBV duct 4 [75], VBV 4 seal [76], and flex duct [63] on the fan frame hub [12].

NOTE: The VBV 4 seal can be orange or gray.

(h) Install the bolts [74] that attach the segment shroud 3 [62] to the retainer ring [10].

(i) Install the screws [73] that attach the segment shroud 3 [62] to the strut 5 pad (right) [71].

(j) Install the screw [72] that attaches the segment shroud 3 [62] to the strut 5 pad (right) [71].

(k) If the segment shroud 2 [32] is in place, install the screws [70] that attach the segment shroud 3 [62] to the junction shroud fitting [68].

(l) If the segment shroud 2 [32] is in place, install the screw [69] that attaches the segment shroud 3 [62] to the junction shroud fitting [68].

(m) Tighten the screws [70] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(n) Tighten the screws [73] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(o) Tighten the screw [69] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(p) Tighten the screw [72] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(q) Tighten the bolts [74] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(r) Tighten the screws [81] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(s) Install the screws [67] that attach the bracket [66] to the segment shroud 3 [62].

(t) Tighten the screws [67] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(u) Install the clamp [64] on the flex duct [63].

(v) Install the flex duct [63] to the air duct [91].

(w) Tighten the screw [65] that attaches the flex duct [63] to the air duct [91].

(x) Tighten the screw [65] to 33.63 in-lb (3.80 N·m) - 37.17 in-lb (4.20 N·m).

(y) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], on screw [69], screws [70], screws [72], screws [73], screws [81], and on bolts [74] with a lint-free cloth, G51328 [CP2674].

SUBTASK 72-23-04-420-002-G00

CAUTION

IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(3) Install the segment shroud 1 [50].

(a) Remove the protective covers from the fan frame hub [12].

(b) If the segment shroud 2 [32] is in place, lubricate the threads and pressure faces of the screw [54] and screws [55] with graphite mineral vaseline, D00601 [CP2101].

(c) Lubricate the threads of the screws [52] with engine oil, D00599 [CP2442].

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(d) Lubricate the threads and pressure faces of the screw [57], screws [58], and bolts [59] with graphite mineral vaseline, D00601 [CP2101].

(e) If it is necessary, lubricate the threads and pressure faces of the screws [80] with graphite mineral vaseline, D00601 [CP2101].

(f) Install the segment shroud 1 [50] with the VBV duct 1 [60] and VBV 1 seal [61] on the fan frame hub [12].

NOTE: The VBV 1 seal can be orange or gray.

(g) Install the bolts [59] that attach the segment shroud 1 [50] to the retainer ring [10].

(h) Install the screws [58] that attach the segment shroud 1 [50] to the strut 1 pad (right) [56].

(i) Install the screw [57] that attaches the segment shroud 1 [50] to the strut 1 pad (right) [56].

(j) If the segment shroud 2 [32] is in place, install the screws [55] that attach the segment shroud 1 [50] to the junction shroud fitting [53].

(k) If the segment shroud 2 [32] is in place, install the screw [54] that attaches the segment shroud 1 [50] to the junction shroud fitting [53].

(l) Tighten the screws [58] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(m) Tighten the screws [55] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(n) Tighten the screw [57] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(o) Tighten the screw [54] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(p) Tighten the bolts [59] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(q) Tighten the screws [80] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(r) Install the screws [52] that attach the bracket [51] to the segment shroud 1 [50].

(s) Tighten the screws [52] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(t) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], on screw [54], screws [55], screw [57], screws [58], screws [80], and on bolts [59] with a lint-free cloth, G51328 [CP2674].

SUBTASK 72-23-04-420-003-G00

CAUTION

IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(4) Install the segment shroud 2 [32].

NOTE: You must install the shroud 1 segment before you install the shroud 2 segment.

(a) Remove the protective covers from the fan frame hub [12].

(b) Lubricate the threads of the bolt [35], screws [36], screws [38], and screws [40] with oil, D00599 [CP2442].

(c) Lubricate the threads and pressure faces of the screw [41], screws [42], screw [43], screws [44], and bolts [45] with graphite mineral vaseline, D00601 [CP2101].

(d) If it is necessary, lubricate the threads and pressure faces of the screws [79] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the segment shroud 2 [32] with the VBV duct 2 [46], VBV 2 seal [47], VBV duct 3 [48], and VBV 3 seal [49] on the fan frame hub [12].

NOTE: The VBV 2 seal and VBV 3 seal can be orange or gray.

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(f) Install the bolts [45] that attach the segment shroud 2 [32] to the retainer ring [10].

(g) Install the screws [44] that attach the segment shroud 2 [32] to the junction shroud fitting [68].

(h) Install the screw [43] that attaches the segment shroud 2 [32] to the junction shroud fitting [68].

(i) Install the screws [42] that attach the segment shroud 2 [32] to the junction shroud fitting [53].

(j) Install the screw [41] that attaches the segment shroud 2 [32] to the junction shroud fitting [53].

(k) Tighten the screws [42] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(l) Tighten the screws [44] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(m) Tighten the screw [41] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(n) Tighten the screw [43] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(o) Tighten the bolts [45] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(p) Tighten the screws [79] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(q) Install the screws [40] that attach the bracket [39] to the segment shroud 2 [32].

(r) Tighten the screws [40] to the 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(s) Install the screws [38] that attach the bracket [37] to the segment shroud 2 [32].

(t) Tighten the screws [38] to the 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(u) Install the screws [36] that attach the bracket [33] to the segment shroud 2 [32].

(v) Tighten the screws [36] to the 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(w) Install the clamp [34] to the bracket [33].

(x) Install the bolt [35] that attaches the clamp [34] to the bracket [33].

(y) Tighten the bolt [35] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

PTF ALL

(z) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], on screw [41], screws [42], screw [43], screws [44], screws [79], and on bolts [45] with a lint-free cloth, G51328 [CP2674].

SUBTASK 72-23-04-420-004-G00

CAUTION IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(5) Install the segment shroud 4 [18].

(a) Remove the protective covers from the fan frame hub [12].

(b) Lubricate the threads and pressure faces of the screw [22] and screws [23] with graphite mineral vaseline, D00601 [CP2101].

(c) Lubricate the threads and pressure faces of the screw [25], screws [26], and bolts [27] with graphite mineral vaseline, D00601 [CP2101].

(d) If it is necessary, lubricate the threads and pressure faces of the screws [78] with graphite mineral vaseline, D00601 [CP2101].

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(e) If the sealant is partially or totally missing, apply the Dapco 2100 sealant, A50483 [CP2927], to these areas (Views X and W, Figure 401):

1) The forward edge of the strut 5 pad (right).

a) Apply firewall sealant at the strut panel.

2) The segment shroud 4 [18].

a) Apply firewall sealant at the shroud segment edge.

(f) Install the segment shroud 4 [18] with the VBV duct 5 [28], VBV 5 seal [29], VBV duct 6 [30], and VBV 6 seal [31] on the fan frame hub [12].

NOTE: The VBV 5 seal and VBV 6 seal can be orange or gray.

(g) Install the bolts [27] that attach the segment shroud 4 [18] to the retainer ring [10].

(h) If the segment shroud 5 [1] is still in place, install the screws [23] that attach the segment shroud 4 [18] to the junction shroud fitting [21].

(i) If the segment shroud 5 [1] is still in place, install the screw [22] that attaches the segment shroud 4 [18] to the junction shroud fitting [21].

(j) Install the screws [26] that attach the segment shroud 4 [18] to the strut 5 pad (left) [24].

(k) Install the screw [25] that attaches the segment shroud 4 [18] to the strut 5 pad (left) [24].

(l) Tighten the screws [23] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(m) Tighten the screws [26] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(n) Tighten the screw [22] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(o) Tighten the screw [25] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(p) Tighten the bolts [27] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(q) Tighten the screws [78] on the VBV duct 6 [30] and VBV duct 5 [28] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(r) Clamp the harness [82] to the bracket [19].

(s) Clamp the harness [84] to the bracket [83].

(t) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], on screw [22], screws [23], screw [25], screws [26], bolts [27], and on screws [78] with a lint-free cloth, G51328 [CP2674].

SUBTASK 72-23-04-420-005-G00

CAUTION

IF YOU USE A SCREW-MACHINE, DO NOT USE IT AT MORE THAN 200 RPM. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SEGMENT SHROUD CAN OCCUR.

(6) Install the segment shroud 5 [1].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Remove the protective covers from the duct WYE [2] and fan frame hub [12].

(b) Lubricate the threads of the bolts [4] with engine oil, D00599 [CP2442].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(c) Remove the protective covers from the duct WYE [2], diffuser plate [93], and fan frame hub [12].

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378 (Continued)

(d) Lubricate the threads of the bolts [4] and bolts [92] with engine oil, D00599 [CP2442].

PTF ALL

(e) If the segment shroud 4 [18] is still in place, lubricate the threads and pressure faces of the screw [5] and screws [6] with graphite mineral vaseline, D00601 [CP2101].

(f) Lubricate the threads and pressure faces of the screw [8], screws [9], and bolts [11] with graphite mineral vaseline, D00601 [CP2101].

(g) If it is necessary, lubricate the threads and pressure faces of the screws [77] with graphite mineral vaseline, D00601 [CP2101].

(h) Install the segment shroud 5 [1] with the inlet duct [13], VBV duct 7 [14], VBV 7 seal [15], VBV duct 8 [16], and VBV 8 seal [17] on the fan frame hub [12].

NOTE: The VBV 7 seal and VBV 8 seal can be orange or gray.

(i) Install the bolts [11] that attach the segment shroud 5 [1] to the retainer ring [10].

(j) Install the screws [9] that attach the segment shroud 5 [1] to the strut 1 pad (left) [7].

(k) Install the screw [8] that attaches the segment shroud 5 [1] to the strut 1 pad (left) [7].

(l) If the segment shroud 4 [18] is still in place, install the screws [6] that attach the segment shroud 5 [1] to the junction shroud fitting [21].

(m) If the segment shroud 4 [18] is still in place, install the screw [5] that attach the segment shroud 5 [1] to the junction shroud fitting [21].

(n) Tighten the screws [9] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(o) Tighten the screws [6] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(p) Tighten the screw [8] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(q) Tighten the screw [5] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

(r) Tighten the bolts [11] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(s) Tighten the screws [77] on the VBV duct 8 [16] and VBV duct 7 [14] to 76.0 in-lb (8.6 N·m) - 83.2 in-lb (9.4 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(t) Install the diffuser plate [93].

1) Install the two bolts [92] that attach the diffuser plate [93] to the VBV duct 7 [14].
2) Tighten the bolts [92] to 106.2 in-lb (12.0 N·m) - 123.9 in-lb (14.0 N·m).

PTF ALL

(u) Install the duct WYE [2].

1) Move the clamp [86] and clamp [88] on the duct WYE [2].
   a) Put it in position so that there is access to the head of the screw [85] and screw [87].
2) Use a back and forth motion to twist and push the duct WYE [2] aft end points on the HPTACC valve [89] and LPTACC valve [90] past the HPTACC valve and LPTACC valve bead a minimum distance of the clamp width.
3) Put the clamp [86] and clamp [88] in position on the rubber joint between the raised HPTACC valve [89] and LPTACC valve [90] bead and the duct WYE [2].

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4) Tighten the screw [85] and screw [87] to 33.63 in-lb (3.80 N·m) - 37.17 in-lb (4.20 N·m).

5) Use a back and forth motion to twist and push the duct WYE [2] forward end point on the inlet duct [13] past the inlet duct bead a minimum distance of the saddle clamps [3] width.

6) Put the saddle clamps [3] in position on the rubber joint between the raised inlet duct [13] bead and the duct WYE [2].

7) Install the bolts [4] on the saddle clamps [3] that attach the duct WYE [2] to the inlet duct [13].

8) Tighten the bolts [4] to the 52.25 in-lb (5.9 N·m) - 57.75 in-lb (6.5 N·m).

(v) Clean the work area and clear the tools and all items used to do the task.

(w) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], on screw [5], screws [6], screw [8], screws [9], screws [77], and on bolts [11] with a lint-free cloth, G51328 [CP2674].

SUBTASK 72-23-04-390-001-G00

(7) Apply the firewall sealant (Figure 402).

(a) Clean the surfaces of the aft edge of the strut pad and the shroud 1 and 5 segments.

CAUTION

MAKE SURE THAT DAPCO 2100 SEALANT, A50483 [CP2927], IS FLUSH BETWEEN ADJACENT SURFACES. THE SEALANT CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Apply Dapco 2100 sealant, A50483 [CP2927], to the aft edge of the left strut pad and the shroud 5 segment.

1) Apply parting agent to the shroud surfaces prior to applying Dapco 2100 sealant, A50483 [CP2927].

(c) Apply Dapco 2100 sealant, A50483 [CP2927], to the aft edge of the right strut pad and the shroud 1 segment.

NOTE: The firewall sealant must be applied at the strut panel and the shroud segment edge.

1) Apply parting agent to the shroud surfaces prior to applying Dapco 2100 sealant, A50483 [CP2927].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-940-001-G00

(1) Remove the protector, SPL-14947.

SUBTASK 72-23-04-080-001-G00

(2) Remove the protective covers from the segment shrouds.

SUBTASK 72-23-04-860-003-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-004-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Fan Frame Module Segment Shrouds Tests

SUBTASK 72-23-04-710-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

NOTE: Refer to Air Ducts (Non LRU items) in the Component Repaired or Replaced list of the Power Plant Test Reference Table.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME MODULE SEGMENT SHROUDS - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fan frame module segment shrouds inspection.

TASK 72-23-04-200-801-G00

2. Fan Frame Module Segment Shrouds Inspection

(Figure 601, Figure 602, Figure 603, Figure 604, Figure 605, Figure 606, Figure 607, Figure 608 or Figure 609, Figure 610, Figure 611, and Figure 612)

A. General

(1) This task gives the instructions to examine the fan frame module segment shrouds.

NOTE: The inner portion of the fan frame module segment shroud should only be inspected when the part is already removed. If the fan frame module segment shroud is not being removed, then this part of the inspection is not required at this time.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-320-801-G00	Fan Frame Shroud Segments 1-5 Blending (P/B 801)
72-23-04-960-801-G00	Fan Frame Shroud Fittings Seal - Replacement (P/B 801)
72-23-04-960-802-G00	Retainer Ring Nut - Replacement (P/B 801)
72-23-04-960-803-G00	Fan Frame Shroud Segment Coating Replacement (P/B 801)
72-23-04-960-805-G00	Shroud Junction Fitting Nut - Replacement (P/B 801)
72-23-04-960-806-G00	Shroud Junction Fitting - Nut Plate Two Lugs - Replacement (P/B 801)
72-23-04-960-807-G00	Retainer Ring Riveted Nuts - Replacement (P/B 801)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-23-04-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

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(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

G. Fan Frame Module Segment Shrouds Inspection

SUBTASK 72-23-04-210-002-G00

(1) Examine the segment 1 shroud [1] (Figure 601 and View A, Figure 612):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

(b) Examine all areas except the area 4 for these conditions:

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(c) Examine the area 1 for these conditions (View D, Figure 601):

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(d) Examine the area 2 for these conditions (Views D and E, Figure 601):

1) Dents

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: In the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(e) Examine the area 3 for these conditions (Views A and C, Figure 601):

1) Fretting, erosion, or wear

a) Any amount up to 0.0007 in. (0.02 mm) in depth.

(f) Examine the area 4 for these conditions (View B, Figure 601):

1) Missing the PTFE (protective coating)

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a) Permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Segment Coating Replacement, TASK 72-23-04-960-803-G00.

2) Nicks, dents, scratches, or wear on the PTFE

a) Any number if the base metal is not impacted.

3) Dents on the base metal

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

4) Nicks and scratches on the base metal

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

5) Fretting on the inner V-groove surface.

a) Permitted up to 0.02 in. (0.5 mm) in depth.

(g) Examine the screws [11] and screws [12] of the segment 1 shroud [1] (View B, Figure 612).

1) Surface oxidation on screw head

a) Permitted.

SUBTASK 72-23-04-210-003-G00

(2) Examine the segment 2 shroud [2] (Figure 602, and View A, Figure 612):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

(b) Examine all areas except the area 4 for these conditions:

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

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<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(c) Examine the area 1 for these conditions (Views D and E, Figure 602):

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(d) Examine the area 2 for these conditions (Views D and E, Figure 602):

1) Dents

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(e) Examine the area 3 for these conditions (Views A and C, Figure 602):

1) Wear

a) Any amount up to 0.0007 in. (0.02 mm) in depth.

(f) Examine the area 4 for these conditions (View B, Figure 602):

1) Missing the PTFE (protective coating)

a) Permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Segment Coating Replacement, TASK 72-23-04-960-803-G00.

2) Nicks, dents, scratches, or wear on the PTFE

a) Any number if the base metal is not impacted.

3) Dents on the base metal

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

4) Nicks and scratches on the base metal

a) Not permitted.

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b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

5) Fretting on the inner V-groove surface.

a) Permitted up to 0.02 in. (0.5 mm) in depth.

(g) Examine the screws [11] and screws [12] of the segment 2 shroud [2] (View B, Figure 612).

1) Surface oxidation on screw head

a) Permitted.

SUBTASK 72-23-04-210-004-G00

(3) Examine the segment 3 shroud [3] (Figure 603 and View A, Figure 612):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

(b) Examine all areas except the area 4 for these conditions:

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(c) Examine the area 1 for these conditions (Views D and E, Figure 603):

1) Dents

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(d) Examine the area 2 for these conditions (Views D and E, Figure 603):

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

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<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(e) Examine the area 3 for these conditions (Views A and C, Figure 603):

1) Wear

a) Any amount up to 0.0007 in. (0.02 mm) in depth.

(f) Examine the area 4 for these conditions (View B, Figure 603):

1) Missing the PTFE (protective coating)

a) Permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Segment Coating Replacement, TASK 72-23-04-960-803-G00.

2) Nicks, dents, scratch, or wear on the PTFE

a) Any number if the base metal is not impacted.

3) Dents on the base metal

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

4) Nicks and scratches on the base metal

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

5) Fretting on the inner V-groove surface.

a) Permitted up to 0.02 in. (0.5 mm) in depth.

(g) Examine the screws [11] and screws [12] of the segment 3 shroud [3] (View B, Figure 612).

1) Surface oxidation on screw head

a) Permitted.

SUBTASK 72-23-04-210-005-G00

(4) Examine the segment 4 shroud [4] (Figure 604 and View A, Figure 612):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

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(b) Examine all areas except the area 4 for these conditions:

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(c) Examine the area 1 for these conditions (View D, Figure 604):

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(d) Examine the area 2 for these conditions (Views D and E, Figure 604):

1) Dents

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(e) Examine the area 3 for these conditions (Views A and C, Figure 604):

1) Wear

a) Any amount up to 0.0007 in. (0.02 mm) in depth.

(f) Examine the area 4 for these conditions (View B, Figure 604):

1) Missing the PTFE (protective coating)

a) Permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Segment Coating Replacement, TASK 72-23-04-960-803-G00.

2) Nicks, dents, scratch, or wear on the PTFE

a) Any number if the base metal is not impacted.

3) Dents on the base metal

a) Any number up to 0.023 in. (0.6 mm) in depth.

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b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

4) Nicks and scratches on the base metal

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

5) Fretting on the inner V-groove surface.

a) Permitted up to 0.02 in. (0.5 mm) in depth.

(g) Examine the screws [11] and screws [12] of the segment 4 shroud [4] (View A, Figure 612).

1) Surface oxidation on screw head

a) Permitted.

SUBTASK 72-23-04-211-001-G00

(5) Examine the segment 5 shroud [5] (Figure 605 and View A, Figure 612):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

(b) Examine all areas except the area 4 for these conditions:

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(c) Examine the area 1 for these conditions (Views D and E, Figure 605):

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

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<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(d) Examine the area 2 for these conditions (Views D and E, Figure 605):

1) Dents

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

(e) Examine the area 3 for these conditions (Views A and C, Figure 605):

1) Wear

a) Any amount up to 0.0007 in. (0.02 mm) in depth.

(f) Examine the area 4 for these conditions (View B, Figure 605):

1) Missing the PTFE (protective coating)

a) Permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Segment Coating Replacement, TASK 72-23-04-960-803-G00.

2) Nicks, dents, scratch, or wear on the PTFE

a) Any number if the base metal is not impacted.

3) Dents on the base metal

a) Any number up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

4) Nicks and scratches on the base metal

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair, if the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm).

c) Repair method.

<1> Do this task: Fan Frame Shroud Segments 1-5 Blending, TASK 72-23-04-320-801-G00.

5) Fretting on the inner V-groove surface.

a) Permitted up to 0.02 in. (0.5 mm) in depth.

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(g) Examine the screws [11] and screws [12] of the segment 5 shroud [5] (View A, Figure 612).

1) Surface oxidation on screw head

a) Permitted.

SUBTASK 72-23-04-210-006-G00

(6) Examine the VBV grids [6] (Figure 606):

(a) Examine all areas for these conditions:

1) Missing.

a) Not permitted.

b) Continue-In-Service limit: Ten flight cycles are permitted with one missing grid and the VBV cavity is free of foreign objects before the first aircraft release of the day.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

2) Cracks

a) Not permitted.

3) Dents

a) Not permitted.

4) Nicks and scratches

a) Not permitted.

5) Missing sulfuric anodization oxidation

a) Not permitted.

(b) Examine the area 3 for these conditions:

1) Wear

a) Any amount up to 0.0019 in. (0.05 mm) in depth.

SUBTASK 72-23-04-210-007-G00

(7) Examine the VBV gaskets [7] (Figure 607):

(a) Examine all areas for these conditions:

1) Wear, textile delamination, or color change

a) Not permitted.

2) Break, cracks, or scratches

a) Not permitted.

3) Bubbles

a) Not permitted.

4) Permanent strain or excessive strain.

a) Not permitted.

(b) Examine the area B (seal rubber pins) for these conditions:

1) Missing

a) Any number, but not two or more consecutive pins.

2) Looseness

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a) Not permitted.

SUBTASK 72-23-04-210-008-G00

(8) Examine the retainer ring [8] (Figure 608 or Figure 609):

(a) Examine all areas for these conditions:

1) Fretting, erosion, or wear

a) Permitted if not more than 0.0007 in. (0.02 mm) in depth or any amount if dim A is more than 0.134 in. (3.4 mm).

2) Cracks

a) Not permitted.

3) Dents

a) Not permitted.

4) Nicks and scratches

a) Not permitted.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0374

(b) Examine the replaceable nuts for these conditions:

1) Missing

a) Not permitted.

b) Repair method.

<1> Install the nut (TASK 72-23-04-960-802-G00).

2) Looseness

a) Not permitted.

b) Repair method.

<1> Do this task: Retainer Ring Nut - Replacement, TASK 72-23-04-960-802-G00.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0374

(c) Examine the replaceable nuts for these conditions:

1) Missing

a) Not permitted.

b) Repair method.

<1> Install the nut (TASK 72-23-04-960-807-G00).

2) Looseness

a) Not permitted.

b) Repair method.

<1> Do this task: Retainer Ring Riveted Nuts - Replacement, TASK 72-23-04-960-807-G00.

PTF ALL

SUBTASK 72-23-04-210-009-G00

(9) Examine the shroud junction fitting [9] (Figure 610):

(a) Examine all areas for these conditions:

1) Cracks

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a) Not permitted.

(b) Examine the area 1 for these conditions (View A, Figure 610):

1) Dents

a) Permitted if not more than 0.023 in. (0.6 mm) in depth.

(c) Examine the area 2 for these conditions (View A, Figure 610):

1) Dents

a) Not permitted.

(d) Examine the area 3 for these conditions (View D, Figure 610):

1) Nicks and scratches

a) Permitted if not more than 0.023 in. (0.6 mm) in depth.

(e) Examine the area 4 for these conditions (View D, Figure 610):

1) Nicks and scratches

a) Not permitted.

(f) Examine the area 5 for these conditions (View C, Figure 610):

1) Fretting, erosion, or wear

a) Any amount if dim B is more than 0.0666 in. (1.69 mm).

(g) Examine the replaceable nuts for these conditions (View A, Figure 610):

1) Missing

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Shroud Junction Fitting Nut - Replacement, TASK 72-23-04-960-805-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Shroud Junction Fitting Nut - Replacement, TASK 72-23-04-960-805-G00.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0373

(h) Examine the replaceable nuts for these conditions (View A, Figure 610):

1) Missing

a) Not serviceable.

b) Maximum repairable limit: any number.

c) To repair, do this task: Shroud Junction Fitting - Nut Plate Two Lugs - Replacement, TASK 72-23-04-960-806-G00.

2) Looseness

a) Not serviceable.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Shroud Junction Fitting - Nut Plate Two Lugs - Replacement, TASK 72-23-04-960-806-G00.

3) Bent, worn, torn

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0373 (Continued)

a) Not serviceable.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Shroud Junction Fitting - Nut Plate Two Lugs - Replacement, TASK 72-23-04-960-806-G00.

PTF ALL

(i) Examine the area 6 (seal) for these conditions (View E, Figure 610):

1) Missing

a) Not permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

3) Break, cracks, scratches, or color change

a) Permitted if not more than 1.18 in. (30 mm) x 0.19 in. (5 mm).

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

4) Bubbles

a) Not permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

5) Permanent strain or excessive strain.

a) Not permitted.

b) Maximum repairable limit: any amount.

c) Repair method.

<1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

SUBTASK 72-23-04-210-010-G00

(10) Examine the shroud junction fitting [10] (Figure 611):

(a) Examine all areas for these conditions:

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1) Cracks
   a) Not permitted.

(b) Examine the area 1 for these conditions (View A, Figure 611):
1) Dents
   a) Permitted if not more than 0.023 in. (0.6 mm) in depth.

(c) Examine the area 2 for these conditions (View A, Figure 611):
1) Dents
   a) Not permitted.

(d) Examine the area 3 for these conditions (View D, Figure 611):
1) Nicks and scratches
   a) Permitted if not more than 0.023 in. (0.6 mm) in depth.

(e) Examine the area 4 for these conditions (View D, Figure 611):
1) Nicks and scratches
   a) Not permitted.

(f) Examine the area 5 for these conditions (View C, Figure 611):
1) Fretting, erosion, or wear
   a) Any amount if dim C is more than 0.0666 in. (1.69 mm).

(g) Examine the replaceable nuts for these conditions (View D, Figure 611):
1) Missing
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Shroud Junction Fitting Nut - Replacement, TASK 72-23-04-960-805-G00.
2) Looseness
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Shroud Junction Fitting Nut - Replacement, TASK 72-23-04-960-805-G00.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0373

(h) Examine the replaceable nuts for these conditions (View D, Figure 611):
1) Missing
   a) Not serviceable.
   b) Maximum repairable limit: any number.
   c) To repair, do this task: Shroud Junction Fitting - Nut Plate Two Lugs - Replacement, TASK 72-23-04-960-806-G00.
2) Looseness
   a) Not serviceable.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Shroud Junction Fitting - Nut Plate Two Lugs - Replacement, TASK 72-23-04-960-806-G00.

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0373 (Continued)

3) Bent, worn, torn
   a) Not serviceable.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Shroud Junction Fitting - Nut Plate Two Lugs - Replacement, TASK 72-23-04-960-806-G00.

PTF ALL

(i) Examine the area 6 (seal) for these conditions (View E, Figure 611):

1) Missing
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) Repair method.
      <1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

2) Looseness
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) Repair method.
      <1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

3) Break, cracks, scratches, or color change
   a) Permitted if not more than 1.18 in. (30 mm) x 0.19 in. (5 mm).
   b) Maximum repairable limit: any amount.
   c) Repair method.
      <1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

4) Bubbles
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) Repair method.
      <1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

5) Permanent strain or excessive strain.
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) Repair method.
      <1> Do this task: Fan Frame Shroud Fittings Seal - Replacement, TASK 72-23-04-960-801-G00.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-006-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

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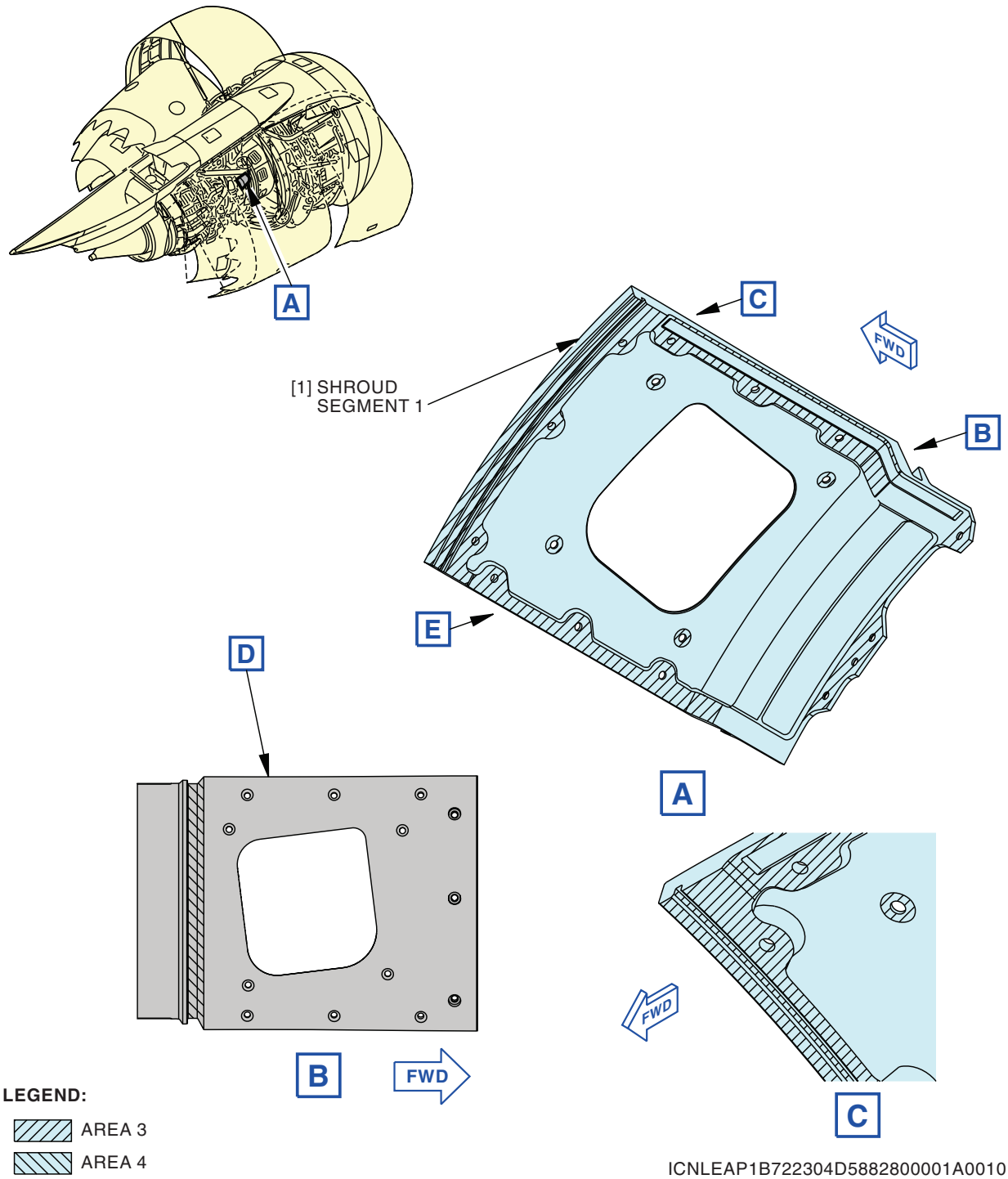

Fan Frame Module Segment 1 Shroud Inspection
Figure 601/72-23-04-990-803-G00 (Sheet 1 of 2)

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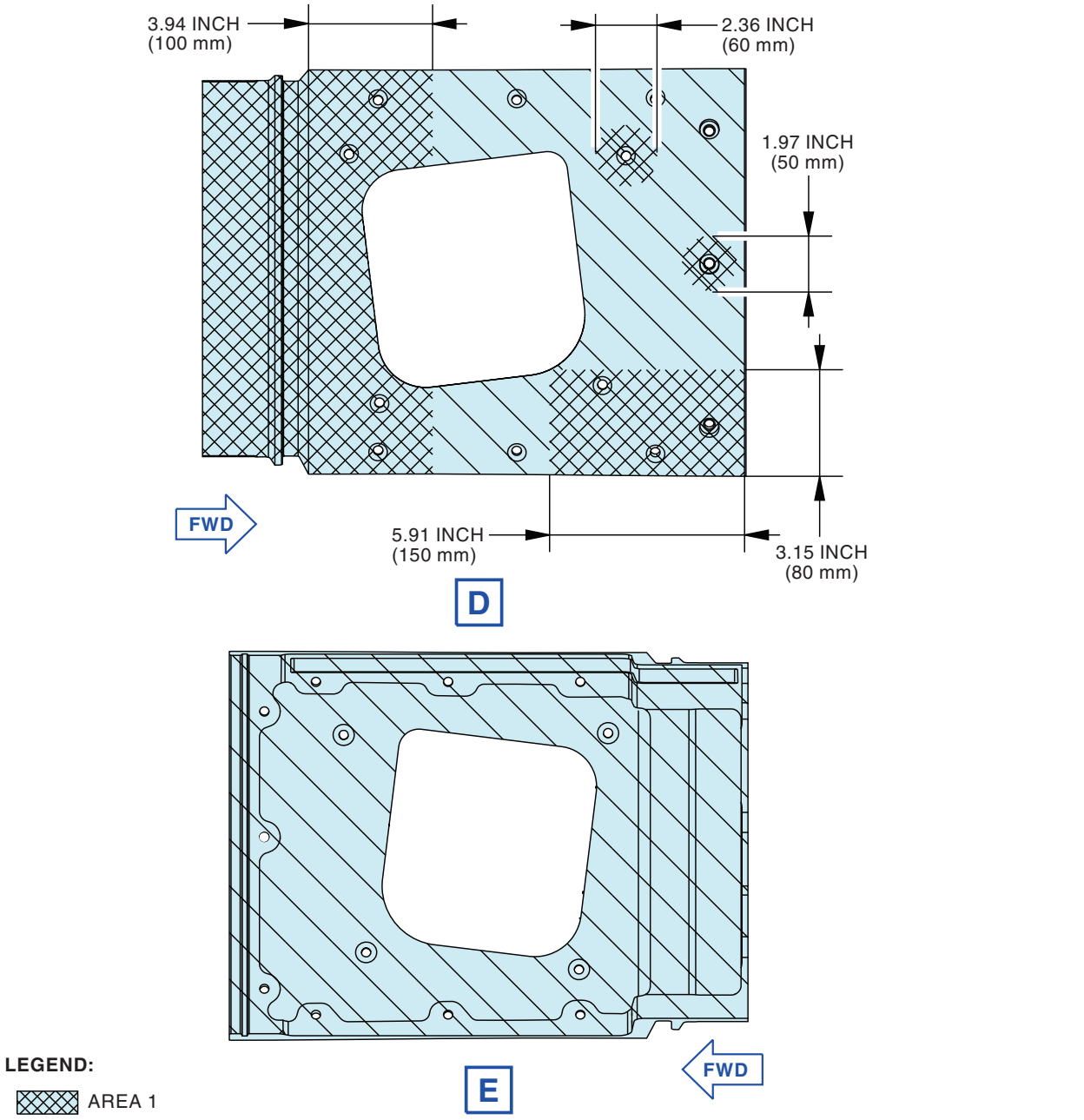

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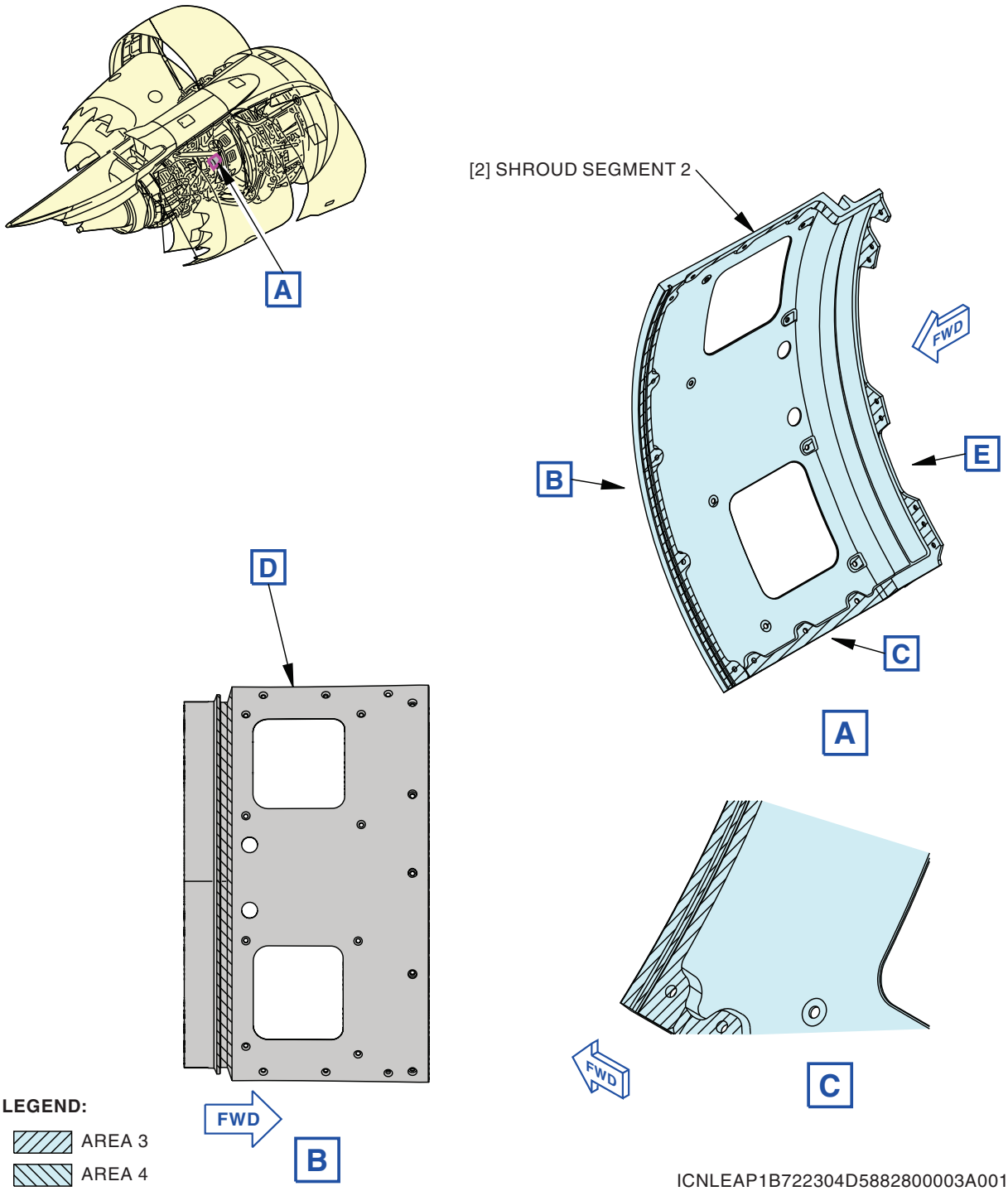
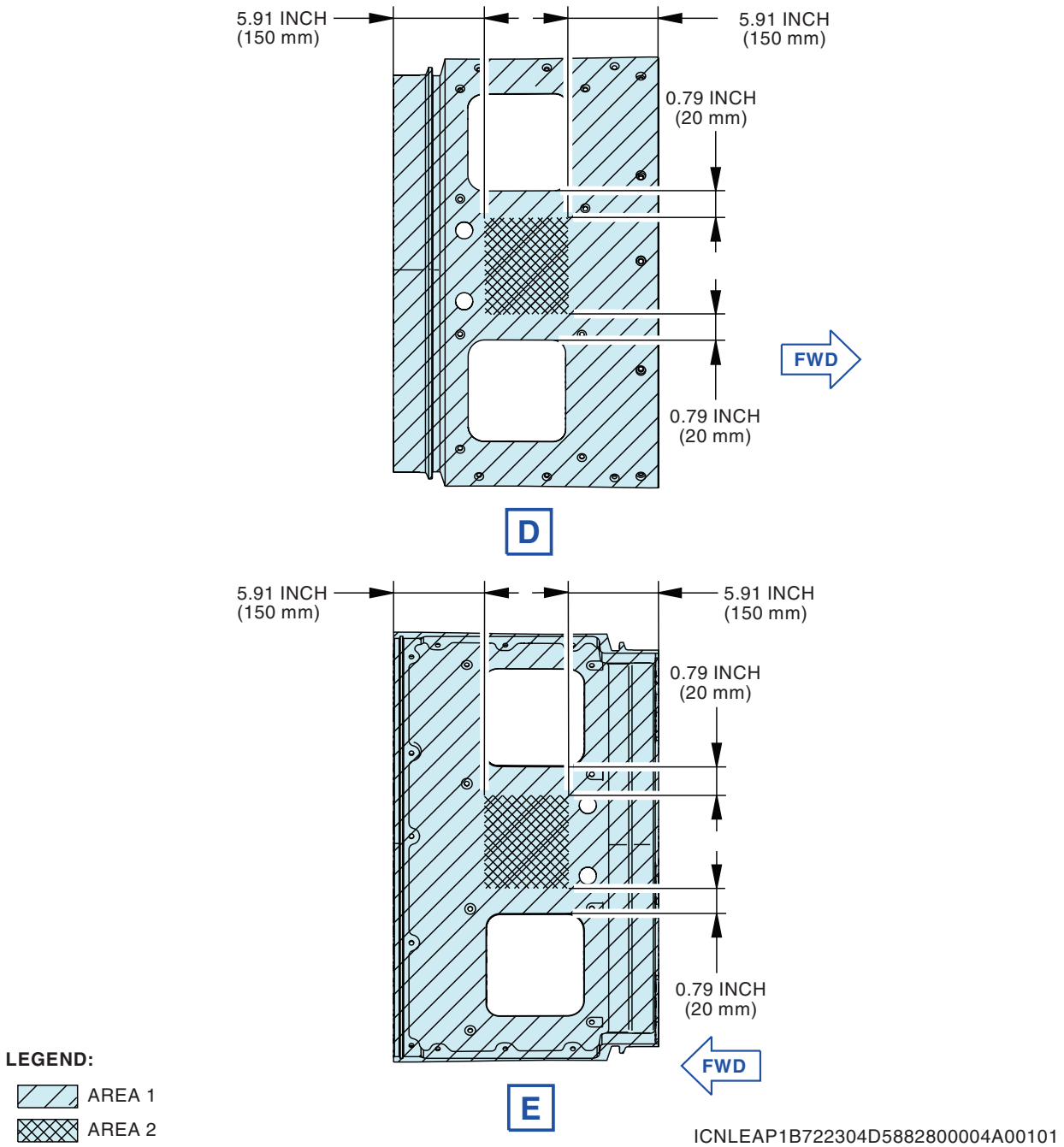
Fan Frame Module Segment 1 Shroud Inspection
Figure 601/72-23-04-990-803-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



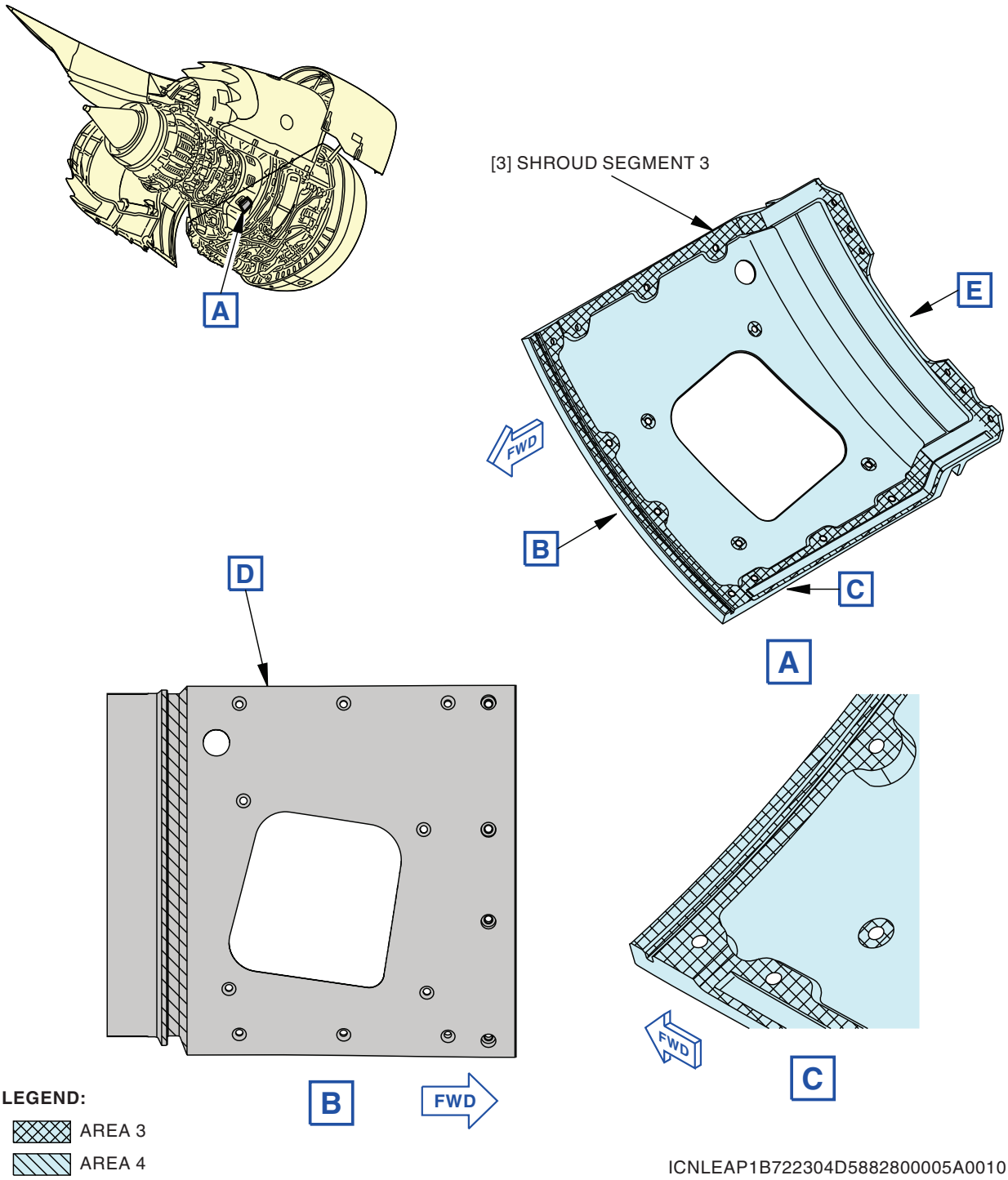
Fan Frame Module Segment 2 Shroud Inspection
Figure 602/72-23-04-990-817-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Module Segment 2 Shroud Inspection
Figure 602/72-23-04-990-817-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Frame Module Segment 3 Shroud Inspection
Figure 603/72-23-04-990-818-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details